T3 candidate UV and EFT1 scalar RGEs

Provenance: saved 16-model operator registry and candidate one-loop RGBeta UV/EFT1 exports. Source beta expressions are algebraically parsed from saved InputForm and typeset in paper coupling notation. No implicit Matchete rescaling. **Status:** candidate tensor basis. General scalar RGBeta–Matchete mapping, quadratic mass-basis matching and full sequential Wilson matching are not certified. *Direct transcription of generated candidate one-loop RGBeta exports; no new symbolic RGE computation is performed.*

Conventions and completeness

All expressions below are parsed from saved native RGBeta InputForm and typeset in mathematical notation without changing field convention or polynomial normalisation:

$$16\pi^2 \frac{dX}{d\ln\mu} = \beta_X^{(1)}.$$

UV: SM+F+S_1+S_2. **EFT1**: fermion integrated out, both scalars retained. These are the saved candidate renormalisable beta functions; no Weinberg or C_{12} Wilson evolution is inferred from them. **Scope**: all scalar quartics, scalar mass squares and holomorphic bilinears, including the original λ_5.

Scalar coupling notation

The original topology interaction remains λ_5. Original scalar and portal interactions retain the notation of the earlier analytical reports. Additional quartic structures use λ_7–λ_{12} where these symbols already exist, followed by $\lambda_{13}, \lambda_{14}, \ldots$ for newly named operators. All labels refer to RGBeta potential coefficients in the saved source convention; the Matchete–RGBeta mapping is not fully certified.

Symbol	Scalar field structure	Models
λ_7	$\left(H^\dagger\right)^2 \left(S_1^\dagger\right)^2$	B-1, C-1
λ_8	$\left(H^\dagger\right)^2 S_2^2$	B-1, C-1
λ_9	$\left(S_1^\dagger\right)^2 \left(S_2^\dagger\right)^2$	B-1, C-1
λ_{10}	$S_1^\dagger S_2 \left(S_2^\dagger\right)^2$	B-1, C-1
λ_{11}	$S_1 \left(S_1^\dagger\right)^2 S_2^\dagger$	B-1, C-1
λ_{12}	$H\, H^\dagger\, S_1^\dagger\, S_2^\dagger$	B-1, C-1
λ_{13}	$\left(S_2^\dagger\right)^4$	A-2, D-2, E-2
λ_{14}	$S_2 \left(S_2^\dagger\right)^3$	A-2, D-2, E-2
λ_{15}	$\left(S_1^\dagger\right)^2 S_2\, S_2^\dagger$	A+0, D+0, E+0
λ_{16}	$\left(S_1^\dagger\right)^2 S_2\, S_2^\dagger$	E+0
λ_{17}	$\left(S_1^\dagger\right)^4$	A+0, D+0, E+0
λ_{18}	$S_1\, S_1^\dagger \left(S_2^\dagger\right)^2$	A-2, D-2, E-2
λ_{19}	$S_1\, S_1^\dagger \left(S_2^\dagger\right)^2$	E-2
λ_{20}	$S_1 \left(S_1^\dagger\right)^3$	A+0, D+0, E+0
λ_{21}	$\left(H^\dagger\right)^2 \left(S_2^\dagger\right)^2$	B-3, C-3
λ_{22}	$\left(H^\dagger\right)^2 S_1^\dagger\, S_2^\dagger$	A-2, D-2, E-2
λ_{23}	$\left(H^\dagger\right)^2 S_1\, S_2$	A+0, D+0, E+0
λ_{24}	$\left(H^\dagger\right)^2 S_1^2$	B+1, C+1
λ_{25}	$H\, H^\dagger \left(S_2^\dagger\right)^2$	A-2, D-2, E-2
λ_{26}	$H\, H^\dagger\, S_1^\dagger\, S_2^\dagger$	B-1, C-1

Symbol	Scalar field structure	Models
λ_{27}	$H\,H^\dagger\,\left(S_1^\dagger\right)^2$	A+0, D+0, E+0

For distinct SU(2) contractions with the same field content, numbered superscripts/index labels remain independent. These symbols denote model-dependent invariant tensors, not universal Clebsch coefficients.

UV candidate one-loop beta functions

T3 − A ($\alpha = -4$)

$$\begin{aligned}\beta^{(1)}_{m_1^2} =& -24 g_1^2 m_1^2 \\ &+ 6\lambda^{(1)}_{12} m_2^2 \\ &+ m_1^2 \left(4\lambda^{(1)}_{S_1} + 4\,\mathrm{Tr}\left[y_1\ (y_1^*)^T\right]\right) \\ &- 8\,\mathrm{Tr}\left[M_F\ (M_F^*)^T\ (y_1)^T\ y_1^*\right]\end{aligned}$$

$$\begin{aligned}\beta^{(1)}_{m_2^2} =& -12 g_2^2 m_2^2 \\ &- 6 g_1^2 m_2^2 \\ &+ 2\lambda^{(1)}_{12} m_1^2 \\ &+ m_2^2 \left(8\lambda^{(1)}_{S_2} + 4\lambda^{(A)}_{S_2} + 2\,\mathrm{Tr}\left[y_2\ (y_2^*)^T\right]\right) \\ &- 4\,\mathrm{Tr}\left[M_F\ (y_2^*)^T\ y_2\ (M_F^*)^T\right]\end{aligned}$$

$$\begin{aligned}\beta^{(1)}_{\lambda_1} =& \frac{9 g_2^4}{4} \\ &- 9 g_2^2 \lambda_1 \\ &+ \frac{3 g_1^4}{4} \\ &+ g_1^2 \left(\frac{3 g_2^2}{2} - 3\lambda_1\right) \\ &+ 12\lambda_1^2 \\ &+ \lambda_1 \left(12\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\right] + 4\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\right] + 12\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\right]\right) \\ &+ 2\left(\lambda^{(1)}_{HS_1}\right)^2 \\ &+ 6\left(\lambda^{(1)}_{HS_2}\right)^2 \\ &+ 3\left(\lambda^{(A)}_{HS_2}\right)^2 \\ &+ 4\lambda_5\lambda_5^* \\ &- 12\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\ Y_d\ (Y_d^*)^T\right] \\ &- 4\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\ Y_e\ (Y_e^*)^T\right] \\ &- 12\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\ Y_u\ (Y_u^*)^T\right]\end{aligned}$$

$$\begin{aligned}\beta^{(1)}_{\lambda^{(1)}_{S_1}} =& 192 g_1^4 \\ &- 48 g_1^2 \lambda^{(1)}_{S_1} \\ &+ 6\left(\lambda^{(1)}_{12}\right)^2 \\ &+ 4\left(\lambda^{(1)}_{HS_1}\right)^2 \\ &+ 10\left(\lambda^{(1)}_{S_1}\right)^2\end{aligned}$$

$$+ \ 8\lambda_{S_1}^{(1)} \operatorname{Tr}\left[y_1 \ (y_1^*)^T\right]$$

$$- \ 8 \operatorname{Tr}\left[y_1 \ (y_1^*)^T \ y_1 \ (y_1^*)^T\right]$$

$$\beta_{\lambda_{S_2}^{(1)}}^{(1)} = 22 g_2^4$$

$$- \ 24 g_2^2 \lambda_{S_2}^{(1)}$$

$$+ \ 12 g_1^4$$

$$+ \ g_1^2 \left(8 g_2^2 - 12 \lambda_{S_2}^{(1)}\right)$$

$$+ \ 2 \left(\lambda_{12}^{(1)}\right)^2$$

$$+ \ 4 \left(\lambda_{HS_2}^{(1)}\right)^2$$

$$+ \ 14 \left(\lambda_{S_2}^{(1)}\right)^2$$

$$+ \ \lambda_{S_2}^{(1)} \left(8 \lambda_{S_2}^{(A)} + 4 \operatorname{Tr}\left[y_2 \ (y_2^*)^T\right]\right)$$

$$+ \ 16 \left(\lambda_{S_2}^{(A)}\right)^2$$

$$- \ \frac{8 \operatorname{Tr}\left[y_2 \ (y_2^*)^T \ y_2 \ (y_2^*)^T\right]}{3}$$

$$\beta_{\lambda_{HS_1}^{(1)}}^{(1)} = - \frac{9 g_2^2 \lambda_{HS_1}^{(1)}}{2}$$

$$+ \ 12 g_1^4$$

$$- \ \frac{51 g_1^2 \lambda_{HS_1}^{(1)}}{2}$$

$$+ \ 6 \lambda_{12}^{(1)} \lambda_{HS_2}^{(1)}$$

$$+ \ 6 \lambda_1 \lambda_{HS_1}^{(1)}$$

$$+ \ 4 \left(\lambda_{HS_1}^{(1)}\right)^2$$

$$+ \ 4 \lambda_{HS_1}^{(1)} \lambda_{S_1}^{(1)}$$

$$+ \ \lambda_{HS_1}^{(1)} \left(4 \operatorname{Tr}\left[y_1 \ (y_1^*)^T\right] + 6 \operatorname{Tr}\left[Y_d \ (Y_d^*)^T\right] + 2 \operatorname{Tr}\left[Y_e \ (Y_e^*)^T\right] + 6 \operatorname{Tr}\left[Y_u \ (Y_u^*)^T\right]\right)$$

$$+ \ 12 \lambda_5 \lambda_5^*$$

$$- \ 4 \operatorname{Tr}\left[y_1 \ (y_1^*)^T \ Y_e \ (Y_e^*)^T\right]$$

$$\beta_{\lambda_{HS_2}^{(1)}}^{(1)} = 6 g_2^4$$

$$- \ \frac{33 g_2^2 \lambda_{HS_2}^{(1)}}{2}$$

$$+ \ 3 g_1^4$$

$$- \ \frac{15 g_1^2 \lambda_{HS_2}^{(1)}}{2}$$

$$
\begin{aligned}
&+ 2\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)} \\
&+ 6\lambda_1\lambda_{HS_2}^{(1)} \\
&+ 4\left(\lambda_{HS_2}^{(1)}\right)^2 \\
&+ 8\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)} \\
&+ \lambda_{HS_2}^{(1)}\left(4\lambda_{S_2}^{(A)} + 2\,\mathrm{Tr}\left[y_2\ \left(y_2^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_d\ \left(Y_d^*\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ \left(Y_e^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ \left(Y_u^*\right)^T\right]\right) \\
&+ 6\left(\lambda_{HS_2}^{(A)}\right)^2 \\
&+ 4\lambda_5\lambda_5^* \\
&- 2\,\mathrm{Tr}\left[y_2\ \left(y_2^*\right)^T\ Y_e\ \left(Y_e^*\right)^T\right]
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{12}^{(1)}}^{(1)} =& - 12g_2^2\lambda_{12}^{(1)} \\
&+ 48g_1^4 \\
&- 30g_1^2\lambda_{12}^{(1)} \\
&+ 4\left(\lambda_{12}^{(1)}\right)^2 \\
&+ 4\lambda_{12}^{(1)}\lambda_{S_1}^{(1)} \\
&+ 8\lambda_{12}^{(1)}\lambda_{S_2}^{(1)} \\
&+ \lambda_{12}^{(1)}\left(4\lambda_{S_2}^{(A)} + 4\,\mathrm{Tr}\left[y_1\ \left(y_1^*\right)^T\right] + 2\,\mathrm{Tr}\left[y_2\ \left(y_2^*\right)^T\right]\right) \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)} \\
&+ 4\lambda_5\lambda_5^* \\
&- 4\,\mathrm{Tr}\left[y_1\ \left(y_1^*\right)^T\ y_2\ \left(y_2^*\right)^T\right]
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_5}^{(1)} =& - \frac{21g_2^2\lambda_5}{2} \\
&- \frac{33g_1^2\lambda_5}{2} \\
&+ 2\lambda_{12}^{(1)}\lambda_5 \\
&+ 2\lambda_1\lambda_5 \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_5 \\
&+ 4\lambda_{HS_2}^{(1)}\lambda_5 \\
&+ \lambda_5\left(4\lambda_{HS_2}^{(A)} + 2\,\mathrm{Tr}\left[y_1\ \left(y_1^*\right)^T\right] + \mathrm{Tr}\left[y_2\ \left(y_2^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_d\ \left(Y_d^*\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ \left(Y_e^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ \left(Y_u^*\right)^T\right]\right)
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{HS_2}^{(A)}}^{(1)} =& - \frac{33g_2^2\lambda_{HS_2}^{(A)}}{2} \\
&+ g_1^2\left(-4g_2^2 - \frac{15\lambda_{HS_2}^{(A)}}{2}\right) \\
&+ 2\lambda_1\lambda_{HS_2}^{(A)} \\
&+ 8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}
\end{aligned}
$$

$$
\begin{aligned}
&+ 2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)} \\
&+ \lambda_{HS_2}^{(A)}\left(16\lambda_{S_2}^{(A)} + 2\,\mathrm{Tr}\left[y_2\ \left(y_2^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_d\ \left(Y_d^*\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ \left(Y_e^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ \left(Y_u^*\right)^T\right]\right) \\
&+ \frac{8\lambda_5\lambda_5^*}{3} \\
&+ \frac{4\,\mathrm{Tr}\left[y_2\ \left(y_2^*\right)^T\ Y_e\ \left(Y_e^*\right)^T\right]}{3}
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda_{S_2}^{(A)}} = &- 2g_2^4 \\
&- 24g_2^2\lambda_{S_2}^{(A)} \\
&+ g_1^2\left(8g_2^2 - 12\lambda_{S_2}^{(A)}\right) \\
&+ 3\left(\lambda_{HS_2}^{(A)}\right)^2 \\
&+ 12\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)} \\
&+ 20\left(\lambda_{S_2}^{(A)}\right)^2 \\
&+ 4\lambda_{S_2}^{(A)}\,\mathrm{Tr}\left[y_2\ \left(y_2^*\right)^T\right] \\
&- \frac{2\,\mathrm{Tr}\left[y_2\ \left(y_2^*\right)^T\ y_2\ \left(y_2^*\right)^T\right]}{3}
\end{aligned}
$$

$\mathrm{T3-A}\,(\alpha = -2)$

$$
\begin{aligned}
\beta^{(1)}_{m_1^2} = &- 6g_1^2m_1^2 \\
&+ 6\lambda_{12}^{(1)}m_2^2 \\
&+ 12\lambda_{18}\left(\mu_{S_2}^2\right)^* \\
&+ m_1^2\left(4\lambda_{S_1}^{(1)} + 4\,\mathrm{Tr}\left[y_1\ \left(y_1^*\right)^T\right]\right) \\
&+ 12\mu_{S_2}^2\lambda_{18}^* \\
&- 8\,\mathrm{Tr}\left[M_F\ \left(M_F^*\right)^T\ \left(y_1\right)^T\ y_1^*\right]
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{m_2^2} = &- 12g_2^2m_2^2 \\
&+ 2\lambda_{12}^{(1)}m_1^2 \\
&+ 20\lambda_{14}\left(\mu_{S_2}^2\right)^* \\
&+ m_2^2\left(8\lambda_{S_2}^{(1)} + 4\lambda_{S_2}^{(A)} + 2\,\mathrm{Tr}\left[y_2\ \left(y_2^*\right)^T\right]\right) \\
&+ 20\mu_{S_2}^2\lambda_{14}^* \\
&- 4\,\mathrm{Tr}\left[M_F\ \left(y_2^*\right)^T\ y_2\ \left(M_F^*\right)^T\right]
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda_1} = &\frac{9g_2^4}{4} \\
&- 9g_2^2\lambda_1 \\
&+ \frac{3g_1^4}{4}
\end{aligned}
$$

$$
\begin{aligned}
&+ \; g_1^2 \left(\frac{3 g_2^2}{2} - 3 \lambda_1 \right) \\
&+ \; 12 \lambda_1^2 \\
&+ \; \lambda_1 \left(12 \, \mathrm{Tr}\Big[Y_d \; (Y_d^*)^T \Big] + 4 \, \mathrm{Tr}\Big[Y_e \; (Y_e^*)^T \Big] + 12 \, \mathrm{Tr}\Big[Y_u \; (Y_u^*)^T \Big] \right) \\
&+ \; 2 \left(\lambda_{HS_1}^{(1)} \right)^2 \\
&+ \; 6 \left(\lambda_{HS_2}^{(1)} \right)^2 \\
&+ \; 3 \left(\lambda_{HS_2}^{(A)} \right)^2 \\
&+ \; 4 \lambda_5 \lambda_5^* \\
&+ \; 4 \lambda_{22} \lambda_{22}^* \\
&+ \; 24 \lambda_{25} \lambda_{25}^* \\
&- \; 12 \, \mathrm{Tr}\Big[Y_d \; (Y_d^*)^T \; Y_d \; (Y_d^*)^T \Big] \\
&- \; 4 \, \mathrm{Tr}\Big[Y_e \; (Y_e^*)^T \; Y_e \; (Y_e^*)^T \Big] \\
&- \; 12 \, \mathrm{Tr}\Big[Y_u \; (Y_u^*)^T \; Y_u \; (Y_u^*)^T \Big]
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{S_1}^{(1)}}^{(1)} &= 12 g_1^4 \\
&- \; 12 g_1^2 \lambda_{S_1}^{(1)} \\
&+ \; 6 \left(\lambda_{12}^{(1)} \right)^2 \\
&+ \; 4 \left(\lambda_{HS_1}^{(1)} \right)^2 \\
&+ \; 10 \left(\lambda_{S_1}^{(1)} \right)^2 \\
&+ \; 8 \lambda_{S_1}^{(1)} \, \mathrm{Tr}\Big[y_1 \; (y_1^*)^T \Big] \\
&+ \; 24 \lambda_{18} \lambda_{18}^* \\
&- \; 8 \, \mathrm{Tr}\Big[y_1 \; (y_1^*)^T \; y_1 \; (y_1^*)^T \Big]
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{S_2}^{(1)}}^{(1)} &= 22 g_2^4 \\
&- \; 24 g_2^2 \lambda_{S_2}^{(1)} \\
&+ \; 2 \left(\lambda_{12}^{(1)} \right)^2 \\
&+ \; 4 \left(\lambda_{HS_2}^{(1)} \right)^2 \\
&+ \; 14 \left(\lambda_{S_2}^{(1)} \right)^2 \\
&+ \; \lambda_{S_2}^{(1)} \left(8 \lambda_{S_2}^{(A)} + 4 \, \mathrm{Tr}\Big[y_2 \; (y_2^*)^T \Big] \right) \\
&+ \; 16 \left(\lambda_{S_2}^{(A)} \right)^2 \\
&+ \; \frac{832 \lambda_{13} \lambda_{13}^*}{3}
\end{aligned}
$$

$$+ \frac{352\lambda_{14}\lambda_{14}^*}{3}$$
$$+ \frac{8\lambda_{18}\lambda_{18}^*}{3}$$
$$+ \frac{16\lambda_{25}\lambda_{25}^*}{3}$$
$$- \frac{8\,\mathrm{Tr}\left[y_2\ (y_2^*)^T\ y_2\ (y_2^*)^T\right]}{3}$$

$$\beta^{(1)}_{\lambda^{(1)}_{HS_1}} = -\frac{9g_2^2\lambda^{(1)}_{HS_1}}{2}$$
$$+ 3g_1^4$$
$$- \frac{15g_1^2\lambda^{(1)}_{HS_1}}{2}$$
$$+ 6\lambda^{(1)}_{12}\lambda^{(1)}_{HS_2}$$
$$+ 6\lambda_1\lambda^{(1)}_{HS_1}$$
$$+ 4\left(\lambda^{(1)}_{HS_1}\right)^2$$
$$+ 4\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$$
$$+ \lambda^{(1)}_{HS_1}\left(4\,\mathrm{Tr}\left[y_1\ (y_1^*)^T\right] + 6\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\right]\right)$$
$$+ 12\lambda_5\lambda_5^*$$
$$+ 12\lambda_{18}\lambda_{25}^*$$
$$+ 12\lambda_{22}\lambda_{22}^*$$
$$+ 12\lambda_{25}\lambda_{18}^*$$
$$- 4\,\mathrm{Tr}\left[y_1\ (y_1^*)^T\ Y_e\ (Y_e^*)^T\right]$$

$$\beta^{(1)}_{\lambda^{(1)}_{HS_2}} = 6g_2^4$$
$$- \frac{33g_2^2\lambda^{(1)}_{HS_2}}{2}$$
$$- \frac{3g_1^2\lambda^{(1)}_{HS_2}}{2}$$
$$+ 2\lambda^{(1)}_{12}\lambda^{(1)}_{HS_1}$$
$$+ 6\lambda_1\lambda^{(1)}_{HS_2}$$
$$+ 4\left(\lambda^{(1)}_{HS_2}\right)^2$$
$$+ 8\lambda^{(1)}_{HS_2}\lambda^{(1)}_{S_2}$$
$$+ \lambda^{(1)}_{HS_2}\left(4\lambda^{(A)}_{S_2} + 2\,\mathrm{Tr}\left[y_2\ (y_2^*)^T\right] + 6\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\right]\right)$$
$$+ 6\left(\lambda^{(A)}_{HS_2}\right)^2$$

$$
\begin{aligned}
&+ 4\lambda_5\lambda_5^* \\
&+ 20\lambda_{14}\lambda_{25}^* \\
&+ 4\lambda_{22}\lambda_{22}^* \\
&+ 4\lambda_{25}\left(5\lambda_{14}^* + 4\lambda_{25}^*\right) \\
&- 2\,\mathrm{Tr}\Big[y_2\ \left(y_2^*\right)^T\ Y_e\ \left(Y_e^*\right)^T\Big]
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda_{12}^{(1)}} =& -12g_2^2\lambda_{12}^{(1)} \\
&- 6g_1^2\lambda_{12}^{(1)} \\
&+ 4\left(\lambda_{12}^{(1)}\right)^2 \\
&+ 4\lambda_{12}^{(1)}\lambda_{S_1}^{(1)} \\
&+ 8\lambda_{12}^{(1)}\lambda_{S_2}^{(1)} \\
&+ \lambda_{12}^{(1)}\left(4\lambda_{S_2}^{(A)} + 4\,\mathrm{Tr}\Big[y_1\ \left(y_1^*\right)^T\Big] + 2\,\mathrm{Tr}\Big[y_2\ \left(y_2^*\right)^T\Big]\right) \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)} \\
&+ 4\lambda_5\lambda_5^* \\
&+ 20\lambda_{14}\lambda_{18}^* \\
&+ 4\lambda_{18}\left(5\lambda_{14}^* + 4\lambda_{18}^*\right) \\
&+ 4\lambda_{22}\lambda_{22}^* \\
&- 4\,\mathrm{Tr}\Big[y_1\ \left(y_1^*\right)^T\ y_2\ \left(y_2^*\right)^T\Big]
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda_5} =& -\frac{21g_2^2\lambda_5}{2} \\
&- \frac{9g_1^2\lambda_5}{2} \\
&+ 2\lambda_{12}^{(1)}\lambda_5 \\
&+ 2\lambda_1\lambda_5 \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_5 \\
&+ 4\lambda_{HS_2}^{(1)}\lambda_5 \\
&+ \lambda_5\left(4\lambda_{HS_2}^{(A)} + 2\,\mathrm{Tr}\Big[y_1\ \left(y_1^*\right)^T\Big] + \mathrm{Tr}\Big[y_2\ \left(y_2^*\right)^T\Big] + 6\,\mathrm{Tr}\Big[Y_d\ \left(Y_d^*\right)^T\Big] + 2\,\mathrm{Tr}\Big[Y_e\ \left(Y_e^*\right)^T\Big] + 6\,\mathrm{Tr}\Big[Y_u\ \left(Y_u^*\right)^T\Big]\right) \\
&+ 4\lambda_{18}\lambda_{22}^* \\
&+ 8\lambda_{25}\lambda_{22}^*
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda_{HS_2}^{(A)}} =& -\frac{33g_2^2\lambda_{HS_2}^{(A)}}{2} \\
&- \frac{3g_1^2\lambda_{HS_2}^{(A)}}{2} \\
&+ 2\lambda_1\lambda_{HS_2}^{(A)}
\end{aligned}
$$

$$
\begin{aligned}
&+ 8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)} \\
&+ 2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)} \\
&+ \lambda_{HS_2}^{(A)}\left(16\lambda_{S_2}^{(A)} + 2\,\mathrm{Tr}\left[y_2\ \left(y_2^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_d\ \left(Y_d^*\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ \left(Y_e^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ \left(Y_u^*\right)^T\right]\right) \\
&+ \frac{8\lambda_5\lambda_5^*}{3} \\
&- \frac{8\lambda_{22}\lambda_{22}^*}{3} \\
&+ \frac{4\,\mathrm{Tr}\left[y_2\ \left(y_2^*\right)^T\ Y_e\ \left(Y_e^*\right)^T\right]}{3}
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{S_2}^{(A)}}^{(1)} = &- 2g_2^4 \\
&- 24g_2^2\lambda_{S_2}^{(A)} \\
&+ 3\left(\lambda_{HS_2}^{(A)}\right)^2 \\
&+ 12\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)} \\
&+ 20\left(\lambda_{S_2}^{(A)}\right)^2 \\
&+ 4\lambda_{S_2}^{(A)}\,\mathrm{Tr}\left[y_2\ \left(y_2^*\right)^T\right] \\
&- \frac{224\lambda_{13}\lambda_{13}^*}{3} \\
&- \frac{44\lambda_{14}\lambda_{14}^*}{3} \\
&- \frac{4\lambda_{18}\lambda_{18}^*}{3} \\
&- \frac{8\lambda_{25}\lambda_{25}^*}{3} \\
&- \frac{2\,\mathrm{Tr}\left[y_2\ \left(y_2^*\right)^T\ y_2\ \left(y_2^*\right)^T\right]}{3}
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{13}}^{(1)} = &- 24g_2^2\lambda_{13} \\
&+ 12\lambda_{S_2}^{(1)}\lambda_{13} \\
&- 8\lambda_{S_2}^{(A)}\lambda_{13} \\
&+ 4\lambda_{13}\,\mathrm{Tr}\left[y_2\ \left(y_2^*\right)^T\right] \\
&+ 11\lambda_{14}^2 \\
&+ \lambda_{18}^2 \\
&+ 2\lambda_{25}^2
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{14}}^{(1)} = &- 24g_2^2\lambda_{14} \\
&+ 2\lambda_{12}^{(1)}\lambda_{18} \\
&+ 4\lambda_{HS_2}^{(1)}\lambda_{25} \\
&+ 22\lambda_{S_2}^{(1)}\lambda_{14} \\
&+ 88\lambda_{13}\lambda_{14}^*
\end{aligned}
$$

$$+ 4\lambda_{14}\,\mathrm{Tr}\Big[y_2\ (y_2^*)^T\Big]$$

$$\beta^{(1)}_{\lambda_{18}} = - 12g_2^2\lambda_{18}$$
$$- 6g_1^2\lambda_{18}$$
$$+ \lambda^{(1)}_{12}\,(10\lambda_{14} + 8\lambda_{18})$$
$$+ 4\lambda^{(1)}_{HS_1}\lambda_{25}$$
$$+ 4\lambda^{(1)}_{S_1}\lambda_{18}$$
$$+ 2\lambda^{(1)}_{S_2}\lambda_{18}$$
$$- 4\lambda^{(A)}_{S_2}\lambda_{18}$$
$$+ 4\lambda_5\lambda_{22}$$
$$+ 40\lambda_{13}\lambda_{18}^*$$
$$+ \lambda_{18}\left(4\,\mathrm{Tr}\Big[y_1\ (y_1^*)^T\Big] + 2\,\mathrm{Tr}\Big[y_2\ (y_2^*)^T\Big]\right)$$

$$\beta^{(1)}_{\lambda_{22}} = - \frac{21g_2^2\lambda_{22}}{2}$$
$$- \frac{9g_1^2\lambda_{22}}{2}$$
$$+ 2\lambda^{(1)}_{12}\lambda_{22}$$
$$+ 2\lambda_1\lambda_{22}$$
$$+ 4\lambda^{(1)}_{HS_1}\lambda_{22}$$
$$+ 4\lambda^{(1)}_{HS_2}\lambda_{22}$$
$$- 4\lambda^{(A)}_{HS_2}\lambda_{22}$$
$$+ 4\lambda_{18}\lambda_5^*$$
$$+ \lambda_{22}\left(2\,\mathrm{Tr}\Big[y_1\ (y_1^*)^T\Big] + \mathrm{Tr}\Big[y_2\ (y_2^*)^T\Big] + 6\,\mathrm{Tr}\Big[Y_d\ (Y_d^*)^T\Big] + 2\,\mathrm{Tr}\Big[Y_e\ (Y_e^*)^T\Big] + 6\,\mathrm{Tr}\Big[Y_u\ (Y_u^*)^T\Big]\right)$$
$$+ 8\lambda_{25}\lambda_5^*$$

$$\beta^{(1)}_{\lambda_{25}} = - \frac{33g_2^2\lambda_{25}}{2}$$
$$- \frac{3g_1^2\lambda_{25}}{2}$$
$$+ 6\lambda_1\lambda_{25}$$
$$+ 2\lambda^{(1)}_{HS_1}\lambda_{18}$$
$$+ \lambda^{(1)}_{HS_2}\,(10\lambda_{14} + 8\lambda_{25})$$
$$+ 2\lambda^{(1)}_{S_2}\lambda_{25}$$
$$- 4\lambda^{(A)}_{S_2}\lambda_{25}$$
$$+ 4\lambda_5\lambda_{22}$$
$$+ 40\lambda_{13}\lambda_{25}^*$$
$$+ \lambda_{25}\left(2\,\mathrm{Tr}\Big[y_2\ (y_2^*)^T\Big] + 6\,\mathrm{Tr}\Big[Y_d\ (Y_d^*)^T\Big] + 2\,\mathrm{Tr}\Big[Y_e\ (Y_e^*)^T\Big] + 6\,\mathrm{Tr}\Big[Y_u\ (Y_u^*)^T\Big]\right)$$

$$\beta^{(1)}_{\mu^2_{S_2}} = - 12g_2^2\mu^2_{S_2}$$
$$+ 2\lambda^{(1)}_{S_2}\mu^2_{S_2}$$

$$
\begin{aligned}
&- 4\lambda_{S_2}^{(A)}\mu_{S_2}^2 \\
&+ 40\lambda_{13}\left(\mu_{S_2}^2\right)^* \\
&+ 10\lambda_{14}m_2^2 \\
&+ 2\lambda_{18}m_1^2 \\
&+ 2\mu_{S_2}^2\,\mathrm{Tr}\Big[y_2\ \left(y_2^*\right)^T\Big]
\end{aligned}
$$

$\mathrm{T3-A}\,(\alpha = 0)$

$$
\begin{aligned}
\beta_{m_1^2}^{(1)} = {} & 6\lambda_{12}^{(1)}m_2^2 \\
&+ 12\lambda_{20}\left(\mu_{S_1}^2\right)^* \\
&+ m_1^2\left(4\lambda_{S_1}^{(1)} + 4\,\mathrm{Tr}\Big[y_1\ \left(y_1^*\right)^T\Big]\right) \\
&+ 12\mu_{S_1}^2\lambda_{20}^* \\
&- 8\,\mathrm{Tr}\Big[M_F\ \left(M_F^*\right)^T\ \left(y_1\right)^T\ y_1^*\Big]
\end{aligned}
$$

$$
\begin{aligned}
\beta_{m_2^2}^{(1)} = {} & -12g_2^2m_2^2 \\
&- 6g_1^2m_2^2 \\
&+ 2\lambda_{12}^{(1)}m_1^2 \\
&+ 4\lambda_{15}\left(\mu_{S_1}^2\right)^* \\
&+ m_2^2\left(8\lambda_{S_2}^{(1)} + 4\lambda_{S_2}^{(A)} + 2\,\mathrm{Tr}\Big[y_2\ \left(y_2^*\right)^T\Big]\right) \\
&+ 4\mu_{S_1}^2\lambda_{15}^* \\
&- 4\,\mathrm{Tr}\Big[M_F\ \left(y_2^*\right)^T\ y_2\ \left(M_F^*\right)^T\Big]
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_1}^{(1)} = {} & \frac{9g_2^4}{4} \\
&- 9g_2^2\lambda_1 \\
&+ \frac{3g_1^4}{4} \\
&+ g_1^2\left(\frac{3g_2^2}{2} - 3\lambda_1\right) \\
&+ 12\lambda_1^2 \\
&+ \lambda_1\left(12\,\mathrm{Tr}\Big[Y_d\ \left(Y_d^*\right)^T\Big] + 4\,\mathrm{Tr}\Big[Y_e\ \left(Y_e^*\right)^T\Big] + 12\,\mathrm{Tr}\Big[Y_u\ \left(Y_u^*\right)^T\Big]\right) \\
&+ 2\left(\lambda_{HS_1}^{(1)}\right)^2 \\
&+ 6\left(\lambda_{HS_2}^{(1)}\right)^2 \\
&+ 3\left(\lambda_{HS_2}^{(A)}\right)^2 \\
&+ 4\lambda_5\lambda_5^* \\
&+ 4\lambda_{23}\lambda_{23}^* \\
&+ 8\lambda_{27}\lambda_{27}^*
\end{aligned}
$$

$$
\begin{aligned}
&- \ 12\,\mathrm{Tr}\Big[Y_d\ (Y_d^*)^T\ Y_d\ (Y_d^*)^T\Big] \\
&- \ 4\,\mathrm{Tr}\Big[Y_e\ (Y_e^*)^T\ Y_e\ (Y_e^*)^T\Big] \\
&- \ 12\,\mathrm{Tr}\Big[Y_u\ (Y_u^*)^T\ Y_u\ (Y_u^*)^T\Big]
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda^{(1)}_{S_1}} &= 6\left(\lambda^{(1)}_{12}\right)^2 \\
&+ \ 4\left(\lambda^{(1)}_{HS_1}\right)^2 \\
&+ \ 10\left(\lambda^{(1)}_{S_1}\right)^2 \\
&+ \ 8\lambda^{(1)}_{S_1}\,\mathrm{Tr}\Big[y_1\ (y_1^*)^T\Big] \\
&+ \ 12\lambda_{15}\lambda_{15}^* \\
&+ \ 288\lambda_{17}\lambda_{17}^* \\
&+ \ 108\lambda_{20}\lambda_{20}^* \\
&+ \ 8\lambda_{27}\lambda_{27}^* \\
&- \ 8\,\mathrm{Tr}\Big[y_1\ (y_1^*)^T\ y_1\ (y_1^*)^T\Big]
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda^{(1)}_{S_2}} &= 22g_2^4 \\
&- \ 24g_2^2\lambda^{(1)}_{S_2} \\
&+ \ 12g_1^4 \\
&+ \ g_1^2\left(8g_2^2 - 12\lambda^{(1)}_{S_2}\right) \\
&+ \ 2\left(\lambda^{(1)}_{12}\right)^2 \\
&+ \ 4\left(\lambda^{(1)}_{HS_2}\right)^2 \\
&+ \ 14\left(\lambda^{(1)}_{S_2}\right)^2 \\
&+ \ \lambda^{(1)}_{S_2}\left(8\lambda^{(A)}_{S_2} + 4\,\mathrm{Tr}\Big[y_2\ (y_2^*)^T\Big]\right) \\
&+ \ 16\left(\lambda^{(A)}_{S_2}\right)^2 \\
&+ \ 8\lambda_{15}\lambda_{15}^* \\
&- \ \frac{8\,\mathrm{Tr}\Big[y_2\ (y_2^*)^T\ y_2\ (y_2^*)^T\Big]}{3}
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda^{(1)}_{HS_1}} &= -\ \frac{9g_2^2\lambda^{(1)}_{HS_1}}{2} \\
&- \ \frac{3g_1^2\lambda^{(1)}_{HS_1}}{2} \\
&+ \ 6\lambda^{(1)}_{12}\lambda^{(1)}_{HS_2} \\
&+ \ 6\lambda_1\lambda^{(1)}_{HS_1}
\end{aligned}
$$

$$
\begin{aligned}
&+ 4\left(\lambda_{HS_1}^{(1)}\right)^2 \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)} \\
&+ \lambda_{HS_1}^{(1)}\left(4\,\mathrm{Tr}\left[y_1\ (y_1^*)^T\right] + 6\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\right]\right) \\
&+ 12\lambda_5\lambda_5^* \\
&+ 12\lambda_{20}\lambda_{27}^* \\
&+ 12\lambda_{23}\lambda_{23}^* \\
&+ 4\lambda_{27}\left(3\lambda_{20}^* + 4\lambda_{27}^*\right) \\
&- 4\,\mathrm{Tr}\left[y_1\ (y_1^*)^T\ Y_e\ (Y_e^*)^T\right]
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda^{(1)}_{HS_2}} &= 6g_2^4 \\
&- \frac{33g_2^2\lambda_{HS_2}^{(1)}}{2} \\
&+ 3g_1^4 \\
&- \frac{15g_1^2\lambda_{HS_2}^{(1)}}{2} \\
&+ 2\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)} \\
&+ 6\lambda_1\lambda_{HS_2}^{(1)} \\
&+ 4\left(\lambda_{HS_2}^{(1)}\right)^2 \\
&+ 8\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)} \\
&+ \lambda_{HS_2}^{(1)}\left(4\lambda_{S_2}^{(A)} + 2\,\mathrm{Tr}\left[y_2\ (y_2^*)^T\right] + 6\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\right]\right) \\
&+ 6\left(\lambda_{HS_2}^{(A)}\right)^2 \\
&+ 4\lambda_5\lambda_5^* \\
&+ 4\lambda_{15}\lambda_{27}^* \\
&+ 4\lambda_{23}\lambda_{23}^* \\
&+ 4\lambda_{27}\lambda_{15}^* \\
&- 2\,\mathrm{Tr}\left[y_2\ (y_2^*)^T\ Y_e\ (Y_e^*)^T\right]
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda^{(1)}_{12}} &= -12g_2^2\lambda_{12}^{(1)} \\
&- 6g_1^2\lambda_{12}^{(1)} \\
&+ 4\left(\lambda_{12}^{(1)}\right)^2 \\
&+ 4\lambda_{12}^{(1)}\lambda_{S_1}^{(1)} \\
&+ 8\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}
\end{aligned}
$$

$$
\begin{aligned}
&+ \lambda_{12}^{(1)} \left(4\lambda_{S_2}^{(A)} + 4\,\mathrm{Tr}\Big[y_1\ (y_1^*)^T\Big] + 2\,\mathrm{Tr}\Big[y_2\ (y_2^*)^T\Big]\right) \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)} \\
&+ 4\lambda_5\lambda_5^* \\
&+ 4\lambda_{15}\left(4\lambda_{15}^* + 3\lambda_{20}^*\right) \\
&+ 12\lambda_{20}\lambda_{15}^* \\
&+ 4\lambda_{23}\lambda_{23}^* \\
&- 4\,\mathrm{Tr}\Big[y_1\ (y_1^*)^T\ y_2\ (y_2^*)^T\Big]
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_5}^{(1)} =& -\frac{21g_2^2\lambda_5}{2} \\
&- \frac{9g_1^2\lambda_5}{2} \\
&+ 2\lambda_{12}^{(1)}\lambda_5 \\
&+ 2\lambda_1\lambda_5 \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_5 \\
&+ 4\lambda_{HS_2}^{(1)}\lambda_5 \\
&+ \lambda_5\left(4\lambda_{HS_2}^{(A)} + 2\,\mathrm{Tr}\Big[y_1\ (y_1^*)^T\Big] + \mathrm{Tr}\Big[y_2\ (y_2^*)^T\Big] + 6\,\mathrm{Tr}\Big[Y_d\ (Y_d^*)^T\Big] + 2\,\mathrm{Tr}\Big[Y_e\ (Y_e^*)^T\Big] + 6\,\mathrm{Tr}\Big[Y_u\ (Y_u^*)^T\Big]\right) \\
&+ 4\left(\lambda_{15}^* + 2\lambda_{27}^*\right)\lambda_{23}^*
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{HS_2}^{(A)}}^{(1)} =& -\frac{33g_2^2\lambda_{HS_2}^{(A)}}{2} \\
&+ g_1^2\left(4g_2^2 - \frac{15\lambda_{HS_2}^{(A)}}{2}\right) \\
&+ 2\lambda_1\lambda_{HS_2}^{(A)} \\
&+ 8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)} \\
&+ 2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)} \\
&+ \lambda_{HS_2}^{(A)}\left(16\lambda_{S_2}^{(A)} + 2\,\mathrm{Tr}\Big[y_2\ (y_2^*)^T\Big] + 6\,\mathrm{Tr}\Big[Y_d\ (Y_d^*)^T\Big] + 2\,\mathrm{Tr}\Big[Y_e\ (Y_e^*)^T\Big] + 6\,\mathrm{Tr}\Big[Y_u\ (Y_u^*)^T\Big]\right) \\
&+ \frac{8\lambda_5\lambda_5^*}{3} \\
&+ \frac{8\lambda_{23}\lambda_{23}^*}{3} \\
&+ \frac{4\,\mathrm{Tr}\Big[y_2\ (y_2^*)^T\ Y_e\ (Y_e^*)^T\Big]}{3}
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{S_2}^{(A)}}^{(1)} =& -2g_2^4 \\
&- 24g_2^2\lambda_{S_2}^{(A)} \\
&+ g_1^2\left(8g_2^2 - 12\lambda_{S_2}^{(A)}\right) \\
&+ 3\left(\lambda_{HS_2}^{(A)}\right)^2
\end{aligned}
$$

$$
\begin{aligned}
&+ 12\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)} \\
&+ 20\left(\lambda_{S_2}^{(A)}\right)^2 \\
&+ 4\lambda_{S_2}^{(A)}\,\mathrm{Tr}\left[y_2\,\left(y_2^*\right)^T\right] \\
&- \frac{2\,\mathrm{Tr}\left[y_2\,\left(y_2^*\right)^T\,y_2\,\left(y_2^*\right)^T\right]}{3}
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{15}}^{(1)} =& -12g_2^2\lambda_{15} \\
&- 6g_1^2\lambda_{15} \\
&+ \lambda_{12}^{(1)}\left(8\lambda_{15} + 6\lambda_{20}\right) \\
&+ 4\lambda_{HS_2}^{(1)}\lambda_{27} \\
&+ 2\lambda_{S_1}^{(1)}\lambda_{15} \\
&+ 8\lambda_{S_2}^{(1)}\lambda_{15} \\
&+ 4\lambda_{S_2}^{(A)}\lambda_{15} \\
&+ \lambda_{15}\left(4\,\mathrm{Tr}\left[y_1\,\left(y_1^*\right)^T\right] + 2\,\mathrm{Tr}\left[y_2\,\left(y_2^*\right)^T\right]\right) \\
&+ 24\lambda_{17}\lambda_{15}^* \\
&+ 4\lambda_5^*\lambda_{23}^*
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{17}}^{(1)} =& 12\lambda_{S_1}^{(1)}\lambda_{17} \\
&+ 3\lambda_{15}^2 \\
&+ 8\lambda_{17}\,\mathrm{Tr}\left[y_1\,\left(y_1^*\right)^T\right] \\
&+ 9\lambda_{20}^2 \\
&+ 2\lambda_{27}^2
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{20}}^{(1)} =& 6\lambda_{12}^{(1)}\lambda_{15} \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_{27} \\
&+ 18\lambda_{S_1}^{(1)}\lambda_{20} \\
&+ 72\lambda_{17}\lambda_{20}^* \\
&+ 8\lambda_{20}\,\mathrm{Tr}\left[y_1\,\left(y_1^*\right)^T\right]
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{23}}^{(1)} =& -\frac{21g_2^2\lambda_{23}}{2} \\
&- \frac{9g_1^2\lambda_{23}}{2} \\
&+ 2\lambda_{12}^{(1)}\lambda_{23} \\
&+ 2\lambda_1\lambda_{23} \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_{23} \\
&+ 4\lambda_{HS_2}^{(1)}\lambda_{23} \\
&+ 4\lambda_{HS_2}^{(A)}\lambda_{23} \\
&+ \lambda_{23}\left(2\,\mathrm{Tr}\left[y_1\,\left(y_1^*\right)^T\right] + \mathrm{Tr}\left[y_2\,\left(y_2^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_d\,\left(Y_d^*\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\,\left(Y_e^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\,\left(Y_u^*\right)^T\right]\right) \\
&+ 4\left(\lambda_{15}^* + 2\lambda_{27}^*\right)\lambda_5^*
\end{aligned}
$$

$$\begin{aligned}\beta^{(1)}_{\lambda_{27}} =& -\frac{9g_2^2\lambda_{27}}{2}\\ &-\frac{3g_1^2\lambda_{27}}{2}\\ &+6\lambda_1\lambda_{27}\\ &+\lambda^{(1)}_{HS_1}\left(6\lambda_{20}+8\lambda_{27}\right)\\ &+6\lambda^{(1)}_{HS_2}\lambda_{15}\\ &+2\lambda^{(1)}_{S_1}\lambda_{27}\\ &+24\lambda_{17}\lambda^*_{27}\\ &+\lambda_{27}\left(4\,\mathrm{Tr}\left[y_1\ \left(y_1^*\right)^T\right]+6\,\mathrm{Tr}\left[Y_d\ \left(Y_d^*\right)^T\right]+2\,\mathrm{Tr}\left[Y_e\ \left(Y_e^*\right)^T\right]+6\,\mathrm{Tr}\left[Y_u\ \left(Y_u^*\right)^T\right]\right)\\ &+12\lambda^*_5\lambda^*_{23}\end{aligned}$$

$$\begin{aligned}\beta^{(1)}_{\mu^2_{S_1}} =& 2\lambda^{(1)}_{S_1}\mu^2_{S_1}\\ &+6\lambda_{15}m_2^2\\ &+24\lambda_{17}\left(\mu^2_{S_1}\right)^*\\ &+6\lambda_{20}m_1^2\\ &+4\mu^2_{S_1}\,\mathrm{Tr}\left[y_1\ \left(y_1^*\right)^T\right]\end{aligned}$$

$\mathrm{T3-B}\,(\alpha=-3)$

$$\begin{aligned}\beta^{(1)}_{m_1^2} =& -\frac{9g_2^2m_1^2}{2}\\ &-\frac{27g_1^2m_1^2}{2}\\ &+4\lambda^{(1)}_{12}m_2^2\\ &+m_1^2\left(6\lambda^{(1)}_{S_1}+2\,\mathrm{Tr}\left[y_1\ \left(y_1^*\right)^T\right]\right)\\ &-4\,\mathrm{Tr}\left[M_F\ \left(M_F^*\right)^T\ \left(y_1\right)^T\ y_1^*\right]\end{aligned}$$

$$\begin{aligned}\beta^{(1)}_{m_2^2} =& -\frac{9g_2^2m_2^2}{2}\\ &-\frac{3g_1^2m_2^2}{2}\\ &+4\lambda^{(1)}_{12}m_1^2\\ &+m_2^2\left(6\lambda^{(1)}_{S_2}+2\,\mathrm{Tr}\left[y_2\ \left(y_2^*\right)^T\right]\right)\\ &-4\,\mathrm{Tr}\left[M_F\ \left(y_2^*\right)^T\ y_2\ \left(M_F^*\right)^T\right]\end{aligned}$$

$$\begin{aligned}\beta^{(1)}_{\lambda_1} =& \frac{9g_2^4}{4}\\ &-9g_2^2\lambda_1\\ &+\frac{3g_1^4}{4}\\ &+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)\\ &+12\lambda_1^2\end{aligned}$$

$$
\begin{aligned}
&+ \lambda_1 \left(12 \operatorname{Tr}\left[Y_d \left(Y_d^*\right)^T\right] + 4 \operatorname{Tr}\left[Y_e \left(Y_e^*\right)^T\right] + 12 \operatorname{Tr}\left[Y_u \left(Y_u^*\right)^T\right]\right) \\
&+ 4 \left(\lambda_{HS_1}^{(1)}\right)^2 \\
&+ \frac{\left(\lambda_{HS_1}^{(A)}\right)^2}{4} \\
&+ 4 \left(\lambda_{HS_2}^{(1)}\right)^2 \\
&+ \frac{\left(\lambda_{HS_2}^{(A)}\right)^2}{4} \\
&+ 4 \lambda_5 \lambda_5^* \\
&+ 8 \lambda_{21} \lambda_{21}^* \\
&- 12 \operatorname{Tr}\left[Y_d \left(Y_d^*\right)^T Y_d \left(Y_d^*\right)^T\right] \\
&- 4 \operatorname{Tr}\left[Y_e \left(Y_e^*\right)^T Y_e \left(Y_e^*\right)^T\right] \\
&- 12 \operatorname{Tr}\left[Y_u \left(Y_u^*\right)^T Y_u \left(Y_u^*\right)^T\right]
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{S_1}^{(1)}}^{(1)} = &\frac{9 g_2^4}{4} \\
&- 9 g_2^2 \lambda_{S_1}^{(1)} \\
&+ \frac{243 g_1^4}{4} \\
&+ g_1^2 \left(\frac{27 g_2^2}{2} - 27 \lambda_{S_1}^{(1)}\right) \\
&+ 4 \left(\lambda_{12}^{(1)}\right)^2 \\
&+ \frac{\left(\lambda_{12}^{(A)}\right)^2}{4} \\
&+ 4 \left(\lambda_{HS_1}^{(1)}\right)^2 \\
&+ \frac{\left(\lambda_{HS_1}^{(A)}\right)^2}{4} \\
&+ 12 \left(\lambda_{S_1}^{(1)}\right)^2 \\
&+ 4 \lambda_{S_1}^{(1)} \operatorname{Tr}\left[y_1 \left(y_1^*\right)^T\right] \\
&- 4 \operatorname{Tr}\left[y_1 \left(y_1^*\right)^T y_1 \left(y_1^*\right)^T\right]
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{S_2}^{(1)}}^{(1)} = &\frac{9 g_2^4}{4} \\
&- 9 g_2^2 \lambda_{S_2}^{(1)} \\
&+ \frac{3 g_1^4}{4} \\
&+ g_1^2 \left(\frac{3 g_2^2}{2} - 3 \lambda_{S_2}^{(1)}\right) \\
&+ 4 \left(\lambda_{12}^{(1)}\right)^2
\end{aligned}
$$

$$
\begin{aligned}
&+ \frac{\left(\lambda_{12}^{(A)}\right)^2}{4} \\
&+ 4\left(\lambda_{HS_2}^{(1)}\right)^2 \\
&+ \frac{\left(\lambda_{HS_2}^{(A)}\right)^2}{4} \\
&+ 12\left(\lambda_{S_2}^{(1)}\right)^2 \\
&+ 4\lambda_{S_2}^{(1)}\,\mathrm{Tr}\Big[y_2\ (y_2^*)^T\Big] \\
&+ 8\lambda_{21}\lambda_{21}^* \\
&- 4\,\mathrm{Tr}\Big[y_2\ (y_2^*)^T\ y_2\ (y_2^*)^T\Big]
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda^{(1)}_{HS_1}} &= \frac{9g_2^4}{4} \\
&- 9g_2^2\lambda_{HS_1}^{(1)} \\
&+ \frac{27g_1^4}{4} \\
&- 15g_1^2\lambda_{HS_1}^{(1)} \\
&+ 4\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)} \\
&+ 6\lambda_1\lambda_{HS_1}^{(1)} \\
&+ 4\left(\lambda_{HS_1}^{(1)}\right)^2 \\
&+ 6\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)} \\
&+ \lambda_{HS_1}^{(1)}\left(2\,\mathrm{Tr}\Big[y_1\ (y_1^*)^T\Big] + 6\,\mathrm{Tr}\Big[Y_d\ (Y_d^*)^T\Big] + 2\,\mathrm{Tr}\Big[Y_e\ (Y_e^*)^T\Big] + 6\,\mathrm{Tr}\Big[Y_u\ (Y_u^*)^T\Big]\right) \\
&+ \frac{3\left(\lambda_{HS_1}^{(A)}\right)^2}{4} \\
&+ 6\lambda_5\lambda_5^* \\
&- 2\,\mathrm{Tr}\Big[y_1\ (y_1^*)^T\ Y_e\ (Y_e^*)^T\Big]
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda^{(1)}_{HS_2}} &= \frac{9g_2^4}{4} \\
&- 9g_2^2\lambda_{HS_2}^{(1)} \\
&+ \frac{3g_1^4}{4} \\
&- 3g_1^2\lambda_{HS_2}^{(1)} \\
&+ 4\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)} \\
&+ 6\lambda_1\lambda_{HS_2}^{(1)} \\
&+ 4\left(\lambda_{HS_2}^{(1)}\right)^2
\end{aligned}
$$

$$
\begin{aligned}
&+ 6\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)} \\
&+ \lambda_{HS_2}^{(1)}\left(2\,\mathrm{Tr}\left[y_2\ (y_2^*)^T\right] + 6\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\right]\right) \\
&+ \frac{3\left(\lambda_{HS_2}^{(A)}\right)^2}{4} \\
&+ 6\lambda_5\lambda_5^* \\
&+ 24\lambda_{21}\lambda_{21}^* \\
&- 2\,\mathrm{Tr}\left[y_2\ (y_2^*)^T\ Y_e\ (Y_e^*)^T\right]
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{12}^{(1)}}^{(1)} = &\frac{9g_2^4}{4} \\
&- 9g_2^2\lambda_{12}^{(1)} \\
&+ \frac{27g_1^4}{4} \\
&- 15g_1^2\lambda_{12}^{(1)} \\
&+ 4\left(\lambda_{12}^{(1)}\right)^2 \\
&+ 6\lambda_{12}^{(1)}\lambda_{S_1}^{(1)} \\
&+ 6\lambda_{12}^{(1)}\lambda_{S_2}^{(1)} \\
&+ \lambda_{12}^{(1)}\left(2\,\mathrm{Tr}\left[y_1\ (y_1^*)^T\right] + 2\,\mathrm{Tr}\left[y_2\ (y_2^*)^T\right]\right) \\
&+ \frac{3\left(\lambda_{12}^{(A)}\right)^2}{4} \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)} \\
&+ 3\lambda_5\lambda_5^* \\
&- 2\,\mathrm{Tr}\left[y_1\ (y_1^*)^T\ y_2\ (y_2^*)^T\right]
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_5}^{(1)} = &- 9g_2^2\lambda_5 \\
&- 9g_1^2\lambda_5 \\
&+ 2\lambda_{12}^{(1)}\lambda_5 \\
&+ 2\lambda_1\lambda_5 \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_5 \\
&+ 4\lambda_{HS_2}^{(1)}\lambda_5 \\
&+ \lambda_5\left(\frac{\lambda_{12}^{(A)}}{6} + \frac{2\lambda_{HS_1}^{(A)}}{3} + 2\lambda_{HS_2}^{(A)} + \mathrm{Tr}\left[y_1\ (y_1^*)^T\right] + \mathrm{Tr}\left[y_2\ (y_2^*)^T\right] + 6\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\right]\right)
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{HS_1}^{(A)}}^{(1)} = &- 9g_2^2\lambda_{HS_1}^{(A)} \\
&+ g_1^2\left(-18g_2^2 - 15\lambda_{HS_1}^{(A)}\right) \\
&+ \lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}
\end{aligned}
$$

$$
\begin{aligned}
&+ 2\lambda_1\lambda_{HS_1}^{(A)} \\
&+ 8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)} \\
&+ 2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)} \\
&+ \lambda_{HS_1}^{(A)}\left(2\,\mathrm{Tr}\left[y_1\ (y_1^*)^T\right] + 6\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\right]\right) \\
&+ \frac{16\lambda_5\lambda_5^*}{3} \\
&+ \frac{8\,\mathrm{Tr}\left[y_1\ (y_1^*)^T\ Y_e\ (Y_e^*)^T\right]}{3}
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda_{HS_2}^{(A)}} =& -9g_2^2\lambda_{HS_2}^{(A)} \\
&+ g_1^2\left(-6g_2^2 - 3\lambda_{HS_2}^{(A)}\right) \\
&+ \lambda_{12}^{(A)}\lambda_{HS_1}^{(A)} \\
&+ 2\lambda_1\lambda_{HS_2}^{(A)} \\
&+ 8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)} \\
&+ 2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)} \\
&+ \lambda_{HS_2}^{(A)}\left(2\,\mathrm{Tr}\left[y_2\ (y_2^*)^T\right] + 6\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\right]\right) \\
&+ 16\lambda_5\lambda_5^* \\
&- 64\lambda_{21}\lambda_{21}^* \\
&+ 8\,\mathrm{Tr}\left[y_2\ (y_2^*)^T\ Y_e\ (Y_e^*)^T\right]
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda_{12}^{(A)}} =& -9g_2^2\lambda_{12}^{(A)} \\
&+ g_1^2\left(18g_2^2 - 15\lambda_{12}^{(A)}\right) \\
&+ 8\lambda_{12}^{(1)}\lambda_{12}^{(A)} \\
&+ 2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)} \\
&+ 2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)} \\
&+ \lambda_{12}^{(A)}\left(2\,\mathrm{Tr}\left[y_1\ (y_1^*)^T\right] + 2\,\mathrm{Tr}\left[y_2\ (y_2^*)^T\right]\right) \\
&+ \lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)} \\
&+ \frac{4\lambda_5\lambda_5^*}{3} \\
&- \frac{8\,\mathrm{Tr}\left[y_1\ (y_1^*)^T\ y_2\ (y_2^*)^T\right]}{3}
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda_{21}} =& -9g_2^2\lambda_{21} \\
&- 3g_1^2\lambda_{21} \\
&+ 2\lambda_1\lambda_{21}
\end{aligned}
$$

$$
\begin{aligned}
&+ 8\lambda_{HS_2}^{(1)}\lambda_{21} \\
&- 4\lambda_{HS_2}^{(A)}\lambda_{21} \\
&+ 2\lambda_{S_2}^{(1)}\lambda_{21} \\
&+ \lambda_{21}\left(2\,\mathrm{Tr}\left[y_2\ (y_2^*)^T\right] + 6\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\right]\right)
\end{aligned}
$$

$\mathrm{T3-B}\,(\alpha=-1)$

$$
\begin{aligned}
\beta_{m_1^2}^{(1)} =& -\frac{9g_2^2m_1^2}{2} \\
&- \frac{3g_1^2m_1^2}{2} \\
&+ 4\lambda_{12}^{(1)}m_2^2 \\
&+ 6\lambda_{11}\left(\mu_{12}^2\right)^* \\
&+ m_1^2\left(6\lambda_{S_1}^{(1)} + 2\,\mathrm{Tr}\left[y_1\ (y_1^*)^T\right]\right) \\
&+ 6\mu_{12}^2\lambda_{11}^* \\
&- 8\,\mathrm{Tr}\left[M_F\,M_F^*\ (y_1)^T\ y_1^*\right] \\
&- 8\,\mathrm{Tr}\left[M_F\ (y_1^*)^T\ y_1\,M_F^*\right]
\end{aligned}
$$

$$
\begin{aligned}
\beta_{m_2^2}^{(1)} =& -\frac{9g_2^2m_2^2}{2} \\
&- \frac{3g_1^2m_2^2}{2} \\
&+ 4\lambda_{12}^{(1)}m_1^2 \\
&+ 6\lambda_{10}\left(\mu_{12}^2\right)^* \\
&+ m_2^2\left(6\lambda_{S_2}^{(1)} + 2\,\mathrm{Tr}\left[y_2\ (y_2^*)^T\right]\right) \\
&+ 6\mu_{12}^2\lambda_{10}^* \\
&- 8\,\mathrm{Tr}\left[M_F\,M_F^*\ (y_2)^T\ y_2^*\right] \\
&- 8\,\mathrm{Tr}\left[M_F\ (y_2^*)^T\ y_2\,M_F^*\right]
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_1}^{(1)} =& \frac{9g_2^4}{4} \\
&- 9g_2^2\lambda_1 \\
&+ \frac{3g_1^4}{4} \\
&+ g_1^2\left(\frac{3g_2^2}{2} - 3\lambda_1\right) \\
&+ 12\lambda_1^2 \\
&+ \lambda_1\left(12\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\right] + 4\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\right] + 12\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\right]\right) \\
&+ 4\left(\lambda_{HS_1}^{(1)}\right)^2 \\
&+ \frac{\left(\lambda_{HS_1}^{(A)}\right)^2}{4}
\end{aligned}
$$

$$
\begin{aligned}
&+ 4\left(\lambda_{HS_2}^{(1)}\right)^2 \\
&+ \frac{\left(\lambda_{HS_2}^{(A)}\right)^2}{4} \\
&+ 8\lambda_7\lambda_7^* \\
&+ 4\lambda_{12}\lambda_{12}^* \\
&+ 8\lambda_8\lambda_8^* \\
&+ 4\lambda_5\lambda_5^* \\
&+ 4\lambda_{26}\lambda_{26}^* \\
&- 12\,\mathrm{Tr}\Big[Y_d\ (Y_d^*)^T\ Y_d\ (Y_d^*)^T\Big] \\
&- 4\,\mathrm{Tr}\Big[Y_e\ (Y_e^*)^T\ Y_e\ (Y_e^*)^T\Big] \\
&- 12\,\mathrm{Tr}\Big[Y_u\ (Y_u^*)^T\ Y_u\ (Y_u^*)^T\Big]
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda_{S_1}^{(1)}} = &\frac{9g_2^4}{4} \\
&- 9g_2^2\lambda_{S_1}^{(1)} \\
&+ \frac{3g_1^4}{4} \\
&+ g_1^2\left(\frac{3g_2^2}{2} - 3\lambda_{S_1}^{(1)}\right) \\
&+ 4\left(\lambda_{12}^{(1)}\right)^2 \\
&+ \frac{\left(\lambda_{12}^{(A)}\right)^2}{4} \\
&+ 4\left(\lambda_{HS_1}^{(1)}\right)^2 \\
&+ \frac{\left(\lambda_{HS_1}^{(A)}\right)^2}{4} \\
&+ 8\lambda_7\lambda_7^* \\
&+ 12\left(\lambda_{S_1}^{(1)}\right)^2 \\
&+ 4\lambda_{S_1}^{(1)}\,\mathrm{Tr}\Big[y_1\ (y_1^*)^T\Big] \\
&+ 24\lambda_{11}\lambda_{11}^* \\
&+ 8\lambda_9\lambda_9^* \\
&- 4\,\mathrm{Tr}\Big[y_1\ (y_1^*)^T\ y_1\ (y_1^*)^T\Big]
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda_{S_2}^{(1)}} = &\frac{9g_2^4}{4} \\
&- 9g_2^2\lambda_{S_2}^{(1)} \\
&+ \frac{3g_1^4}{4}
\end{aligned}
$$

$$
\begin{aligned}
&+ g_1^2 \left(\frac{3g_2^2}{2} - 3\lambda_{S_2}^{(1)}\right) \\
&+ 4\left(\lambda_{12}^{(1)}\right)^2 \\
&+ \frac{\left(\lambda_{12}^{(A)}\right)^2}{4} \\
&+ 4\left(\lambda_{HS_2}^{(1)}\right)^2 \\
&+ \frac{\left(\lambda_{HS_2}^{(A)}\right)^2}{4} \\
&+ 8\lambda_8\lambda_8^* \\
&+ 8\lambda_9\lambda_9^* \\
&+ 24\lambda_{10}\lambda_{10}^* \\
&+ 12\left(\lambda_{S_2}^{(1)}\right)^2 \\
&+ 4\lambda_{S_2}^{(1)} \operatorname{Tr}\left[y_2\ (y_2^*)^T\right] \\
&- 4\operatorname{Tr}\left[y_2\ (y_2^*)^T\ y_2\ (y_2^*)^T\right]
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{HS_1}^{(1)}}^{(1)} &= \frac{9g_2^4}{4} \\
&- 9g_2^2\lambda_{HS_1}^{(1)} \\
&+ \frac{3g_1^4}{4} \\
&- 3g_1^2\lambda_{HS_1}^{(1)} \\
&+ 4\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)} \\
&+ 6\lambda_1\lambda_{HS_1}^{(1)} \\
&+ 4\left(\lambda_{HS_1}^{(1)}\right)^2 \\
&+ 6\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)} \\
&+ \lambda_{HS_1}^{(1)}\left(2\operatorname{Tr}\left[y_1\ (y_1^*)^T\right] + 6\operatorname{Tr}\left[Y_d\ (Y_d^*)^T\right] + 2\operatorname{Tr}\left[Y_e\ (Y_e^*)^T\right] + 6\operatorname{Tr}\left[Y_u\ (Y_u^*)^T\right]\right) \\
&+ \frac{3\left(\lambda_{HS_1}^{(A)}\right)^2}{4} \\
&+ 24\lambda_7\lambda_7^* \\
&+ \lambda_{12}\left(4\lambda_{12}^* + 3\lambda_{11}^* + 2\lambda_{26}^*\right) \\
&+ 3\lambda_{11}\left(\lambda_{12}^* - \lambda_{26}^*\right) \\
&+ 6\lambda_5\lambda_5^* \\
&+ \lambda_{26}\left(2\lambda_{12}^* - 3\lambda_{11}^* + 4\lambda_{26}^*\right)
\end{aligned}
$$

$$
\begin{aligned}
&- \ 2\,\mathrm{Tr}\Big[y_1\ (y_1^*)^T\ Y_e\ (Y_e^*)^T\Big]
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda^{(1)}_{HS_2}} &= \frac{9g_2^4}{4} \\
&- \ 9g_2^2\lambda^{(1)}_{HS_2} \\
&+ \ \frac{3g_1^4}{4} \\
&- \ 3g_1^2\lambda^{(1)}_{HS_2} \\
&+ \ 4\lambda^{(1)}_{12}\lambda^{(1)}_{HS_1} \\
&+ \ 6\lambda_1\lambda^{(1)}_{HS_2} \\
&+ \ 4\left(\lambda^{(1)}_{HS_2}\right)^2 \\
&+ \ 6\lambda^{(1)}_{HS_2}\lambda^{(1)}_{S_2} \\
&+ \ \lambda^{(1)}_{HS_2}\left(2\,\mathrm{Tr}\Big[y_2\ (y_2^*)^T\Big] + 6\,\mathrm{Tr}\Big[Y_d\ (Y_d^*)^T\Big] + 2\,\mathrm{Tr}\Big[Y_e\ (Y_e^*)^T\Big] + 6\,\mathrm{Tr}\Big[Y_u\ (Y_u^*)^T\Big]\right) \\
&+ \ \frac{3\left(\lambda^{(A)}_{HS_2}\right)^2}{4} \\
&+ \ \lambda_{12}\left(4\lambda^*_{12} + 3\lambda^*_{10} + 2\lambda^*_{26}\right) \\
&+ \ 24\lambda_8\lambda^*_8 \\
&+ \ 3\lambda_{10}\left(\lambda^*_{12} - \lambda^*_{26}\right) \\
&+ \ 6\lambda_5\lambda^*_5 \\
&+ \ \lambda_{26}\left(2\lambda^*_{12} - 3\lambda^*_{10} + 4\lambda^*_{26}\right) \\
&- \ 2\,\mathrm{Tr}\Big[y_2\ (y_2^*)^T\ Y_e\ (Y_e^*)^T\Big]
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda^{(1)}_{12}} &= \frac{9g_2^4}{4} \\
&- \ 9g_2^2\lambda^{(1)}_{12} \\
&+ \ \frac{3g_1^4}{4} \\
&- \ 3g_1^2\lambda^{(1)}_{12} \\
&+ \ 4\left(\lambda^{(1)}_{12}\right)^2 \\
&+ \ 6\lambda^{(1)}_{12}\lambda^{(1)}_{S_1} \\
&+ \ 6\lambda^{(1)}_{12}\lambda^{(1)}_{S_2} \\
&+ \ \lambda^{(1)}_{12}\left(2\,\mathrm{Tr}\Big[y_1\ (y_1^*)^T\Big] + 2\,\mathrm{Tr}\Big[y_2\ (y_2^*)^T\Big]\right) \\
&+ \ \frac{3\left(\lambda^{(A)}_{12}\right)^2}{4} \\
&+ \ 4\lambda^{(1)}_{HS_1}\lambda^{(1)}_{HS_2}
\end{aligned}
$$

$$
\begin{aligned}
&+ \lambda_{12} \left(2\lambda_{12}^* + \lambda_{26}^*\right) \\
&+ 9\lambda_{11} \left(\lambda_{11}^* + \lambda_{10}^*\right) \\
&+ 24\lambda_9\lambda_9^* \\
&+ 9\lambda_{10} \left(\lambda_{11}^* + \lambda_{10}^*\right) \\
&+ 3\lambda_5\lambda_5^* \\
&+ \lambda_{26} \left(\lambda_{12}^* + 2\lambda_{26}^*\right) \\
&- 2\,\mathrm{Tr}\left[y_1 \left(y_1^*\right)^T y_2 \left(y_2^*\right)^T\right] \\
&- 4\,\mathrm{Tr}\left[y_1 \left(y_2^*\right)^T y_2 \left(y_1^*\right)^T\right]
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_5}^{(1)} = &- 9g_2^2\lambda_5 \\
&- 3g_1^2\lambda_5 \\
&+ 2\lambda_{12}^{(1)}\lambda_5 \\
&+ 2\lambda_1\lambda_5 \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_5 \\
&+ 4\lambda_{HS_2}^{(1)}\lambda_5 \\
&+ \lambda_5 \left(\frac{\lambda_{12}^{(A)}}{6} + \frac{2\lambda_{HS_1}^{(A)}}{3} + 2\lambda_{HS_2}^{(A)} + \mathrm{Tr}\left[y_1 \left(y_1^*\right)^T\right] + \mathrm{Tr}\left[y_2 \left(y_2^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_d \left(Y_d^*\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e \left(Y_e^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u \left(Y_u^*\right)^T\right]\right) \\
&- \frac{2\lambda_7^*\,\mathrm{Tr}\left[y_2 \left(y_1^*\right)^T\right]}{3} \\
&+ 2\lambda_8^*\,\mathrm{Tr}\left[y_1 \left(y_2^*\right)^T\right]
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{HS_1}^{(A)}}^{(1)} = &- 9g_2^2\lambda_{HS_1}^{(A)} \\
&+ g_1^2 \left(-6g_2^2 - 3\lambda_{HS_1}^{(A)}\right) \\
&+ \lambda_{12}^{(A)}\lambda_{HS_2}^{(A)} \\
&+ 2\lambda_1\lambda_{HS_1}^{(A)} \\
&+ 8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)} \\
&+ 2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)} \\
&+ \lambda_{HS_1}^{(A)} \left(2\,\mathrm{Tr}\left[y_1 \left(y_1^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_d \left(Y_d^*\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e \left(Y_e^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u \left(Y_u^*\right)^T\right]\right) \\
&- 64\lambda_7\lambda_7^* \\
&+ 4\lambda_{12} \left(2\lambda_{12}^* + \lambda_{11}^*\right) \\
&+ 4\lambda_{11} \left(\lambda_{12}^* + \lambda_{26}^*\right) \\
&+ \frac{16\lambda_5\lambda_5^*}{3} \\
&+ 4\lambda_{26} \left(\lambda_{11}^* - 2\lambda_{26}^*\right)
\end{aligned}
$$

$$+ \frac{8\,\mathrm{Tr}\left[y_1\ (y_1^*)^T\ Y_e\ (Y_e^*)^T\right]}{3}$$

$$\begin{aligned}
\beta^{(1)}_{\lambda^{(A)}_{HS_2}} =& -9g_2^2\lambda^{(A)}_{HS_2} \\
&+ g_1^2\left(6g_2^2 - 3\lambda^{(A)}_{HS_2}\right) \\
&+ \lambda^{(A)}_{12}\lambda^{(A)}_{HS_1} \\
&+ 2\lambda_1\lambda^{(A)}_{HS_2} \\
&+ 8\lambda^{(1)}_{HS_2}\lambda^{(A)}_{HS_2} \\
&+ 2\lambda^{(A)}_{HS_2}\lambda^{(1)}_{S_2} \\
&+ \lambda^{(A)}_{HS_2}\left(2\,\mathrm{Tr}\left[y_2\ (y_2^*)^T\right] + 6\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\right]\right) \\
&- 4\lambda_{12}\left(2\lambda^*_{12} + \lambda^*_{10}\right) \\
&+ 64\lambda_8\lambda^*_8 \\
&- 4\lambda_{10}\left(\lambda^*_{12} + \lambda^*_{26}\right) \\
&+ 16\lambda_5\lambda^*_5 \\
&+ \lambda_{26}\left(-4\lambda^*_{10} + 8\lambda^*_{26}\right) \\
&+ 8\,\mathrm{Tr}\left[y_2\ (y_2^*)^T\ Y_e\ (Y_e^*)^T\right]
\end{aligned}$$

$$\begin{aligned}
\beta^{(1)}_{\lambda^{(A)}_{12}} =& -9g_2^2\lambda^{(A)}_{12} \\
&+ g_1^2\left(-6g_2^2 - 3\lambda^{(A)}_{12}\right) \\
&+ 8\lambda^{(1)}_{12}\lambda^{(A)}_{12} \\
&+ 2\lambda^{(A)}_{12}\lambda^{(1)}_{S_1} \\
&+ 2\lambda^{(A)}_{12}\lambda^{(1)}_{S_2} \\
&+ \lambda^{(A)}_{12}\left(2\,\mathrm{Tr}\left[y_1\ (y_1^*)^T\right] + 2\,\mathrm{Tr}\left[y_2\ (y_2^*)^T\right]\right) \\
&+ \lambda^{(A)}_{HS_1}\lambda^{(A)}_{HS_2} \\
&+ 4\lambda_{12}\lambda^*_{26} \\
&- 4\lambda_{11}\left(5\lambda^*_{11} + \lambda^*_{10}\right) \\
&- 64\lambda_9\lambda^*_9 \\
&- 4\lambda_{10}\left(\lambda^*_{11} + 5\lambda^*_{10}\right) \\
&+ \frac{4\lambda_5\lambda^*_5}{3} \\
&+ 4\lambda_{26}\lambda^*_{12} \\
&- \frac{8\,\mathrm{Tr}\left[y_1\ (y_1^*)^T\ y_2\ (y_2^*)^T\right]}{3}
\end{aligned}$$

$$\begin{aligned}
\beta^{(1)}_{\lambda_8} =& -9g_2^2\lambda_8 \\
&- 3g_1^2\lambda_8 \\
&+ 2\lambda_1\lambda_8 \\
&+ 8\lambda^{(1)}_{HS_2}\lambda_8 \\
&+ 4\lambda^{(A)}_{HS_2}\lambda_8 \\
&+ 4\lambda_7\lambda_9^* \\
&+ 2\lambda_8\lambda^{(1)}_{S_2} \\
&+ \lambda_8\left(2\,\mathrm{Tr}\left[y_2\ (y_2^*)^T\right] + 6\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\right]\right) \\
&+ \lambda_5^*\,\mathrm{Tr}\left[y_1\ (y_2^*)^T\right]
\end{aligned}$$

$$\begin{aligned}
\beta^{(1)}_{\lambda_7} =& -9g_2^2\lambda_7 \\
&- 3g_1^2\lambda_7 \\
&+ 2\lambda_1\lambda_7 \\
&+ 8\lambda^{(1)}_{HS_1}\lambda_7 \\
&- 4\lambda^{(A)}_{HS_1}\lambda_7 \\
&+ 2\lambda_7\lambda^{(1)}_{S_1} \\
&+ \lambda_7\left(2\,\mathrm{Tr}\left[y_1\ (y_1^*)^T\right] + 6\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\right]\right) \\
&+ 4\lambda_8\lambda_9 \\
&- \frac{\lambda_5^*\,\mathrm{Tr}\left[y_2\ (y_1^*)^T\right]}{3}
\end{aligned}$$

$$\begin{aligned}
\beta^{(1)}_{\lambda_9} =& -9g_2^2\lambda_9 \\
&- 3g_1^2\lambda_9 \\
&+ 8\lambda^{(1)}_{12}\lambda_9 \\
&- 4\lambda^{(A)}_{12}\lambda_9 \\
&+ 4\lambda_7\lambda_8^* \\
&+ \lambda_{12}^2 \\
&+ 2\lambda^{(1)}_{S_1}\lambda_9 \\
&+ 5\lambda_{11}^2 \\
&+ 2\lambda_{11}\lambda_{10} \\
&+ 2\lambda_9\lambda^{(1)}_{S_2} \\
&+ \lambda_9\left(2\,\mathrm{Tr}\left[y_1\ (y_1^*)^T\right] + 2\,\mathrm{Tr}\left[y_2\ (y_2^*)^T\right]\right) \\
&+ 5\lambda_{10}^2 \\
&+ \lambda_{26}^2 \\
&+ \frac{2\,\mathrm{Tr}\left[y_2\ (y_1^*)^T\ y_2\ (y_1^*)^T\right]}{3}
\end{aligned}$$

$$\begin{aligned}
\beta^{(1)}_{\lambda_{10}} =& -9g_2^2\lambda_{10} \\
&- 3g_1^2\lambda_{10}
\end{aligned}$$

$$
\begin{aligned}
&+ \lambda_{12}^{(1)} \left(6\lambda_{11} + 6\lambda_{10}\right) \\
&+ \lambda_{12}^{(A)} \left(-\frac{\lambda_{11}}{2} - \frac{5\lambda_{10}}{2}\right) \\
&+ \lambda_{HS_2}^{(1)} \left(2\lambda_{12} - 2\lambda_{26}\right) \\
&+ \lambda_{HS_2}^{(A)} \left(-\frac{\lambda_{12}}{2} - \frac{\lambda_{26}}{2}\right) \\
&+ 4\lambda_9 \left(\lambda_{11}^* + 5\lambda_{10}^*\right) \\
&+ 12\lambda_{10}\lambda_{S_2}^{(1)} \\
&+ \lambda_{10} \left(\mathrm{Tr}\left[y_1 \ (y_1^*)^T\right] + 3\,\mathrm{Tr}\left[y_2 \ (y_2^*)^T\right]\right)
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{11}}^{(1)} = &- 9g_2^2\lambda_{11} \\
&- 3g_1^2\lambda_{11} \\
&+ \lambda_{12}^{(1)} \left(6\lambda_{11} + 6\lambda_{10}\right) \\
&+ \lambda_{12}^{(A)} \left(-\frac{5\lambda_{11}}{2} - \frac{\lambda_{10}}{2}\right) \\
&+ \lambda_{HS_1}^{(1)} \left(2\lambda_{12} - 2\lambda_{26}\right) \\
&+ \lambda_{HS_1}^{(A)} \left(\frac{\lambda_{12}}{2} + \frac{\lambda_{26}}{2}\right) \\
&+ 12\lambda_{S_1}^{(1)}\lambda_{11} \\
&+ \lambda_{11} \left(3\,\mathrm{Tr}\left[y_1 \ (y_1^*)^T\right] + \mathrm{Tr}\left[y_2 \ (y_2^*)^T\right]\right) \\
&+ 4\lambda_9 \left(5\lambda_{11}^* + \lambda_{10}^*\right)
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{12}}^{(1)} = &- 9g_2^2\lambda_{12} \\
&- 3g_1^2\lambda_{12} \\
&+ 2\lambda_{12}^{(1)}\lambda_{12} \\
&+ \lambda_{12}^{(A)} \left(-\frac{\lambda_{12}}{2} + \lambda_{26}\right) \\
&+ \lambda_1 \left(4\lambda_{12} - 2\lambda_{26}\right) \\
&+ \lambda_{HS_1}^{(1)} \left(4\lambda_{12} + 6\lambda_{11}\right) \\
&+ \lambda_{HS_1}^{(A)} \left(2\lambda_{12} + \frac{\lambda_{11}}{2} + \lambda_{26}\right) \\
&+ \lambda_{HS_2}^{(1)} \left(4\lambda_{12} + 6\lambda_{10}\right) \\
&+ \lambda_{HS_2}^{(A)} \left(-2\lambda_{12} - \frac{\lambda_{10}}{2} - \lambda_{26}\right) \\
&+ \lambda_{12} \left(\mathrm{Tr}\left[y_1 \ (y_1^*)^T\right] + \mathrm{Tr}\left[y_2 \ (y_2^*)^T\right] + 6\,\mathrm{Tr}\left[Y_d \ (Y_d^*)^T\right] + 2\,\mathrm{Tr}\left[Y_e \ (Y_e^*)^T\right] + 6\,\mathrm{Tr}\left[Y_u \ (Y_u^*)^T\right]\right) \\
&+ \lambda_9 \left(8\lambda_{12}^* - 4\lambda_{26}^*\right)
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{26}}^{(1)} = &- 9g_2^2\lambda_{26} \\
&- 3g_1^2\lambda_{26} \\
&+ 2\lambda_{12}^{(1)}\lambda_{26} \\
&+ \lambda_{12}^{(A)} \left(\lambda_{12} - \frac{\lambda_{26}}{2}\right) \\
&+ \lambda_1 \left(-2\lambda_{12} + 4\lambda_{26}\right)
\end{aligned}
$$

$$
\begin{aligned}
&+ \lambda_{HS_1}^{(1)} \left(-6\lambda_{11} + 4\lambda_{26}\right) \\
&+ \lambda_{HS_1}^{(A)} \left(-\lambda_{12} + \frac{\lambda_{11}}{2} - 2\lambda_{26}\right) \\
&+ \lambda_{HS_2}^{(1)} \left(-6\lambda_{10} + 4\lambda_{26}\right) \\
&+ \lambda_{HS_2}^{(A)} \left(\lambda_{12} - \frac{\lambda_{10}}{2} + 2\lambda_{26}\right) \\
&+ \lambda_9 \left(-4\lambda_{12}^* + 8\lambda_{26}^*\right) \\
&+ \lambda_{26} \left(\mathrm{Tr}\left[y_1 \left(y_1^*\right)^T\right] + \mathrm{Tr}\left[y_2 \left(y_2^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_d \left(Y_d^*\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e \left(Y_e^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u \left(Y_u^*\right)^T\right]\right)
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\mu_{12}^2}^{(1)} = &- \frac{9g_2^2\mu_{12}^2}{2} \\
&- \frac{3g_1^2\mu_{12}^2}{2} \\
&+ 2\lambda_{12}^{(1)}\mu_{12}^2 \\
&- \frac{3\lambda_{12}^{(A)}\mu_{12}^2}{2} \\
&+ 6\lambda_{11}m_1^2 \\
&+ 12\lambda_9 \left(\mu_{12}^2\right)^* \\
&+ 6\lambda_{10}m_2^2 \\
&+ \mu_{12}^2 \left(\mathrm{Tr}\left[y_1 \left(y_1^*\right)^T\right] + \mathrm{Tr}\left[y_2 \left(y_2^*\right)^T\right]\right)
\end{aligned}
$$

$\mathrm{T3-B}\,(\alpha = 1)$

$$
\begin{aligned}
\beta_{m_1^2}^{(1)} = &- \frac{9g_2^2m_1^2}{2} \\
&- \frac{3g_1^2m_1^2}{2} \\
&+ 4\lambda_{12}^{(1)}m_2^2 \\
&+ m_1^2 \left(6\lambda_{S_1}^{(1)} + 2\,\mathrm{Tr}\left[y_1 \left(y_1^*\right)^T\right]\right) \\
&- 4\,\mathrm{Tr}\left[M_F \left(M_F^*\right)^T \left(y_1\right)^T y_1^*\right]
\end{aligned}
$$

$$
\begin{aligned}
\beta_{m_2^2}^{(1)} = &- \frac{9g_2^2m_2^2}{2} \\
&- \frac{27g_1^2m_2^2}{2} \\
&+ 4\lambda_{12}^{(1)}m_1^2 \\
&+ m_2^2 \left(6\lambda_{S_2}^{(1)} + 2\,\mathrm{Tr}\left[y_2 \left(y_2^*\right)^T\right]\right) \\
&- 4\,\mathrm{Tr}\left[M_F \left(y_2^*\right)^T y_2 \left(M_F^*\right)^T\right]
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_1}^{(1)} = &\frac{9g_2^4}{4} \\
&- 9g_2^2\lambda_1 \\
&+ \frac{3g_1^4}{4} \\
&+ g_1^2 \left(\frac{3g_2^2}{2} - 3\lambda_1\right)
\end{aligned}
$$

$$
\begin{aligned}
&+ 12\lambda_1^2 \\
&+ \lambda_1 \left(12 \operatorname{Tr}\left[Y_d \left(Y_d^*\right)^T\right] + 4 \operatorname{Tr}\left[Y_e \left(Y_e^*\right)^T\right] + 12 \operatorname{Tr}\left[Y_u \left(Y_u^*\right)^T\right]\right) \\
&+ 4 \left(\lambda_{HS_1}^{(1)}\right)^2 \\
&+ \frac{\left(\lambda_{HS_1}^{(A)}\right)^2}{4} \\
&+ 4 \left(\lambda_{HS_2}^{(1)}\right)^2 \\
&+ \frac{\left(\lambda_{HS_2}^{(A)}\right)^2}{4} \\
&+ 4\lambda_5\lambda_5^* \\
&+ 8\lambda_{24}\lambda_{24}^* \\
&- 12 \operatorname{Tr}\left[Y_d \left(Y_d^*\right)^T Y_d \left(Y_d^*\right)^T\right] \\
&- 4 \operatorname{Tr}\left[Y_e \left(Y_e^*\right)^T Y_e \left(Y_e^*\right)^T\right] \\
&- 12 \operatorname{Tr}\left[Y_u \left(Y_u^*\right)^T Y_u \left(Y_u^*\right)^T\right]
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{S_1}^{(1)}}^{(1)} = &\frac{9g_2^4}{4} \\
&- 9g_2^2\lambda_{S_1}^{(1)} \\
&+ \frac{3g_1^4}{4} \\
&+ g_1^2 \left(\frac{3g_2^2}{2} - 3\lambda_{S_1}^{(1)}\right) \\
&+ 4 \left(\lambda_{12}^{(1)}\right)^2 \\
&+ \frac{\left(\lambda_{12}^{(A)}\right)^2}{4} \\
&+ 4 \left(\lambda_{HS_1}^{(1)}\right)^2 \\
&+ \frac{\left(\lambda_{HS_1}^{(A)}\right)^2}{4} \\
&+ 12 \left(\lambda_{S_1}^{(1)}\right)^2 \\
&+ 4\lambda_{S_1}^{(1)} \operatorname{Tr}\left[y_1 \left(y_1^*\right)^T\right] \\
&+ 8\lambda_{24}\lambda_{24}^* \\
&- 4 \operatorname{Tr}\left[y_1 \left(y_1^*\right)^T y_1 \left(y_1^*\right)^T\right]
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{S_2}^{(1)}}^{(1)} = &\frac{9g_2^4}{4} \\
&- 9g_2^2\lambda_{S_2}^{(1)} \\
&+ \frac{243g_1^4}{4}
\end{aligned}
$$

$$
\begin{aligned}
&+ g_1^2 \left(\frac{27 g_2^2}{2} - 27 \lambda_{S_2}^{(1)}\right) \\
&+ 4 \left(\lambda_{12}^{(1)}\right)^2 \\
&+ \frac{\left(\lambda_{12}^{(A)}\right)^2}{4} \\
&+ 4 \left(\lambda_{HS_2}^{(1)}\right)^2 \\
&+ \frac{\left(\lambda_{HS_2}^{(A)}\right)^2}{4} \\
&+ 12 \left(\lambda_{S_2}^{(1)}\right)^2 \\
&+ 4 \lambda_{S_2}^{(1)} \operatorname{Tr}\left[y_2 \ (y_2^*)^T\right] \\
&- 4 \operatorname{Tr}\left[y_2 \ (y_2^*)^T \ y_2 \ (y_2^*)^T\right]
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{HS_1}^{(1)}}^{(1)} &= \frac{9 g_2^4}{4} \\
&- 9 g_2^2 \lambda_{HS_1}^{(1)} \\
&+ \frac{3 g_1^4}{4} \\
&- 3 g_1^2 \lambda_{HS_1}^{(1)} \\
&+ 4 \lambda_{12}^{(1)} \lambda_{HS_2}^{(1)} \\
&+ 6 \lambda_1 \lambda_{HS_1}^{(1)} \\
&+ 4 \left(\lambda_{HS_1}^{(1)}\right)^2 \\
&+ 6 \lambda_{HS_1}^{(1)} \lambda_{S_1}^{(1)} \\
&+ \lambda_{HS_1}^{(1)} \left(2 \operatorname{Tr}\left[y_1 \ (y_1^*)^T\right] + 6 \operatorname{Tr}\left[Y_d \ (Y_d^*)^T\right] + 2 \operatorname{Tr}\left[Y_e \ (Y_e^*)^T\right] + 6 \operatorname{Tr}\left[Y_u \ (Y_u^*)^T\right]\right) \\
&+ \frac{3 \left(\lambda_{HS_1}^{(A)}\right)^2}{4} \\
&+ 6 \lambda_5 \lambda_5^* \\
&+ 24 \lambda_{24} \lambda_{24}^* \\
&- 2 \operatorname{Tr}\left[y_1 \ (y_1^*)^T \ Y_e \ (Y_e^*)^T\right]
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{HS_2}^{(1)}}^{(1)} &= \frac{9 g_2^4}{4} \\
&- 9 g_2^2 \lambda_{HS_2}^{(1)} \\
&+ \frac{27 g_1^4}{4} \\
&- 15 g_1^2 \lambda_{HS_2}^{(1)}
\end{aligned}
$$

$$
\begin{aligned}
&+ 4\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)} \\
&+ 6\lambda_1\lambda_{HS_2}^{(1)} \\
&+ 4\left(\lambda_{HS_2}^{(1)}\right)^2 \\
&+ 6\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)} \\
&+ \lambda_{HS_2}^{(1)}\left(2\,\mathrm{Tr}\left[y_2\ (y_2^*)^T\right] + 6\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\right]\right) \\
&+ \frac{3\left(\lambda_{HS_2}^{(A)}\right)^2}{4} \\
&+ 6\lambda_5\lambda_5^* \\
&- 2\,\mathrm{Tr}\left[y_2\ (y_2^*)^T\ Y_e\ (Y_e^*)^T\right]
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{12}^{(1)}}^{(1)} = &\frac{9g_2^4}{4} \\
&- 9g_2^2\lambda_{12}^{(1)} \\
&+ \frac{27g_1^4}{4} \\
&- 15g_1^2\lambda_{12}^{(1)} \\
&+ 4\left(\lambda_{12}^{(1)}\right)^2 \\
&+ 6\lambda_{12}^{(1)}\lambda_{S_1}^{(1)} \\
&+ 6\lambda_{12}^{(1)}\lambda_{S_2}^{(1)} \\
&+ \lambda_{12}^{(1)}\left(2\,\mathrm{Tr}\left[y_1\ (y_1^*)^T\right] + 2\,\mathrm{Tr}\left[y_2\ (y_2^*)^T\right]\right) \\
&+ \frac{3\left(\lambda_{12}^{(A)}\right)^2}{4} \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)} \\
&+ 3\lambda_5\lambda_5^* \\
&- 2\,\mathrm{Tr}\left[y_1\ (y_1^*)^T\ y_2\ (y_2^*)^T\right]
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_5}^{(1)} = &- 9g_2^2\lambda_5 \\
&- 9g_1^2\lambda_5 \\
&+ 2\lambda_{12}^{(1)}\lambda_5 \\
&+ 2\lambda_1\lambda_5 \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_5 \\
&+ 4\lambda_{HS_2}^{(1)}\lambda_5 \\
&+ \lambda_5\left(\frac{\lambda_{12}^{(A)}}{6} + \frac{2\lambda_{HS_1}^{(A)}}{3} + 2\lambda_{HS_2}^{(A)} + \mathrm{Tr}\left[y_1\ (y_1^*)^T\right] + \mathrm{Tr}\left[y_2\ (y_2^*)^T\right] + 6\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\right]\right)
\end{aligned}
$$

$$
\beta_{\lambda_{HS_1}^{(A)}}^{(1)} = - 9g_2^2\lambda_{HS_1}^{(A)}
$$

$$
\begin{aligned}
&+ g_1^2 \left(6 g_2^2 - 3 \lambda_{HS_1}^{(A)}\right) \\
&+ \lambda_{12}^{(A)} \lambda_{HS_2}^{(A)} \\
&+ 2 \lambda_1 \lambda_{HS_1}^{(A)} \\
&+ 8 \lambda_{HS_1}^{(1)} \lambda_{HS_1}^{(A)} \\
&+ 2 \lambda_{HS_1}^{(A)} \lambda_{S_1}^{(1)} \\
&+ \lambda_{HS_1}^{(A)} \left(2 \operatorname{Tr}\left[y_1 \ (y_1^*)^T\right] + 6 \operatorname{Tr}\left[Y_d \ (Y_d^*)^T\right] + 2 \operatorname{Tr}\left[Y_e \ (Y_e^*)^T\right] + 6 \operatorname{Tr}\left[Y_u \ (Y_u^*)^T\right]\right) \\
&+ \frac{16 \lambda_5 \lambda_5^*}{3} \\
&+ 64 \lambda_{24} \lambda_{24}^* \\
&+ \frac{8 \operatorname{Tr}\left[y_1 \ (y_1^*)^T \ Y_e \ (Y_e^*)^T\right]}{3}
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{HS_2}^{(A)}}^{(1)} = &- 9 g_2^2 \lambda_{HS_2}^{(A)} \\
&+ g_1^2 \left(18 g_2^2 - 15 \lambda_{HS_2}^{(A)}\right) \\
&+ \lambda_{12}^{(A)} \lambda_{HS_1}^{(A)} \\
&+ 2 \lambda_1 \lambda_{HS_2}^{(A)} \\
&+ 8 \lambda_{HS_2}^{(1)} \lambda_{HS_2}^{(A)} \\
&+ 2 \lambda_{HS_2}^{(A)} \lambda_{S_2}^{(1)} \\
&+ \lambda_{HS_2}^{(A)} \left(2 \operatorname{Tr}\left[y_2 \ (y_2^*)^T\right] + 6 \operatorname{Tr}\left[Y_d \ (Y_d^*)^T\right] + 2 \operatorname{Tr}\left[Y_e \ (Y_e^*)^T\right] + 6 \operatorname{Tr}\left[Y_u \ (Y_u^*)^T\right]\right) \\
&+ 16 \lambda_5 \lambda_5^* \\
&+ 8 \operatorname{Tr}\left[y_2 \ (y_2^*)^T \ Y_e \ (Y_e^*)^T\right]
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{12}^{(A)}}^{(1)} = &- 9 g_2^2 \lambda_{12}^{(A)} \\
&+ g_1^2 \left(18 g_2^2 - 15 \lambda_{12}^{(A)}\right) \\
&+ 8 \lambda_{12}^{(1)} \lambda_{12}^{(A)} \\
&+ 2 \lambda_{12}^{(A)} \lambda_{S_1}^{(1)} \\
&+ 2 \lambda_{12}^{(A)} \lambda_{S_2}^{(1)} \\
&+ \lambda_{12}^{(A)} \left(2 \operatorname{Tr}\left[y_1 \ (y_1^*)^T\right] + 2 \operatorname{Tr}\left[y_2 \ (y_2^*)^T\right]\right) \\
&+ \lambda_{HS_1}^{(A)} \lambda_{HS_2}^{(A)} \\
&+ \frac{4 \lambda_5 \lambda_5^*}{3} \\
&- \frac{8 \operatorname{Tr}\left[y_1 \ (y_1^*)^T \ y_2 \ (y_2^*)^T\right]}{3}
\end{aligned}
$$

$$
\beta_{\lambda_{24}}^{(1)} = - 9 g_2^2 \lambda_{24}
$$

$$
\begin{aligned}
&- 3g_1^2\lambda_{24} \\
&+ 2\lambda_1\lambda_{24} \\
&+ 8\lambda_{HS_1}^{(1)}\lambda_{24} \\
&+ 4\lambda_{HS_1}^{(A)}\lambda_{24} \\
&+ 2\lambda_{S_1}^{(1)}\lambda_{24} \\
&+ \lambda_{24}\left(2\,\mathrm{Tr}\left[y_1\ (y_1^*)^T\right] + 6\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\right]\right)
\end{aligned}
$$

$\mathrm{T3-C}\,(\alpha=-3)$

$$
\begin{aligned}
\beta_{m_1^2}^{(1)} = &- \frac{9g_2^2m_1^2}{2} \\
&- \frac{27g_1^2m_1^2}{2} \\
&+ 4\lambda_{12}^{(1)}m_2^2 \\
&+ m_1^2\left(6\lambda_{S_1}^{(1)} + 3\,\mathrm{Tr}\left[y_1\ (y_1^*)^T\right]\right) \\
&- 6\,\mathrm{Tr}\left[M_F\ (M_F^*)^T\ (y_1)^T\ y_1^*\right]
\end{aligned}
$$

$$
\begin{aligned}
\beta_{m_2^2}^{(1)} = &- \frac{9g_2^2m_2^2}{2} \\
&- \frac{3g_1^2m_2^2}{2} \\
&+ 4\lambda_{12}^{(1)}m_1^2 \\
&+ m_2^2\left(6\lambda_{S_2}^{(1)} + 3\,\mathrm{Tr}\left[y_2\ (y_2^*)^T\right]\right) \\
&- 6\,\mathrm{Tr}\left[M_F\ (y_2^*)^T\ y_2\ (M_F^*)^T\right]
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_1}^{(1)} = &\frac{9g_2^4}{4} \\
&- 9g_2^2\lambda_1 \\
&+ \frac{3g_1^4}{4} \\
&+ g_1^2\left(\frac{3g_2^2}{2} - 3\lambda_1\right) \\
&+ 12\lambda_1^2 \\
&+ \lambda_1\left(12\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\right] + 4\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\right] + 12\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\right]\right) \\
&+ 4\left(\lambda_{HS_1}^{(1)}\right)^2 \\
&+ \frac{\left(\lambda_{HS_1}^{(A)}\right)^2}{4} \\
&+ 4\left(\lambda_{HS_2}^{(1)}\right)^2 \\
&+ \frac{\left(\lambda_{HS_2}^{(A)}\right)^2}{4} \\
&+ 4\lambda_5\lambda_5^* \\
&+ 8\lambda_{21}\lambda_{21}^*
\end{aligned}
$$

$$
\begin{aligned}
&- 12\,\mathrm{Tr}\Big[Y_d\ (Y_d^*)^T\ Y_d\ (Y_d^*)^T\Big] \\
&- 4\,\mathrm{Tr}\Big[Y_e\ (Y_e^*)^T\ Y_e\ (Y_e^*)^T\Big] \\
&- 12\,\mathrm{Tr}\Big[Y_u\ (Y_u^*)^T\ Y_u\ (Y_u^*)^T\Big]
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda^{(1)}_{S_1}} = &\frac{9g_2^4}{4} \\
&- 9g_2^2\lambda^{(1)}_{S_1} \\
&+ \frac{243g_1^4}{4} \\
&+ g_1^2\left(\frac{27g_2^2}{2} - 27\lambda^{(1)}_{S_1}\right) \\
&+ 4\left(\lambda^{(1)}_{12}\right)^2 \\
&+ \frac{\left(\lambda^{(A)}_{12}\right)^2}{4} \\
&+ 4\left(\lambda^{(1)}_{HS_1}\right)^2 \\
&+ \frac{\left(\lambda^{(A)}_{HS_1}\right)^2}{4} \\
&+ 12\left(\lambda^{(1)}_{S_1}\right)^2 \\
&+ 6\lambda^{(1)}_{S_1}\,\mathrm{Tr}\Big[y_1\ (y_1^*)^T\Big] \\
&- 5\,\mathrm{Tr}\Big[y_1\ (y_1^*)^T\ y_1\ (y_1^*)^T\Big]
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda^{(1)}_{S_2}} = &\frac{9g_2^4}{4} \\
&- 9g_2^2\lambda^{(1)}_{S_2} \\
&+ \frac{3g_1^4}{4} \\
&+ g_1^2\left(\frac{3g_2^2}{2} - 3\lambda^{(1)}_{S_2}\right) \\
&+ 4\left(\lambda^{(1)}_{12}\right)^2 \\
&+ \frac{\left(\lambda^{(A)}_{12}\right)^2}{4} \\
&+ 4\left(\lambda^{(1)}_{HS_2}\right)^2 \\
&+ \frac{\left(\lambda^{(A)}_{HS_2}\right)^2}{4} \\
&+ 12\left(\lambda^{(1)}_{S_2}\right)^2 \\
&+ 6\lambda^{(1)}_{S_2}\,\mathrm{Tr}\Big[y_2\ (y_2^*)^T\Big] \\
&+ 8\lambda_{21}\lambda_{21}^*
\end{aligned}
$$

$$- \ 5 \operatorname{Tr}\left[y_2 \ (y_2^*)^T \ y_2 \ (y_2^*)^T\right]$$

$$\beta^{(1)}_{\lambda^{(1)}_{HS_1}} = \frac{9 g_2^4}{4}$$
$$- \ 9 g_2^2 \lambda^{(1)}_{HS_1}$$
$$+ \ \frac{27 g_1^4}{4}$$
$$- \ 15 g_1^2 \lambda^{(1)}_{HS_1}$$
$$+ \ 4 \lambda^{(1)}_{12} \lambda^{(1)}_{HS_2}$$
$$+ \ 6 \lambda_1 \lambda^{(1)}_{HS_1}$$
$$+ \ 4 \left(\lambda^{(1)}_{HS_1}\right)^2$$
$$+ \ 6 \lambda^{(1)}_{HS_1} \lambda^{(1)}_{S_1}$$
$$+ \ \lambda^{(1)}_{HS_1} \left(3 \operatorname{Tr}\left[y_1 \ (y_1^*)^T\right] + 6 \operatorname{Tr}\left[Y_d \ (Y_d^*)^T\right] + 2 \operatorname{Tr}\left[Y_e \ (Y_e^*)^T\right] + 6 \operatorname{Tr}\left[Y_u \ (Y_u^*)^T\right]\right)$$
$$+ \ \frac{3 \left(\lambda^{(A)}_{HS_1}\right)^2}{4}$$
$$+ \ 6 \lambda_5 \lambda_5^*$$
$$- \ 3 \operatorname{Tr}\left[y_1 \ (y_1^*)^T \ Y_e \ (Y_e^*)^T\right]$$

$$\beta^{(1)}_{\lambda^{(1)}_{HS_2}} = \frac{9 g_2^4}{4}$$
$$- \ 9 g_2^2 \lambda^{(1)}_{HS_2}$$
$$+ \ \frac{3 g_1^4}{4}$$
$$- \ 3 g_1^2 \lambda^{(1)}_{HS_2}$$
$$+ \ 4 \lambda^{(1)}_{12} \lambda^{(1)}_{HS_1}$$
$$+ \ 6 \lambda_1 \lambda^{(1)}_{HS_2}$$
$$+ \ 4 \left(\lambda^{(1)}_{HS_2}\right)^2$$
$$+ \ 6 \lambda^{(1)}_{HS_2} \lambda^{(1)}_{S_2}$$
$$+ \ \lambda^{(1)}_{HS_2} \left(3 \operatorname{Tr}\left[y_2 \ (y_2^*)^T\right] + 6 \operatorname{Tr}\left[Y_d \ (Y_d^*)^T\right] + 2 \operatorname{Tr}\left[Y_e \ (Y_e^*)^T\right] + 6 \operatorname{Tr}\left[Y_u \ (Y_u^*)^T\right]\right)$$
$$+ \ \frac{3 \left(\lambda^{(A)}_{HS_2}\right)^2}{4}$$
$$+ \ 6 \lambda_5 \lambda_5^*$$
$$+ \ 24 \lambda_{21} \lambda_{21}^*$$
$$- \ 3 \operatorname{Tr}\left[y_2 \ (y_2^*)^T \ Y_e \ (Y_e^*)^T\right]$$

$$\begin{aligned}
\beta^{(1)}_{\lambda^{(1)}_{12}} =& \frac{9g_2^4}{4} \\
&- 9g_2^2\lambda^{(1)}_{12} \\
&+ \frac{27g_1^4}{4} \\
&- 15g_1^2\lambda^{(1)}_{12} \\
&+ 4\left(\lambda^{(1)}_{12}\right)^2 \\
&+ 6\lambda^{(1)}_{12}\lambda^{(1)}_{S_1} \\
&+ 6\lambda^{(1)}_{12}\lambda^{(1)}_{S_2} \\
&+ \lambda^{(1)}_{12}\left(3\,\mathrm{Tr}\left[y_1\ (y_1^*)^T\right] + 3\,\mathrm{Tr}\left[y_2\ (y_2^*)^T\right]\right) \\
&+ \frac{3\left(\lambda^{(A)}_{12}\right)^2}{4} \\
&+ 4\lambda^{(1)}_{HS_1}\lambda^{(1)}_{HS_2} \\
&+ 3\lambda_5\lambda_5^* \\
&- \frac{9\,\mathrm{Tr}\left[y_1\ (y_1^*)^T\ y_2\ (y_2^*)^T\right]}{2}
\end{aligned}$$

$$\begin{aligned}
\beta^{(1)}_{\lambda_5} =& -9g_2^2\lambda_5 \\
&- 9g_1^2\lambda_5 \\
&+ 2\lambda^{(1)}_{12}\lambda_5 \\
&+ 2\lambda_1\lambda_5 \\
&+ 4\lambda^{(1)}_{HS_1}\lambda_5 \\
&+ 4\lambda^{(1)}_{HS_2}\lambda_5 \\
&+ \lambda_5\left(\frac{\lambda^{(A)}_{12}}{6} + \frac{2\lambda^{(A)}_{HS_1}}{3} + 2\lambda^{(A)}_{HS_2} + \frac{3\,\mathrm{Tr}\left[y_1\ (y_1^*)^T\right]}{2} + \frac{3\,\mathrm{Tr}\left[y_2\ (y_2^*)^T\right]}{2} + 6\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\right]\right)
\end{aligned}$$

$$\begin{aligned}
\beta^{(1)}_{\lambda^{(A)}_{HS_1}} =& -9g_2^2\lambda^{(A)}_{HS_1} \\
&+ g_1^2\left(-18g_2^2 - 15\lambda^{(A)}_{HS_1}\right) \\
&+ \lambda^{(A)}_{12}\lambda^{(A)}_{HS_2} \\
&+ 2\lambda_1\lambda^{(A)}_{HS_1} \\
&+ 8\lambda^{(1)}_{HS_1}\lambda^{(A)}_{HS_1} \\
&+ 2\lambda^{(A)}_{HS_1}\lambda^{(1)}_{S_1} \\
&+ \lambda^{(A)}_{HS_1}\left(3\,\mathrm{Tr}\left[y_1\ (y_1^*)^T\right] + 6\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\right]\right) \\
&+ \frac{16\lambda_5\lambda_5^*}{3} \\
&- \frac{4\,\mathrm{Tr}\left[y_1\ (y_1^*)^T\ Y_e\ (Y_e^*)^T\right]}{3}
\end{aligned}$$

$$\begin{aligned}\beta^{(1)}_{\lambda^{(A)}_{HS_2}} =& -9g_2^2\lambda^{(A)}_{HS_2} \\ &+ g_1^2\left(-6g_2^2 - 3\lambda^{(A)}_{HS_2}\right) \\ &+ \lambda^{(A)}_{12}\lambda^{(A)}_{HS_1} \\ &+ 2\lambda_1\lambda^{(A)}_{HS_2} \\ &+ 8\lambda^{(1)}_{HS_2}\lambda^{(A)}_{HS_2} \\ &+ 2\lambda^{(A)}_{HS_2}\lambda^{(1)}_{S_2} \\ &+ \lambda^{(A)}_{HS_2}\left(3\,\mathrm{Tr}\left[y_2\ (y_2^*)^T\right] + 6\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\right]\right) \\ &+ 16\lambda_5\lambda_5^* \\ &- 64\lambda_{21}\lambda_{21}^* \\ &- 4\,\mathrm{Tr}\left[y_2\ (y_2^*)^T\ Y_e\ (Y_e^*)^T\right]\end{aligned}$$

$$\begin{aligned}\beta^{(1)}_{\lambda^{(A)}_{12}} =& -9g_2^2\lambda^{(A)}_{12} \\ &+ g_1^2\left(18g_2^2 - 15\lambda^{(A)}_{12}\right) \\ &+ 8\lambda^{(1)}_{12}\lambda^{(A)}_{12} \\ &+ 2\lambda^{(A)}_{12}\lambda^{(1)}_{S_1} \\ &+ 2\lambda^{(A)}_{12}\lambda^{(1)}_{S_2} \\ &+ \lambda^{(A)}_{12}\left(3\,\mathrm{Tr}\left[y_1\ (y_1^*)^T\right] + 3\,\mathrm{Tr}\left[y_2\ (y_2^*)^T\right]\right) \\ &+ \lambda^{(A)}_{HS_1}\lambda^{(A)}_{HS_2} \\ &+ \frac{4\lambda_5\lambda_5^*}{3} \\ &- \frac{2\,\mathrm{Tr}\left[y_1\ (y_1^*)^T\ y_2\ (y_2^*)^T\right]}{3}\end{aligned}$$

$$\begin{aligned}\beta^{(1)}_{\lambda_{21}} =& -9g_2^2\lambda_{21} \\ &- 3g_1^2\lambda_{21} \\ &+ 2\lambda_1\lambda_{21} \\ &+ 8\lambda^{(1)}_{HS_2}\lambda_{21} \\ &- 4\lambda^{(A)}_{HS_2}\lambda_{21} \\ &+ 2\lambda^{(1)}_{S_2}\lambda_{21} \\ &+ \lambda_{21}\left(3\,\mathrm{Tr}\left[y_2\ (y_2^*)^T\right] + 6\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\right]\right)\end{aligned}$$

$\mathrm{T3-C}\,(\alpha = -1)$

$$\begin{aligned}\beta^{(1)}_{m_1^2} =& -\frac{9g_2^2m_1^2}{2} \\ &- \frac{3g_1^2m_1^2}{2}\end{aligned}$$

$$
\begin{aligned}
&+ 4\lambda_{12}^{(1)} m_2^2 \\
&+ 6\lambda_{11} \left(\mu_{12}^2\right)^* \\
&+ m_1^2 \left(6\lambda_{S_1}^{(1)} + 3\,\mathrm{Tr}\left[y_1\ \left(y_1^*\right)^T\right]\right) \\
&+ 6\mu_{12}^2 \lambda_{11}^* \\
&- 12\,\mathrm{Tr}\left[M_F\ M_F^*\ \left(y_1\right)^T\ y_1^*\right] \\
&- 12\,\mathrm{Tr}\left[M_F\ \left(y_1^*\right)^T\ y_1\ M_F^*\right]
\end{aligned}
$$

$$
\begin{aligned}
\beta_{m_2^2}^{(1)} = &- \frac{9g_2^2 m_2^2}{2} \\
&- \frac{3g_1^2 m_2^2}{2} \\
&+ 4\lambda_{12}^{(1)} m_1^2 \\
&+ 6\lambda_{10} \left(\mu_{12}^2\right)^* \\
&+ m_2^2 \left(6\lambda_{S_2}^{(1)} + 3\,\mathrm{Tr}\left[y_2\ \left(y_2^*\right)^T\right]\right) \\
&+ 6\mu_{12}^2 \lambda_{10}^* \\
&- 12\,\mathrm{Tr}\left[M_F\ M_F^*\ \left(y_2\right)^T\ y_2^*\right] \\
&- 12\,\mathrm{Tr}\left[M_F\ \left(y_2^*\right)^T\ y_2\ M_F^*\right]
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_1}^{(1)} = &\frac{9g_2^4}{4} \\
&- 9g_2^2 \lambda_1 \\
&+ \frac{3g_1^4}{4} \\
&+ g_1^2 \left(\frac{3g_2^2}{2} - 3\lambda_1\right) \\
&+ 12\lambda_1^2 \\
&+ \lambda_1 \left(12\,\mathrm{Tr}\left[Y_d\ \left(Y_d^*\right)^T\right] + 4\,\mathrm{Tr}\left[Y_e\ \left(Y_e^*\right)^T\right] + 12\,\mathrm{Tr}\left[Y_u\ \left(Y_u^*\right)^T\right]\right) \\
&+ 4\left(\lambda_{HS_1}^{(1)}\right)^2 \\
&+ \frac{\left(\lambda_{HS_1}^{(A)}\right)^2}{4} \\
&+ 4\left(\lambda_{HS_2}^{(1)}\right)^2 \\
&+ \frac{\left(\lambda_{HS_2}^{(A)}\right)^2}{4} \\
&+ 8\lambda_7 \lambda_7^* \\
&+ 4\lambda_{12} \lambda_{12}^* \\
&+ 8\lambda_8 \lambda_8^* \\
&+ 4\lambda_5 \lambda_5^* \\
&+ 4\lambda_{26} \lambda_{26}^* \\
&- 12\,\mathrm{Tr}\left[Y_d\ \left(Y_d^*\right)^T\ Y_d\ \left(Y_d^*\right)^T\right]
\end{aligned}
$$

$$
\begin{aligned}
&- 4\,\mathrm{Tr}\Big[Y_e\ (Y_e^*)^T\ Y_e\ (Y_e^*)^T\Big] \\
&- 12\,\mathrm{Tr}\Big[Y_u\ (Y_u^*)^T\ Y_u\ (Y_u^*)^T\Big]
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda^{(1)}_{S_1}} = &\frac{9g_2^4}{4} \\
&- 9g_2^2\lambda^{(1)}_{S_1} \\
&+ \frac{3g_1^4}{4} \\
&+ g_1^2\left(\frac{3g_2^2}{2} - 3\lambda^{(1)}_{S_1}\right) \\
&+ 4\left(\lambda^{(1)}_{12}\right)^2 \\
&+ \frac{\left(\lambda^{(A)}_{12}\right)^2}{4} \\
&+ 4\left(\lambda^{(1)}_{HS_1}\right)^2 \\
&+ \frac{\left(\lambda^{(A)}_{HS_1}\right)^2}{4} \\
&+ 8\lambda_7\lambda_7^* \\
&+ 12\left(\lambda^{(1)}_{S_1}\right)^2 \\
&+ 6\lambda^{(1)}_{S_1}\,\mathrm{Tr}\Big[y_1\ (y_1^*)^T\Big] \\
&+ 24\lambda_{11}\lambda_{11}^* \\
&+ 8\lambda_9\lambda_9^* \\
&- 5\,\mathrm{Tr}\Big[y_1\ (y_1^*)^T\ y_1\ (y_1^*)^T\Big]
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda^{(1)}_{S_2}} = &\frac{9g_2^4}{4} \\
&- 9g_2^2\lambda^{(1)}_{S_2} \\
&+ \frac{3g_1^4}{4} \\
&+ g_1^2\left(\frac{3g_2^2}{2} - 3\lambda^{(1)}_{S_2}\right) \\
&+ 4\left(\lambda^{(1)}_{12}\right)^2 \\
&+ \frac{\left(\lambda^{(A)}_{12}\right)^2}{4} \\
&+ 4\left(\lambda^{(1)}_{HS_2}\right)^2 \\
&+ \frac{\left(\lambda^{(A)}_{HS_2}\right)^2}{4} \\
&+ 8\lambda_8\lambda_8^*
\end{aligned}
$$

$$
\begin{aligned}
&+ 8\lambda_9\lambda_9^* \\
&+ 24\lambda_{10}\lambda_{10}^* \\
&+ 12\left(\lambda_{S_2}^{(1)}\right)^2 \\
&+ 6\lambda_{S_2}^{(1)}\,\mathrm{Tr}\Big[y_2\ (y_2^*)^T\Big] \\
&- 5\,\mathrm{Tr}\Big[y_2\ (y_2^*)^T\ y_2\ (y_2^*)^T\Big]
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda^{(1)}_{HS_1}} &= \frac{9g_2^4}{4} \\
&- 9g_2^2\lambda_{HS_1}^{(1)} \\
&+ \frac{3g_1^4}{4} \\
&- 3g_1^2\lambda_{HS_1}^{(1)} \\
&+ 4\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)} \\
&+ 6\lambda_1\lambda_{HS_1}^{(1)} \\
&+ 4\left(\lambda_{HS_1}^{(1)}\right)^2 \\
&+ 6\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)} \\
&+ \lambda_{HS_1}^{(1)}\left(3\,\mathrm{Tr}\Big[y_1\ (y_1^*)^T\Big] + 6\,\mathrm{Tr}\Big[Y_d\ (Y_d^*)^T\Big] + 2\,\mathrm{Tr}\Big[Y_e\ (Y_e^*)^T\Big] + 6\,\mathrm{Tr}\Big[Y_u\ (Y_u^*)^T\Big]\right) \\
&+ \frac{3\left(\lambda_{HS_1}^{(A)}\right)^2}{4} \\
&+ 24\lambda_7\lambda_7^* \\
&+ \lambda_{12}\left(4\lambda_{12}^* + 3\lambda_{11}^* + 2\lambda_{26}^*\right) \\
&+ 3\lambda_{11}\left(\lambda_{12}^* - \lambda_{26}^*\right) \\
&+ 6\lambda_5\lambda_5^* \\
&+ \lambda_{26}\left(2\lambda_{12}^* - 3\lambda_{11}^* + 4\lambda_{26}^*\right) \\
&- 3\,\mathrm{Tr}\Big[y_1\ (y_1^*)^T\ Y_e\ (Y_e^*)^T\Big]
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda^{(1)}_{HS_2}} &= \frac{9g_2^4}{4} \\
&- 9g_2^2\lambda_{HS_2}^{(1)} \\
&+ \frac{3g_1^4}{4} \\
&- 3g_1^2\lambda_{HS_2}^{(1)} \\
&+ 4\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)} \\
&+ 6\lambda_1\lambda_{HS_2}^{(1)}
\end{aligned}
$$

$$
\begin{aligned}
&+ 4\left(\lambda_{HS_2}^{(1)}\right)^2 \\
&+ 6\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)} \\
&+ \lambda_{HS_2}^{(1)}\left(3\,\mathrm{Tr}\left[y_2\ (y_2^*)^T\right] + 6\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\right]\right) \\
&+ \frac{3\left(\lambda_{HS_2}^{(A)}\right)^2}{4} \\
&+ \lambda_{12}\left(4\lambda_{12}^* + 3\lambda_{10}^* + 2\lambda_{26}^*\right) \\
&+ 24\lambda_8\lambda_8^* \\
&+ 3\lambda_{10}\left(\lambda_{12}^* - \lambda_{26}^*\right) \\
&+ 6\lambda_5\lambda_5^* \\
&+ \lambda_{26}\left(2\lambda_{12}^* - 3\lambda_{10}^* + 4\lambda_{26}^*\right) \\
&- 3\,\mathrm{Tr}\left[y_2\ (y_2^*)^T\ Y_e\ (Y_e^*)^T\right]
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{12}^{(1)}}^{(1)} = &\frac{9g_2^4}{4} \\
&- 9g_2^2\lambda_{12}^{(1)} \\
&+ \frac{3g_1^4}{4} \\
&- 3g_1^2\lambda_{12}^{(1)} \\
&+ 4\left(\lambda_{12}^{(1)}\right)^2 \\
&+ 6\lambda_{12}^{(1)}\lambda_{S_1}^{(1)} \\
&+ 6\lambda_{12}^{(1)}\lambda_{S_2}^{(1)} \\
&+ \lambda_{12}^{(1)}\left(3\,\mathrm{Tr}\left[y_1\ (y_1^*)^T\right] + 3\,\mathrm{Tr}\left[y_2\ (y_2^*)^T\right]\right) \\
&+ \frac{3\left(\lambda_{12}^{(A)}\right)^2}{4} \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)} \\
&+ \lambda_{12}\left(2\lambda_{12}^* + \lambda_{26}^*\right) \\
&+ 9\lambda_{11}\left(\lambda_{11}^* + \lambda_{10}^*\right) \\
&+ 24\lambda_9\lambda_9^* \\
&+ 9\lambda_{10}\left(\lambda_{11}^* + \lambda_{10}^*\right) \\
&+ 3\lambda_5\lambda_5^* \\
&+ \lambda_{26}\left(\lambda_{12}^* + 2\lambda_{26}^*\right) \\
&- \frac{9\,\mathrm{Tr}\left[y_1\ (y_1^*)^T\ y_2\ (y_2^*)^T\right]}{2} \\
&- 3\,\mathrm{Tr}\left[y_1\ (y_2^*)^T\ y_2\ (y_1^*)^T\right]
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda_5} =& -9g_2^2\lambda_5 \\
&- 3g_1^2\lambda_5 \\
&+ 2\lambda_{12}^{(1)}\lambda_5 \\
&+ 2\lambda_1\lambda_5 \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_5 \\
&+ 4\lambda_{HS_2}^{(1)}\lambda_5 \\
&+ \lambda_5\left(\frac{\lambda_{12}^{(A)}}{6} + \frac{2\lambda_{HS_1}^{(A)}}{3} + 2\lambda_{HS_2}^{(A)} + \frac{3\,\mathrm{Tr}\left[y_1\,\left(y_1^*\right)^T\right]}{2} + \frac{3\,\mathrm{Tr}\left[y_2\,\left(y_2^*\right)^T\right]}{2} + 6\,\mathrm{Tr}\left[Y_d\,\left(Y_d^*\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\,\left(Y_e^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\,\left(Y_u^*\right)^T\right]\right) \\
&- \lambda_7^*\,\mathrm{Tr}\left[y_2\,\left(y_1^*\right)^T\right] \\
&+ 3\lambda_8^*\,\mathrm{Tr}\left[y_1\,\left(y_2^*\right)^T\right]
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda_{HS_1}^{(A)}} =& -9g_2^2\lambda_{HS_1}^{(A)} \\
&+ g_1^2\left(-6g_2^2 - 3\lambda_{HS_1}^{(A)}\right) \\
&+ \lambda_{12}^{(A)}\lambda_{HS_2}^{(A)} \\
&+ 2\lambda_1\lambda_{HS_1}^{(A)} \\
&+ 8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)} \\
&+ 2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)} \\
&+ \lambda_{HS_1}^{(A)}\left(3\,\mathrm{Tr}\left[y_1\,\left(y_1^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_d\,\left(Y_d^*\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\,\left(Y_e^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\,\left(Y_u^*\right)^T\right]\right) \\
&- 64\lambda_7\lambda_7^* \\
&+ 4\lambda_{12}\left(2\lambda_{12}^* + \lambda_{11}^*\right) \\
&+ 4\lambda_{11}\left(\lambda_{12}^* + \lambda_{26}^*\right) \\
&+ \frac{16\lambda_5\lambda_5^*}{3} \\
&+ 4\lambda_{26}\left(\lambda_{11}^* - 2\lambda_{26}^*\right) \\
&- \frac{4\,\mathrm{Tr}\left[y_1\,\left(y_1^*\right)^T\,Y_e\,\left(Y_e^*\right)^T\right]}{3}
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda_{HS_2}^{(A)}} =& -9g_2^2\lambda_{HS_2}^{(A)} \\
&+ g_1^2\left(6g_2^2 - 3\lambda_{HS_2}^{(A)}\right) \\
&+ \lambda_{12}^{(A)}\lambda_{HS_1}^{(A)} \\
&+ 2\lambda_1\lambda_{HS_2}^{(A)} \\
&+ 8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)} \\
&+ 2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)} \\
&+ \lambda_{HS_2}^{(A)}\left(3\,\mathrm{Tr}\left[y_2\,\left(y_2^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_d\,\left(Y_d^*\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\,\left(Y_e^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\,\left(Y_u^*\right)^T\right]\right)
\end{aligned}
$$

$$
\begin{aligned}
&- 4\lambda_{12}\left(2\lambda_{12}^{*} + \lambda_{10}^{*}\right) \\
&+ 64\lambda_{8}\lambda_{8}^{*} \\
&- 4\lambda_{10}\left(\lambda_{12}^{*} + \lambda_{26}^{*}\right) \\
&+ 16\lambda_{5}\lambda_{5}^{*} \\
&+ \lambda_{26}\left(-4\lambda_{10}^{*} + 8\lambda_{26}^{*}\right) \\
&- 4\,\mathrm{Tr}\left[y_2\ \left(y_2^{*}\right)^{T}\ Y_e\ \left(Y_e^{*}\right)^{T}\right]
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda_{12}^{(A)}} =& - 9g_2^2\lambda_{12}^{(A)} \\
&+ g_1^2\left(-6g_2^2 - 3\lambda_{12}^{(A)}\right) \\
&+ 8\lambda_{12}^{(1)}\lambda_{12}^{(A)} \\
&+ 2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)} \\
&+ 2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)} \\
&+ \lambda_{12}^{(A)}\left(3\,\mathrm{Tr}\left[y_1\ \left(y_1^{*}\right)^{T}\right] + 3\,\mathrm{Tr}\left[y_2\ \left(y_2^{*}\right)^{T}\right]\right) \\
&+ \lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)} \\
&+ 4\lambda_{12}\lambda_{26}^{*} \\
&- 4\lambda_{11}\left(5\lambda_{11}^{*} + \lambda_{10}^{*}\right) \\
&- 64\lambda_{9}\lambda_{9}^{*} \\
&- 4\lambda_{10}\left(\lambda_{11}^{*} + 5\lambda_{10}^{*}\right) \\
&+ \frac{4\lambda_{5}\lambda_{5}^{*}}{3} \\
&+ 4\lambda_{26}\lambda_{12}^{*} \\
&- \frac{2\,\mathrm{Tr}\left[y_1\ \left(y_1^{*}\right)^{T}\ y_2\ \left(y_2^{*}\right)^{T}\right]}{3} \\
&- \frac{8\,\mathrm{Tr}\left[y_1\ \left(y_2^{*}\right)^{T}\ y_2\ \left(y_1^{*}\right)^{T}\right]}{3}
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda_8} =& - 9g_2^2\lambda_8 \\
&- 3g_1^2\lambda_8 \\
&+ 2\lambda_1\lambda_8 \\
&+ 8\lambda_{HS_2}^{(1)}\lambda_8 \\
&+ 4\lambda_{HS_2}^{(A)}\lambda_8 \\
&+ 4\lambda_7\lambda_9^{*} \\
&+ 2\lambda_8\lambda_{S_2}^{(1)} \\
&+ \lambda_8\left(3\,\mathrm{Tr}\left[y_2\ \left(y_2^{*}\right)^{T}\right] + 6\,\mathrm{Tr}\left[Y_d\ \left(Y_d^{*}\right)^{T}\right] + 2\,\mathrm{Tr}\left[Y_e\ \left(Y_e^{*}\right)^{T}\right] + 6\,\mathrm{Tr}\left[Y_u\ \left(Y_u^{*}\right)^{T}\right]\right) \\
&+ \frac{3\lambda_5^{*}\,\mathrm{Tr}\left[y_1\ \left(y_2^{*}\right)^{T}\right]}{2}
\end{aligned}
$$

$$\begin{aligned}\beta^{(1)}_{\lambda_7} =& -9g_2^2\lambda_7 \\ &- 3g_1^2\lambda_7 \\ &+ 2\lambda_1\lambda_7 \\ &+ 8\lambda^{(1)}_{HS_1}\lambda_7 \\ &- 4\lambda^{(A)}_{HS_1}\lambda_7 \\ &+ 2\lambda_7\lambda^{(1)}_{S_1} \\ &+ \lambda_7\left(3\,\mathrm{Tr}\left[y_1\ (y_1^*)^T\right] + 6\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\right]\right) \\ &+ 4\lambda_8\lambda_9 \\ &- \frac{\lambda_5^*\,\mathrm{Tr}\left[y_2\ (y_1^*)^T\right]}{2}\end{aligned}$$

$$\begin{aligned}\beta^{(1)}_{\lambda_9} =& -9g_2^2\lambda_9 \\ &- 3g_1^2\lambda_9 \\ &+ 8\lambda^{(1)}_{12}\lambda_9 \\ &- 4\lambda^{(A)}_{12}\lambda_9 \\ &+ 4\lambda_7\lambda_8^* \\ &+ \lambda_{12}^2 \\ &+ 2\lambda^{(1)}_{S_1}\lambda_9 \\ &+ 5\lambda_{11}^2 \\ &+ 2\lambda_{11}\lambda_{10} \\ &+ 2\lambda_9\lambda^{(1)}_{S_2} \\ &+ \lambda_9\left(3\,\mathrm{Tr}\left[y_1\ (y_1^*)^T\right] + 3\,\mathrm{Tr}\left[y_2\ (y_2^*)^T\right]\right) \\ &+ 5\lambda_{10}^2 \\ &+ \lambda_{26}^2 \\ &+ \frac{5\,\mathrm{Tr}\left[y_2\ (y_1^*)^T\ y_2\ (y_1^*)^T\right]}{6}\end{aligned}$$

$$\begin{aligned}\beta^{(1)}_{\lambda_{10}} =& -9g_2^2\lambda_{10} \\ &- 3g_1^2\lambda_{10} \\ &+ \lambda^{(1)}_{12}\left(6\lambda_{11} + 6\lambda_{10}\right) \\ &+ \lambda^{(A)}_{12}\left(-\frac{\lambda_{11}}{2} - \frac{5\lambda_{10}}{2}\right) \\ &+ \lambda^{(1)}_{HS_2}\left(2\lambda_{12} - 2\lambda_{26}\right) \\ &+ \lambda^{(A)}_{HS_2}\left(-\frac{\lambda_{12}}{2} - \frac{\lambda_{26}}{2}\right) \\ &+ 4\lambda_9\left(\lambda_{11}^* + 5\lambda_{10}^*\right) \\ &+ 12\lambda_{10}\lambda^{(1)}_{S_2} \\ &+ \lambda_{10}\left(\frac{3\,\mathrm{Tr}\left[y_1\ (y_1^*)^T\right]}{2} + \frac{9\,\mathrm{Tr}\left[y_2\ (y_2^*)^T\right]}{2}\right)\end{aligned}$$

$$\begin{aligned}
\beta^{(1)}_{\lambda_{11}} =& -9g_2^2\lambda_{11} \\
&- 3g_1^2\lambda_{11} \\
&+ \lambda_{12}^{(1)}\left(6\lambda_{11} + 6\lambda_{10}\right) \\
&+ \lambda_{12}^{(A)}\left(-\frac{5\lambda_{11}}{2} - \frac{\lambda_{10}}{2}\right) \\
&+ \lambda_{HS_1}^{(1)}\left(2\lambda_{12} - 2\lambda_{26}\right) \\
&+ \lambda_{HS_1}^{(A)}\left(\frac{\lambda_{12}}{2} + \frac{\lambda_{26}}{2}\right) \\
&+ 12\lambda_{S_1}^{(1)}\lambda_{11} \\
&+ \lambda_{11}\left(\frac{9\,\mathrm{Tr}\left[y_1\ \left(y_1^*\right)^T\right]}{2} + \frac{3\,\mathrm{Tr}\left[y_2\ \left(y_2^*\right)^T\right]}{2}\right) \\
&+ 4\lambda_9\left(5\lambda_{11}^* + \lambda_{10}^*\right)
\end{aligned}$$

$$\begin{aligned}
\beta^{(1)}_{\lambda_{12}} =& -9g_2^2\lambda_{12} \\
&- 3g_1^2\lambda_{12} \\
&+ 2\lambda_{12}^{(1)}\lambda_{12} \\
&+ \lambda_{12}^{(A)}\left(-\frac{\lambda_{12}}{2} + \lambda_{26}\right) \\
&+ \lambda_1\left(4\lambda_{12} - 2\lambda_{26}\right) \\
&+ \lambda_{HS_1}^{(1)}\left(4\lambda_{12} + 6\lambda_{11}\right) \\
&+ \lambda_{HS_1}^{(A)}\left(2\lambda_{12} + \frac{\lambda_{11}}{2} + \lambda_{26}\right) \\
&+ \lambda_{HS_2}^{(1)}\left(4\lambda_{12} + 6\lambda_{10}\right) \\
&+ \lambda_{HS_2}^{(A)}\left(-2\lambda_{12} - \frac{\lambda_{10}}{2} - \lambda_{26}\right) \\
&+ \lambda_{12}\left(\frac{3\,\mathrm{Tr}\left[y_1\ \left(y_1^*\right)^T\right]}{2} + \frac{3\,\mathrm{Tr}\left[y_2\ \left(y_2^*\right)^T\right]}{2} + 6\,\mathrm{Tr}\left[Y_d\ \left(Y_d^*\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ \left(Y_e^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ \left(Y_u^*\right)^T\right]\right) \\
&+ \lambda_9\left(8\lambda_{12}^* - 4\lambda_{26}^*\right)
\end{aligned}$$

$$\begin{aligned}
\beta^{(1)}_{\lambda_{26}} =& -9g_2^2\lambda_{26} \\
&- 3g_1^2\lambda_{26} \\
&+ 2\lambda_{12}^{(1)}\lambda_{26} \\
&+ \lambda_{12}^{(A)}\left(\lambda_{12} - \frac{\lambda_{26}}{2}\right) \\
&+ \lambda_1\left(-2\lambda_{12} + 4\lambda_{26}\right) \\
&+ \lambda_{HS_1}^{(1)}\left(-6\lambda_{11} + 4\lambda_{26}\right) \\
&+ \lambda_{HS_1}^{(A)}\left(-\lambda_{12} + \frac{\lambda_{11}}{2} - 2\lambda_{26}\right) \\
&+ \lambda_{HS_2}^{(1)}\left(-6\lambda_{10} + 4\lambda_{26}\right) \\
&+ \lambda_{HS_2}^{(A)}\left(\lambda_{12} - \frac{\lambda_{10}}{2} + 2\lambda_{26}\right) \\
&+ \lambda_9\left(-4\lambda_{12}^* + 8\lambda_{26}^*\right)
\end{aligned}$$

$$+ \lambda_{26} \left(\frac{3 \operatorname{Tr}\left[y_1 \left(y_1^*\right)^T\right]}{2} + \frac{3 \operatorname{Tr}\left[y_2 \left(y_2^*\right)^T\right]}{2} + 6 \operatorname{Tr}\left[Y_d \left(Y_d^*\right)^T\right] + 2 \operatorname{Tr}\left[Y_e \left(Y_e^*\right)^T\right] + 6 \operatorname{Tr}\left[Y_u \left(Y_u^*\right)^T\right] \right)$$

$$\begin{aligned}
\beta^{(1)}_{\mu_{12}^2} =& - \frac{9 g_2^2 \mu_{12}^2}{2} \\
& - \frac{3 g_1^2 \mu_{12}^2}{2} \\
& + 2 \lambda_{12}^{(1)} \mu_{12}^2 \\
& - \frac{3 \lambda_{12}^{(A)} \mu_{12}^2}{2} \\
& + 6 \lambda_{11} m_1^2 \\
& + 12 \lambda_9 \left(\mu_{12}^2\right)^* \\
& + 6 \lambda_{10} m_2^2 \\
& + \mu_{12}^2 \left(\frac{3 \operatorname{Tr}\left[y_1 \left(y_1^*\right)^T\right]}{2} + \frac{3 \operatorname{Tr}\left[y_2 \left(y_2^*\right)^T\right]}{2} \right)
\end{aligned}$$

$\mathrm{T3-C}\,(\alpha = 1)$

$$\begin{aligned}
\beta^{(1)}_{m_1^2} =& - \frac{9 g_2^2 m_1^2}{2} \\
& - \frac{3 g_1^2 m_1^2}{2} \\
& + 4 \lambda_{12}^{(1)} m_2^2 \\
& + m_1^2 \left(6 \lambda_{S_1}^{(1)} + 3 \operatorname{Tr}\left[y_1 \left(y_1^*\right)^T\right]\right) \\
& - 6 \operatorname{Tr}\left[M_F \left(M_F^*\right)^T \left(y_1\right)^T y_1^*\right]
\end{aligned}$$

$$\begin{aligned}
\beta^{(1)}_{m_2^2} =& - \frac{9 g_2^2 m_2^2}{2} \\
& - \frac{27 g_1^2 m_2^2}{2} \\
& + 4 \lambda_{12}^{(1)} m_1^2 \\
& + m_2^2 \left(6 \lambda_{S_2}^{(1)} + 3 \operatorname{Tr}\left[y_2 \left(y_2^*\right)^T\right]\right) \\
& - 6 \operatorname{Tr}\left[M_F \left(y_2^*\right)^T y_2 \left(M_F^*\right)^T\right]
\end{aligned}$$

$$\begin{aligned}
\beta^{(1)}_{\lambda_1} =& \frac{9 g_2^4}{4} \\
& - 9 g_2^2 \lambda_1 \\
& + \frac{3 g_1^4}{4} \\
& + g_1^2 \left(\frac{3 g_2^2}{2} - 3 \lambda_1\right) \\
& + 12 \lambda_1^2 \\
& + \lambda_1 \left(12 \operatorname{Tr}\left[Y_d \left(Y_d^*\right)^T\right] + 4 \operatorname{Tr}\left[Y_e \left(Y_e^*\right)^T\right] + 12 \operatorname{Tr}\left[Y_u \left(Y_u^*\right)^T\right]\right) \\
& + 4 \left(\lambda_{HS_1}^{(1)}\right)^2
\end{aligned}$$

$$
\begin{aligned}
&+ \frac{\left(\lambda_{HS_1}^{(A)}\right)^2}{4} \\
&+ 4\left(\lambda_{HS_2}^{(1)}\right)^2 \\
&+ \frac{\left(\lambda_{HS_2}^{(A)}\right)^2}{4} \\
&+ 4\lambda_5\lambda_5^* \\
&+ 8\lambda_{24}\lambda_{24}^* \\
&- 12\,\mathrm{Tr}\Big[Y_d\ (Y_d^*)^T\ Y_d\ (Y_d^*)^T\Big] \\
&- 4\,\mathrm{Tr}\Big[Y_e\ (Y_e^*)^T\ Y_e\ (Y_e^*)^T\Big] \\
&- 12\,\mathrm{Tr}\Big[Y_u\ (Y_u^*)^T\ Y_u\ (Y_u^*)^T\Big]
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda^{(1)}_{S_1}} = &\frac{9g_2^4}{4} \\
&- 9g_2^2\lambda_{S_1}^{(1)} \\
&+ \frac{3g_1^4}{4} \\
&+ g_1^2\left(\frac{3g_2^2}{2} - 3\lambda_{S_1}^{(1)}\right) \\
&+ 4\left(\lambda_{12}^{(1)}\right)^2 \\
&+ \frac{\left(\lambda_{12}^{(A)}\right)^2}{4} \\
&+ 4\left(\lambda_{HS_1}^{(1)}\right)^2 \\
&+ \frac{\left(\lambda_{HS_1}^{(A)}\right)^2}{4} \\
&+ 12\left(\lambda_{S_1}^{(1)}\right)^2 \\
&+ 6\lambda_{S_1}^{(1)}\,\mathrm{Tr}\Big[y_1\ (y_1^*)^T\Big] \\
&+ 8\lambda_{24}\lambda_{24}^* \\
&- 5\,\mathrm{Tr}\Big[y_1\ (y_1^*)^T\ y_1\ (y_1^*)^T\Big]
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda^{(1)}_{S_2}} = &\frac{9g_2^4}{4} \\
&- 9g_2^2\lambda_{S_2}^{(1)} \\
&+ \frac{243g_1^4}{4} \\
&+ g_1^2\left(\frac{27g_2^2}{2} - 27\lambda_{S_2}^{(1)}\right) \\
&+ 4\left(\lambda_{12}^{(1)}\right)^2
\end{aligned}
$$

$$
\begin{aligned}
&+ \frac{\left(\lambda_{12}^{(A)}\right)^2}{4} \\
&+ 4\left(\lambda_{HS_2}^{(1)}\right)^2 \\
&+ \frac{\left(\lambda_{HS_2}^{(A)}\right)^2}{4} \\
&+ 12\left(\lambda_{S_2}^{(1)}\right)^2 \\
&+ 6\lambda_{S_2}^{(1)}\,\mathrm{Tr}\Big[y_2\ (y_2^*)^T\Big] \\
&- 5\,\mathrm{Tr}\Big[y_2\ (y_2^*)^T\ y_2\ (y_2^*)^T\Big]
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda^{(1)}_{HS_1}} &= \frac{9g_2^4}{4} \\
&- 9g_2^2\lambda_{HS_1}^{(1)} \\
&+ \frac{3g_1^4}{4} \\
&- 3g_1^2\lambda_{HS_1}^{(1)} \\
&+ 4\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)} \\
&+ 6\lambda_1\lambda_{HS_1}^{(1)} \\
&+ 4\left(\lambda_{HS_1}^{(1)}\right)^2 \\
&+ 6\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)} \\
&+ \lambda_{HS_1}^{(1)}\left(3\,\mathrm{Tr}\Big[y_1\ (y_1^*)^T\Big] + 6\,\mathrm{Tr}\Big[Y_d\ (Y_d^*)^T\Big] + 2\,\mathrm{Tr}\Big[Y_e\ (Y_e^*)^T\Big] + 6\,\mathrm{Tr}\Big[Y_u\ (Y_u^*)^T\Big]\right) \\
&+ \frac{3\left(\lambda_{HS_1}^{(A)}\right)^2}{4} \\
&+ 6\lambda_5\lambda_5^* \\
&+ 24\lambda_{24}\lambda_{24}^* \\
&- 3\,\mathrm{Tr}\Big[y_1\ (y_1^*)^T\ Y_e\ (Y_e^*)^T\Big]
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda^{(1)}_{HS_2}} &= \frac{9g_2^4}{4} \\
&- 9g_2^2\lambda_{HS_2}^{(1)} \\
&+ \frac{27g_1^4}{4} \\
&- 15g_1^2\lambda_{HS_2}^{(1)} \\
&+ 4\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)} \\
&+ 6\lambda_1\lambda_{HS_2}^{(1)} \\
&+ 4\left(\lambda_{HS_2}^{(1)}\right)^2
\end{aligned}
$$

$$
\begin{aligned}
&+ 6\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)} \\
&+ \lambda_{HS_2}^{(1)}\left(3\,\mathrm{Tr}\left[y_2\ (y_2^*)^T\right] + 6\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\right]\right) \\
&+ \frac{3\left(\lambda_{HS_2}^{(A)}\right)^2}{4} \\
&+ 6\lambda_5\lambda_5^* \\
&- 3\,\mathrm{Tr}\left[y_2\ (y_2^*)^T\ Y_e\ (Y_e^*)^T\right]
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{12}^{(1)}}^{(1)} = &\frac{9g_2^4}{4} \\
&- 9g_2^2\lambda_{12}^{(1)} \\
&+ \frac{27g_1^4}{4} \\
&- 15g_1^2\lambda_{12}^{(1)} \\
&+ 4\left(\lambda_{12}^{(1)}\right)^2 \\
&+ 6\lambda_{12}^{(1)}\lambda_{S_1}^{(1)} \\
&+ 6\lambda_{12}^{(1)}\lambda_{S_2}^{(1)} \\
&+ \lambda_{12}^{(1)}\left(3\,\mathrm{Tr}\left[y_1\ (y_1^*)^T\right] + 3\,\mathrm{Tr}\left[y_2\ (y_2^*)^T\right]\right) \\
&+ \frac{3\left(\lambda_{12}^{(A)}\right)^2}{4} \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)} \\
&+ 3\lambda_5\lambda_5^* \\
&- \frac{9\,\mathrm{Tr}\left[y_1\ (y_1^*)^T\ y_2\ (y_2^*)^T\right]}{2}
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_5}^{(1)} = &- 9g_2^2\lambda_5 \\
&- 9g_1^2\lambda_5 \\
&+ 2\lambda_{12}^{(1)}\lambda_5 \\
&+ 2\lambda_1\lambda_5 \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_5 \\
&+ 4\lambda_{HS_2}^{(1)}\lambda_5 \\
&+ \lambda_5\left(\frac{\lambda_{12}^{(A)}}{6} + \frac{2\lambda_{HS_1}^{(A)}}{3} + 2\lambda_{HS_2}^{(A)} + \frac{3\,\mathrm{Tr}\left[y_1\ (y_1^*)^T\right]}{2} + \frac{3\,\mathrm{Tr}\left[y_2\ (y_2^*)^T\right]}{2} + 6\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\right]\right)
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{HS_1}^{(A)}}^{(1)} = &- 9g_2^2\lambda_{HS_1}^{(A)} \\
&+ g_1^2\left(6g_2^2 - 3\lambda_{HS_1}^{(A)}\right) \\
&+ \lambda_{12}^{(A)}\lambda_{HS_2}^{(A)} \\
&+ 2\lambda_1\lambda_{HS_1}^{(A)}
\end{aligned}
$$

$$
\begin{aligned}
&+ 8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)} \\
&+ 2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)} \\
&+ \lambda_{HS_1}^{(A)}\left(3\,\mathrm{Tr}\left[y_1\ (y_1^*)^T\right] + 6\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\right]\right) \\
&+ \frac{16\lambda_5\lambda_5^*}{3} \\
&+ 64\lambda_{24}\lambda_{24}^* \\
&- \frac{4\,\mathrm{Tr}\left[y_1\ (y_1^*)^T\ Y_e\ (Y_e^*)^T\right]}{3}
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda_{HS_2}^{(A)}} =& -9g_2^2\lambda_{HS_2}^{(A)} \\
&+ g_1^2\left(18g_2^2 - 15\lambda_{HS_2}^{(A)}\right) \\
&+ \lambda_{12}^{(A)}\lambda_{HS_1}^{(A)} \\
&+ 2\lambda_1\lambda_{HS_2}^{(A)} \\
&+ 8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)} \\
&+ 2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)} \\
&+ \lambda_{HS_2}^{(A)}\left(3\,\mathrm{Tr}\left[y_2\ (y_2^*)^T\right] + 6\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\right]\right) \\
&+ 16\lambda_5\lambda_5^* \\
&- 4\,\mathrm{Tr}\left[y_2\ (y_2^*)^T\ Y_e\ (Y_e^*)^T\right]
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda_{12}^{(A)}} =& -9g_2^2\lambda_{12}^{(A)} \\
&+ g_1^2\left(18g_2^2 - 15\lambda_{12}^{(A)}\right) \\
&+ 8\lambda_{12}^{(1)}\lambda_{12}^{(A)} \\
&+ 2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)} \\
&+ 2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)} \\
&+ \lambda_{12}^{(A)}\left(3\,\mathrm{Tr}\left[y_1\ (y_1^*)^T\right] + 3\,\mathrm{Tr}\left[y_2\ (y_2^*)^T\right]\right) \\
&+ \lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)} \\
&+ \frac{4\lambda_5\lambda_5^*}{3} \\
&- \frac{2\,\mathrm{Tr}\left[y_1\ (y_1^*)^T\ y_2\ (y_2^*)^T\right]}{3}
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda_{24}} =& -9g_2^2\lambda_{24} \\
&- 3g_1^2\lambda_{24} \\
&+ 2\lambda_1\lambda_{24} \\
&+ 8\lambda_{HS_1}^{(1)}\lambda_{24} \\
&+ 4\lambda_{HS_1}^{(A)}\lambda_{24}
\end{aligned}
$$

$$+\ 2\lambda_{S_1}^{(1)}\lambda_{24}$$
$$+\ \lambda_{24}\left(3\,\mathrm{Tr}\left[y_1\ \left(y_1^*\right)^T\right]+6\,\mathrm{Tr}\left[Y_d\ \left(Y_d^*\right)^T\right]+2\,\mathrm{Tr}\left[Y_e\ \left(Y_e^*\right)^T\right]+6\,\mathrm{Tr}\left[Y_u\ \left(Y_u^*\right)^T\right]\right)$$

T3 − D ($\alpha = -2$)

$$\beta_{m_1^2}^{(1)} = -\ 12g_2^2m_1^2$$
$$-\ 6g_1^2m_1^2$$
$$+\ 2\lambda_{12}^{(1)}m_2^2$$
$$+\ 4\lambda_{18}\left(\mu_{S_2}^2\right)^*$$
$$+\ m_1^2\left(8\lambda_{S_1}^{(1)}+4\lambda_{S_1}^{(A)}+2\,\mathrm{Tr}\left[y_1\ \left(y_1^*\right)^T\right]\right)$$
$$+\ 4\mu_{S_2}^2\lambda_{18}^*$$
$$-\ 4\,\mathrm{Tr}\left[M_F\ \left(M_F^*\right)^T\ \left(y_1\right)^T\ y_1^*\right]$$

$$\beta_{m_2^2}^{(1)} = 6\lambda_{12}^{(1)}m_1^2$$
$$+\ 12\lambda_{14}\left(\mu_{S_2}^2\right)^*$$
$$+\ m_2^2\left(4\lambda_{S_2}^{(1)}+4\,\mathrm{Tr}\left[y_2\ \left(y_2^*\right)^T\right]\right)$$
$$+\ 12\mu_{S_2}^2\lambda_{14}^*$$
$$-\ 8\,\mathrm{Tr}\left[M_F\ \left(y_2^*\right)^T\ y_2\ \left(M_F^*\right)^T\right]$$

$$\beta_{\lambda_1}^{(1)} = \frac{9g_2^4}{4}$$
$$-\ 9g_2^2\lambda_1$$
$$+\ \frac{3g_1^4}{4}$$
$$+\ g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$$
$$+\ 12\lambda_1^2$$
$$+\ \lambda_1\left(12\,\mathrm{Tr}\left[Y_d\ \left(Y_d^*\right)^T\right]+4\,\mathrm{Tr}\left[Y_e\ \left(Y_e^*\right)^T\right]+12\,\mathrm{Tr}\left[Y_u\ \left(Y_u^*\right)^T\right]\right)$$
$$+\ 6\left(\lambda_{HS_1}^{(1)}\right)^2$$
$$+\ 3\left(\lambda_{HS_1}^{(A)}\right)^2$$
$$+\ 2\left(\lambda_{HS_2}^{(1)}\right)^2$$
$$+\ 4\lambda_5\lambda_5^*$$
$$+\ 4\lambda_{22}\lambda_{22}^*$$
$$+\ 8\lambda_{25}\lambda_{25}^*$$
$$-\ 12\,\mathrm{Tr}\left[Y_d\ \left(Y_d^*\right)^T\ Y_d\ \left(Y_d^*\right)^T\right]$$
$$-\ 4\,\mathrm{Tr}\left[Y_e\ \left(Y_e^*\right)^T\ Y_e\ \left(Y_e^*\right)^T\right]$$
$$-\ 12\,\mathrm{Tr}\left[Y_u\ \left(Y_u^*\right)^T\ Y_u\ \left(Y_u^*\right)^T\right]$$

$$\beta^{(1)}_{\lambda^{(1)}_{S_1}} = 22 g_2^4$$
$$- \ 24 g_2^2 \lambda^{(1)}_{S_1}$$
$$+ \ 12 g_1^4$$
$$+ \ g_1^2 \left(8 g_2^2 - 12 \lambda^{(1)}_{S_1}\right)$$
$$+ \ 2 \left(\lambda^{(1)}_{12}\right)^2$$
$$+ \ 4 \left(\lambda^{(1)}_{HS_1}\right)^2$$
$$+ \ 14 \left(\lambda^{(1)}_{S_1}\right)^2$$
$$+ \ \lambda^{(1)}_{S_1} \left(8 \lambda^{(A)}_{S_1} + 4 \, \mathrm{Tr}\left[y_1 \ (y_1^*)^T\right]\right)$$
$$+ \ 16 \left(\lambda^{(A)}_{S_1}\right)^2$$
$$+ \ 8 \lambda_{18} \lambda^*_{18}$$
$$- \ \frac{8 \, \mathrm{Tr}\left[y_1 \ (y_1^*)^T \ y_1 \ (y_1^*)^T\right]}{3}$$

$$\beta^{(1)}_{\lambda^{(1)}_{S_2}} = 6 \left(\lambda^{(1)}_{12}\right)^2$$
$$+ \ 4 \left(\lambda^{(1)}_{HS_2}\right)^2$$
$$+ \ 10 \left(\lambda^{(1)}_{S_2}\right)^2$$
$$+ \ 8 \lambda^{(1)}_{S_2} \, \mathrm{Tr}\left[y_2 \ (y_2^*)^T\right]$$
$$+ \ 288 \lambda_{13} \lambda^*_{13}$$
$$+ \ 108 \lambda_{14} \lambda^*_{14}$$
$$+ \ 12 \lambda_{18} \lambda^*_{18}$$
$$+ \ 8 \lambda_{25} \lambda^*_{25}$$
$$- \ 8 \, \mathrm{Tr}\left[y_2 \ (y_2^*)^T \ y_2 \ (y_2^*)^T\right]$$

$$\beta^{(1)}_{\lambda^{(1)}_{HS_1}} = 6 g_2^4$$
$$- \ \frac{33 g_2^2 \lambda^{(1)}_{HS_1}}{2}$$
$$+ \ 3 g_1^4$$
$$- \ \frac{15 g_1^2 \lambda^{(1)}_{HS_1}}{2}$$
$$+ \ 2 \lambda^{(1)}_{12} \lambda^{(1)}_{HS_2}$$
$$+ \ 6 \lambda_1 \lambda^{(1)}_{HS_1}$$

$$
\begin{aligned}
&+ 4\left(\lambda_{HS_1}^{(1)}\right)^2 \\
&+ 8\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)} \\
&+ \lambda_{HS_1}^{(1)}\left(4\lambda_{S_1}^{(A)} + 2\,\mathrm{Tr}\Big[y_1\ (y_1^*)^T\Big] + 6\,\mathrm{Tr}\Big[Y_d\ (Y_d^*)^T\Big] + 2\,\mathrm{Tr}\Big[Y_e\ (Y_e^*)^T\Big] + 6\,\mathrm{Tr}\Big[Y_u\ (Y_u^*)^T\Big]\right) \\
&+ 6\left(\lambda_{HS_1}^{(A)}\right)^2 \\
&+ 4\lambda_5\lambda_5^* \\
&+ 4\lambda_{18}\lambda_{25}^* \\
&+ 4\lambda_{22}\lambda_{22}^* \\
&+ 4\lambda_{25}\lambda_{18}^* \\
&- 2\,\mathrm{Tr}\Big[y_1\ (y_1^*)^T\ Y_e\ (Y_e^*)^T\Big]
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda^{(1)}_{HS_2}} =& -\frac{9g_2^2\lambda_{HS_2}^{(1)}}{2} \\
&- \frac{3g_1^2\lambda_{HS_2}^{(1)}}{2} \\
&+ 6\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)} \\
&+ 6\lambda_1\lambda_{HS_2}^{(1)} \\
&+ 4\left(\lambda_{HS_2}^{(1)}\right)^2 \\
&+ 4\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)} \\
&+ \lambda_{HS_2}^{(1)}\left(4\,\mathrm{Tr}\Big[y_2\ (y_2^*)^T\Big] + 6\,\mathrm{Tr}\Big[Y_d\ (Y_d^*)^T\Big] + 2\,\mathrm{Tr}\Big[Y_e\ (Y_e^*)^T\Big] + 6\,\mathrm{Tr}\Big[Y_u\ (Y_u^*)^T\Big]\right) \\
&+ 12\lambda_5\lambda_5^* \\
&+ 12\lambda_{14}\lambda_{25}^* \\
&+ 12\lambda_{22}\lambda_{22}^* \\
&+ 4\lambda_{25}\left(3\lambda_{14}^* + 4\lambda_{25}^*\right) \\
&- 4\,\mathrm{Tr}\Big[y_2\ (y_2^*)^T\ Y_e\ (Y_e^*)^T\Big]
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda^{(1)}_{12}} =& -12g_2^2\lambda_{12}^{(1)} \\
&- 6g_1^2\lambda_{12}^{(1)} \\
&+ 4\left(\lambda_{12}^{(1)}\right)^2 \\
&+ 8\lambda_{12}^{(1)}\lambda_{S_1}^{(1)} \\
&+ 4\lambda_{12}^{(1)}\lambda_{S_2}^{(1)} \\
&+ \lambda_{12}^{(1)}\left(4\lambda_{S_1}^{(A)} + 2\,\mathrm{Tr}\Big[y_1\ (y_1^*)^T\Big] + 4\,\mathrm{Tr}\Big[y_2\ (y_2^*)^T\Big]\right) \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}
\end{aligned}
$$

$$
\begin{aligned}
&+ 4\lambda_5\lambda_5^* \\
&+ 12\lambda_{14}\lambda_{18}^* \\
&+ 4\lambda_{18}\left(3\lambda_{14}^* + 4\lambda_{18}^*\right) \\
&+ 4\lambda_{22}\lambda_{22}^* \\
&- 4\,\mathrm{Tr}\left[y_1\ \left(y_1^*\right)^T\ y_2\ \left(y_2^*\right)^T\right]
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda_5} =& -\frac{21g_2^2\lambda_5}{2} \\
&- \frac{9g_1^2\lambda_5}{2} \\
&+ 2\lambda_{12}^{(1)}\lambda_5 \\
&+ 2\lambda_1\lambda_5 \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_5 \\
&+ 4\lambda_{HS_2}^{(1)}\lambda_5 \\
&+ \lambda_5\left(-4\lambda_{HS_1}^{(A)} + \mathrm{Tr}\left[y_1\ \left(y_1^*\right)^T\right] + 2\,\mathrm{Tr}\left[y_2\ \left(y_2^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_d\ \left(Y_d^*\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ \left(Y_e^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ \left(Y_u^*\right)^T\right]\right) \\
&+ 4\lambda_{18}\lambda_{22}^* \\
&+ 8\lambda_{25}\lambda_{22}^*
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda_{HS_1}^{(A)}} =& -\frac{33g_2^2\lambda_{HS_1}^{(A)}}{2} \\
&+ g_1^2\left(-4g_2^2 - \frac{15\lambda_{HS_1}^{(A)}}{2}\right) \\
&+ 2\lambda_1\lambda_{HS_1}^{(A)} \\
&+ 8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)} \\
&+ 2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)} \\
&+ \lambda_{HS_1}^{(A)}\left(16\lambda_{S_1}^{(A)} + 2\,\mathrm{Tr}\left[y_1\ \left(y_1^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_d\ \left(Y_d^*\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ \left(Y_e^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ \left(Y_u^*\right)^T\right]\right) \\
&- \frac{8\lambda_5\lambda_5^*}{3} \\
&- \frac{8\lambda_{22}\lambda_{22}^*}{3} \\
&- \frac{4\,\mathrm{Tr}\left[y_1\ \left(y_1^*\right)^T\ Y_e\ \left(Y_e^*\right)^T\right]}{3}
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda_{S_1}^{(A)}} =& -2g_2^4 \\
&- 24g_2^2\lambda_{S_1}^{(A)} \\
&+ g_1^2\left(8g_2^2 - 12\lambda_{S_1}^{(A)}\right) \\
&+ 3\left(\lambda_{HS_1}^{(A)}\right)^2 \\
&+ 12\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)}
\end{aligned}
$$

$$
\begin{aligned}
&+ 20 \left(\lambda_{S_1}^{(A)}\right)^2 \\
&+ 4\lambda_{S_1}^{(A)} \operatorname{Tr}\left[y_1 \left(y_1^*\right)^T\right] \\
&- \frac{2 \operatorname{Tr}\left[y_1 \left(y_1^*\right)^T y_1 \left(y_1^*\right)^T\right]}{3}
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{13}}^{(1)} = & 12\lambda_{S_2}^{(1)}\lambda_{13} \\
&+ 8\lambda_{13} \operatorname{Tr}\left[y_2 \left(y_2^*\right)^T\right] \\
&+ 9\lambda_{14}^2 \\
&+ 3\lambda_{18}^2 \\
&+ 2\lambda_{25}^2
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{14}}^{(1)} = & 6\lambda_{12}^{(1)}\lambda_{18} \\
&+ 4\lambda_{HS_2}^{(1)}\lambda_{25} \\
&+ 18\lambda_{S_2}^{(1)}\lambda_{14} \\
&+ 72\lambda_{13}\lambda_{14}^* \\
&+ 8\lambda_{14} \operatorname{Tr}\left[y_2 \left(y_2^*\right)^T\right]
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{18}}^{(1)} = & -12g_2^2\lambda_{18} \\
&- 6g_1^2\lambda_{18} \\
&+ \lambda_{12}^{(1)} \left(6\lambda_{14} + 8\lambda_{18}\right) \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_{25} \\
&+ 8\lambda_{S_1}^{(1)}\lambda_{18} \\
&+ 4\lambda_{S_1}^{(A)}\lambda_{18} \\
&+ 2\lambda_{S_2}^{(1)}\lambda_{18} \\
&+ 4\lambda_5\lambda_{22} \\
&+ 24\lambda_{13}\lambda_{18}^* \\
&+ \lambda_{18} \left(2 \operatorname{Tr}\left[y_1 \left(y_1^*\right)^T\right] + 4 \operatorname{Tr}\left[y_2 \left(y_2^*\right)^T\right]\right)
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{22}}^{(1)} = & -\frac{21g_2^2\lambda_{22}}{2} \\
&- \frac{9g_1^2\lambda_{22}}{2} \\
&+ 2\lambda_{12}^{(1)}\lambda_{22} \\
&+ 2\lambda_1\lambda_{22} \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_{22} \\
&- 4\lambda_{HS_1}^{(A)}\lambda_{22} \\
&+ 4\lambda_{HS_2}^{(1)}\lambda_{22} \\
&+ 4\lambda_{18}\lambda_5^* \\
&+ \lambda_{22} \left(\operatorname{Tr}\left[y_1 \left(y_1^*\right)^T\right] + 2 \operatorname{Tr}\left[y_2 \left(y_2^*\right)^T\right] + 6 \operatorname{Tr}\left[Y_d \left(Y_d^*\right)^T\right] + 2 \operatorname{Tr}\left[Y_e \left(Y_e^*\right)^T\right] + 6 \operatorname{Tr}\left[Y_u \left(Y_u^*\right)^T\right]\right) \\
&+ 8\lambda_{25}\lambda_5^*
\end{aligned}
$$

$$\begin{aligned}\beta^{(1)}_{\lambda_{25}} =& -\frac{9g_2^2\lambda_{25}}{2}\\ &-\frac{3g_1^2\lambda_{25}}{2}\\ &+6\lambda_1\lambda_{25}\\ &+6\lambda^{(1)}_{HS_1}\lambda_{18}\\ &+\lambda^{(1)}_{HS_2}\left(6\lambda_{14}+8\lambda_{25}\right)\\ &+2\lambda^{(1)}_{S_2}\lambda_{25}\\ &+12\lambda_5\lambda_{22}\\ &+24\lambda_{13}\lambda^*_{25}\\ &+\lambda_{25}\left(4\,\mathrm{Tr}\left[y_2\ \left(y_2^*\right)^T\right]+6\,\mathrm{Tr}\left[Y_d\ \left(Y_d^*\right)^T\right]+2\,\mathrm{Tr}\left[Y_e\ \left(Y_e^*\right)^T\right]+6\,\mathrm{Tr}\left[Y_u\ \left(Y_u^*\right)^T\right]\right)\end{aligned}$$

$$\begin{aligned}\beta^{(1)}_{\mu^2_{S_2}} =& 2\lambda^{(1)}_{S_2}\mu^2_{S_2}\\ &+24\lambda_{13}\left(\mu^2_{S_2}\right)^*\\ &+6\lambda_{14}m_2^2\\ &+6\lambda_{18}m_1^2\\ &+4\mu^2_{S_2}\,\mathrm{Tr}\left[y_2\ \left(y_2^*\right)^T\right]\end{aligned}$$

$\mathrm{T3-D}\,(\alpha=0)$

$$\begin{aligned}\beta^{(1)}_{m_1^2} =& -12g_2^2m_1^2\\ &+2\lambda^{(1)}_{12}m_2^2\\ &+20\lambda_{20}\left(\mu^2_{S_1}\right)^*\\ &+m_1^2\left(8\lambda^{(1)}_{S_1}+4\lambda^{(A)}_{S_1}+2\,\mathrm{Tr}\left[y_1\ \left(y_1^*\right)^T\right]\right)\\ &+20\mu^2_{S_1}\lambda^*_{20}\\ &-4\,\mathrm{Tr}\left[M_F\ \left(M_F^*\right)^T\ \left(y_1\right)^T\ y_1^*\right]\end{aligned}$$

$$\begin{aligned}\beta^{(1)}_{m_2^2} =& -6g_1^2m_2^2\\ &+6\lambda^{(1)}_{12}m_1^2\\ &+12\lambda_{15}\left(\mu^2_{S_1}\right)^*\\ &+m_2^2\left(4\lambda^{(1)}_{S_2}+4\,\mathrm{Tr}\left[y_2\ \left(y_2^*\right)^T\right]\right)\\ &+12\mu^2_{S_1}\lambda^*_{15}\\ &-8\,\mathrm{Tr}\left[M_F\ \left(y_2^*\right)^T\ y_2\ \left(M_F^*\right)^T\right]\end{aligned}$$

$$\begin{aligned}\beta^{(1)}_{\lambda_1} =& \frac{9g_2^4}{4}\\ &-9g_2^2\lambda_1\\ &+\frac{3g_1^4}{4}\\ &+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)\end{aligned}$$

$$
\begin{aligned}
&+ 12\lambda_1^2 \\
&+ \lambda_1 \left(12 \operatorname{Tr}\left[Y_d \left(Y_d^*\right)^T\right] + 4 \operatorname{Tr}\left[Y_e \left(Y_e^*\right)^T\right] + 12 \operatorname{Tr}\left[Y_u \left(Y_u^*\right)^T\right]\right) \\
&+ 6 \left(\lambda_{HS_1}^{(1)}\right)^2 \\
&+ 3 \left(\lambda_{HS_1}^{(A)}\right)^2 \\
&+ 2 \left(\lambda_{HS_2}^{(1)}\right)^2 \\
&+ 4\lambda_5\lambda_5^* \\
&+ 4\lambda_{23}\lambda_{23}^* \\
&+ 24\lambda_{27}\lambda_{27}^* \\
&- 12 \operatorname{Tr}\left[Y_d \left(Y_d^*\right)^T Y_d \left(Y_d^*\right)^T\right] \\
&- 4 \operatorname{Tr}\left[Y_e \left(Y_e^*\right)^T Y_e \left(Y_e^*\right)^T\right] \\
&- 12 \operatorname{Tr}\left[Y_u \left(Y_u^*\right)^T Y_u \left(Y_u^*\right)^T\right]
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{S_1}^{(1)}}^{(1)} = {} & 22g_2^4 \\
&- 24g_2^2\lambda_{S_1}^{(1)} \\
&+ 2 \left(\lambda_{12}^{(1)}\right)^2 \\
&+ 4 \left(\lambda_{HS_1}^{(1)}\right)^2 \\
&+ 14 \left(\lambda_{S_1}^{(1)}\right)^2 \\
&+ \lambda_{S_1}^{(1)} \left(8\lambda_{S_1}^{(A)} + 4 \operatorname{Tr}\left[y_1 \left(y_1^*\right)^T\right]\right) \\
&+ 16 \left(\lambda_{S_1}^{(A)}\right)^2 \\
&+ \frac{8\lambda_{15}\lambda_{15}^*}{3} \\
&+ \frac{832\lambda_{17}\lambda_{17}^*}{3} \\
&+ \frac{352\lambda_{20}\lambda_{20}^*}{3} \\
&+ \frac{16\lambda_{27}\lambda_{27}^*}{3} \\
&- \frac{8 \operatorname{Tr}\left[y_1 \left(y_1^*\right)^T y_1 \left(y_1^*\right)^T\right]}{3}
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{S_2}^{(1)}}^{(1)} = {} & 12g_1^4 \\
&- 12g_1^2\lambda_{S_2}^{(1)} \\
&+ 6 \left(\lambda_{12}^{(1)}\right)^2 \\
&+ 4 \left(\lambda_{HS_2}^{(1)}\right)^2 \\
&+ 10 \left(\lambda_{S_2}^{(1)}\right)^2
\end{aligned}
$$

$$+ \ 8\lambda_{S_2}^{(1)} \operatorname{Tr}\left[y_2 \ (y_2^*)^T\right]$$

$$+ \ 24\lambda_{15}\lambda_{15}^*$$

$$- \ 8 \operatorname{Tr}\left[y_2 \ (y_2^*)^T \ y_2 \ (y_2^*)^T\right]$$

$$\beta^{(1)}_{\lambda^{(1)}_{HS_1}} = 6g_2^4$$

$$- \ \frac{33g_2^2\lambda_{HS_1}^{(1)}}{2}$$

$$- \ \frac{3g_1^2\lambda_{HS_1}^{(1)}}{2}$$

$$+ \ 2\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$$

$$+ \ 6\lambda_1\lambda_{HS_1}^{(1)}$$

$$+ \ 4\left(\lambda_{HS_1}^{(1)}\right)^2$$

$$+ \ 8\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$$

$$+ \ \lambda_{HS_1}^{(1)}\left(4\lambda_{S_1}^{(A)} + 2\operatorname{Tr}\left[y_1 \ (y_1^*)^T\right] + 6\operatorname{Tr}\left[Y_d \ (Y_d^*)^T\right] + 2\operatorname{Tr}\left[Y_e \ (Y_e^*)^T\right] + 6\operatorname{Tr}\left[Y_u \ (Y_u^*)^T\right]\right)$$

$$+ \ 6\left(\lambda_{HS_1}^{(A)}\right)^2$$

$$+ \ 4\lambda_5\lambda_5^*$$

$$+ \ 20\lambda_{20}\lambda_{27}^*$$

$$+ \ 4\lambda_{23}\lambda_{23}^*$$

$$+ \ 4\lambda_{27}\left(5\lambda_{20}^* + 4\lambda_{27}^*\right)$$

$$- \ 2\operatorname{Tr}\left[y_1 \ (y_1^*)^T \ Y_e \ (Y_e^*)^T\right]$$

$$\beta^{(1)}_{\lambda^{(1)}_{HS_2}} = - \ \frac{9g_2^2\lambda_{HS_2}^{(1)}}{2}$$

$$+ \ 3g_1^4$$

$$- \ \frac{15g_1^2\lambda_{HS_2}^{(1)}}{2}$$

$$+ \ 6\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$$

$$+ \ 6\lambda_1\lambda_{HS_2}^{(1)}$$

$$+ \ 4\left(\lambda_{HS_2}^{(1)}\right)^2$$

$$+ \ 4\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$$

$$+ \ \lambda_{HS_2}^{(1)}\left(4\operatorname{Tr}\left[y_2 \ (y_2^*)^T\right] + 6\operatorname{Tr}\left[Y_d \ (Y_d^*)^T\right] + 2\operatorname{Tr}\left[Y_e \ (Y_e^*)^T\right] + 6\operatorname{Tr}\left[Y_u \ (Y_u^*)^T\right]\right)$$

$$+ \ 12\lambda_5\lambda_5^*$$

$$+ \ 12\lambda_{15}\lambda_{27}^*$$

$$+ 12\lambda_{23}\lambda_{23}^{*}$$

$$+ 12\lambda_{27}\lambda_{15}^{*}$$

$$- 4\,\mathrm{Tr}\left[y_2\ \left(y_2^{*}\right)^{T}\ Y_e\ \left(Y_e^{*}\right)^{T}\right]$$

$$\beta^{(1)}_{\lambda_{12}^{(1)}} = - 12g_2^2\lambda_{12}^{(1)}$$

$$- 6g_1^2\lambda_{12}^{(1)}$$

$$+ 4\left(\lambda_{12}^{(1)}\right)^2$$

$$+ 8\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$$

$$+ 4\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$$

$$+ \lambda_{12}^{(1)}\left(4\lambda_{S_1}^{(A)} + 2\,\mathrm{Tr}\left[y_1\ \left(y_1^{*}\right)^{T}\right] + 4\,\mathrm{Tr}\left[y_2\ \left(y_2^{*}\right)^{T}\right]\right)$$

$$+ 4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$$

$$+ 4\lambda_5\lambda_5^{*}$$

$$+ 4\lambda_{15}\left(4\lambda_{15}^{*} + 5\lambda_{20}^{*}\right)$$

$$+ 20\lambda_{20}\lambda_{15}^{*}$$

$$+ 4\lambda_{23}\lambda_{23}^{*}$$

$$- 4\,\mathrm{Tr}\left[y_1\ \left(y_1^{*}\right)^{T}\ y_2\ \left(y_2^{*}\right)^{T}\right]$$

$$\beta^{(1)}_{\lambda_5} = - \frac{21g_2^2\lambda_5}{2}$$

$$- \frac{9g_1^2\lambda_5}{2}$$

$$+ 2\lambda_{12}^{(1)}\lambda_5$$

$$+ 2\lambda_1\lambda_5$$

$$+ 4\lambda_{HS_1}^{(1)}\lambda_5$$

$$+ 4\lambda_{HS_2}^{(1)}\lambda_5$$

$$+ \lambda_5\left(-4\lambda_{HS_1}^{(A)} + \mathrm{Tr}\left[y_1\ \left(y_1^{*}\right)^{T}\right] + 2\,\mathrm{Tr}\left[y_2\ \left(y_2^{*}\right)^{T}\right] + 6\,\mathrm{Tr}\left[Y_d\ \left(Y_d^{*}\right)^{T}\right] + 2\,\mathrm{Tr}\left[Y_e\ \left(Y_e^{*}\right)^{T}\right] + 6\,\mathrm{Tr}\left[Y_u\ \left(Y_u^{*}\right)^{T}\right]\right)$$

$$+ 4\left(\lambda_{15}^{*} + 2\lambda_{27}^{*}\right)\lambda_{23}^{*}$$

$$\beta^{(1)}_{\lambda_{HS_1}^{(A)}} = - \frac{33g_2^2\lambda_{HS_1}^{(A)}}{2}$$

$$- \frac{3g_1^2\lambda_{HS_1}^{(A)}}{2}$$

$$+ 2\lambda_1\lambda_{HS_1}^{(A)}$$

$$+ 8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$$

$$+ 2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$$

$$+ \lambda_{HS_1}^{(A)}\left(16\lambda_{S_1}^{(A)} + 2\,\mathrm{Tr}\left[y_1\ \left(y_1^{*}\right)^{T}\right] + 6\,\mathrm{Tr}\left[Y_d\ \left(Y_d^{*}\right)^{T}\right] + 2\,\mathrm{Tr}\left[Y_e\ \left(Y_e^{*}\right)^{T}\right] + 6\,\mathrm{Tr}\left[Y_u\ \left(Y_u^{*}\right)^{T}\right]\right)$$

$$
\begin{aligned}
&- \frac{8\lambda_5\lambda_5^*}{3} \\
&+ \frac{8\lambda_{23}\lambda_{23}^*}{3} \\
&- \frac{4\,\mathrm{Tr}\left[y_1\ (y_1^*)^T\ Y_e\ (Y_e^*)^T\right]}{3}
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda_{S_1}^{(A)}} =& -2g_2^4 \\
&- 24g_2^2\lambda_{S_1}^{(A)} \\
&+ 3\left(\lambda_{HS_1}^{(A)}\right)^2 \\
&+ 12\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)} \\
&+ 20\left(\lambda_{S_1}^{(A)}\right)^2 \\
&+ 4\lambda_{S_1}^{(A)}\,\mathrm{Tr}\left[y_1\ (y_1^*)^T\right] \\
&- \frac{4\lambda_{15}\lambda_{15}^*}{3} \\
&- \frac{224\lambda_{17}\lambda_{17}^*}{3} \\
&- \frac{44\lambda_{20}\lambda_{20}^*}{3} \\
&- \frac{8\lambda_{27}\lambda_{27}^*}{3} \\
&- \frac{2\,\mathrm{Tr}\left[y_1\ (y_1^*)^T\ y_1\ (y_1^*)^T\right]}{3}
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda_{15}} =& -12g_2^2\lambda_{15} \\
&- 6g_1^2\lambda_{15} \\
&+ \lambda_{12}^{(1)}\left(8\lambda_{15} + 10\lambda_{20}\right) \\
&+ 4\lambda_{HS_2}^{(1)}\lambda_{27} \\
&+ 2\lambda_{S_1}^{(1)}\lambda_{15} \\
&- 4\lambda_{S_1}^{(A)}\lambda_{15} \\
&+ 4\lambda_{S_2}^{(1)}\lambda_{15} \\
&+ \lambda_{15}\left(2\,\mathrm{Tr}\left[y_1\ (y_1^*)^T\right] + 4\,\mathrm{Tr}\left[y_2\ (y_2^*)^T\right]\right) \\
&+ 40\lambda_{17}\lambda_{15}^* \\
&+ 4\lambda_5^*\lambda_{23}^*
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda_{17}} =& -24g_2^2\lambda_{17} \\
&+ 12\lambda_{S_1}^{(1)}\lambda_{17} \\
&- 8\lambda_{S_1}^{(A)}\lambda_{17} \\
&+ \lambda_{15}^2 \\
&+ 4\lambda_{17}\,\mathrm{Tr}\left[y_1\ (y_1^*)^T\right]
\end{aligned}
$$

$$
\begin{aligned}
&+ 11\lambda_{20}^2 \\
&+ 2\lambda_{27}^2
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{20}}^{(1)} =& - 24 g_2^2 \lambda_{20} \\
&+ 2\lambda_{12}^{(1)} \lambda_{15} \\
&+ 4\lambda_{HS_1}^{(1)} \lambda_{27} \\
&+ 22\lambda_{S_1}^{(1)} \lambda_{20} \\
&+ 88\lambda_{17}\lambda_{20}^* \\
&+ 4\lambda_{20} \operatorname{Tr}\left[y_1 \ (y_1^*)^T\right]
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{23}}^{(1)} =& - \frac{21 g_2^2 \lambda_{23}}{2} \\
&- \frac{9 g_1^2 \lambda_{23}}{2} \\
&+ 2\lambda_{12}^{(1)} \lambda_{23} \\
&+ 2\lambda_1 \lambda_{23} \\
&+ 4\lambda_{HS_1}^{(1)} \lambda_{23} \\
&+ 4\lambda_{HS_1}^{(A)} \lambda_{23} \\
&+ 4\lambda_{HS_2}^{(1)} \lambda_{23} \\
&+ \lambda_{23} \left(\operatorname{Tr}\left[y_1 \ (y_1^*)^T\right] + 2 \operatorname{Tr}\left[y_2 \ (y_2^*)^T\right] + 6 \operatorname{Tr}\left[Y_d \ (Y_d^*)^T\right] + 2 \operatorname{Tr}\left[Y_e \ (Y_e^*)^T\right] + 6 \operatorname{Tr}\left[Y_u \ (Y_u^*)^T\right]\right) \\
&+ 4 \left(\lambda_{15}^* + 2\lambda_{27}^*\right) \lambda_5^*
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{27}}^{(1)} =& - \frac{33 g_2^2 \lambda_{27}}{2} \\
&- \frac{3 g_1^2 \lambda_{27}}{2} \\
&+ 6\lambda_1 \lambda_{27} \\
&+ \lambda_{HS_1}^{(1)} \left(10\lambda_{20} + 8\lambda_{27}\right) \\
&+ 2\lambda_{HS_2}^{(1)} \lambda_{15} \\
&+ 2\lambda_{S_1}^{(1)} \lambda_{27} \\
&- 4\lambda_{S_1}^{(A)} \lambda_{27} \\
&+ 40\lambda_{17}\lambda_{27}^* \\
&+ \lambda_{27} \left(2 \operatorname{Tr}\left[y_1 \ (y_1^*)^T\right] + 6 \operatorname{Tr}\left[Y_d \ (Y_d^*)^T\right] + 2 \operatorname{Tr}\left[Y_e \ (Y_e^*)^T\right] + 6 \operatorname{Tr}\left[Y_u \ (Y_u^*)^T\right]\right) \\
&+ 4\lambda_5^* \lambda_{23}^*
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\mu_{S_1}^2}^{(1)} =& - 12 g_2^2 \mu_{S_1}^2 \\
&+ 2\lambda_{S_1}^{(1)} \mu_{S_1}^2 \\
&- 4\lambda_{S_1}^{(A)} \mu_{S_1}^2 \\
&+ 2\lambda_{15} m_2^2 \\
&+ 40\lambda_{17} \left(\mu_{S_1}^2\right)^* \\
&+ 10\lambda_{20} m_1^2
\end{aligned}
$$

$$+\ 2\mu_{S_1}^2\ \mathrm{Tr}\Big[y_1\ \left(y_1^*\right)^T\Big]$$

$\mathrm{T3-D}\,(\alpha=2)$

$$\beta^{(1)}_{m_1^2} = -\ 12g_2^2m_1^2$$
$$-\ 6g_1^2m_1^2$$
$$+\ 2\lambda_{12}^{(1)}m_2^2$$
$$+\ m_1^2\left(8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)} + 2\,\mathrm{Tr}\Big[y_1\ \left(y_1^*\right)^T\Big]\right)$$
$$-\ 4\,\mathrm{Tr}\Big[M_F\ \left(M_F^*\right)^T\ \left(y_1\right)^T\ y_1^*\Big]$$

$$\beta^{(1)}_{m_2^2} = -\ 24g_1^2m_2^2$$
$$+\ 6\lambda_{12}^{(1)}m_1^2$$
$$+\ m_2^2\left(4\lambda_{S_2}^{(1)} + 4\,\mathrm{Tr}\Big[y_2\ \left(y_2^*\right)^T\Big]\right)$$
$$-\ 8\,\mathrm{Tr}\Big[M_F\ \left(y_2^*\right)^T\ y_2\ \left(M_F^*\right)^T\Big]$$

$$\beta^{(1)}_{\lambda_1} = \frac{9g_2^4}{4}$$
$$-\ 9g_2^2\lambda_1$$
$$+\ \frac{3g_1^4}{4}$$
$$+\ g_1^2\left(\frac{3g_2^2}{2} - 3\lambda_1\right)$$
$$+\ 12\lambda_1^2$$
$$+\ \lambda_1\left(12\,\mathrm{Tr}\Big[Y_d\ \left(Y_d^*\right)^T\Big] + 4\,\mathrm{Tr}\Big[Y_e\ \left(Y_e^*\right)^T\Big] + 12\,\mathrm{Tr}\Big[Y_u\ \left(Y_u^*\right)^T\Big]\right)$$
$$+\ 6\left(\lambda_{HS_1}^{(1)}\right)^2$$
$$+\ 3\left(\lambda_{HS_1}^{(A)}\right)^2$$
$$+\ 2\left(\lambda_{HS_2}^{(1)}\right)^2$$
$$+\ 4\lambda_5\lambda_5^*$$
$$-\ 12\,\mathrm{Tr}\Big[Y_d\ \left(Y_d^*\right)^T\ Y_d\ \left(Y_d^*\right)^T\Big]$$
$$-\ 4\,\mathrm{Tr}\Big[Y_e\ \left(Y_e^*\right)^T\ Y_e\ \left(Y_e^*\right)^T\Big]$$
$$-\ 12\,\mathrm{Tr}\Big[Y_u\ \left(Y_u^*\right)^T\ Y_u\ \left(Y_u^*\right)^T\Big]$$

$$\beta^{(1)}_{\lambda_{S_1}^{(1)}} = 22g_2^4$$
$$-\ 24g_2^2\lambda_{S_1}^{(1)}$$
$$+\ 12g_1^4$$
$$+\ g_1^2\left(8g_2^2 - 12\lambda_{S_1}^{(1)}\right)$$
$$+\ 2\left(\lambda_{12}^{(1)}\right)^2$$

$$
\begin{aligned}
&+ 4\left(\lambda_{HS_1}^{(1)}\right)^2 \\
&+ 14\left(\lambda_{S_1}^{(1)}\right)^2 \\
&+ \lambda_{S_1}^{(1)}\left(8\lambda_{S_1}^{(A)} + 4\,\mathrm{Tr}\left[y_1\ (y_1^*)^T\right]\right) \\
&+ 16\left(\lambda_{S_1}^{(A)}\right)^2 \\
&- \frac{8\,\mathrm{Tr}\left[y_1\ (y_1^*)^T\ y_1\ (y_1^*)^T\right]}{3}
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda^{(1)}_{S_2}} &= 192g_1^4 \\
&- 48g_1^2\lambda_{S_2}^{(1)} \\
&+ 6\left(\lambda_{12}^{(1)}\right)^2 \\
&+ 4\left(\lambda_{HS_2}^{(1)}\right)^2 \\
&+ 10\left(\lambda_{S_2}^{(1)}\right)^2 \\
&+ 8\lambda_{S_2}^{(1)}\,\mathrm{Tr}\left[y_2\ (y_2^*)^T\right] \\
&- 8\,\mathrm{Tr}\left[y_2\ (y_2^*)^T\ y_2\ (y_2^*)^T\right]
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda^{(1)}_{HS_1}} &= 6g_2^4 \\
&- \frac{33g_2^2\lambda_{HS_1}^{(1)}}{2} \\
&+ 3g_1^4 \\
&- \frac{15g_1^2\lambda_{HS_1}^{(1)}}{2} \\
&+ 2\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)} \\
&+ 6\lambda_1\lambda_{HS_1}^{(1)} \\
&+ 4\left(\lambda_{HS_1}^{(1)}\right)^2 \\
&+ 8\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)} \\
&+ \lambda_{HS_1}^{(1)}\left(4\lambda_{S_1}^{(A)} + 2\,\mathrm{Tr}\left[y_1\ (y_1^*)^T\right] + 6\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\right]\right) \\
&+ 6\left(\lambda_{HS_1}^{(A)}\right)^2 \\
&+ 4\lambda_5\lambda_5^* \\
&- 2\,\mathrm{Tr}\left[y_1\ (y_1^*)^T\ Y_e\ (Y_e^*)^T\right]
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda^{(1)}_{HS_2}} &= -\frac{9g_2^2\lambda_{HS_2}^{(1)}}{2} \\
&+ 12g_1^4
\end{aligned}
$$

$$
\begin{aligned}
&- \frac{51 g_1^2 \lambda_{HS_2}^{(1)}}{2} \\
&+ 6 \lambda_{12}^{(1)} \lambda_{HS_1}^{(1)} \\
&+ 6 \lambda_1 \lambda_{HS_2}^{(1)} \\
&+ 4 \left(\lambda_{HS_2}^{(1)}\right)^2 \\
&+ 4 \lambda_{HS_2}^{(1)} \lambda_{S_2}^{(1)} \\
&+ \lambda_{HS_2}^{(1)} \left(4 \operatorname{Tr}\left[y_2 \, (y_2^*)^T\right] + 6 \operatorname{Tr}\left[Y_d \, (Y_d^*)^T\right] + 2 \operatorname{Tr}\left[Y_e \, (Y_e^*)^T\right] + 6 \operatorname{Tr}\left[Y_u \, (Y_u^*)^T\right]\right) \\
&+ 12 \lambda_5 \lambda_5^* \\
&- 4 \operatorname{Tr}\left[y_2 \, (y_2^*)^T \, Y_e \, (Y_e^*)^T\right]
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{12}^{(1)}}^{(1)} = &- 12 g_2^2 \lambda_{12}^{(1)} \\
&+ 48 g_1^4 \\
&- 30 g_1^2 \lambda_{12}^{(1)} \\
&+ 4 \left(\lambda_{12}^{(1)}\right)^2 \\
&+ 8 \lambda_{12}^{(1)} \lambda_{S_1}^{(1)} \\
&+ 4 \lambda_{12}^{(1)} \lambda_{S_2}^{(1)} \\
&+ \lambda_{12}^{(1)} \left(4 \lambda_{S_1}^{(A)} + 2 \operatorname{Tr}\left[y_1 \, (y_1^*)^T\right] + 4 \operatorname{Tr}\left[y_2 \, (y_2^*)^T\right]\right) \\
&+ 4 \lambda_{HS_1}^{(1)} \lambda_{HS_2}^{(1)} \\
&+ 4 \lambda_5 \lambda_5^* \\
&- 4 \operatorname{Tr}\left[y_1 \, (y_1^*)^T \, y_2 \, (y_2^*)^T\right]
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_5}^{(1)} = &- \frac{21 g_2^2 \lambda_5}{2} \\
&- \frac{33 g_1^2 \lambda_5}{2} \\
&+ 2 \lambda_{12}^{(1)} \lambda_5 \\
&+ 2 \lambda_1 \lambda_5 \\
&+ 4 \lambda_{HS_1}^{(1)} \lambda_5 \\
&+ 4 \lambda_{HS_2}^{(1)} \lambda_5 \\
&+ \lambda_5 \left(-4 \lambda_{HS_1}^{(A)} + \operatorname{Tr}\left[y_1 \, (y_1^*)^T\right] + 2 \operatorname{Tr}\left[y_2 \, (y_2^*)^T\right] + 6 \operatorname{Tr}\left[Y_d \, (Y_d^*)^T\right] + 2 \operatorname{Tr}\left[Y_e \, (Y_e^*)^T\right] + 6 \operatorname{Tr}\left[Y_u \, (Y_u^*)^T\right]\right)
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{HS_1}^{(A)}}^{(1)} = &- \frac{33 g_2^2 \lambda_{HS_1}^{(A)}}{2} \\
&+ g_1^2 \left(4 g_2^2 - \frac{15 \lambda_{HS_1}^{(A)}}{2}\right) \\
&+ 2 \lambda_1 \lambda_{HS_1}^{(A)}
\end{aligned}
$$

$$
\begin{aligned}
&+ 8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)} \\
&+ 2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)} \\
&+ \lambda_{HS_1}^{(A)}\left(16\lambda_{S_1}^{(A)} + 2\,\mathrm{Tr}\left[y_1\ \left(y_1^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_d\ \left(Y_d^*\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ \left(Y_e^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ \left(Y_u^*\right)^T\right]\right) \\
&- \frac{8\lambda_5\lambda_5^*}{3} \\
&- \frac{4\,\mathrm{Tr}\left[y_1\ \left(y_1^*\right)^T\ Y_e\ \left(Y_e^*\right)^T\right]}{3}
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda_{S_1}^{(A)}} =& -2g_2^4 \\
&- 24g_2^2\lambda_{S_1}^{(A)} \\
&+ g_1^2\left(8g_2^2 - 12\lambda_{S_1}^{(A)}\right) \\
&+ 3\left(\lambda_{HS_1}^{(A)}\right)^2 \\
&+ 12\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)} \\
&+ 20\left(\lambda_{S_1}^{(A)}\right)^2 \\
&+ 4\lambda_{S_1}^{(A)}\,\mathrm{Tr}\left[y_1\ \left(y_1^*\right)^T\right] \\
&- \frac{2\,\mathrm{Tr}\left[y_1\ \left(y_1^*\right)^T\ y_1\ \left(y_1^*\right)^T\right]}{3}
\end{aligned}
$$

$\mathrm{T3-E}\,(\alpha = -4)$

$$
\begin{aligned}
\beta^{(1)}_{m_1^2} =& -12g_2^2m_1^2 \\
&- 24g_1^2m_1^2 \\
&+ m_1^2\left(8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)} + 2\,\mathrm{Tr}\left[y_1\ \left(y_1^*\right)^T\right]\right) \\
&+ m_2^2\left(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)}\right) \\
&- 4\,\mathrm{Tr}\left[M_F\ \left(M_F^*\right)^T\ \left(y_1\right)^T\ y_1^*\right]
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{m_2^2} =& -12g_2^2m_2^2 \\
&- 6g_1^2m_2^2 \\
&+ m_1^2\left(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)}\right) \\
&+ m_2^2\left(8\lambda_{S_2}^{(1)} + 4\lambda_{S_2}^{(A)} + 2\,\mathrm{Tr}\left[y_2\ \left(y_2^*\right)^T\right]\right) \\
&- 4\,\mathrm{Tr}\left[M_F\ \left(y_2^*\right)^T\ y_2\ \left(M_F^*\right)^T\right]
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda_1} =& \frac{9g_2^4}{4} \\
&- 9g_2^2\lambda_1 \\
&+ \frac{3g_1^4}{4}
\end{aligned}
$$

$$
\begin{aligned}
&+ g_1^2 \left(\frac{3g_2^2}{2} - 3\lambda_1\right) \\
&+ 12\lambda_1^2 \\
&+ \lambda_1 \left(12\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\right] + 4\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\right] + 12\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\right]\right) \\
&+ 6\left(\lambda_{HS_1}^{(1)}\right)^2 \\
&+ 3\left(\lambda_{HS_1}^{(A)}\right)^2 \\
&+ 6\left(\lambda_{HS_2}^{(1)}\right)^2 \\
&+ 3\left(\lambda_{HS_2}^{(A)}\right)^2 \\
&+ 4\lambda_5\lambda_5^* \\
&- 12\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\ Y_d\ (Y_d^*)^T\right] \\
&- 4\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\ Y_e\ (Y_e^*)^T\right] \\
&- 12\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\ Y_u\ (Y_u^*)^T\right]
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{S_1}^{(1)}}^{(1)} &= 22g_2^4 \\
&- 24g_2^2\lambda_{S_1}^{(1)} \\
&+ 192g_1^4 \\
&+ g_1^2\left(32g_2^2 - 48\lambda_{S_1}^{(1)}\right) \\
&+ 6\left(\lambda_{12}^{(1)}\right)^2 \\
&+ 8\lambda_{12}^{(1)}\lambda_{12}^{(\times)} \\
&+ \frac{20\left(\lambda_{12}^{(\times)}\right)^2}{3} \\
&+ 4\left(\lambda_{HS_1}^{(1)}\right)^2 \\
&+ 14\left(\lambda_{S_1}^{(1)}\right)^2 \\
&+ \lambda_{S_1}^{(1)}\left(8\lambda_{S_1}^{(A)} + 4\,\mathrm{Tr}\left[y_1\ (y_1^*)^T\right]\right) \\
&+ 16\left(\lambda_{S_1}^{(A)}\right)^2 \\
&- \frac{8\,\mathrm{Tr}\left[y_1\ (y_1^*)^T\ y_1\ (y_1^*)^T\right]}{3}
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{S_2}^{(1)}}^{(1)} &= 22g_2^4 \\
&- 24g_2^2\lambda_{S_2}^{(1)} \\
&+ 12g_1^4 \\
&+ g_1^2\left(8g_2^2 - 12\lambda_{S_2}^{(1)}\right)
\end{aligned}
$$

$$
\begin{aligned}
&+ 6\left(\lambda_{12}^{(1)}\right)^2 \\
&+ 8\lambda_{12}^{(1)}\lambda_{12}^{(\times)} \\
&+ \frac{20\left(\lambda_{12}^{(\times)}\right)^2}{3} \\
&+ 4\left(\lambda_{HS_2}^{(1)}\right)^2 \\
&+ 14\left(\lambda_{S_2}^{(1)}\right)^2 \\
&+ \lambda_{S_2}^{(1)}\left(8\lambda_{S_2}^{(A)} + 4\,\mathrm{Tr}\left[y_2\ \left(y_2^*\right)^T\right]\right) \\
&+ 16\left(\lambda_{S_2}^{(A)}\right)^2 \\
&- \frac{8\,\mathrm{Tr}\left[y_2\ \left(y_2^*\right)^T\ y_2\ \left(y_2^*\right)^T\right]}{3}
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda^{(1)}_{HS_1}} &= 6g_2^4 \\
&- \frac{33g_2^2\lambda_{HS_1}^{(1)}}{2} \\
&+ 12g_1^4 \\
&- \frac{51g_1^2\lambda_{HS_1}^{(1)}}{2} \\
&+ 6\lambda_1\lambda_{HS_1}^{(1)} \\
&+ 4\left(\lambda_{HS_1}^{(1)}\right)^2 \\
&+ 8\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)} \\
&+ \lambda_{HS_1}^{(1)}\left(4\lambda_{S_1}^{(A)} + 2\,\mathrm{Tr}\left[y_1\ \left(y_1^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_d\ \left(Y_d^*\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ \left(Y_e^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ \left(Y_u^*\right)^T\right]\right) \\
&+ 6\left(\lambda_{HS_1}^{(A)}\right)^2 \\
&+ \lambda_{HS_2}^{(1)}\left(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)}\right) \\
&+ 4\lambda_5\lambda_5^* \\
&- 2\,\mathrm{Tr}\left[y_1\ \left(y_1^*\right)^T\ Y_e\ \left(Y_e^*\right)^T\right]
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda^{(1)}_{HS_2}} &= 6g_2^4 \\
&- \frac{33g_2^2\lambda_{HS_2}^{(1)}}{2} \\
&+ 3g_1^4 \\
&- \frac{15g_1^2\lambda_{HS_2}^{(1)}}{2} \\
&+ 6\lambda_1\lambda_{HS_2}^{(1)}
\end{aligned}
$$

$$
\begin{aligned}
&+ \lambda_{HS_1}^{(1)} \left(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)}\right) \\
&+ 4 \left(\lambda_{HS_2}^{(1)}\right)^2 \\
&+ 8\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)} \\
&+ \lambda_{HS_2}^{(1)} \left(4\lambda_{S_2}^{(A)} + 2\,\mathrm{Tr}\left[y_2\ \left(y_2^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_d\ \left(Y_d^*\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ \left(Y_e^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ \left(Y_u^*\right)^T\right]\right) \\
&+ 6 \left(\lambda_{HS_2}^{(A)}\right)^2 \\
&+ 4\lambda_5\lambda_5^* \\
&- 2\,\mathrm{Tr}\left[y_2\ \left(y_2^*\right)^T\ Y_e\ \left(Y_e^*\right)^T\right]
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{12}^{(1)}}^{(1)} = &\frac{63g_2^4}{5} \\
&- 24g_2^2\lambda_{12}^{(1)} \\
&+ 48g_1^4 \\
&- 30g_1^2\lambda_{12}^{(1)} \\
&+ 4 \left(\lambda_{12}^{(1)}\right)^2 \\
&+ \lambda_{12}^{(1)} \left(4\lambda_{S_1}^{(A)} + 4\lambda_{S_2}^{(A)} + 2\,\mathrm{Tr}\left[y_1\ \left(y_1^*\right)^T\right] + 2\,\mathrm{Tr}\left[y_2\ \left(y_2^*\right)^T\right]\right) \\
&+ 16 \left(\lambda_{12}^{(A)}\right)^2 \\
&+ 4 \left(\lambda_{12}^{(\times)}\right)^2 \\
&+ 4\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)} \\
&+ 4\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)} \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)} \\
&+ \lambda_{S_1}^{(1)} \left(8\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)}\right) \\
&+ \lambda_{S_2}^{(1)} \left(8\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)}\right) \\
&+ 2\lambda_5\lambda_5^* \\
&- \frac{22\,\mathrm{Tr}\left[y_1\ \left(y_1^*\right)^T\ y_2\ \left(y_2^*\right)^T\right]}{15}
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_5}^{(1)} = &- \frac{33g_2^2\lambda_5}{2} \\
&- \frac{33g_1^2\lambda_5}{2} \\
&+ 2\lambda_{12}^{(1)}\lambda_5 \\
&+ 2\lambda_1\lambda_5 \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_5 \\
&+ 4\lambda_{HS_2}^{(1)}\lambda_5 \\
&+ \lambda_5 \left(-\frac{4\lambda_{12}^{(A)}}{3} - 2\lambda_{12}^{(\times)} - 2\lambda_{HS_1}^{(A)} - \frac{4\lambda_{HS_2}^{(A)}}{3} + \mathrm{Tr}\left[y_1\ \left(y_1^*\right)^T\right] + \mathrm{Tr}\left[y_2\ \left(y_2^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_d\ \left(Y_d^*\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ \left(Y_e^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ \left(Y_u^*\right)^T\right]\right)
\end{aligned}
$$

$$\begin{aligned}\beta^{(1)}_{\lambda^{(A)}_{HS_1}} =& -\frac{33g_2^2\lambda^{(A)}_{HS_1}}{2}\\ &+ g_1^2\left(-8g_2^2-\frac{51\lambda^{(A)}_{HS_1}}{2}\right)\\ &+ 12\lambda^{(A)}_{12}\lambda^{(A)}_{HS_2}\\ &+ 2\lambda_1\lambda^{(A)}_{HS_1}\\ &+ 8\lambda^{(1)}_{HS_1}\lambda^{(A)}_{HS_1}\\ &+ 2\lambda^{(A)}_{HS_1}\lambda^{(1)}_{S_1}\\ &+ \lambda^{(A)}_{HS_1}\left(16\lambda^{(A)}_{S_1}+2\,\mathrm{Tr}\left[y_1\ (y_1^*)^T\right]+6\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\right]+2\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\right]+6\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\right]\right)\\ &- \frac{4\lambda_5\lambda_5^*}{3}\\ &- \frac{4\,\mathrm{Tr}\left[y_1\ (y_1^*)^T\ Y_e\ (Y_e^*)^T\right]}{3}\end{aligned}$$

$$\begin{aligned}\beta^{(1)}_{\lambda^{(A)}_{HS_2}} =& -\frac{33g_2^2\lambda^{(A)}_{HS_2}}{2}\\ &+ g_1^2\left(-4g_2^2-\frac{15\lambda^{(A)}_{HS_2}}{2}\right)\\ &+ 12\lambda^{(A)}_{12}\lambda^{(A)}_{HS_1}\\ &+ 2\lambda_1\lambda^{(A)}_{HS_2}\\ &+ 8\lambda^{(1)}_{HS_2}\lambda^{(A)}_{HS_2}\\ &+ 2\lambda^{(A)}_{HS_2}\lambda^{(1)}_{S_2}\\ &+ \lambda^{(A)}_{HS_2}\left(16\lambda^{(A)}_{S_2}+2\,\mathrm{Tr}\left[y_2\ (y_2^*)^T\right]+6\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\right]+2\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\right]+6\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\right]\right)\\ &- \frac{8\lambda_5\lambda_5^*}{9}\\ &+ \frac{4\,\mathrm{Tr}\left[y_2\ (y_2^*)^T\ Y_e\ (Y_e^*)^T\right]}{3}\end{aligned}$$

$$\begin{aligned}\beta^{(1)}_{\lambda^{(A)}_{S_1}} =& -2g_2^4\\ &- 24g_2^2\lambda^{(A)}_{S_1}\\ &+ g_1^2\left(32g_2^2-48\lambda^{(A)}_{S_1}\right)\\ &+ 12\left(\lambda^{(A)}_{12}\right)^2\\ &- \frac{4\left(\lambda^{(\times)}_{12}\right)^2}{3}\\ &+ 3\left(\lambda^{(A)}_{HS_1}\right)^2\\ &+ 12\lambda^{(1)}_{S_1}\lambda^{(A)}_{S_1}\end{aligned}$$

$$
\begin{aligned}
&+ 20 \left(\lambda_{S_1}^{(A)}\right)^2 \\
&+ 4\lambda_{S_1}^{(A)} \operatorname{Tr}\left[y_1 \ (y_1^*)^T\right] \\
&- \frac{2 \operatorname{Tr}\left[y_1 \ (y_1^*)^T \ y_1 \ (y_1^*)^T\right]}{3}
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda_{S_2}^{(A)}} = &- 2g_2^4 \\
&- 24g_2^2\lambda_{S_2}^{(A)} \\
&+ g_1^2 \left(8g_2^2 - 12\lambda_{S_2}^{(A)}\right) \\
&+ 12 \left(\lambda_{12}^{(A)}\right)^2 \\
&- \frac{4 \left(\lambda_{12}^{(\times)}\right)^2}{3} \\
&+ 3 \left(\lambda_{HS_2}^{(A)}\right)^2 \\
&+ 12\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)} \\
&+ 20 \left(\lambda_{S_2}^{(A)}\right)^2 \\
&+ 4\lambda_{S_2}^{(A)} \operatorname{Tr}\left[y_2 \ (y_2^*)^T\right] \\
&- \frac{2 \operatorname{Tr}\left[y_2 \ (y_2^*)^T \ y_2 \ (y_2^*)^T\right]}{3}
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda_{12}^{(A)}} = &- 24g_2^2\lambda_{12}^{(A)} \\
&+ g_1^2 \left(16g_2^2 - 30\lambda_{12}^{(A)}\right) \\
&+ 8\lambda_{12}^{(1)}\lambda_{12}^{(A)} \\
&+ 2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)} \\
&+ 16\lambda_{12}^{(A)}\lambda_{S_1}^{(A)} \\
&+ 2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)} \\
&+ 16\lambda_{12}^{(A)}\lambda_{S_2}^{(A)} \\
&+ \lambda_{12}^{(A)} \left(\frac{16\lambda_{12}^{(\times)}}{3} + 2 \operatorname{Tr}\left[y_1 \ (y_1^*)^T\right] + 2 \operatorname{Tr}\left[y_2 \ (y_2^*)^T\right]\right) \\
&+ 3\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)} \\
&- \frac{2\lambda_5\lambda_5^*}{9} \\
&+ \frac{2 \operatorname{Tr}\left[y_1 \ (y_1^*)^T \ y_2 \ (y_2^*)^T\right]}{3}
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda_{12}^{(\times)}} = &\frac{51g_2^4}{10} \\
&- 24g_2^2\lambda_{12}^{(\times)}
\end{aligned}
$$

$$
\begin{aligned}
&- 30 g_1^2 \lambda_{12}^{(\times)} \\
&+ 8 \lambda_{12}^{(1)} \lambda_{12}^{(\times)} \\
&+ 10 \left(\lambda_{12}^{(\times)}\right)^2 \\
&+ 2 \lambda_{12}^{(\times)} \lambda_{S_1}^{(1)} \\
&- 2 \lambda_{12}^{(\times)} \lambda_{S_1}^{(A)} \\
&+ 2 \lambda_{12}^{(\times)} \lambda_{S_2}^{(1)} \\
&- 2 \lambda_{12}^{(\times)} \lambda_{S_2}^{(A)} \\
&+ \lambda_{12}^{(\times)} \left(2 \operatorname{Tr}\left[y_1 \ (y_1^*)^T\right] + 2 \operatorname{Tr}\left[y_2 \ (y_2^*)^T\right]\right) \\
&- \lambda_5 \lambda_5^* \\
&- \frac{4 \operatorname{Tr}\left[y_1 \ (y_1^*)^T \ y_2 \ (y_2^*)^T\right]}{5}
\end{aligned}
$$

$\mathrm{T3-E}\,(\alpha = -2)$

$$
\begin{aligned}
\beta_{m_1^2}^{(1)} = &- 12 g_2^2 m_1^2 \\
&- 6 g_1^2 m_1^2 \\
&+ 12 \lambda_{18} \left(\mu_{S_2}^2\right)^* \\
&+ 4 \lambda_{19} \left(\mu_{S_2}^2\right)^* \\
&+ m_1^2 \left(8 \lambda_{S_1}^{(1)} + 4 \lambda_{S_1}^{(A)} + 2 \operatorname{Tr}\left[y_1 \ (y_1^*)^T\right]\right) \\
&+ m_2^2 \left(6 \lambda_{12}^{(1)} + 4 \lambda_{12}^{(\times)}\right) \\
&+ 4 \mu_{S_2}^2 \left(3 \lambda_{18}^* + \lambda_{19}^*\right) \\
&- 4 \operatorname{Tr}\left[M_F \ (M_F^*)^T \ (y_1)^T \ y_1^*\right]
\end{aligned}
$$

$$
\begin{aligned}
\beta_{m_2^2}^{(1)} = &- 12 g_2^2 m_2^2 \\
&+ 20 \lambda_{14} \left(\mu_{S_2}^2\right)^* \\
&+ m_1^2 \left(6 \lambda_{12}^{(1)} + 4 \lambda_{12}^{(\times)}\right) \\
&+ m_2^2 \left(8 \lambda_{S_2}^{(1)} + 4 \lambda_{S_2}^{(A)} + 2 \operatorname{Tr}\left[y_2 \ (y_2^*)^T\right]\right) \\
&+ 20 \mu_{S_2}^2 \lambda_{14}^* \\
&- 4 \operatorname{Tr}\left[M_F \ (y_2^*)^T \ y_2 \ (M_F^*)^T\right]
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_1}^{(1)} = &\frac{9 g_2^4}{4} \\
&- 9 g_2^2 \lambda_1 \\
&+ \frac{3 g_1^4}{4} \\
&+ g_1^2 \left(\frac{3 g_2^2}{2} - 3 \lambda_1\right) \\
&+ 12 \lambda_1^2
\end{aligned}
$$

$$
\begin{aligned}
&+ \lambda_1 \left(12 \operatorname{Tr}\left[Y_d \left(Y_d^*\right)^T\right] + 4 \operatorname{Tr}\left[Y_e \left(Y_e^*\right)^T\right] + 12 \operatorname{Tr}\left[Y_u \left(Y_u^*\right)^T\right]\right) \\
&+ 6 \left(\lambda_{HS_1}^{(1)}\right)^2 \\
&+ 3 \left(\lambda_{HS_1}^{(A)}\right)^2 \\
&+ 6 \left(\lambda_{HS_2}^{(1)}\right)^2 \\
&+ 3 \left(\lambda_{HS_2}^{(A)}\right)^2 \\
&+ 4\lambda_5\lambda_5^* \\
&+ 4\lambda_{22}\lambda_{22}^* \\
&+ 24\lambda_{25}\lambda_{25}^* \\
&- 12 \operatorname{Tr}\left[Y_d \left(Y_d^*\right)^T Y_d \left(Y_d^*\right)^T\right] \\
&- 4 \operatorname{Tr}\left[Y_e \left(Y_e^*\right)^T Y_e \left(Y_e^*\right)^T\right] \\
&- 12 \operatorname{Tr}\left[Y_u \left(Y_u^*\right)^T Y_u \left(Y_u^*\right)^T\right]
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{S_1}^{(1)}}^{(1)} = &22g_2^4 \\
&- 24g_2^2\lambda_{S_1}^{(1)} \\
&+ 12g_1^4 \\
&+ g_1^2 \left(8g_2^2 - 12\lambda_{S_1}^{(1)}\right) \\
&+ 6 \left(\lambda_{12}^{(1)}\right)^2 \\
&+ 8\lambda_{12}^{(1)}\lambda_{12}^{(\times)} \\
&+ \frac{20 \left(\lambda_{12}^{(\times)}\right)^2}{3} \\
&+ 4 \left(\lambda_{HS_1}^{(1)}\right)^2 \\
&+ 14 \left(\lambda_{S_1}^{(1)}\right)^2 \\
&+ \lambda_{S_1}^{(1)} \left(8\lambda_{S_1}^{(A)} + 4 \operatorname{Tr}\left[y_1 \left(y_1^*\right)^T\right]\right) \\
&+ 16 \left(\lambda_{S_1}^{(A)}\right)^2 \\
&+ 8\lambda_{18} \left(3\lambda_{18}^* + \lambda_{19}^*\right) \\
&+ \lambda_{19} \left(8\lambda_{18}^* + \frac{20\lambda_{19}^*}{3}\right) \\
&- \frac{8 \operatorname{Tr}\left[y_1 \left(y_1^*\right)^T y_1 \left(y_1^*\right)^T\right]}{3}
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{S_2}^{(1)}}^{(1)} = &22g_2^4 \\
&- 24g_2^2\lambda_{S_2}^{(1)} \\
&+ 6 \left(\lambda_{12}^{(1)}\right)^2
\end{aligned}
$$

$$
\begin{aligned}
&+ 8\lambda_{12}^{(1)}\lambda_{12}^{(\times)} \\
&+ \frac{20\left(\lambda_{12}^{(\times)}\right)^2}{3} \\
&+ 4\left(\lambda_{HS_2}^{(1)}\right)^2 \\
&+ 14\left(\lambda_{S_2}^{(1)}\right)^2 \\
&+ \lambda_{S_2}^{(1)}\left(8\lambda_{S_2}^{(A)} + 4\,\mathrm{Tr}\left[y_2\ \left(y_2^*\right)^T\right]\right) \\
&+ 16\left(\lambda_{S_2}^{(A)}\right)^2 \\
&+ \frac{832\lambda_{13}\lambda_{13}^*}{3} \\
&+ \frac{352\lambda_{14}\lambda_{14}^*}{3} \\
&+ \frac{8\lambda_{18}\left(3\lambda_{18}^* + \lambda_{19}^*\right)}{3} \\
&+ \lambda_{19}\left(\frac{8\lambda_{18}^*}{3} + 4\lambda_{19}^*\right) \\
&+ \frac{16\lambda_{25}\lambda_{25}^*}{3} \\
&- \frac{8\,\mathrm{Tr}\left[y_2\ \left(y_2^*\right)^T\ y_2\ \left(y_2^*\right)^T\right]}{3}
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{HS_1}^{(1)}} &= 6g_2^4 \\
&- \frac{33g_2^2\lambda_{HS_1}^{(1)}}{2} \\
&+ 3g_1^4 \\
&- \frac{15g_1^2\lambda_{HS_1}^{(1)}}{2} \\
&+ 6\lambda_1\lambda_{HS_1}^{(1)} \\
&+ 4\left(\lambda_{HS_1}^{(1)}\right)^2 \\
&+ 8\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)} \\
&+ \lambda_{HS_1}^{(1)}\left(4\lambda_{S_1}^{(A)} + 2\,\mathrm{Tr}\left[y_1\ \left(y_1^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_d\ \left(Y_d^*\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ \left(Y_e^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ \left(Y_u^*\right)^T\right]\right) \\
&+ 6\left(\lambda_{HS_1}^{(A)}\right)^2 \\
&+ \lambda_{HS_2}^{(1)}\left(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)}\right) \\
&+ 4\lambda_5\lambda_5^* \\
&+ 12\lambda_{18}\lambda_{25}^* \\
&+ 4\lambda_{19}\lambda_{25}^* \\
&+ 4\lambda_{22}\lambda_{22}^*
\end{aligned}
$$

$$
\begin{aligned}
&+\ 4\lambda_{25}\left(3\lambda_{18}^{*}+\lambda_{19}^{*}\right)\\
&-\ 2\,\mathrm{Tr}\left[y_1\ \left(y_1^{*}\right)^{T}\ Y_e\ \left(Y_e^{*}\right)^{T}\right]
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda^{(1)}_{HS_2}} &= 6g_2^4\\
&-\ \frac{33g_2^2\lambda^{(1)}_{HS_2}}{2}\\
&-\ \frac{3g_1^2\lambda^{(1)}_{HS_2}}{2}\\
&+\ 6\lambda_1\lambda^{(1)}_{HS_2}\\
&+\ \lambda^{(1)}_{HS_1}\left(6\lambda^{(1)}_{12}+4\lambda^{(\times)}_{12}\right)\\
&+\ 4\left(\lambda^{(1)}_{HS_2}\right)^2\\
&+\ 8\lambda^{(1)}_{HS_2}\lambda^{(1)}_{S_2}\\
&+\ \lambda^{(1)}_{HS_2}\left(4\lambda^{(A)}_{S_2}+2\,\mathrm{Tr}\left[y_2\ \left(y_2^{*}\right)^{T}\right]+6\,\mathrm{Tr}\left[Y_d\ \left(Y_d^{*}\right)^{T}\right]+2\,\mathrm{Tr}\left[Y_e\ \left(Y_e^{*}\right)^{T}\right]+6\,\mathrm{Tr}\left[Y_u\ \left(Y_u^{*}\right)^{T}\right]\right)\\
&+\ 6\left(\lambda^{(A)}_{HS_2}\right)^2\\
&+\ 4\lambda_5\lambda_5^{*}\\
&+\ 20\lambda_{14}\lambda_{25}^{*}\\
&+\ 4\lambda_{22}\lambda_{22}^{*}\\
&+\ 4\lambda_{25}\left(5\lambda_{14}^{*}+4\lambda_{25}^{*}\right)\\
&-\ 2\,\mathrm{Tr}\left[y_2\ \left(y_2^{*}\right)^{T}\ Y_e\ \left(Y_e^{*}\right)^{T}\right]
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda^{(1)}_{12}} &= \frac{63g_2^4}{5}\\
&-\ 24g_2^2\lambda^{(1)}_{12}\\
&-\ 6g_1^2\lambda^{(1)}_{12}\\
&+\ 4\left(\lambda^{(1)}_{12}\right)^2\\
&+\ \lambda^{(1)}_{12}\left(4\lambda^{(A)}_{S_1}+4\lambda^{(A)}_{S_2}+2\,\mathrm{Tr}\left[y_1\ \left(y_1^{*}\right)^{T}\right]+2\,\mathrm{Tr}\left[y_2\ \left(y_2^{*}\right)^{T}\right]\right)\\
&+\ 16\left(\lambda^{(A)}_{12}\right)^2\\
&+\ 4\left(\lambda^{(\times)}_{12}\right)^2\\
&+\ 4\lambda^{(\times)}_{12}\lambda^{(A)}_{S_1}\\
&+\ 4\lambda^{(\times)}_{12}\lambda^{(A)}_{S_2}\\
&+\ 4\lambda^{(1)}_{HS_1}\lambda^{(1)}_{HS_2}\\
&+\ \lambda^{(1)}_{S_1}\left(8\lambda^{(1)}_{12}+4\lambda^{(\times)}_{12}\right)\\
&+\ \lambda^{(1)}_{S_2}\left(8\lambda^{(1)}_{12}+4\lambda^{(\times)}_{12}\right)
\end{aligned}
$$

$$
\begin{aligned}
&+ 2\lambda_5\lambda_5^* \\
&+ 4\lambda_{14}\left(5\lambda_{18}^* + \lambda_{19}^*\right) \\
&+ 4\lambda_{18}\left(5\lambda_{14}^* + 4\lambda_{18}^*\right) \\
&+ 4\lambda_{19}\left(\lambda_{14}^* + \lambda_{19}^*\right) \\
&+ 2\lambda_{22}\lambda_{22}^* \\
&- \frac{22\,\mathrm{Tr}\left[y_1\ \left(y_1^*\right)^T\ y_2\ \left(y_2^*\right)^T\right]}{15}
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda_5} =& -\frac{33g_2^2\lambda_5}{2} \\
&- \frac{9g_1^2\lambda_5}{2} \\
&+ 2\lambda_{12}^{(1)}\lambda_5 \\
&+ 2\lambda_1\lambda_5 \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_5 \\
&+ 4\lambda_{HS_2}^{(1)}\lambda_5 \\
&+ \lambda_5\left(-\frac{4\lambda_{12}^{(A)}}{3} - 2\lambda_{12}^{(\times)} - 2\lambda_{HS_1}^{(A)} - \frac{4\lambda_{HS_2}^{(A)}}{3} + \mathrm{Tr}\left[y_1\ \left(y_1^*\right)^T\right] + \mathrm{Tr}\left[y_2\ \left(y_2^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_d\ \left(Y_d^*\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ \left(Y_e^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ \left(Y_u^*\right)^T\right]\right) \\
&+ 4\lambda_{18}\lambda_{22}^* \\
&- 2\lambda_{19}\lambda_{22}^* \\
&+ 8\lambda_{25}\lambda_{22}^*
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda_{HS_1}^{(A)}} =& -\frac{33g_2^2\lambda_{HS_1}^{(A)}}{2} \\
&+ g_1^2\left(-4g_2^2 - \frac{15\lambda_{HS_1}^{(A)}}{2}\right) \\
&+ 12\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)} \\
&+ 2\lambda_1\lambda_{HS_1}^{(A)} \\
&+ 8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)} \\
&+ 2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)} \\
&+ \lambda_{HS_1}^{(A)}\left(16\lambda_{S_1}^{(A)} + 2\,\mathrm{Tr}\left[y_1\ \left(y_1^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_d\ \left(Y_d^*\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ \left(Y_e^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ \left(Y_u^*\right)^T\right]\right) \\
&- \frac{4\lambda_5\lambda_5^*}{3} \\
&- \frac{4\lambda_{22}\lambda_{22}^*}{3} \\
&- \frac{4\,\mathrm{Tr}\left[y_1\ \left(y_1^*\right)^T\ Y_e\ \left(Y_e^*\right)^T\right]}{3}
\end{aligned}
$$

$$
\beta^{(1)}_{\lambda_{HS_2}^{(A)}} = -\frac{33g_2^2\lambda_{HS_2}^{(A)}}{2}
$$

$$
\begin{aligned}
&- \frac{3 g_1^2 \lambda_{HS_2}^{(A)}}{2} \\
&+ 12 \lambda_{12}^{(A)} \lambda_{HS_1}^{(A)} \\
&+ 2 \lambda_1 \lambda_{HS_2}^{(A)} \\
&+ 8 \lambda_{HS_2}^{(1)} \lambda_{HS_2}^{(A)} \\
&+ 2 \lambda_{HS_2}^{(A)} \lambda_{S_2}^{(1)} \\
&+ \lambda_{HS_2}^{(A)} \left(16 \lambda_{S_2}^{(A)} + 2 \operatorname{Tr}\left[y_2 \, (y_2^*)^T \right] + 6 \operatorname{Tr}\left[Y_d \, (Y_d^*)^T \right] + 2 \operatorname{Tr}\left[Y_e \, (Y_e^*)^T \right] + 6 \operatorname{Tr}\left[Y_u \, (Y_u^*)^T \right] \right) \\
&- \frac{8 \lambda_5 \lambda_5^*}{9} \\
&- \frac{4 \lambda_{22} \lambda_{22}^*}{3} \\
&+ \frac{4 \operatorname{Tr}\left[y_2 \, (y_2^*)^T \, Y_e \, (Y_e^*)^T \right]}{3}
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{S_1}^{(A)}}^{(1)} = &- 2 g_2^4 \\
&- 24 g_2^2 \lambda_{S_1}^{(A)} \\
&+ g_1^2 \left(8 g_2^2 - 12 \lambda_{S_1}^{(A)} \right) \\
&+ 12 \left(\lambda_{12}^{(A)} \right)^2 \\
&- \frac{4 \left(\lambda_{12}^{(\times)} \right)^2}{3} \\
&+ 3 \left(\lambda_{HS_1}^{(A)} \right)^2 \\
&+ 12 \lambda_{S_1}^{(1)} \lambda_{S_1}^{(A)} \\
&+ 20 \left(\lambda_{S_1}^{(A)} \right)^2 \\
&+ 4 \lambda_{S_1}^{(A)} \operatorname{Tr}\left[y_1 \, (y_1^*)^T \right] \\
&- \frac{4 \lambda_{19} \lambda_{19}^*}{3} \\
&- \frac{2 \operatorname{Tr}\left[y_1 \, (y_1^*)^T \, y_1 \, (y_1^*)^T \right]}{3}
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{S_2}^{(A)}}^{(1)} = &- 2 g_2^4 \\
&- 24 g_2^2 \lambda_{S_2}^{(A)} \\
&+ 12 \left(\lambda_{12}^{(A)} \right)^2 \\
&- \frac{4 \left(\lambda_{12}^{(\times)} \right)^2}{3} \\
&+ 3 \left(\lambda_{HS_2}^{(A)} \right)^2
\end{aligned}
$$

$$
\begin{aligned}
&+ 12\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)} \\
&+ 20\left(\lambda_{S_2}^{(A)}\right)^2 \\
&+ 4\lambda_{S_2}^{(A)}\,\mathrm{Tr}\left[y_2\ \left(y_2^*\right)^T\right] \\
&- \frac{224\lambda_{13}\lambda_{13}^*}{3} \\
&- \frac{44\lambda_{14}\lambda_{14}^*}{3} \\
&- \frac{4\lambda_{18}\left(3\lambda_{18}^* + \lambda_{19}^*\right)}{3} \\
&- \frac{4\lambda_{19}\lambda_{18}^*}{3} \\
&- \frac{8\lambda_{25}\lambda_{25}^*}{3} \\
&- \frac{2\,\mathrm{Tr}\left[y_2\ \left(y_2^*\right)^T\ y_2\ \left(y_2^*\right)^T\right]}{3}
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{12}^{(A)}}^{(1)} = &- 24g_2^2\lambda_{12}^{(A)} \\
&- 6g_1^2\lambda_{12}^{(A)} \\
&+ 8\lambda_{12}^{(1)}\lambda_{12}^{(A)} \\
&+ 2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)} \\
&+ 16\lambda_{12}^{(A)}\lambda_{S_1}^{(A)} \\
&+ 2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)} \\
&+ 16\lambda_{12}^{(A)}\lambda_{S_2}^{(A)} \\
&+ \lambda_{12}^{(A)}\left(\frac{16\lambda_{12}^{(\times)}}{3} + 2\,\mathrm{Tr}\left[y_1\ \left(y_1^*\right)^T\right] + 2\,\mathrm{Tr}\left[y_2\ \left(y_2^*\right)^T\right]\right) \\
&+ 3\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)} \\
&- \frac{2\lambda_5\lambda_5^*}{9} \\
&- \frac{\lambda_{22}\lambda_{22}^*}{3} \\
&+ \frac{2\,\mathrm{Tr}\left[y_1\ \left(y_1^*\right)^T\ y_2\ \left(y_2^*\right)^T\right]}{3}
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{12}^{(\times)}}^{(1)} = &\ \frac{51g_2^4}{10} \\
&- 24g_2^2\lambda_{12}^{(\times)} \\
&- 6g_1^2\lambda_{12}^{(\times)} \\
&+ 8\lambda_{12}^{(1)}\lambda_{12}^{(\times)} \\
&+ 10\left(\lambda_{12}^{(\times)}\right)^2 \\
&+ 2\lambda_{12}^{(\times)}\lambda_{S_1}^{(1)}
\end{aligned}
$$

$$
\begin{aligned}
&- 2\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)} \\
&+ 2\lambda_{12}^{(\times)}\lambda_{S_2}^{(1)} \\
&- 2\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)} \\
&+ \lambda_{12}^{(\times)}\left(2\,\mathrm{Tr}\left[y_1\ (y_1^*)^T\right] + 2\,\mathrm{Tr}\left[y_2\ (y_2^*)^T\right]\right) \\
&- \lambda_5\lambda_5^* \\
&+ 4\lambda_{14}\lambda_{19}^* \\
&+ 8\lambda_{18}\lambda_{19}^* \\
&+ \lambda_{19}\left(4\lambda_{14}^* + 8\lambda_{18}^* + 10\lambda_{19}^*\right) \\
&- \lambda_{22}\lambda_{22}^* \\
&- \frac{4\,\mathrm{Tr}\left[y_1\ (y_1^*)^T\ y_2\ (y_2^*)^T\right]}{5}
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{13}}^{(1)} = &- 24g_2^2\lambda_{13} \\
&+ 12\lambda_{S_2}^{(1)}\lambda_{13} \\
&- 8\lambda_{S_2}^{(A)}\lambda_{13} \\
&+ 4\lambda_{13}\,\mathrm{Tr}\left[y_2\ (y_2^*)^T\right] \\
&+ 11\lambda_{14}^2 \\
&+ 3\lambda_{18}^2 \\
&+ 2\lambda_{18}\lambda_{19} \\
&+ \lambda_{19}^2 \\
&+ 2\lambda_{25}^2
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{14}}^{(1)} = &- 24g_2^2\lambda_{14} \\
&+ \lambda_{12}^{(1)}\left(6\lambda_{18} + 2\lambda_{19}\right) \\
&+ \lambda_{12}^{(\times)}\left(4\lambda_{18} + 4\lambda_{19}\right) \\
&+ 4\lambda_{HS_2}^{(1)}\lambda_{25} \\
&+ 22\lambda_{S_2}^{(1)}\lambda_{14} \\
&+ 88\lambda_{13}\lambda_{14}^* \\
&+ 4\lambda_{14}\,\mathrm{Tr}\left[y_2\ (y_2^*)^T\right]
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{18}}^{(1)} = &- 24g_2^2\lambda_{18} \\
&- 6g_1^2\lambda_{18} \\
&+ \lambda_{12}^{(1)}\left(10\lambda_{14} + 8\lambda_{18}\right) \\
&+ \lambda_{12}^{(\times)}\left(4\lambda_{14} + 4\lambda_{19}\right) \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_{25} \\
&+ \lambda_{S_1}^{(1)}\left(8\lambda_{18} + 2\lambda_{19}\right) \\
&+ \lambda_{S_1}^{(A)}\left(4\lambda_{18} + 2\lambda_{19}\right)
\end{aligned}
$$

$$
\begin{aligned}
&+ 2\lambda_{S_2}^{(1)}\lambda_{18} \\
&+ \lambda_{S_2}^{(A)}\left(-4\lambda_{18} - 2\lambda_{19}\right) \\
&+ 2\lambda_5\lambda_{22} \\
&+ 8\lambda_{13}\left(5\lambda_{18}^* + \lambda_{19}^*\right) \\
&+ \lambda_{18}\left(2\,\mathrm{Tr}\left[y_1\ \left(y_1^*\right)^T\right] + 2\,\mathrm{Tr}\left[y_2\ \left(y_2^*\right)^T\right]\right)
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{19}}^{(1)} =& - 24g_2^2\lambda_{19} \\
&- 6g_1^2\lambda_{19} \\
&+ 8\lambda_{12}^{(1)}\lambda_{19} \\
&+ \lambda_{12}^{(\times)}\left(8\lambda_{14} + 16\lambda_{18} + 20\lambda_{19}\right) \\
&+ 2\lambda_{S_1}^{(1)}\lambda_{19} \\
&- 2\lambda_{S_1}^{(A)}\lambda_{19} \\
&+ 2\lambda_{S_2}^{(1)}\lambda_{19} \\
&+ 2\lambda_{S_2}^{(A)}\lambda_{19} \\
&- 2\lambda_5\lambda_{22} \\
&+ 16\lambda_{13}\lambda_{19}^* \\
&+ \lambda_{19}\left(2\,\mathrm{Tr}\left[y_1\ \left(y_1^*\right)^T\right] + 2\,\mathrm{Tr}\left[y_2\ \left(y_2^*\right)^T\right]\right)
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{22}}^{(1)} =& - \frac{33g_2^2\lambda_{22}}{2} \\
&- \frac{9g_1^2\lambda_{22}}{2} \\
&+ 2\lambda_{12}^{(1)}\lambda_{22} \\
&- 2\lambda_{12}^{(A)}\lambda_{22} \\
&- 2\lambda_{12}^{(\times)}\lambda_{22} \\
&+ 2\lambda_1\lambda_{22} \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_{22} \\
&- 2\lambda_{HS_1}^{(A)}\lambda_{22} \\
&+ 4\lambda_{HS_2}^{(1)}\lambda_{22} \\
&- 2\lambda_{HS_2}^{(A)}\lambda_{22} \\
&+ 4\lambda_{18}\lambda_5^* \\
&- 2\lambda_{19}\lambda_5^* \\
&+ \lambda_{22}\left(\mathrm{Tr}\left[y_1\ \left(y_1^*\right)^T\right] + \mathrm{Tr}\left[y_2\ \left(y_2^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_d\ \left(Y_d^*\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ \left(Y_e^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ \left(Y_u^*\right)^T\right]\right) \\
&+ 8\lambda_{25}\lambda_5^*
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{25}}^{(1)} =& - \frac{33g_2^2\lambda_{25}}{2} \\
&- \frac{3g_1^2\lambda_{25}}{2} \\
&+ 6\lambda_1\lambda_{25} \\
&+ \lambda_{HS_1}^{(1)}\left(6\lambda_{18} + 2\lambda_{19}\right)
\end{aligned}
$$

$$
\begin{aligned}
&+ \lambda_{HS_2}^{(1)} \left(10\lambda_{14} + 8\lambda_{25}\right) \\
&+ 2\lambda_{S_2}^{(1)} \lambda_{25} \\
&- 4\lambda_{S_2}^{(A)} \lambda_{25} \\
&+ 4\lambda_5 \lambda_{22} \\
&+ 40\lambda_{13} \lambda_{25}^* \\
&+ \lambda_{25} \left(2\,\mathrm{Tr}\left[y_2\ \left(y_2^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_d\ \left(Y_d^*\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ \left(Y_e^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ \left(Y_u^*\right)^T\right]\right)
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\mu_{S_2}^2}^{(1)} = &- 12 g_2^2 \mu_{S_2}^2 \\
&+ 2\lambda_{S_2}^{(1)} \mu_{S_2}^2 \\
&- 4\lambda_{S_2}^{(A)} \mu_{S_2}^2 \\
&+ 40\lambda_{13} \left(\mu_{S_2}^2\right)^* \\
&+ 10\lambda_{14} m_2^2 \\
&+ m_1^2 \left(6\lambda_{18} + 2\lambda_{19}\right) \\
&+ 2\mu_{S_2}^2\,\mathrm{Tr}\left[y_2\ \left(y_2^*\right)^T\right]
\end{aligned}
$$

$\mathrm{T3} - \mathrm{E}\,(\alpha = 0)$

$$
\begin{aligned}
\beta_{m_1^2}^{(1)} = &- 12 g_2^2 m_1^2 \\
&+ 20\lambda_{20} \left(\mu_{S_1}^2\right)^* \\
&+ m_1^2 \left(8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)} + 2\,\mathrm{Tr}\left[y_1\ \left(y_1^*\right)^T\right]\right) \\
&+ m_2^2 \left(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)}\right) \\
&+ 20\mu_{S_1}^2 \lambda_{20}^* \\
&- 4\,\mathrm{Tr}\left[M_F\ \left(M_F^*\right)^T\ \left(y_1\right)^T\ y_1^*\right]
\end{aligned}
$$

$$
\begin{aligned}
\beta_{m_2^2}^{(1)} = &- 12 g_2^2 m_2^2 \\
&- 6 g_1^2 m_2^2 \\
&+ 12\lambda_{15} \left(\mu_{S_1}^2\right)^* \\
&+ 4\lambda_{16} \left(\mu_{S_1}^2\right)^* \\
&+ m_1^2 \left(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)}\right) \\
&+ m_2^2 \left(8\lambda_{S_2}^{(1)} + 4\lambda_{S_2}^{(A)} + 2\,\mathrm{Tr}\left[y_2\ \left(y_2^*\right)^T\right]\right) \\
&+ 4\mu_{S_1}^2 \left(3\lambda_{15}^* + \lambda_{16}^*\right) \\
&- 4\,\mathrm{Tr}\left[M_F\ \left(y_2^*\right)^T\ y_2\ \left(M_F^*\right)^T\right]
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_1}^{(1)} = &\ \frac{9 g_2^4}{4} \\
&- 9 g_2^2 \lambda_1 \\
&+ \frac{3 g_1^4}{4}
\end{aligned}
$$

$$
\begin{aligned}
&+ g_1^2 \left(\frac{3g_2^2}{2} - 3\lambda_1\right) \\
&+ 12\lambda_1^2 \\
&+ \lambda_1 \left(12\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\right] + 4\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\right] + 12\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\right]\right) \\
&+ 6\left(\lambda_{HS_1}^{(1)}\right)^2 \\
&+ 3\left(\lambda_{HS_1}^{(A)}\right)^2 \\
&+ 6\left(\lambda_{HS_2}^{(1)}\right)^2 \\
&+ 3\left(\lambda_{HS_2}^{(A)}\right)^2 \\
&+ 4\lambda_5\lambda_5^* \\
&+ 4\lambda_{23}\lambda_{23}^* \\
&+ 24\lambda_{27}\lambda_{27}^* \\
&- 12\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\ Y_d\ (Y_d^*)^T\right] \\
&- 4\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\ Y_e\ (Y_e^*)^T\right] \\
&- 12\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\ Y_u\ (Y_u^*)^T\right]
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{S_1}^{(1)}}^{(1)} = {} & 22g_2^4 \\
&- 24g_2^2\lambda_{S_1}^{(1)} \\
&+ 6\left(\lambda_{12}^{(1)}\right)^2 \\
&+ 8\lambda_{12}^{(1)}\lambda_{12}^{(\times)} \\
&+ \frac{20\left(\lambda_{12}^{(\times)}\right)^2}{3} \\
&+ 4\left(\lambda_{HS_1}^{(1)}\right)^2 \\
&+ 14\left(\lambda_{S_1}^{(1)}\right)^2 \\
&+ \lambda_{S_1}^{(1)}\left(8\lambda_{S_1}^{(A)} + 4\,\mathrm{Tr}\left[y_1\ (y_1^*)^T\right]\right) \\
&+ 16\left(\lambda_{S_1}^{(A)}\right)^2 \\
&+ \frac{8\lambda_{15}\left(3\lambda_{15}^* + \lambda_{16}^*\right)}{3} \\
&+ \lambda_{16}\left(\frac{8\lambda_{15}^*}{3} + 4\lambda_{16}^*\right) \\
&+ \frac{832\lambda_{17}\lambda_{17}^*}{3} \\
&+ \frac{352\lambda_{20}\lambda_{20}^*}{3} \\
&+ \frac{16\lambda_{27}\lambda_{27}^*}{3}
\end{aligned}
$$

$$
\begin{aligned}
&- \frac{8\,\mathrm{Tr}\left[y_1\ (y_1^*)^T\ y_1\ (y_1^*)^T\right]}{3}
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda^{(1)}_{S_2}} &= 22 g_2^4 \\
&- 24 g_2^2 \lambda^{(1)}_{S_2} \\
&+ 12 g_1^4 \\
&+ g_1^2 \left(8 g_2^2 - 12 \lambda^{(1)}_{S_2}\right) \\
&+ 6 \left(\lambda^{(1)}_{12}\right)^2 \\
&+ 8 \lambda^{(1)}_{12} \lambda^{(\times)}_{12} \\
&+ \frac{20 \left(\lambda^{(\times)}_{12}\right)^2}{3} \\
&+ 4 \left(\lambda^{(1)}_{HS_2}\right)^2 \\
&+ 14 \left(\lambda^{(1)}_{S_2}\right)^2 \\
&+ \lambda^{(1)}_{S_2} \left(8 \lambda^{(A)}_{S_2} + 4\,\mathrm{Tr}\left[y_2\ (y_2^*)^T\right]\right) \\
&+ 16 \left(\lambda^{(A)}_{S_2}\right)^2 \\
&+ 8 \lambda_{15} \left(3 \lambda^*_{15} + \lambda^*_{16}\right) \\
&+ \lambda_{16} \left(8 \lambda^*_{15} + \frac{20 \lambda^*_{16}}{3}\right) \\
&- \frac{8\,\mathrm{Tr}\left[y_2\ (y_2^*)^T\ y_2\ (y_2^*)^T\right]}{3}
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda^{(1)}_{HS_1}} &= 6 g_2^4 \\
&- \frac{33 g_2^2 \lambda^{(1)}_{HS_1}}{2} \\
&- \frac{3 g_1^2 \lambda^{(1)}_{HS_1}}{2} \\
&+ 6 \lambda_1 \lambda^{(1)}_{HS_1} \\
&+ 4 \left(\lambda^{(1)}_{HS_1}\right)^2 \\
&+ 8 \lambda^{(1)}_{HS_1} \lambda^{(1)}_{S_1} \\
&+ \lambda^{(1)}_{HS_1} \left(4 \lambda^{(A)}_{S_1} + 2\,\mathrm{Tr}\left[y_1\ (y_1^*)^T\right] + 6\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\right]\right) \\
&+ 6 \left(\lambda^{(A)}_{HS_1}\right)^2 \\
&+ \lambda^{(1)}_{HS_2} \left(6 \lambda^{(1)}_{12} + 4 \lambda^{(\times)}_{12}\right) \\
&+ 4 \lambda_5 \lambda^*_5
\end{aligned}
$$

$$
\begin{aligned}
&+ 20\lambda_{20}\lambda_{27}^{*} \\
&+ 4\lambda_{23}\lambda_{23}^{*} \\
&+ 4\lambda_{27}\left(5\lambda_{20}^{*} + 4\lambda_{27}^{*}\right) \\
&- 2\,\mathrm{Tr}\Big[y_1\ \left(y_1^{*}\right)^{T}\ Y_e\ \left(Y_e^{*}\right)^{T}\Big]
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda^{(1)}_{HS_2}} &= 6g_2^4 \\
&- \frac{33g_2^2\lambda^{(1)}_{HS_2}}{2} \\
&+ 3g_1^4 \\
&- \frac{15g_1^2\lambda^{(1)}_{HS_2}}{2} \\
&+ 6\lambda_1\lambda^{(1)}_{HS_2} \\
&+ \lambda^{(1)}_{HS_1}\left(6\lambda^{(1)}_{12} + 4\lambda^{(\times)}_{12}\right) \\
&+ 4\left(\lambda^{(1)}_{HS_2}\right)^2 \\
&+ 8\lambda^{(1)}_{HS_2}\lambda^{(1)}_{S_2} \\
&+ \lambda^{(1)}_{HS_2}\left(4\lambda^{(A)}_{S_2} + 2\,\mathrm{Tr}\Big[y_2\ \left(y_2^{*}\right)^{T}\Big] + 6\,\mathrm{Tr}\Big[Y_d\ \left(Y_d^{*}\right)^{T}\Big] + 2\,\mathrm{Tr}\Big[Y_e\ \left(Y_e^{*}\right)^{T}\Big] + 6\,\mathrm{Tr}\Big[Y_u\ \left(Y_u^{*}\right)^{T}\Big]\right) \\
&+ 6\left(\lambda^{(A)}_{HS_2}\right)^2 \\
&+ 4\lambda_5\lambda_5^{*} \\
&+ 12\lambda_{15}\lambda_{27}^{*} \\
&+ 4\lambda_{16}\lambda_{27}^{*} \\
&+ 4\lambda_{23}\lambda_{23}^{*} \\
&+ 4\lambda_{27}\left(3\lambda_{15}^{*} + \lambda_{16}^{*}\right) \\
&- 2\,\mathrm{Tr}\Big[y_2\ \left(y_2^{*}\right)^{T}\ Y_e\ \left(Y_e^{*}\right)^{T}\Big]
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda^{(1)}_{12}} &= \frac{63g_2^4}{5} \\
&- 24g_2^2\lambda^{(1)}_{12} \\
&- 6g_1^2\lambda^{(1)}_{12} \\
&+ 4\left(\lambda^{(1)}_{12}\right)^2 \\
&+ \lambda^{(1)}_{12}\left(4\lambda^{(A)}_{S_1} + 4\lambda^{(A)}_{S_2} + 2\,\mathrm{Tr}\Big[y_1\ \left(y_1^{*}\right)^{T}\Big] + 2\,\mathrm{Tr}\Big[y_2\ \left(y_2^{*}\right)^{T}\Big]\right) \\
&+ 16\left(\lambda^{(A)}_{12}\right)^2 \\
&+ 4\left(\lambda^{(\times)}_{12}\right)^2
\end{aligned}
$$

$$
\begin{aligned}
&+ 4\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)} \\
&+ 4\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)} \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)} \\
&+ \lambda_{S_1}^{(1)}\left(8\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)}\right) \\
&+ \lambda_{S_2}^{(1)}\left(8\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)}\right) \\
&+ 2\lambda_5\lambda_5^* \\
&+ 4\lambda_{15}\left(4\lambda_{15}^* + 5\lambda_{20}^*\right) \\
&+ 4\lambda_{16}\left(\lambda_{16}^* + \lambda_{20}^*\right) \\
&+ 4\lambda_{20}\left(5\lambda_{15}^* + \lambda_{16}^*\right) \\
&+ 2\lambda_{23}\lambda_{23}^* \\
&- \frac{22\,\mathrm{Tr}\left[y_1\ \left(y_1^*\right)^T\ y_2\ \left(y_2^*\right)^T\right]}{15}
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_5}^{(1)} =& -\frac{33g_2^2\lambda_5}{2} \\
&- \frac{9g_1^2\lambda_5}{2} \\
&+ 2\lambda_{12}^{(1)}\lambda_5 \\
&+ 2\lambda_1\lambda_5 \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_5 \\
&+ 4\lambda_{HS_2}^{(1)}\lambda_5 \\
&+ \lambda_5\left(-\frac{4\lambda_{12}^{(A)}}{3} - 2\lambda_{12}^{(\times)} - 2\lambda_{HS_1}^{(A)} - \frac{4\lambda_{HS_2}^{(A)}}{3} + \mathrm{Tr}\left[y_1\ \left(y_1^*\right)^T\right] + \mathrm{Tr}\left[y_2\ \left(y_2^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_d\ \left(Y_d^*\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ \left(Y_e^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ \left(Y_u^*\right)^T\right]\right) \\
&+ 2\left(2\lambda_{15}^* - \lambda_{16}^* + 4\lambda_{27}^*\right)\lambda_{23}^*
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{HS_1}^{(A)}}^{(1)} =& -\frac{33g_2^2\lambda_{HS_1}^{(A)}}{2} \\
&- \frac{3g_1^2\lambda_{HS_1}^{(A)}}{2} \\
&+ 12\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)} \\
&+ 2\lambda_1\lambda_{HS_1}^{(A)} \\
&+ 8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)} \\
&+ 2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)} \\
&+ \lambda_{HS_1}^{(A)}\left(16\lambda_{S_1}^{(A)} + 2\,\mathrm{Tr}\left[y_1\ \left(y_1^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_d\ \left(Y_d^*\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ \left(Y_e^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ \left(Y_u^*\right)^T\right]\right) \\
&- \frac{4\lambda_5\lambda_5^*}{3} \\
&- \frac{8\lambda_{23}\lambda_{23}^*}{9}
\end{aligned}
$$

$$
\begin{aligned}
&- \frac{4\,\mathrm{Tr}\left[y_1\ (y_1^*)^T\ Y_e\ (Y_e^*)^T\right]}{3}
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda^{(A)}_{HS_2}} =& -\frac{33 g_2^2 \lambda^{(A)}_{HS_2}}{2} \\
&+ g_1^2 \left(4 g_2^2 - \frac{15 \lambda^{(A)}_{HS_2}}{2}\right) \\
&+ 12 \lambda^{(A)}_{12} \lambda^{(A)}_{HS_1} \\
&+ 2 \lambda_1 \lambda^{(A)}_{HS_2} \\
&+ 8 \lambda^{(1)}_{HS_2} \lambda^{(A)}_{HS_2} \\
&+ 2 \lambda^{(A)}_{HS_2} \lambda^{(1)}_{S_2} \\
&+ \lambda^{(A)}_{HS_2} \left(16 \lambda^{(A)}_{S_2} + 2\,\mathrm{Tr}\left[y_2\ (y_2^*)^T\right] + 6\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\right]\right) \\
&- \frac{8 \lambda_5 \lambda_5^*}{9} \\
&- \frac{8 \lambda_{23} \lambda_{23}^*}{9} \\
&+ \frac{4\,\mathrm{Tr}\left[y_2\ (y_2^*)^T\ Y_e\ (Y_e^*)^T\right]}{3}
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda^{(A)}_{S_1}} =& -2 g_2^4 \\
&- 24 g_2^2 \lambda^{(A)}_{S_1} \\
&+ 12 \left(\lambda^{(A)}_{12}\right)^2 \\
&- \frac{4 \left(\lambda^{(\times)}_{12}\right)^2}{3} \\
&+ 3 \left(\lambda^{(A)}_{HS_1}\right)^2 \\
&+ 12 \lambda^{(1)}_{S_1} \lambda^{(A)}_{S_1} \\
&+ 20 \left(\lambda^{(A)}_{S_1}\right)^2 \\
&+ 4 \lambda^{(A)}_{S_1}\,\mathrm{Tr}\left[y_1\ (y_1^*)^T\right] \\
&- \frac{4 \lambda_{15} \left(3 \lambda_{15}^* + \lambda_{16}^*\right)}{3} \\
&- \frac{4 \lambda_{16} \lambda_{15}^*}{3} \\
&- \frac{224 \lambda_{17} \lambda_{17}^*}{3} \\
&- \frac{44 \lambda_{20} \lambda_{20}^*}{3} \\
&- \frac{8 \lambda_{27} \lambda_{27}^*}{3} \\
&- \frac{2\,\mathrm{Tr}\left[y_1\ (y_1^*)^T\ y_1\ (y_1^*)^T\right]}{3}
\end{aligned}
$$

$$\beta^{(1)}_{\lambda^{(A)}_{S_2}} = - 2g_2^4$$
$$- 24g_2^2\lambda^{(A)}_{S_2}$$
$$+ g_1^2\left(8g_2^2 - 12\lambda^{(A)}_{S_2}\right)$$
$$+ 12\left(\lambda^{(A)}_{12}\right)^2$$
$$- \frac{4\left(\lambda^{(\times)}_{12}\right)^2}{3}$$
$$+ 3\left(\lambda^{(A)}_{HS_2}\right)^2$$
$$+ 12\lambda^{(1)}_{S_2}\lambda^{(A)}_{S_2}$$
$$+ 20\left(\lambda^{(A)}_{S_2}\right)^2$$
$$+ 4\lambda^{(A)}_{S_2}\,\mathrm{Tr}\left[y_2\ \left(y_2^*\right)^T\right]$$
$$- \frac{4\lambda_{16}\lambda^*_{16}}{3}$$
$$- \frac{2\,\mathrm{Tr}\left[y_2\ \left(y_2^*\right)^T\ y_2\ \left(y_2^*\right)^T\right]}{3}$$

$$\beta^{(1)}_{\lambda^{(A)}_{12}} = - 24g_2^2\lambda^{(A)}_{12}$$
$$- 6g_1^2\lambda^{(A)}_{12}$$
$$+ 8\lambda^{(1)}_{12}\lambda^{(A)}_{12}$$
$$+ 2\lambda^{(A)}_{12}\lambda^{(1)}_{S_1}$$
$$+ 16\lambda^{(A)}_{12}\lambda^{(A)}_{S_1}$$
$$+ 2\lambda^{(A)}_{12}\lambda^{(1)}_{S_2}$$
$$+ 16\lambda^{(A)}_{12}\lambda^{(A)}_{S_2}$$
$$+ \lambda^{(A)}_{12}\left(\frac{16\lambda^{(\times)}_{12}}{3} + 2\,\mathrm{Tr}\left[y_1\ \left(y_1^*\right)^T\right] + 2\,\mathrm{Tr}\left[y_2\ \left(y_2^*\right)^T\right]\right)$$
$$+ 3\lambda^{(A)}_{HS_1}\lambda^{(A)}_{HS_2}$$
$$- \frac{2\lambda_5\lambda^*_5}{9}$$
$$- \frac{\lambda_{23}\lambda^*_{23}}{3}$$
$$+ \frac{2\,\mathrm{Tr}\left[y_1\ \left(y_1^*\right)^T\ y_2\ \left(y_2^*\right)^T\right]}{3}$$

$$\beta^{(1)}_{\lambda^{(\times)}_{12}} = \frac{51g_2^4}{10}$$
$$- 24g_2^2\lambda^{(\times)}_{12}$$
$$- 6g_1^2\lambda^{(\times)}_{12}$$
$$+ 8\lambda^{(1)}_{12}\lambda^{(\times)}_{12}$$

$$
\begin{aligned}
&+ 10\left(\lambda_{12}^{(\times)}\right)^2 \\
&+ 2\lambda_{12}^{(\times)}\lambda_{S_1}^{(1)} \\
&- 2\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)} \\
&+ 2\lambda_{12}^{(\times)}\lambda_{S_2}^{(1)} \\
&- 2\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)} \\
&+ \lambda_{12}^{(\times)}\left(2\,\mathrm{Tr}\left[y_1\ (y_1^*)^T\right] + 2\,\mathrm{Tr}\left[y_2\ (y_2^*)^T\right]\right) \\
&- \lambda_5\lambda_5^* \\
&+ 8\lambda_{15}\lambda_{16}^* \\
&+ \lambda_{16}\left(8\lambda_{15}^* + 10\lambda_{16}^* + 4\lambda_{20}^*\right) \\
&+ 4\lambda_{20}\lambda_{16}^* \\
&- \lambda_{23}\lambda_{23}^* \\
&- \frac{4\,\mathrm{Tr}\left[y_1\ (y_1^*)^T\ y_2\ (y_2^*)^T\right]}{5}
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{15}}^{(1)} = &- 24g_2^2\lambda_{15} \\
&- 6g_1^2\lambda_{15} \\
&+ \lambda_{12}^{(1)}\left(8\lambda_{15} + 10\lambda_{20}\right) \\
&+ \lambda_{12}^{(\times)}\left(4\lambda_{16} + 4\lambda_{20}\right) \\
&+ 4\lambda_{HS_2}^{(1)}\lambda_{27} \\
&+ 2\lambda_{S_1}^{(1)}\lambda_{15} \\
&+ \lambda_{S_1}^{(A)}\left(-4\lambda_{15} - 2\lambda_{16}\right) \\
&+ \lambda_{S_2}^{(1)}\left(8\lambda_{15} + 2\lambda_{16}\right) \\
&+ \lambda_{S_2}^{(A)}\left(4\lambda_{15} + 2\lambda_{16}\right) \\
&+ \lambda_{15}\left(2\,\mathrm{Tr}\left[y_1\ (y_1^*)^T\right] + 2\,\mathrm{Tr}\left[y_2\ (y_2^*)^T\right]\right) \\
&+ 8\lambda_{17}\left(5\lambda_{15}^* + \lambda_{16}^*\right) \\
&+ 2\lambda_5^*\lambda_{23}^*
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{16}}^{(1)} = &- 24g_2^2\lambda_{16} \\
&- 6g_1^2\lambda_{16} \\
&+ 8\lambda_{12}^{(1)}\lambda_{16} \\
&+ \lambda_{12}^{(\times)}\left(16\lambda_{15} + 20\lambda_{16} + 8\lambda_{20}\right) \\
&+ 2\lambda_{S_1}^{(1)}\lambda_{16} \\
&+ 2\lambda_{S_1}^{(A)}\lambda_{16} \\
&+ 2\lambda_{S_2}^{(1)}\lambda_{16} \\
&- 2\lambda_{S_2}^{(A)}\lambda_{16} \\
&+ \lambda_{16}\left(2\,\mathrm{Tr}\left[y_1\ (y_1^*)^T\right] + 2\,\mathrm{Tr}\left[y_2\ (y_2^*)^T\right]\right)
\end{aligned}
$$

$$
\begin{aligned}
&+ 16\lambda_{17}\lambda_{16}^{*} \\
&- 2\lambda_{5}^{*}\lambda_{23}^{*}
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{17}}^{(1)} = &- 24g_{2}^{2}\lambda_{17} \\
&+ 12\lambda_{S_{1}}^{(1)}\lambda_{17} \\
&- 8\lambda_{S_{1}}^{(A)}\lambda_{17} \\
&+ 3\lambda_{15}^{2} \\
&+ 2\lambda_{15}\lambda_{16} \\
&+ \lambda_{16}^{2} \\
&+ 4\lambda_{17}\,\mathrm{Tr}\left[y_{1}\ \left(y_{1}^{*}\right)^{T}\right] \\
&+ 11\lambda_{20}^{2} \\
&+ 2\lambda_{27}^{2}
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{20}}^{(1)} = &- 24g_{2}^{2}\lambda_{20} \\
&+ \lambda_{12}^{(1)}\left(6\lambda_{15} + 2\lambda_{16}\right) \\
&+ \lambda_{12}^{(\times)}\left(4\lambda_{15} + 4\lambda_{16}\right) \\
&+ 4\lambda_{HS_{1}}^{(1)}\lambda_{27} \\
&+ 22\lambda_{S_{1}}^{(1)}\lambda_{20} \\
&+ 88\lambda_{17}\lambda_{20}^{*} \\
&+ 4\lambda_{20}\,\mathrm{Tr}\left[y_{1}\ \left(y_{1}^{*}\right)^{T}\right]
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{23}}^{(1)} = &- \frac{33g_{2}^{2}\lambda_{23}}{2} \\
&- \frac{9g_{1}^{2}\lambda_{23}}{2} \\
&+ 2\lambda_{12}^{(1)}\lambda_{23} \\
&- 2\lambda_{12}^{(A)}\lambda_{23} \\
&- 2\lambda_{12}^{(\times)}\lambda_{23} \\
&+ 2\lambda_{1}\lambda_{23} \\
&+ 4\lambda_{HS_{1}}^{(1)}\lambda_{23} \\
&- \frac{4\lambda_{HS_{1}}^{(A)}\lambda_{23}}{3} \\
&+ 4\lambda_{HS_{2}}^{(1)}\lambda_{23} \\
&- \frac{4\lambda_{HS_{2}}^{(A)}\lambda_{23}}{3} \\
&+ \lambda_{23}\left(\mathrm{Tr}\left[y_{1}\ \left(y_{1}^{*}\right)^{T}\right] + \mathrm{Tr}\left[y_{2}\ \left(y_{2}^{*}\right)^{T}\right] + 6\,\mathrm{Tr}\left[Y_{d}\ \left(Y_{d}^{*}\right)^{T}\right] + 2\,\mathrm{Tr}\left[Y_{e}\ \left(Y_{e}^{*}\right)^{T}\right] + 6\,\mathrm{Tr}\left[Y_{u}\ \left(Y_{u}^{*}\right)^{T}\right]\right) \\
&+ 2\left(2\lambda_{15}^{*} - \lambda_{16}^{*} + 4\lambda_{27}^{*}\right)\lambda_{5}^{*}
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{27}}^{(1)} = &- \frac{33g_{2}^{2}\lambda_{27}}{2} \\
&- \frac{3g_{1}^{2}\lambda_{27}}{2} \\
&+ 6\lambda_{1}\lambda_{27}
\end{aligned}
$$

$$
\begin{aligned}
&+\ \lambda_{HS_1}^{(1)}\left(10\lambda_{20}+8\lambda_{27}\right)\\
&+\ \lambda_{HS_2}^{(1)}\left(6\lambda_{15}+2\lambda_{16}\right)\\
&+\ 2\lambda_{S_1}^{(1)}\lambda_{27}\\
&-\ 4\lambda_{S_1}^{(A)}\lambda_{27}\\
&+\ 40\lambda_{17}\lambda_{27}^{*}\\
&+\ \lambda_{27}\left(2\,\mathrm{Tr}\left[y_1\ \left(y_1^{*}\right)^{T}\right]+6\,\mathrm{Tr}\left[Y_d\ \left(Y_d^{*}\right)^{T}\right]+2\,\mathrm{Tr}\left[Y_e\ \left(Y_e^{*}\right)^{T}\right]+6\,\mathrm{Tr}\left[Y_u\ \left(Y_u^{*}\right)^{T}\right]\right)\\
&+\ 4\lambda_5^{*}\lambda_{23}^{*}
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\mu_{S_1}^2}^{(1)} =&-\ 12g_2^2\mu_{S_1}^2\\
&+\ 2\lambda_{S_1}^{(1)}\mu_{S_1}^2\\
&-\ 4\lambda_{S_1}^{(A)}\mu_{S_1}^2\\
&+\ 40\lambda_{17}\left(\mu_{S_1}^2\right)^{*}\\
&+\ 10\lambda_{20}m_1^2\\
&+\ m_2^2\left(6\lambda_{15}+2\lambda_{16}\right)\\
&+\ 2\mu_{S_1}^2\,\mathrm{Tr}\left[y_1\ \left(y_1^{*}\right)^{T}\right]
\end{aligned}
$$

$\mathrm{T3}-\mathrm{E}\,(\alpha=2)$

$$
\begin{aligned}
\beta_{m_1^2}^{(1)} =&-\ 12g_2^2m_1^2\\
&-\ 6g_1^2m_1^2\\
&+\ m_1^2\left(8\lambda_{S_1}^{(1)}+4\lambda_{S_1}^{(A)}+2\,\mathrm{Tr}\left[y_1\ \left(y_1^{*}\right)^{T}\right]\right)\\
&+\ m_2^2\left(6\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)}\right)\\
&-\ 4\,\mathrm{Tr}\left[M_F\ \left(M_F^{*}\right)^{T}\ \left(y_1\right)^{T}\ y_1^{*}\right]
\end{aligned}
$$

$$
\begin{aligned}
\beta_{m_2^2}^{(1)} =&-\ 12g_2^2m_2^2\\
&-\ 24g_1^2m_2^2\\
&+\ m_1^2\left(6\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)}\right)\\
&+\ m_2^2\left(8\lambda_{S_2}^{(1)}+4\lambda_{S_2}^{(A)}+2\,\mathrm{Tr}\left[y_2\ \left(y_2^{*}\right)^{T}\right]\right)\\
&-\ 4\,\mathrm{Tr}\left[M_F\ \left(y_2^{*}\right)^{T}\ y_2\ \left(M_F^{*}\right)^{T}\right]
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_1}^{(1)} =&\ \frac{9g_2^4}{4}\\
&-\ 9g_2^2\lambda_1\\
&+\ \frac{3g_1^4}{4}\\
&+\ g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)\\
&+\ 12\lambda_1^2\\
&+\ \lambda_1\left(12\,\mathrm{Tr}\left[Y_d\ \left(Y_d^{*}\right)^{T}\right]+4\,\mathrm{Tr}\left[Y_e\ \left(Y_e^{*}\right)^{T}\right]+12\,\mathrm{Tr}\left[Y_u\ \left(Y_u^{*}\right)^{T}\right]\right)
\end{aligned}
$$

$$
\begin{aligned}
&+ 6\left(\lambda_{HS_1}^{(1)}\right)^2 \\
&+ 3\left(\lambda_{HS_1}^{(A)}\right)^2 \\
&+ 6\left(\lambda_{HS_2}^{(1)}\right)^2 \\
&+ 3\left(\lambda_{HS_2}^{(A)}\right)^2 \\
&+ 4\lambda_5\lambda_5^* \\
&- 12\,\mathrm{Tr}\Big[Y_d\ (Y_d^*)^T\ Y_d\ (Y_d^*)^T\Big] \\
&- 4\,\mathrm{Tr}\Big[Y_e\ (Y_e^*)^T\ Y_e\ (Y_e^*)^T\Big] \\
&- 12\,\mathrm{Tr}\Big[Y_u\ (Y_u^*)^T\ Y_u\ (Y_u^*)^T\Big]
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda^{(1)}_{S_1}} &= 22g_2^4 \\
&- 24g_2^2\lambda_{S_1}^{(1)} \\
&+ 12g_1^4 \\
&+ g_1^2\left(8g_2^2 - 12\lambda_{S_1}^{(1)}\right) \\
&+ 6\left(\lambda_{12}^{(1)}\right)^2 \\
&+ 8\lambda_{12}^{(1)}\lambda_{12}^{(\times)} \\
&+ \frac{20\left(\lambda_{12}^{(\times)}\right)^2}{3} \\
&+ 4\left(\lambda_{HS_1}^{(1)}\right)^2 \\
&+ 14\left(\lambda_{S_1}^{(1)}\right)^2 \\
&+ \lambda_{S_1}^{(1)}\left(8\lambda_{S_1}^{(A)} + 4\,\mathrm{Tr}\Big[y_1\ (y_1^*)^T\Big]\right) \\
&+ 16\left(\lambda_{S_1}^{(A)}\right)^2 \\
&- \frac{8\,\mathrm{Tr}\Big[y_1\ (y_1^*)^T\ y_1\ (y_1^*)^T\Big]}{3}
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda^{(1)}_{S_2}} &= 22g_2^4 \\
&- 24g_2^2\lambda_{S_2}^{(1)} \\
&+ 192g_1^4 \\
&+ g_1^2\left(32g_2^2 - 48\lambda_{S_2}^{(1)}\right) \\
&+ 6\left(\lambda_{12}^{(1)}\right)^2 \\
&+ 8\lambda_{12}^{(1)}\lambda_{12}^{(\times)} \\
&+ \frac{20\left(\lambda_{12}^{(\times)}\right)^2}{3}
\end{aligned}
$$

$$
\begin{aligned}
&+ 4\left(\lambda_{HS_2}^{(1)}\right)^2 \\
&+ 14\left(\lambda_{S_2}^{(1)}\right)^2 \\
&+ \lambda_{S_2}^{(1)}\left(8\lambda_{S_2}^{(A)} + 4\,\mathrm{Tr}\left[y_2\ \left(y_2^*\right)^T\right]\right) \\
&+ 16\left(\lambda_{S_2}^{(A)}\right)^2 \\
&- \frac{8\,\mathrm{Tr}\left[y_2\ \left(y_2^*\right)^T\ y_2\ \left(y_2^*\right)^T\right]}{3}
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda^{(1)}_{HS_1}} &= 6g_2^4 \\
&- \frac{33g_2^2\lambda_{HS_1}^{(1)}}{2} \\
&+ 3g_1^4 \\
&- \frac{15g_1^2\lambda_{HS_1}^{(1)}}{2} \\
&+ 6\lambda_1\lambda_{HS_1}^{(1)} \\
&+ 4\left(\lambda_{HS_1}^{(1)}\right)^2 \\
&+ 8\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)} \\
&+ \lambda_{HS_1}^{(1)}\left(4\lambda_{S_1}^{(A)} + 2\,\mathrm{Tr}\left[y_1\ \left(y_1^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_d\ \left(Y_d^*\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ \left(Y_e^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ \left(Y_u^*\right)^T\right]\right) \\
&+ 6\left(\lambda_{HS_1}^{(A)}\right)^2 \\
&+ \lambda_{HS_2}^{(1)}\left(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)}\right) \\
&+ 4\lambda_5\lambda_5^* \\
&- 2\,\mathrm{Tr}\left[y_1\ \left(y_1^*\right)^T\ Y_e\ \left(Y_e^*\right)^T\right]
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda^{(1)}_{HS_2}} &= 6g_2^4 \\
&- \frac{33g_2^2\lambda_{HS_2}^{(1)}}{2} \\
&+ 12g_1^4 \\
&- \frac{51g_1^2\lambda_{HS_2}^{(1)}}{2} \\
&+ 6\lambda_1\lambda_{HS_2}^{(1)} \\
&+ \lambda_{HS_1}^{(1)}\left(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)}\right) \\
&+ 4\left(\lambda_{HS_2}^{(1)}\right)^2 \\
&+ 8\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)} \\
&+ \lambda_{HS_2}^{(1)}\left(4\lambda_{S_2}^{(A)} + 2\,\mathrm{Tr}\left[y_2\ \left(y_2^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_d\ \left(Y_d^*\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ \left(Y_e^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ \left(Y_u^*\right)^T\right]\right)
\end{aligned}
$$

$$+ 6\left(\lambda_{HS_2}^{(A)}\right)^2$$

$$+ 4\lambda_5\lambda_5^*$$

$$- 2\,\mathrm{Tr}\Big[y_2\ (y_2^*)^T\ Y_e\ (Y_e^*)^T\Big]$$

$$\beta_{\lambda_{12}^{(1)}}^{(1)} = \frac{63g_2^4}{5}$$

$$- 24g_2^2\lambda_{12}^{(1)}$$

$$+ 48g_1^4$$

$$- 30g_1^2\lambda_{12}^{(1)}$$

$$+ 4\left(\lambda_{12}^{(1)}\right)^2$$

$$+ \lambda_{12}^{(1)}\left(4\lambda_{S_1}^{(A)} + 4\lambda_{S_2}^{(A)} + 2\,\mathrm{Tr}\Big[y_1\ (y_1^*)^T\Big] + 2\,\mathrm{Tr}\Big[y_2\ (y_2^*)^T\Big]\right)$$

$$+ 16\left(\lambda_{12}^{(A)}\right)^2$$

$$+ 4\left(\lambda_{12}^{(\times)}\right)^2$$

$$+ 4\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}$$

$$+ 4\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}$$

$$+ 4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$$

$$+ \lambda_{S_1}^{(1)}\left(8\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)}\right)$$

$$+ \lambda_{S_2}^{(1)}\left(8\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)}\right)$$

$$+ 2\lambda_5\lambda_5^*$$

$$- \frac{22\,\mathrm{Tr}\Big[y_1\ (y_1^*)^T\ y_2\ (y_2^*)^T\Big]}{15}$$

$$\beta_{\lambda_5}^{(1)} = -\frac{33g_2^2\lambda_5}{2}$$

$$- \frac{33g_1^2\lambda_5}{2}$$

$$+ 2\lambda_{12}^{(1)}\lambda_5$$

$$+ 2\lambda_1\lambda_5$$

$$+ 4\lambda_{HS_1}^{(1)}\lambda_5$$

$$+ 4\lambda_{HS_2}^{(1)}\lambda_5$$

$$+ \lambda_5\left(-\frac{4\lambda_{12}^{(A)}}{3} - 2\lambda_{12}^{(\times)} - 2\lambda_{HS_1}^{(A)} - \frac{4\lambda_{HS_2}^{(A)}}{3} + \mathrm{Tr}\Big[y_1\ (y_1^*)^T\Big] + \mathrm{Tr}\Big[y_2\ (y_2^*)^T\Big] + 6\,\mathrm{Tr}\Big[Y_d\ (Y_d^*)^T\Big] + 2\,\mathrm{Tr}\Big[Y_e\ (Y_e^*)^T\Big] + 6\,\mathrm{Tr}\Big[Y_u\ (Y_u^*)^T\Big]\right)$$

$$\beta_{\lambda_{HS_1}^{(A)}}^{(1)} = -\frac{33g_2^2\lambda_{HS_1}^{(A)}}{2}$$

$$+ g_1^2\left(4g_2^2 - \frac{15\lambda_{HS_1}^{(A)}}{2}\right)$$

$$+ 12\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$$

$$
\begin{aligned}
&+ 2\lambda_1\lambda_{HS_1}^{(A)} \\
&+ 8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)} \\
&+ 2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)} \\
&+ \lambda_{HS_1}^{(A)}\left(16\lambda_{S_1}^{(A)} + 2\,\mathrm{Tr}\left[y_1\ (y_1^*)^T\right] + 6\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\right]\right) \\
&- \frac{4\lambda_5\lambda_5^*}{3} \\
&- \frac{4\,\mathrm{Tr}\left[y_1\ (y_1^*)^T\ Y_e\ (Y_e^*)^T\right]}{3}
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda_{HS_2}^{(A)}} =& -\frac{33g_2^2\lambda_{HS_2}^{(A)}}{2} \\
&+ g_1^2\left(8g_2^2 - \frac{51\lambda_{HS_2}^{(A)}}{2}\right) \\
&+ 12\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)} \\
&+ 2\lambda_1\lambda_{HS_2}^{(A)} \\
&+ 8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)} \\
&+ 2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)} \\
&+ \lambda_{HS_2}^{(A)}\left(16\lambda_{S_2}^{(A)} + 2\,\mathrm{Tr}\left[y_2\ (y_2^*)^T\right] + 6\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\right]\right) \\
&- \frac{8\lambda_5\lambda_5^*}{9} \\
&+ \frac{4\,\mathrm{Tr}\left[y_2\ (y_2^*)^T\ Y_e\ (Y_e^*)^T\right]}{3}
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda_{S_1}^{(A)}} =& -2g_2^4 \\
&- 24g_2^2\lambda_{S_1}^{(A)} \\
&+ g_1^2\left(8g_2^2 - 12\lambda_{S_1}^{(A)}\right) \\
&+ 12\left(\lambda_{12}^{(A)}\right)^2 \\
&- \frac{4\left(\lambda_{12}^{(\times)}\right)^2}{3} \\
&+ 3\left(\lambda_{HS_1}^{(A)}\right)^2 \\
&+ 12\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)} \\
&+ 20\left(\lambda_{S_1}^{(A)}\right)^2 \\
&+ 4\lambda_{S_1}^{(A)}\,\mathrm{Tr}\left[y_1\ (y_1^*)^T\right] \\
&- \frac{2\,\mathrm{Tr}\left[y_1\ (y_1^*)^T\ y_1\ (y_1^*)^T\right]}{3}
\end{aligned}
$$

$$\beta^{(1)}_{\lambda^{(A)}_{S_2}} = -2g_2^4$$
$$- 24g_2^2\lambda^{(A)}_{S_2}$$
$$+ g_1^2\left(32g_2^2 - 48\lambda^{(A)}_{S_2}\right)$$
$$+ 12\left(\lambda^{(A)}_{12}\right)^2$$
$$- \frac{4\left(\lambda^{(\times)}_{12}\right)^2}{3}$$
$$+ 3\left(\lambda^{(A)}_{HS_2}\right)^2$$
$$+ 12\lambda^{(1)}_{S_2}\lambda^{(A)}_{S_2}$$
$$+ 20\left(\lambda^{(A)}_{S_2}\right)^2$$
$$+ 4\lambda^{(A)}_{S_2}\,\mathrm{Tr}\left[y_2\ \left(y_2^*\right)^T\right]$$
$$- \frac{2\,\mathrm{Tr}\left[y_2\ \left(y_2^*\right)^T\ y_2\ \left(y_2^*\right)^T\right]}{3}$$

$$\beta^{(1)}_{\lambda^{(A)}_{12}} = -24g_2^2\lambda^{(A)}_{12}$$
$$+ g_1^2\left(16g_2^2 - 30\lambda^{(A)}_{12}\right)$$
$$+ 8\lambda^{(1)}_{12}\lambda^{(A)}_{12}$$
$$+ 2\lambda^{(A)}_{12}\lambda^{(1)}_{S_1}$$
$$+ 16\lambda^{(A)}_{12}\lambda^{(A)}_{S_1}$$
$$+ 2\lambda^{(A)}_{12}\lambda^{(1)}_{S_2}$$
$$+ 16\lambda^{(A)}_{12}\lambda^{(A)}_{S_2}$$
$$+ \lambda^{(A)}_{12}\left(\frac{16\lambda^{(\times)}_{12}}{3} + 2\,\mathrm{Tr}\left[y_1\ \left(y_1^*\right)^T\right] + 2\,\mathrm{Tr}\left[y_2\ \left(y_2^*\right)^T\right]\right)$$
$$+ 3\lambda^{(A)}_{HS_1}\lambda^{(A)}_{HS_2}$$
$$- \frac{2\lambda_5\lambda_5^*}{9}$$
$$+ \frac{2\,\mathrm{Tr}\left[y_1\ \left(y_1^*\right)^T\ y_2\ \left(y_2^*\right)^T\right]}{3}$$

$$\beta^{(1)}_{\lambda^{(\times)}_{12}} = \frac{51g_2^4}{10}$$
$$- 24g_2^2\lambda^{(\times)}_{12}$$
$$- 30g_1^2\lambda^{(\times)}_{12}$$
$$+ 8\lambda^{(1)}_{12}\lambda^{(\times)}_{12}$$
$$+ 10\left(\lambda^{(\times)}_{12}\right)^2$$
$$+ 2\lambda^{(\times)}_{12}\lambda^{(1)}_{S_1}$$

$$
\begin{aligned}
&- \ 2\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)} \\
&+ \ 2\lambda_{12}^{(\times)}\lambda_{S_2}^{(1)} \\
&- \ 2\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)} \\
&+ \ \lambda_{12}^{(\times)}\left(2\,\mathrm{Tr}\Big[y_1\ (y_1^*)^T\Big] + 2\,\mathrm{Tr}\Big[y_2\ (y_2^*)^T\Big]\right) \\
&- \ \lambda_5\lambda_5^* \\
&- \ \frac{4\,\mathrm{Tr}\Big[y_1\ (y_1^*)^T\ y_2\ (y_2^*)^T\Big]}{5}
\end{aligned}
$$

EFT1 candidate one-loop beta functions

T3 − A ($\alpha = -4$)

$$\begin{aligned}\beta^{(1)}_{m_1^2} =& -24 g_1^2 m_1^2 \\ &+ 6\lambda^{(1)}_{12} m_2^2 \\ &+ 4\lambda^{(1)}_{S_1} m_1^2\end{aligned}$$

$$\begin{aligned}\beta^{(1)}_{m_2^2} =& -12 g_2^2 m_2^2 \\ &- 6 g_1^2 m_2^2 \\ &+ 2\lambda^{(1)}_{12} m_1^2 \\ &+ m_2^2 \left(8\lambda^{(1)}_{S_2} + 4\lambda^{(A)}_{S_2}\right)\end{aligned}$$

$$\begin{aligned}\beta^{(1)}_{\lambda_1} =& \frac{9 g_2^4}{4} \\ &- 9 g_2^2 \lambda_1 \\ &+ \frac{3 g_1^4}{4} \\ &+ g_1^2 \left(\frac{3 g_2^2}{2} - 3\lambda_1\right) \\ &+ 12\lambda_1^2 \\ &+ \lambda_1 \left(12 \operatorname{Tr}\left[Y_d \left(Y_d^*\right)^T\right] + 4 \operatorname{Tr}\left[Y_e \left(Y_e^*\right)^T\right] + 12 \operatorname{Tr}\left[Y_u \left(Y_u^*\right)^T\right]\right) \\ &+ 2\left(\lambda^{(1)}_{HS_1}\right)^2 \\ &+ 6\left(\lambda^{(1)}_{HS_2}\right)^2 \\ &+ 3\left(\lambda^{(A)}_{HS_2}\right)^2 \\ &+ 4\lambda_5 \lambda_5^* \\ &- 12 \operatorname{Tr}\left[Y_d \left(Y_d^*\right)^T Y_d \left(Y_d^*\right)^T\right] \\ &- 4 \operatorname{Tr}\left[Y_e \left(Y_e^*\right)^T Y_e \left(Y_e^*\right)^T\right] \\ &- 12 \operatorname{Tr}\left[Y_u \left(Y_u^*\right)^T Y_u \left(Y_u^*\right)^T\right]\end{aligned}$$

$$\begin{aligned}\beta^{(1)}_{\lambda^{(1)}_{S_1}} =& 192 g_1^4 \\ &- 48 g_1^2 \lambda^{(1)}_{S_1} \\ &+ 6\left(\lambda^{(1)}_{12}\right)^2 \\ &+ 4\left(\lambda^{(1)}_{HS_1}\right)^2 \\ &+ 10\left(\lambda^{(1)}_{S_1}\right)^2\end{aligned}$$

$$\begin{aligned}\beta^{(1)}_{\lambda^{(1)}_{S_2}} =& 22 g_2^4 \\ &- 24 g_2^2 \lambda^{(1)}_{S_2}\end{aligned}$$

$$+ \ 12 g_1^4$$

$$+ \ g_1^2 \left(8 g_2^2 - 12 \lambda_{S_2}^{(1)}\right)$$

$$+ \ 2 \left(\lambda_{12}^{(1)}\right)^2$$

$$+ \ 4 \left(\lambda_{HS_2}^{(1)}\right)^2$$

$$+ \ 14 \left(\lambda_{S_2}^{(1)}\right)^2$$

$$+ \ 8 \lambda_{S_2}^{(1)} \lambda_{S_2}^{(A)}$$

$$+ \ 16 \left(\lambda_{S_2}^{(A)}\right)^2$$

$$\beta_{\lambda_{HS_1}^{(1)}}^{(1)} = - \frac{9 g_2^2 \lambda_{HS_1}^{(1)}}{2}$$

$$+ \ 12 g_1^4$$

$$- \ \frac{51 g_1^2 \lambda_{HS_1}^{(1)}}{2}$$

$$+ \ 6 \lambda_{12}^{(1)} \lambda_{HS_2}^{(1)}$$

$$+ \ 6 \lambda_1 \lambda_{HS_1}^{(1)}$$

$$+ \ 4 \left(\lambda_{HS_1}^{(1)}\right)^2$$

$$+ \ 4 \lambda_{HS_1}^{(1)} \lambda_{S_1}^{(1)}$$

$$+ \ \lambda_{HS_1}^{(1)} \left(6 \operatorname{Tr}\left[Y_d \ \left(Y_d^*\right)^T\right] + 2 \operatorname{Tr}\left[Y_e \ \left(Y_e^*\right)^T\right] + 6 \operatorname{Tr}\left[Y_u \ \left(Y_u^*\right)^T\right]\right)$$

$$+ \ 12 \lambda_5 \lambda_5^*$$

$$\beta_{\lambda_{HS_2}^{(1)}}^{(1)} = 6 g_2^4$$

$$- \ \frac{33 g_2^2 \lambda_{HS_2}^{(1)}}{2}$$

$$+ \ 3 g_1^4$$

$$- \ \frac{15 g_1^2 \lambda_{HS_2}^{(1)}}{2}$$

$$+ \ 2 \lambda_{12}^{(1)} \lambda_{HS_1}^{(1)}$$

$$+ \ 6 \lambda_1 \lambda_{HS_2}^{(1)}$$

$$+ \ 4 \left(\lambda_{HS_2}^{(1)}\right)^2$$

$$+ \ 8 \lambda_{HS_2}^{(1)} \lambda_{S_2}^{(1)}$$

$$+ \ \lambda_{HS_2}^{(1)} \left(4 \lambda_{S_2}^{(A)} + 6 \operatorname{Tr}\left[Y_d \ \left(Y_d^*\right)^T\right] + 2 \operatorname{Tr}\left[Y_e \ \left(Y_e^*\right)^T\right] + 6 \operatorname{Tr}\left[Y_u \ \left(Y_u^*\right)^T\right]\right)$$

$$+ \ 6 \left(\lambda_{HS_2}^{(A)}\right)^2$$

$$+ 4\lambda_5\lambda_5^*$$

$$\beta^{(1)}_{\lambda_{12}^{(1)}} = - 12g_2^2\lambda_{12}^{(1)}$$
$$+ 48g_1^4$$
$$- 30g_1^2\lambda_{12}^{(1)}$$
$$+ 4\left(\lambda_{12}^{(1)}\right)^2$$
$$+ 4\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$$
$$+ 8\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$$
$$+ 4\lambda_{12}^{(1)}\lambda_{S_2}^{(A)}$$
$$+ 4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$$
$$+ 4\lambda_5\lambda_5^*$$

$$\beta^{(1)}_{\lambda_5} = - \frac{21g_2^2\lambda_5}{2}$$
$$- \frac{33g_1^2\lambda_5}{2}$$
$$+ 2\lambda_{12}^{(1)}\lambda_5$$
$$+ 2\lambda_1\lambda_5$$
$$+ 4\lambda_{HS_1}^{(1)}\lambda_5$$
$$+ 4\lambda_{HS_2}^{(1)}\lambda_5$$
$$+ \lambda_5\left(4\lambda_{HS_2}^{(A)} + 6\,\mathrm{Tr}\left[Y_d\ \left(Y_d^*\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ \left(Y_e^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ \left(Y_u^*\right)^T\right]\right)$$

$$\beta^{(1)}_{\lambda_{HS_2}^{(A)}} = - \frac{33g_2^2\lambda_{HS_2}^{(A)}}{2}$$
$$+ g_1^2\left(-4g_2^2 - \frac{15\lambda_{HS_2}^{(A)}}{2}\right)$$
$$+ 2\lambda_1\lambda_{HS_2}^{(A)}$$
$$+ 8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$$
$$+ 2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$$
$$+ \lambda_{HS_2}^{(A)}\left(16\lambda_{S_2}^{(A)} + 6\,\mathrm{Tr}\left[Y_d\ \left(Y_d^*\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ \left(Y_e^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ \left(Y_u^*\right)^T\right]\right)$$
$$+ \frac{8\lambda_5\lambda_5^*}{3}$$

$$\beta^{(1)}_{\lambda_{S_2}^{(A)}} = - 2g_2^4$$
$$- 24g_2^2\lambda_{S_2}^{(A)}$$
$$+ g_1^2\left(8g_2^2 - 12\lambda_{S_2}^{(A)}\right)$$
$$+ 3\left(\lambda_{HS_2}^{(A)}\right)^2$$

$$+ \ 12\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)}$$

$$+ \ 20\left(\lambda_{S_2}^{(A)}\right)^2$$

$\mathrm{T3-A}\,(\alpha=-2)$

$$\beta_{m_1^2}^{(1)} = -\ 6g_1^2m_1^2$$
$$+ \ 6\lambda_{12}^{(1)}m_2^2$$
$$+ \ 4\lambda_{S_1}^{(1)}m_1^2$$
$$+ \ 12\lambda_{18}\left(\mu_{S_2}^2\right)^*$$
$$+ \ 12\mu_{S_2}^2\lambda_{18}^*$$

$$\beta_{m_2^2}^{(1)} = -\ 12g_2^2m_2^2$$
$$+ \ 2\lambda_{12}^{(1)}m_1^2$$
$$+ \ 20\lambda_{14}\left(\mu_{S_2}^2\right)^*$$
$$+ \ m_2^2\left(8\lambda_{S_2}^{(1)} + 4\lambda_{S_2}^{(A)}\right)$$
$$+ \ 20\mu_{S_2}^2\lambda_{14}^*$$

$$\beta_{\lambda_1}^{(1)} = \frac{9g_2^4}{4}$$
$$- \ 9g_2^2\lambda_1$$
$$+ \ \frac{3g_1^4}{4}$$
$$+ \ g_1^2\left(\frac{3g_2^2}{2} - 3\lambda_1\right)$$
$$+ \ 12\lambda_1^2$$
$$+ \ \lambda_1\left(12\,\mathrm{Tr}\left[Y_d\ \left(Y_d^*\right)^T\right] + 4\,\mathrm{Tr}\left[Y_e\ \left(Y_e^*\right)^T\right] + 12\,\mathrm{Tr}\left[Y_u\ \left(Y_u^*\right)^T\right]\right)$$
$$+ \ 2\left(\lambda_{HS_1}^{(1)}\right)^2$$
$$+ \ 6\left(\lambda_{HS_2}^{(1)}\right)^2$$
$$+ \ 3\left(\lambda_{HS_2}^{(A)}\right)^2$$
$$+ \ 4\lambda_5\lambda_5^*$$
$$+ \ 4\lambda_{22}\lambda_{22}^*$$
$$+ \ 24\lambda_{25}\lambda_{25}^*$$
$$- \ 12\,\mathrm{Tr}\left[Y_d\ \left(Y_d^*\right)^T\ Y_d\ \left(Y_d^*\right)^T\right]$$
$$- \ 4\,\mathrm{Tr}\left[Y_e\ \left(Y_e^*\right)^T\ Y_e\ \left(Y_e^*\right)^T\right]$$
$$- \ 12\,\mathrm{Tr}\left[Y_u\ \left(Y_u^*\right)^T\ Y_u\ \left(Y_u^*\right)^T\right]$$

$$\beta_{\lambda_{S_1}^{(1)}}^{(1)} = 12g_1^4$$
$$- \ 12g_1^2\lambda_{S_1}^{(1)}$$

$$
\begin{aligned}
&+ 6\left(\lambda_{12}^{(1)}\right)^2 \\
&+ 4\left(\lambda_{HS_1}^{(1)}\right)^2 \\
&+ 10\left(\lambda_{S_1}^{(1)}\right)^2 \\
&+ 24\lambda_{18}\lambda_{18}^{*}
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda^{(1)}_{S_2}} = {} & 22g_2^4 \\
&- 24g_2^2\lambda_{S_2}^{(1)} \\
&+ 2\left(\lambda_{12}^{(1)}\right)^2 \\
&+ 4\left(\lambda_{HS_2}^{(1)}\right)^2 \\
&+ 14\left(\lambda_{S_2}^{(1)}\right)^2 \\
&+ 8\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)} \\
&+ 16\left(\lambda_{S_2}^{(A)}\right)^2 \\
&+ \frac{832\lambda_{13}\lambda_{13}^{*}}{3} \\
&+ \frac{352\lambda_{14}\lambda_{14}^{*}}{3} \\
&+ \frac{8\lambda_{18}\lambda_{18}^{*}}{3} \\
&+ \frac{16\lambda_{25}\lambda_{25}^{*}}{3}
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda^{(1)}_{HS_1}} = {} & -\frac{9g_2^2\lambda_{HS_1}^{(1)}}{2} \\
&+ 3g_1^4 \\
&- \frac{15g_1^2\lambda_{HS_1}^{(1)}}{2} \\
&+ 6\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)} \\
&+ 6\lambda_1\lambda_{HS_1}^{(1)} \\
&+ 4\left(\lambda_{HS_1}^{(1)}\right)^2 \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)} \\
&+ \lambda_{HS_1}^{(1)}\left(6\,\mathrm{Tr}\left[Y_d\ (Y_d^{*})^T\right] + 2\,\mathrm{Tr}\left[Y_e\ (Y_e^{*})^T\right] + 6\,\mathrm{Tr}\left[Y_u\ (Y_u^{*})^T\right]\right) \\
&+ 12\lambda_5\lambda_5^{*} \\
&+ 12\lambda_{18}\lambda_{25}^{*} \\
&+ 12\lambda_{22}\lambda_{22}^{*}
\end{aligned}
$$

$$+ 12\lambda_{25}\lambda_{18}^*$$

$$\beta^{(1)}_{\lambda^{(1)}_{HS_2}} = 6g_2^4$$
$$- \frac{33g_2^2\lambda^{(1)}_{HS_2}}{2}$$
$$- \frac{3g_1^2\lambda^{(1)}_{HS_2}}{2}$$
$$+ 2\lambda^{(1)}_{12}\lambda^{(1)}_{HS_1}$$
$$+ 6\lambda_1\lambda^{(1)}_{HS_2}$$
$$+ 4\left(\lambda^{(1)}_{HS_2}\right)^2$$
$$+ 8\lambda^{(1)}_{HS_2}\lambda^{(1)}_{S_2}$$
$$+ \lambda^{(1)}_{HS_2}\left(4\lambda^{(A)}_{S_2} + 6\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\right]\right)$$
$$+ 6\left(\lambda^{(A)}_{HS_2}\right)^2$$
$$+ 4\lambda_5\lambda_5^*$$
$$+ 20\lambda_{14}\lambda_{25}^*$$
$$+ 4\lambda_{22}\lambda_{22}^*$$
$$+ 4\lambda_{25}\left(5\lambda_{14}^* + 4\lambda_{25}^*\right)$$

$$\beta^{(1)}_{\lambda^{(1)}_{12}} = - 12g_2^2\lambda^{(1)}_{12}$$
$$- 6g_1^2\lambda^{(1)}_{12}$$
$$+ 4\left(\lambda^{(1)}_{12}\right)^2$$
$$+ 4\lambda^{(1)}_{12}\lambda^{(1)}_{S_1}$$
$$+ 8\lambda^{(1)}_{12}\lambda^{(1)}_{S_2}$$
$$+ 4\lambda^{(1)}_{12}\lambda^{(A)}_{S_2}$$
$$+ 4\lambda^{(1)}_{HS_1}\lambda^{(1)}_{HS_2}$$
$$+ 4\lambda_5\lambda_5^*$$
$$+ 20\lambda_{14}\lambda_{18}^*$$
$$+ 4\lambda_{18}\left(5\lambda_{14}^* + 4\lambda_{18}^*\right)$$
$$+ 4\lambda_{22}\lambda_{22}^*$$

$$\beta^{(1)}_{\lambda_5} = - \frac{21g_2^2\lambda_5}{2}$$
$$- \frac{9g_1^2\lambda_5}{2}$$
$$+ 2\lambda^{(1)}_{12}\lambda_5$$

$$
\begin{aligned}
&+ 2\lambda_1\lambda_5 \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_5 \\
&+ 4\lambda_{HS_2}^{(1)}\lambda_5 \\
&+ \lambda_5\left(4\lambda_{HS_2}^{(A)} + 6\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\right]\right) \\
&+ 4\lambda_{18}\lambda_{22}^* \\
&+ 8\lambda_{25}\lambda_{22}^*
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda_{HS_2}^{(A)}} =& -\frac{33g_2^2\lambda_{HS_2}^{(A)}}{2} \\
&- \frac{3g_1^2\lambda_{HS_2}^{(A)}}{2} \\
&+ 2\lambda_1\lambda_{HS_2}^{(A)} \\
&+ 8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)} \\
&+ 2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)} \\
&+ \lambda_{HS_2}^{(A)}\left(16\lambda_{S_2}^{(A)} + 6\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\right]\right) \\
&+ \frac{8\lambda_5\lambda_5^*}{3} \\
&- \frac{8\lambda_{22}\lambda_{22}^*}{3}
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda_{S_2}^{(A)}} =& -2g_2^4 \\
&- 24g_2^2\lambda_{S_2}^{(A)} \\
&+ 3\left(\lambda_{HS_2}^{(A)}\right)^2 \\
&+ 12\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)} \\
&+ 20\left(\lambda_{S_2}^{(A)}\right)^2 \\
&- \frac{224\lambda_{13}\lambda_{13}^*}{3} \\
&- \frac{44\lambda_{14}\lambda_{14}^*}{3} \\
&- \frac{4\lambda_{18}\lambda_{18}^*}{3} \\
&- \frac{8\lambda_{25}\lambda_{25}^*}{3}
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda_{13}} =& -24g_2^2\lambda_{13} \\
&+ 12\lambda_{S_2}^{(1)}\lambda_{13} \\
&- 8\lambda_{S_2}^{(A)}\lambda_{13} \\
&+ 11\lambda_{14}^2 \\
&+ \lambda_{18}^2 \\
&+ 2\lambda_{25}^2
\end{aligned}
$$

$$\beta^{(1)}_{\lambda_{14}} = -24g_2^2\lambda_{14}$$
$$+ 2\lambda^{(1)}_{12}\lambda_{18}$$
$$+ 4\lambda^{(1)}_{HS_2}\lambda_{25}$$
$$+ 22\lambda^{(1)}_{S_2}\lambda_{14}$$
$$+ 88\lambda_{13}\lambda^*_{14}$$

$$\beta^{(1)}_{\lambda_{18}} = -12g_2^2\lambda_{18}$$
$$- 6g_1^2\lambda_{18}$$
$$+ \lambda^{(1)}_{12}\left(10\lambda_{14} + 8\lambda_{18}\right)$$
$$+ 4\lambda^{(1)}_{HS_1}\lambda_{25}$$
$$+ 4\lambda^{(1)}_{S_1}\lambda_{18}$$
$$+ 2\lambda^{(1)}_{S_2}\lambda_{18}$$
$$- 4\lambda^{(A)}_{S_2}\lambda_{18}$$
$$+ 4\lambda_5\lambda_{22}$$
$$+ 40\lambda_{13}\lambda^*_{18}$$

$$\beta^{(1)}_{\lambda_{22}} = -\frac{21g_2^2\lambda_{22}}{2}$$
$$- \frac{9g_1^2\lambda_{22}}{2}$$
$$+ 2\lambda^{(1)}_{12}\lambda_{22}$$
$$+ 2\lambda_1\lambda_{22}$$
$$+ 4\lambda^{(1)}_{HS_1}\lambda_{22}$$
$$+ 4\lambda^{(1)}_{HS_2}\lambda_{22}$$
$$- 4\lambda^{(A)}_{HS_2}\lambda_{22}$$
$$+ 4\lambda_{18}\lambda^*_5$$
$$+ \lambda_{22}\left(6\,\mathrm{Tr}\left[Y_d\ \left(Y^*_d\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ \left(Y^*_e\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ \left(Y^*_u\right)^T\right]\right)$$
$$+ 8\lambda_{25}\lambda^*_5$$

$$\beta^{(1)}_{\lambda_{25}} = -\frac{33g_2^2\lambda_{25}}{2}$$
$$- \frac{3g_1^2\lambda_{25}}{2}$$
$$+ 6\lambda_1\lambda_{25}$$
$$+ 2\lambda^{(1)}_{HS_1}\lambda_{18}$$
$$+ \lambda^{(1)}_{HS_2}\left(10\lambda_{14} + 8\lambda_{25}\right)$$
$$+ 2\lambda^{(1)}_{S_2}\lambda_{25}$$
$$- 4\lambda^{(A)}_{S_2}\lambda_{25}$$
$$+ 4\lambda_5\lambda_{22}$$
$$+ 40\lambda_{13}\lambda^*_{25}$$
$$+ \lambda_{25}\left(6\,\mathrm{Tr}\left[Y_d\ \left(Y^*_d\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ \left(Y^*_e\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ \left(Y^*_u\right)^T\right]\right)$$

$$\beta^{(1)}_{\mu^2_{S_2}} = -12 g_2^2 \mu^2_{S_2} + 2\lambda^{(1)}_{S_2} \mu^2_{S_2} - 4\lambda^{(A)}_{S_2} \mu^2_{S_2} + 40\lambda_{13} \left(\mu^2_{S_2}\right)^* + 10\lambda_{14} m_2^2 + 2\lambda_{18} m_1^2$$

T3 − A ($\alpha = 0$)

$$\beta^{(1)}_{m_1^2} = 6\lambda^{(1)}_{12} m_2^2 + 4\lambda^{(1)}_{S_1} m_1^2 + 12\lambda_{20} \left(\mu^2_{S_1}\right)^* + 12\mu^2_{S_1} \lambda^*_{20}$$

$$\beta^{(1)}_{m_2^2} = -12 g_2^2 m_2^2 - 6 g_1^2 m_2^2 + 2\lambda^{(1)}_{12} m_1^2 + 4\lambda_{15} \left(\mu^2_{S_1}\right)^* + m_2^2 \left(8\lambda^{(1)}_{S_2} + 4\lambda^{(A)}_{S_2}\right) + 4\mu^2_{S_1} \lambda^*_{15}$$

$$\beta^{(1)}_{\lambda_1} = \frac{9 g_2^4}{4} - 9 g_2^2 \lambda_1 + \frac{3 g_1^4}{4} + g_1^2 \left(\frac{3 g_2^2}{2} - 3\lambda_1\right) + 12\lambda_1^2 + \lambda_1 \left(12 \operatorname{Tr}\left[Y_d \left(Y_d^*\right)^T\right] + 4 \operatorname{Tr}\left[Y_e \left(Y_e^*\right)^T\right] + 12 \operatorname{Tr}\left[Y_u \left(Y_u^*\right)^T\right]\right) + 2\left(\lambda^{(1)}_{HS_1}\right)^2 + 6\left(\lambda^{(1)}_{HS_2}\right)^2 + 3\left(\lambda^{(A)}_{HS_2}\right)^2 + 4\lambda_5 \lambda_5^* + 4\lambda_{23} \lambda_{23}^* + 8\lambda_{27} \lambda_{27}^* - 12 \operatorname{Tr}\left[Y_d \left(Y_d^*\right)^T Y_d \left(Y_d^*\right)^T\right] - 4 \operatorname{Tr}\left[Y_e \left(Y_e^*\right)^T Y_e \left(Y_e^*\right)^T\right]$$

$$- \ 12 \operatorname{Tr}\left[Y_u \ \left(Y_u^*\right)^T \ Y_u \ \left(Y_u^*\right)^T\right]$$

$$\beta^{(1)}_{\lambda^{(1)}_{S_1}} = 6 \left(\lambda^{(1)}_{12}\right)^2$$
$$+ \ 4 \left(\lambda^{(1)}_{HS_1}\right)^2$$
$$+ \ 10 \left(\lambda^{(1)}_{S_1}\right)^2$$
$$+ \ 12 \lambda_{15} \lambda^*_{15}$$
$$+ \ 288 \lambda_{17} \lambda^*_{17}$$
$$+ \ 108 \lambda_{20} \lambda^*_{20}$$
$$+ \ 8 \lambda_{27} \lambda^*_{27}$$

$$\beta^{(1)}_{\lambda^{(1)}_{S_2}} = 22 g_2^4$$
$$- \ 24 g_2^2 \lambda^{(1)}_{S_2}$$
$$+ \ 12 g_1^4$$
$$+ \ g_1^2 \left(8 g_2^2 - 12 \lambda^{(1)}_{S_2}\right)$$
$$+ \ 2 \left(\lambda^{(1)}_{12}\right)^2$$
$$+ \ 4 \left(\lambda^{(1)}_{HS_2}\right)^2$$
$$+ \ 14 \left(\lambda^{(1)}_{S_2}\right)^2$$
$$+ \ 8 \lambda^{(1)}_{S_2} \lambda^{(A)}_{S_2}$$
$$+ \ 16 \left(\lambda^{(A)}_{S_2}\right)^2$$
$$+ \ 8 \lambda_{15} \lambda^*_{15}$$

$$\beta^{(1)}_{\lambda^{(1)}_{HS_1}} = - \frac{9 g_2^2 \lambda^{(1)}_{HS_1}}{2}$$
$$- \ \frac{3 g_1^2 \lambda^{(1)}_{HS_1}}{2}$$
$$+ \ 6 \lambda^{(1)}_{12} \lambda^{(1)}_{HS_2}$$
$$+ \ 6 \lambda_1 \lambda^{(1)}_{HS_1}$$
$$+ \ 4 \left(\lambda^{(1)}_{HS_1}\right)^2$$
$$+ \ 4 \lambda^{(1)}_{HS_1} \lambda^{(1)}_{S_1}$$
$$+ \ \lambda^{(1)}_{HS_1} \left(6 \operatorname{Tr}\left[Y_d \ \left(Y_d^*\right)^T\right] + 2 \operatorname{Tr}\left[Y_e \ \left(Y_e^*\right)^T\right] + 6 \operatorname{Tr}\left[Y_u \ \left(Y_u^*\right)^T\right]\right)$$
$$+ \ 12 \lambda_5 \lambda^*_5$$
$$+ \ 12 \lambda_{20} \lambda^*_{27}$$

$$
\begin{aligned}
&+ 12\lambda_{23}\lambda_{23}^{*} \\
&+ 4\lambda_{27}\left(3\lambda_{20}^{*} + 4\lambda_{27}^{*}\right)
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda^{(1)}_{HS_2}} &= 6g_2^4 \\
&- \frac{33g_2^2\lambda^{(1)}_{HS_2}}{2} \\
&+ 3g_1^4 \\
&- \frac{15g_1^2\lambda^{(1)}_{HS_2}}{2} \\
&+ 2\lambda^{(1)}_{12}\lambda^{(1)}_{HS_1} \\
&+ 6\lambda_1\lambda^{(1)}_{HS_2} \\
&+ 4\left(\lambda^{(1)}_{HS_2}\right)^2 \\
&+ 8\lambda^{(1)}_{HS_2}\lambda^{(1)}_{S_2} \\
&+ \lambda^{(1)}_{HS_2}\left(4\lambda^{(A)}_{S_2} + 6\,\mathrm{Tr}\left[Y_d\,\left(Y_d^{*}\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\,\left(Y_e^{*}\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\,\left(Y_u^{*}\right)^T\right]\right) \\
&+ 6\left(\lambda^{(A)}_{HS_2}\right)^2 \\
&+ 4\lambda_5\lambda_5^{*} \\
&+ 4\lambda_{15}\lambda_{27}^{*} \\
&+ 4\lambda_{23}\lambda_{23}^{*} \\
&+ 4\lambda_{27}\lambda_{15}^{*}
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda^{(1)}_{12}} &= -12g_2^2\lambda^{(1)}_{12} \\
&- 6g_1^2\lambda^{(1)}_{12} \\
&+ 4\left(\lambda^{(1)}_{12}\right)^2 \\
&+ 4\lambda^{(1)}_{12}\lambda^{(1)}_{S_1} \\
&+ 8\lambda^{(1)}_{12}\lambda^{(1)}_{S_2} \\
&+ 4\lambda^{(1)}_{12}\lambda^{(A)}_{S_2} \\
&+ 4\lambda^{(1)}_{HS_1}\lambda^{(1)}_{HS_2} \\
&+ 4\lambda_5\lambda_5^{*} \\
&+ 4\lambda_{15}\left(4\lambda_{15}^{*} + 3\lambda_{20}^{*}\right) \\
&+ 12\lambda_{20}\lambda_{15}^{*} \\
&+ 4\lambda_{23}\lambda_{23}^{*}
\end{aligned}
$$

$$
\beta^{(1)}_{\lambda_5} = -\frac{21g_2^2\lambda_5}{2}
$$

$$
\begin{aligned}
&- \frac{9g_1^2\lambda_5}{2} \\
&+ 2\lambda_{12}^{(1)}\lambda_5 \\
&+ 2\lambda_1\lambda_5 \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_5 \\
&+ 4\lambda_{HS_2}^{(1)}\lambda_5 \\
&+ \lambda_5\left(4\lambda_{HS_2}^{(A)} + 6\,\mathrm{Tr}\left[Y_d\,\left(Y_d^*\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\,\left(Y_e^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\,\left(Y_u^*\right)^T\right]\right) \\
&+ 4\left(\lambda_{15}^* + 2\lambda_{27}^*\right)\lambda_{23}^*
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda_{HS_2}^{(A)}} =& - \frac{33g_2^2\lambda_{HS_2}^{(A)}}{2} \\
&+ g_1^2\left(4g_2^2 - \frac{15\lambda_{HS_2}^{(A)}}{2}\right) \\
&+ 2\lambda_1\lambda_{HS_2}^{(A)} \\
&+ 8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)} \\
&+ 2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)} \\
&+ \lambda_{HS_2}^{(A)}\left(16\lambda_{S_2}^{(A)} + 6\,\mathrm{Tr}\left[Y_d\,\left(Y_d^*\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\,\left(Y_e^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\,\left(Y_u^*\right)^T\right]\right) \\
&+ \frac{8\lambda_5\lambda_5^*}{3} \\
&+ \frac{8\lambda_{23}\lambda_{23}^*}{3}
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda_{S_2}^{(A)}} =& - 2g_2^4 \\
&- 24g_2^2\lambda_{S_2}^{(A)} \\
&+ g_1^2\left(8g_2^2 - 12\lambda_{S_2}^{(A)}\right) \\
&+ 3\left(\lambda_{HS_2}^{(A)}\right)^2 \\
&+ 12\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)} \\
&+ 20\left(\lambda_{S_2}^{(A)}\right)^2
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda_{15}} =& - 12g_2^2\lambda_{15} \\
&- 6g_1^2\lambda_{15} \\
&+ \lambda_{12}^{(1)}\left(8\lambda_{15} + 6\lambda_{20}\right) \\
&+ 4\lambda_{HS_2}^{(1)}\lambda_{27} \\
&+ 2\lambda_{S_1}^{(1)}\lambda_{15} \\
&+ 8\lambda_{S_2}^{(1)}\lambda_{15} \\
&+ 4\lambda_{S_2}^{(A)}\lambda_{15} \\
&+ 24\lambda_{17}\lambda_{15}^* \\
&+ 4\lambda_5^*\lambda_{23}^*
\end{aligned}
$$

$$\beta^{(1)}_{\lambda_{17}} = 12\lambda^{(1)}_{S_1}\lambda_{17} + 3\lambda^2_{15} + 9\lambda^2_{20} + 2\lambda^2_{27}$$

$$\beta^{(1)}_{\lambda_{20}} = 6\lambda^{(1)}_{12}\lambda_{15} + 4\lambda^{(1)}_{HS_1}\lambda_{27} + 18\lambda^{(1)}_{S_1}\lambda_{20} + 72\lambda_{17}\lambda^*_{20}$$

$$\beta^{(1)}_{\lambda_{23}} = -\frac{21g_2^2\lambda_{23}}{2} - \frac{9g_1^2\lambda_{23}}{2} + 2\lambda^{(1)}_{12}\lambda_{23} + 2\lambda_1\lambda_{23} + 4\lambda^{(1)}_{HS_1}\lambda_{23} + 4\lambda^{(1)}_{HS_2}\lambda_{23} + 4\lambda^{(A)}_{HS_2}\lambda_{23} + \lambda_{23}\left(6\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\right]\right) + 4\left(\lambda^*_{15} + 2\lambda^*_{27}\right)\lambda^*_5$$

$$\beta^{(1)}_{\lambda_{27}} = -\frac{9g_2^2\lambda_{27}}{2} - \frac{3g_1^2\lambda_{27}}{2} + 6\lambda_1\lambda_{27} + \lambda^{(1)}_{HS_1}\left(6\lambda_{20} + 8\lambda_{27}\right) + 6\lambda^{(1)}_{HS_2}\lambda_{15} + 2\lambda^{(1)}_{S_1}\lambda_{27} + 24\lambda_{17}\lambda^*_{27} + \lambda_{27}\left(6\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\right]\right) + 12\lambda^*_5\lambda^*_{23}$$

$$\beta^{(1)}_{\mu^2_{S_1}} = 2\lambda^{(1)}_{S_1}\mu^2_{S_1} + 6\lambda_{15}m_2^2 + 24\lambda_{17}\left(\mu^2_{S_1}\right)^* + 6\lambda_{20}m_1^2$$

T3 − B ($\alpha = -3$)

$$\beta^{(1)}_{m_1^2} = -\frac{9g_2^2m_1^2}{2}$$

$$
\begin{aligned}
&- \frac{27 g_1^2 m_1^2}{2} \\
&+ 4 \lambda_{12}^{(1)} m_2^2 \\
&+ 6 \lambda_{S_1}^{(1)} m_1^2
\end{aligned}
$$

$$
\begin{aligned}
\beta_{m_2^2}^{(1)} = &- \frac{9 g_2^2 m_2^2}{2} \\
&- \frac{3 g_1^2 m_2^2}{2} \\
&+ 4 \lambda_{12}^{(1)} m_1^2 \\
&+ 6 \lambda_{S_2}^{(1)} m_2^2
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_1}^{(1)} = &\frac{9 g_2^4}{4} \\
&- 9 g_2^2 \lambda_1 \\
&+ \frac{3 g_1^4}{4} \\
&+ g_1^2 \left(\frac{3 g_2^2}{2} - 3 \lambda_1 \right) \\
&+ 12 \lambda_1^2 \\
&+ \lambda_1 \left(12 \operatorname{Tr}\left[Y_d \, (Y_d^*)^T \right] + 4 \operatorname{Tr}\left[Y_e \, (Y_e^*)^T \right] + 12 \operatorname{Tr}\left[Y_u \, (Y_u^*)^T \right] \right) \\
&+ 4 \left(\lambda_{HS_1}^{(1)} \right)^2 \\
&+ \frac{\left(\lambda_{HS_1}^{(A)} \right)^2}{4} \\
&+ 4 \left(\lambda_{HS_2}^{(1)} \right)^2 \\
&+ \frac{\left(\lambda_{HS_2}^{(A)} \right)^2}{4} \\
&+ 4 \lambda_5 \lambda_5^* \\
&+ 8 \lambda_{21} \lambda_{21}^* \\
&- 12 \operatorname{Tr}\left[Y_d \, (Y_d^*)^T \, Y_d \, (Y_d^*)^T \right] \\
&- 4 \operatorname{Tr}\left[Y_e \, (Y_e^*)^T \, Y_e \, (Y_e^*)^T \right] \\
&- 12 \operatorname{Tr}\left[Y_u \, (Y_u^*)^T \, Y_u \, (Y_u^*)^T \right]
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{S_1}^{(1)}}^{(1)} = &\frac{9 g_2^4}{4} \\
&- 9 g_2^2 \lambda_{S_1}^{(1)} \\
&+ \frac{243 g_1^4}{4} \\
&+ g_1^2 \left(\frac{27 g_2^2}{2} - 27 \lambda_{S_1}^{(1)} \right) \\
&+ 4 \left(\lambda_{12}^{(1)} \right)^2
\end{aligned}
$$

$$
\begin{aligned}
&+ \frac{\left(\lambda_{12}^{(A)}\right)^2}{4} \\
&+ 4\left(\lambda_{HS_1}^{(1)}\right)^2 \\
&+ \frac{\left(\lambda_{HS_1}^{(A)}\right)^2}{4} \\
&+ 12\left(\lambda_{S_1}^{(1)}\right)^2
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda^{(1)}_{S_2}} = &\frac{9g_2^4}{4} \\
&- 9g_2^2\lambda_{S_2}^{(1)} \\
&+ \frac{3g_1^4}{4} \\
&+ g_1^2\left(\frac{3g_2^2}{2} - 3\lambda_{S_2}^{(1)}\right) \\
&+ 4\left(\lambda_{12}^{(1)}\right)^2 \\
&+ \frac{\left(\lambda_{12}^{(A)}\right)^2}{4} \\
&+ 4\left(\lambda_{HS_2}^{(1)}\right)^2 \\
&+ \frac{\left(\lambda_{HS_2}^{(A)}\right)^2}{4} \\
&+ 12\left(\lambda_{S_2}^{(1)}\right)^2 \\
&+ 8\lambda_{21}\lambda_{21}^*
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda^{(1)}_{HS_1}} = &\frac{9g_2^4}{4} \\
&- 9g_2^2\lambda_{HS_1}^{(1)} \\
&+ \frac{27g_1^4}{4} \\
&- 15g_1^2\lambda_{HS_1}^{(1)} \\
&+ 4\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)} \\
&+ 6\lambda_1\lambda_{HS_1}^{(1)} \\
&+ 4\left(\lambda_{HS_1}^{(1)}\right)^2 \\
&+ 6\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)} \\
&+ \lambda_{HS_1}^{(1)}\left(6\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\right]\right) \\
&+ \frac{3\left(\lambda_{HS_1}^{(A)}\right)^2}{4}
\end{aligned}
$$

$$+ 6\lambda_5\lambda_5^*$$

$$\beta^{(1)}_{\lambda^{(1)}_{HS_2}} = \frac{9g_2^4}{4}$$
$$- 9g_2^2\lambda^{(1)}_{HS_2}$$
$$+ \frac{3g_1^4}{4}$$
$$- 3g_1^2\lambda^{(1)}_{HS_2}$$
$$+ 4\lambda^{(1)}_{12}\lambda^{(1)}_{HS_1}$$
$$+ 6\lambda_1\lambda^{(1)}_{HS_2}$$
$$+ 4\left(\lambda^{(1)}_{HS_2}\right)^2$$
$$+ 6\lambda^{(1)}_{HS_2}\lambda^{(1)}_{S_2}$$
$$+ \lambda^{(1)}_{HS_2}\left(6\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\right]\right)$$
$$+ \frac{3\left(\lambda^{(A)}_{HS_2}\right)^2}{4}$$
$$+ 6\lambda_5\lambda_5^*$$
$$+ 24\lambda_{21}\lambda_{21}^*$$

$$\beta^{(1)}_{\lambda^{(1)}_{12}} = \frac{9g_2^4}{4}$$
$$- 9g_2^2\lambda^{(1)}_{12}$$
$$+ \frac{27g_1^4}{4}$$
$$- 15g_1^2\lambda^{(1)}_{12}$$
$$+ 4\left(\lambda^{(1)}_{12}\right)^2$$
$$+ 6\lambda^{(1)}_{12}\lambda^{(1)}_{S_1}$$
$$+ 6\lambda^{(1)}_{12}\lambda^{(1)}_{S_2}$$
$$+ \frac{3\left(\lambda^{(A)}_{12}\right)^2}{4}$$
$$+ 4\lambda^{(1)}_{HS_1}\lambda^{(1)}_{HS_2}$$
$$+ 3\lambda_5\lambda_5^*$$

$$\beta^{(1)}_{\lambda_5} = - 9g_2^2\lambda_5$$
$$- 9g_1^2\lambda_5$$
$$+ 2\lambda^{(1)}_{12}\lambda_5$$
$$+ 2\lambda_1\lambda_5$$
$$+ 4\lambda^{(1)}_{HS_1}\lambda_5$$

$$+ 4\lambda_{HS_2}^{(1)}\lambda_5$$
$$+ \lambda_5\left(\frac{\lambda_{12}^{(A)}}{6} + \frac{2\lambda_{HS_1}^{(A)}}{3} + 2\lambda_{HS_2}^{(A)} + 6\,\mathrm{Tr}\left[Y_d\ \left(Y_d^*\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ \left(Y_e^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ \left(Y_u^*\right)^T\right]\right)$$

$$\beta^{(1)}_{\lambda_{HS_1}^{(A)}} = -9g_2^2\lambda_{HS_1}^{(A)}$$
$$+ g_1^2\left(-18g_2^2 - 15\lambda_{HS_1}^{(A)}\right)$$
$$+ \lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$$
$$+ 2\lambda_1\lambda_{HS_1}^{(A)}$$
$$+ 8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$$
$$+ 2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$$
$$+ \lambda_{HS_1}^{(A)}\left(6\,\mathrm{Tr}\left[Y_d\ \left(Y_d^*\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ \left(Y_e^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ \left(Y_u^*\right)^T\right]\right)$$
$$+ \frac{16\lambda_5\lambda_5^*}{3}$$

$$\beta^{(1)}_{\lambda_{HS_2}^{(A)}} = -9g_2^2\lambda_{HS_2}^{(A)}$$
$$+ g_1^2\left(-6g_2^2 - 3\lambda_{HS_2}^{(A)}\right)$$
$$+ \lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$$
$$+ 2\lambda_1\lambda_{HS_2}^{(A)}$$
$$+ 8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$$
$$+ 2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$$
$$+ \lambda_{HS_2}^{(A)}\left(6\,\mathrm{Tr}\left[Y_d\ \left(Y_d^*\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ \left(Y_e^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ \left(Y_u^*\right)^T\right]\right)$$
$$+ 16\lambda_5\lambda_5^*$$
$$- 64\lambda_{21}\lambda_{21}^*$$

$$\beta^{(1)}_{\lambda_{12}^{(A)}} = -9g_2^2\lambda_{12}^{(A)}$$
$$+ g_1^2\left(18g_2^2 - 15\lambda_{12}^{(A)}\right)$$
$$+ 8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$$
$$+ 2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$$
$$+ 2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$$
$$+ \lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$$
$$+ \frac{4\lambda_5\lambda_5^*}{3}$$

$$\beta^{(1)}_{\lambda_{21}} = -9g_2^2\lambda_{21}$$
$$- 3g_1^2\lambda_{21}$$
$$+ 2\lambda_1\lambda_{21}$$

$$
\begin{aligned}
&+ 8\lambda_{HS_2}^{(1)}\lambda_{21} \\
&- 4\lambda_{HS_2}^{(A)}\lambda_{21} \\
&+ 2\lambda_{S_2}^{(1)}\lambda_{21} \\
&+ \lambda_{21}\left(6\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\right]\right)
\end{aligned}
$$

$\mathrm{T3-B}\,(\alpha=-1)$

$$
\begin{aligned}
\beta_{m_1^2}^{(1)} =& -\frac{9g_2^2m_1^2}{2} \\
&- \frac{3g_1^2m_1^2}{2} \\
&+ 4\lambda_{12}^{(1)}m_2^2 \\
&+ 6\lambda_{S_1}^{(1)}m_1^2 \\
&+ 6\lambda_{11}\left(\mu_{12}^2\right)^* \\
&+ 6\mu_{12}^2\lambda_{11}^*
\end{aligned}
$$

$$
\begin{aligned}
\beta_{m_2^2}^{(1)} =& -\frac{9g_2^2m_2^2}{2} \\
&- \frac{3g_1^2m_2^2}{2} \\
&+ 4\lambda_{12}^{(1)}m_1^2 \\
&+ 6\lambda_{10}\left(\mu_{12}^2\right)^* \\
&+ 6\lambda_{S_2}^{(1)}m_2^2 \\
&+ 6\mu_{12}^2\lambda_{10}^*
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_1}^{(1)} =& \frac{9g_2^4}{4} \\
&- 9g_2^2\lambda_1 \\
&+ \frac{3g_1^4}{4} \\
&+ g_1^2\left(\frac{3g_2^2}{2} - 3\lambda_1\right) \\
&+ 12\lambda_1^2 \\
&+ \lambda_1\left(12\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\right] + 4\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\right] + 12\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\right]\right) \\
&+ 4\left(\lambda_{HS_1}^{(1)}\right)^2 \\
&+ \frac{\left(\lambda_{HS_1}^{(A)}\right)^2}{4} \\
&+ 4\left(\lambda_{HS_2}^{(1)}\right)^2 \\
&+ \frac{\left(\lambda_{HS_2}^{(A)}\right)^2}{4} \\
&+ 8\lambda_7\lambda_7^* \\
&+ 4\lambda_{12}\lambda_{12}^*
\end{aligned}
$$

$$
\begin{aligned}
&+ 8\lambda_8\lambda_8^* \\
&+ 4\lambda_5\lambda_5^* \\
&+ 4\lambda_{26}\lambda_{26}^* \\
&- 12\,\mathrm{Tr}\Big[Y_d\ (Y_d^*)^T\ Y_d\ (Y_d^*)^T\Big] \\
&- 4\,\mathrm{Tr}\Big[Y_e\ (Y_e^*)^T\ Y_e\ (Y_e^*)^T\Big] \\
&- 12\,\mathrm{Tr}\Big[Y_u\ (Y_u^*)^T\ Y_u\ (Y_u^*)^T\Big]
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda^{(1)}_{S_1}} = &\frac{9g_2^4}{4} \\
&- 9g_2^2\lambda^{(1)}_{S_1} \\
&+ \frac{3g_1^4}{4} \\
&+ g_1^2\left(\frac{3g_2^2}{2} - 3\lambda^{(1)}_{S_1}\right) \\
&+ 4\left(\lambda^{(1)}_{12}\right)^2 \\
&+ \frac{\left(\lambda^{(A)}_{12}\right)^2}{4} \\
&+ 4\left(\lambda^{(1)}_{HS_1}\right)^2 \\
&+ \frac{\left(\lambda^{(A)}_{HS_1}\right)^2}{4} \\
&+ 8\lambda_7\lambda_7^* \\
&+ 12\left(\lambda^{(1)}_{S_1}\right)^2 \\
&+ 24\lambda_{11}\lambda_{11}^* \\
&+ 8\lambda_9\lambda_9^*
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda^{(1)}_{S_2}} = &\frac{9g_2^4}{4} \\
&- 9g_2^2\lambda^{(1)}_{S_2} \\
&+ \frac{3g_1^4}{4} \\
&+ g_1^2\left(\frac{3g_2^2}{2} - 3\lambda^{(1)}_{S_2}\right) \\
&+ 4\left(\lambda^{(1)}_{12}\right)^2 \\
&+ \frac{\left(\lambda^{(A)}_{12}\right)^2}{4} \\
&+ 4\left(\lambda^{(1)}_{HS_2}\right)^2 \\
&+ \frac{\left(\lambda^{(A)}_{HS_2}\right)^2}{4}
\end{aligned}
$$

$$+ 8\lambda_8\lambda_8^*$$

$$+ 8\lambda_9\lambda_9^*$$

$$+ 24\lambda_{10}\lambda_{10}^*$$

$$+ 12\left(\lambda_{S_2}^{(1)}\right)^2$$

$$\beta^{(1)}_{\lambda^{(1)}_{HS_1}} = \frac{9g_2^4}{4}$$

$$- 9g_2^2\lambda^{(1)}_{HS_1}$$

$$+ \frac{3g_1^4}{4}$$

$$- 3g_1^2\lambda^{(1)}_{HS_1}$$

$$+ 4\lambda_{12}^{(1)}\lambda^{(1)}_{HS_2}$$

$$+ 6\lambda_1\lambda^{(1)}_{HS_1}$$

$$+ 4\left(\lambda^{(1)}_{HS_1}\right)^2$$

$$+ 6\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$$

$$+ \lambda^{(1)}_{HS_1}\left(6\,\mathrm{Tr}\left[Y_d\ \left(Y_d^*\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ \left(Y_e^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ \left(Y_u^*\right)^T\right]\right)$$

$$+ \frac{3\left(\lambda^{(A)}_{HS_1}\right)^2}{4}$$

$$+ 24\lambda_7\lambda_7^*$$

$$+ \lambda_{12}\left(4\lambda_{12}^* + 3\lambda_{11}^* + 2\lambda_{26}^*\right)$$

$$+ 3\lambda_{11}\left(\lambda_{12}^* - \lambda_{26}^*\right)$$

$$+ 6\lambda_5\lambda_5^*$$

$$+ \lambda_{26}\left(2\lambda_{12}^* - 3\lambda_{11}^* + 4\lambda_{26}^*\right)$$

$$\beta^{(1)}_{\lambda^{(1)}_{HS_2}} = \frac{9g_2^4}{4}$$

$$- 9g_2^2\lambda^{(1)}_{HS_2}$$

$$+ \frac{3g_1^4}{4}$$

$$- 3g_1^2\lambda^{(1)}_{HS_2}$$

$$+ 4\lambda_{12}^{(1)}\lambda^{(1)}_{HS_1}$$

$$+ 6\lambda_1\lambda^{(1)}_{HS_2}$$

$$+ 4\left(\lambda^{(1)}_{HS_2}\right)^2$$

$$+ 6\lambda^{(1)}_{HS_2}\lambda^{(1)}_{S_2}$$

$$
\begin{aligned}
&+ \lambda_{HS_2}^{(1)} \left(6 \operatorname{Tr}\left[Y_d \left(Y_d^*\right)^T\right] + 2 \operatorname{Tr}\left[Y_e \left(Y_e^*\right)^T\right] + 6 \operatorname{Tr}\left[Y_u \left(Y_u^*\right)^T\right]\right) \\
&+ \frac{3 \left(\lambda_{HS_2}^{(A)}\right)^2}{4} \\
&+ \lambda_{12} \left(4\lambda_{12}^* + 3\lambda_{10}^* + 2\lambda_{26}^*\right) \\
&+ 24\lambda_8\lambda_8^* \\
&+ 3\lambda_{10} \left(\lambda_{12}^* - \lambda_{26}^*\right) \\
&+ 6\lambda_5\lambda_5^* \\
&+ \lambda_{26} \left(2\lambda_{12}^* - 3\lambda_{10}^* + 4\lambda_{26}^*\right)
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{12}^{(1)}}^{(1)} = &\frac{9g_2^4}{4} \\
&- 9g_2^2\lambda_{12}^{(1)} \\
&+ \frac{3g_1^4}{4} \\
&- 3g_1^2\lambda_{12}^{(1)} \\
&+ 4 \left(\lambda_{12}^{(1)}\right)^2 \\
&+ 6\lambda_{12}^{(1)}\lambda_{S_1}^{(1)} \\
&+ 6\lambda_{12}^{(1)}\lambda_{S_2}^{(1)} \\
&+ \frac{3 \left(\lambda_{12}^{(A)}\right)^2}{4} \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)} \\
&+ \lambda_{12} \left(2\lambda_{12}^* + \lambda_{26}^*\right) \\
&+ 9\lambda_{11} \left(\lambda_{11}^* + \lambda_{10}^*\right) \\
&+ 24\lambda_9\lambda_9^* \\
&+ 9\lambda_{10} \left(\lambda_{11}^* + \lambda_{10}^*\right) \\
&+ 3\lambda_5\lambda_5^* \\
&+ \lambda_{26} \left(\lambda_{12}^* + 2\lambda_{26}^*\right)
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_5}^{(1)} = &- 9g_2^2\lambda_5 \\
&- 3g_1^2\lambda_5 \\
&+ 2\lambda_{12}^{(1)}\lambda_5 \\
&+ 2\lambda_1\lambda_5 \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_5 \\
&+ 4\lambda_{HS_2}^{(1)}\lambda_5 \\
&+ \lambda_5 \left(\frac{\lambda_{12}^{(A)}}{6} + \frac{2\lambda_{HS_1}^{(A)}}{3} + 2\lambda_{HS_2}^{(A)} + 6 \operatorname{Tr}\left[Y_d \left(Y_d^*\right)^T\right] + 2 \operatorname{Tr}\left[Y_e \left(Y_e^*\right)^T\right] + 6 \operatorname{Tr}\left[Y_u \left(Y_u^*\right)^T\right]\right)
\end{aligned}
$$

$$\begin{aligned}\beta^{(1)}_{\lambda^{(A)}_{HS_1}} =& -9g_2^2\lambda^{(A)}_{HS_1}\\ &+ g_1^2\left(-6g_2^2 - 3\lambda^{(A)}_{HS_1}\right)\\ &+ \lambda^{(A)}_{12}\lambda^{(A)}_{HS_2}\\ &+ 2\lambda_1\lambda^{(A)}_{HS_1}\\ &+ 8\lambda^{(1)}_{HS_1}\lambda^{(A)}_{HS_1}\\ &+ 2\lambda^{(A)}_{HS_1}\lambda^{(1)}_{S_1}\\ &+ \lambda^{(A)}_{HS_1}\left(6\,\mathrm{Tr}\left[Y_d\ \left(Y_d^*\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ \left(Y_e^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ \left(Y_u^*\right)^T\right]\right)\\ &- 64\lambda_7\lambda_7^*\\ &+ 4\lambda_{12}\left(2\lambda_{12}^* + \lambda_{11}^*\right)\\ &+ 4\lambda_{11}\left(\lambda_{12}^* + \lambda_{26}^*\right)\\ &+ \frac{16\lambda_5\lambda_5^*}{3}\\ &+ 4\lambda_{26}\left(\lambda_{11}^* - 2\lambda_{26}^*\right)\end{aligned}$$

$$\begin{aligned}\beta^{(1)}_{\lambda^{(A)}_{HS_2}} =& -9g_2^2\lambda^{(A)}_{HS_2}\\ &+ g_1^2\left(6g_2^2 - 3\lambda^{(A)}_{HS_2}\right)\\ &+ \lambda^{(A)}_{12}\lambda^{(A)}_{HS_1}\\ &+ 2\lambda_1\lambda^{(A)}_{HS_2}\\ &+ 8\lambda^{(1)}_{HS_2}\lambda^{(A)}_{HS_2}\\ &+ 2\lambda^{(A)}_{HS_2}\lambda^{(1)}_{S_2}\\ &+ \lambda^{(A)}_{HS_2}\left(6\,\mathrm{Tr}\left[Y_d\ \left(Y_d^*\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ \left(Y_e^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ \left(Y_u^*\right)^T\right]\right)\\ &- 4\lambda_{12}\left(2\lambda_{12}^* + \lambda_{10}^*\right)\\ &+ 64\lambda_8\lambda_8^*\\ &- 4\lambda_{10}\left(\lambda_{12}^* + \lambda_{26}^*\right)\\ &+ 16\lambda_5\lambda_5^*\\ &+ \lambda_{26}\left(-4\lambda_{10}^* + 8\lambda_{26}^*\right)\end{aligned}$$

$$\begin{aligned}\beta^{(1)}_{\lambda^{(A)}_{12}} =& -9g_2^2\lambda^{(A)}_{12}\\ &+ g_1^2\left(-6g_2^2 - 3\lambda^{(A)}_{12}\right)\\ &+ 8\lambda^{(1)}_{12}\lambda^{(A)}_{12}\\ &+ 2\lambda^{(A)}_{12}\lambda^{(1)}_{S_1}\\ &+ 2\lambda^{(A)}_{12}\lambda^{(1)}_{S_2}\end{aligned}$$

$$
\begin{aligned}
&+ \lambda_{HS_1}^{(A)} \lambda_{HS_2}^{(A)} \\
&+ 4\lambda_{12}\lambda_{26}^{*} \\
&- 4\lambda_{11}\left(5\lambda_{11}^{*} + \lambda_{10}^{*}\right) \\
&- 64\lambda_{9}\lambda_{9}^{*} \\
&- 4\lambda_{10}\left(\lambda_{11}^{*} + 5\lambda_{10}^{*}\right) \\
&+ \frac{4\lambda_{5}\lambda_{5}^{*}}{3} \\
&+ 4\lambda_{26}\lambda_{12}^{*}
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_8}^{(1)} = &- 9g_2^2\lambda_8 \\
&- 3g_1^2\lambda_8 \\
&+ 2\lambda_1\lambda_8 \\
&+ 8\lambda_{HS_2}^{(1)}\lambda_8 \\
&+ 4\lambda_{HS_2}^{(A)}\lambda_8 \\
&+ 4\lambda_7\lambda_9^{*} \\
&+ 2\lambda_8\lambda_{S_2}^{(1)} \\
&+ \lambda_8\left(6\,\mathrm{Tr}\left[Y_d\ (Y_d^{*})^{T}\right] + 2\,\mathrm{Tr}\left[Y_e\ (Y_e^{*})^{T}\right] + 6\,\mathrm{Tr}\left[Y_u\ (Y_u^{*})^{T}\right]\right)
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_7}^{(1)} = &- 9g_2^2\lambda_7 \\
&- 3g_1^2\lambda_7 \\
&+ 2\lambda_1\lambda_7 \\
&+ 8\lambda_{HS_1}^{(1)}\lambda_7 \\
&- 4\lambda_{HS_1}^{(A)}\lambda_7 \\
&+ 2\lambda_7\lambda_{S_1}^{(1)} \\
&+ \lambda_7\left(6\,\mathrm{Tr}\left[Y_d\ (Y_d^{*})^{T}\right] + 2\,\mathrm{Tr}\left[Y_e\ (Y_e^{*})^{T}\right] + 6\,\mathrm{Tr}\left[Y_u\ (Y_u^{*})^{T}\right]\right) \\
&+ 4\lambda_8\lambda_9
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_9}^{(1)} = &- 9g_2^2\lambda_9 \\
&- 3g_1^2\lambda_9 \\
&+ 8\lambda_{12}^{(1)}\lambda_9 \\
&- 4\lambda_{12}^{(A)}\lambda_9 \\
&+ 4\lambda_7\lambda_8^{*} \\
&+ \lambda_{12}^2 \\
&+ 2\lambda_{S_1}^{(1)}\lambda_9 \\
&+ 5\lambda_{11}^2 \\
&+ 2\lambda_{11}\lambda_{10} \\
&+ 2\lambda_9\lambda_{S_2}^{(1)} \\
&+ 5\lambda_{10}^2
\end{aligned}
$$

$$+\ \lambda_{26}^2$$

$$\beta_{\lambda_{10}}^{(1)} = -\ 9g_2^2\lambda_{10}$$
$$-\ 3g_1^2\lambda_{10}$$
$$+\ \lambda_{12}^{(1)}\left(6\lambda_{11}+6\lambda_{10}\right)$$
$$+\ \lambda_{12}^{(A)}\left(-\frac{\lambda_{11}}{2}-\frac{5\lambda_{10}}{2}\right)$$
$$+\ \lambda_{HS_2}^{(1)}\left(2\lambda_{12}-2\lambda_{26}\right)$$
$$+\ \lambda_{HS_2}^{(A)}\left(-\frac{\lambda_{12}}{2}-\frac{\lambda_{26}}{2}\right)$$
$$+\ 4\lambda_9\left(\lambda_{11}^*+5\lambda_{10}^*\right)$$
$$+\ 12\lambda_{10}\lambda_{S_2}^{(1)}$$

$$\beta_{\lambda_{11}}^{(1)} = -\ 9g_2^2\lambda_{11}$$
$$-\ 3g_1^2\lambda_{11}$$
$$+\ \lambda_{12}^{(1)}\left(6\lambda_{11}+6\lambda_{10}\right)$$
$$+\ \lambda_{12}^{(A)}\left(-\frac{5\lambda_{11}}{2}-\frac{\lambda_{10}}{2}\right)$$
$$+\ \lambda_{HS_1}^{(1)}\left(2\lambda_{12}-2\lambda_{26}\right)$$
$$+\ \lambda_{HS_1}^{(A)}\left(\frac{\lambda_{12}}{2}+\frac{\lambda_{26}}{2}\right)$$
$$+\ 12\lambda_{S_1}^{(1)}\lambda_{11}$$
$$+\ 4\lambda_9\left(5\lambda_{11}^*+\lambda_{10}^*\right)$$

$$\beta_{\lambda_{12}}^{(1)} = -\ 9g_2^2\lambda_{12}$$
$$-\ 3g_1^2\lambda_{12}$$
$$+\ 2\lambda_{12}^{(1)}\lambda_{12}$$
$$+\ \lambda_{12}^{(A)}\left(-\frac{\lambda_{12}}{2}+\lambda_{26}\right)$$
$$+\ \lambda_1\left(4\lambda_{12}-2\lambda_{26}\right)$$
$$+\ \lambda_{HS_1}^{(1)}\left(4\lambda_{12}+6\lambda_{11}\right)$$
$$+\ \lambda_{HS_1}^{(A)}\left(2\lambda_{12}+\frac{\lambda_{11}}{2}+\lambda_{26}\right)$$
$$+\ \lambda_{HS_2}^{(1)}\left(4\lambda_{12}+6\lambda_{10}\right)$$
$$+\ \lambda_{HS_2}^{(A)}\left(-2\lambda_{12}-\frac{\lambda_{10}}{2}-\lambda_{26}\right)$$
$$+\ \lambda_{12}\left(6\,\mathrm{Tr}\left[Y_d\ \left(Y_d^*\right)^T\right]+2\,\mathrm{Tr}\left[Y_e\ \left(Y_e^*\right)^T\right]+6\,\mathrm{Tr}\left[Y_u\ \left(Y_u^*\right)^T\right]\right)$$
$$+\ \lambda_9\left(8\lambda_{12}^*-4\lambda_{26}^*\right)$$

$$\beta_{\lambda_{26}}^{(1)} = -\ 9g_2^2\lambda_{26}$$
$$-\ 3g_1^2\lambda_{26}$$
$$+\ 2\lambda_{12}^{(1)}\lambda_{26}$$
$$+\ \lambda_{12}^{(A)}\left(\lambda_{12}-\frac{\lambda_{26}}{2}\right)$$

$$
\begin{aligned}
&+ \lambda_1 \left(-2\lambda_{12} + 4\lambda_{26}\right) \\
&+ \lambda_{HS_1}^{(1)} \left(-6\lambda_{11} + 4\lambda_{26}\right) \\
&+ \lambda_{HS_1}^{(A)} \left(-\lambda_{12} + \frac{\lambda_{11}}{2} - 2\lambda_{26}\right) \\
&+ \lambda_{HS_2}^{(1)} \left(-6\lambda_{10} + 4\lambda_{26}\right) \\
&+ \lambda_{HS_2}^{(A)} \left(\lambda_{12} - \frac{\lambda_{10}}{2} + 2\lambda_{26}\right) \\
&+ \lambda_9 \left(-4\lambda_{12}^* + 8\lambda_{26}^*\right) \\
&+ \lambda_{26} \left(6 \operatorname{Tr}\left[Y_d \left(Y_d^*\right)^T\right] + 2 \operatorname{Tr}\left[Y_e \left(Y_e^*\right)^T\right] + 6 \operatorname{Tr}\left[Y_u \left(Y_u^*\right)^T\right]\right)
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\mu_{12}^2}^{(1)} = &- \frac{9 g_2^2 \mu_{12}^2}{2} \\
&- \frac{3 g_1^2 \mu_{12}^2}{2} \\
&+ 2\lambda_{12}^{(1)} \mu_{12}^2 \\
&- \frac{3 \lambda_{12}^{(A)} \mu_{12}^2}{2} \\
&+ 6\lambda_{11} m_1^2 \\
&+ 12\lambda_9 \left(\mu_{12}^2\right)^* \\
&+ 6\lambda_{10} m_2^2
\end{aligned}
$$

$\mathrm{T3-B}\,(\alpha = 1)$

$$
\begin{aligned}
\beta_{m_1^2}^{(1)} = &- \frac{9 g_2^2 m_1^2}{2} \\
&- \frac{3 g_1^2 m_1^2}{2} \\
&+ 4\lambda_{12}^{(1)} m_2^2 \\
&+ 6\lambda_{S_1}^{(1)} m_1^2
\end{aligned}
$$

$$
\begin{aligned}
\beta_{m_2^2}^{(1)} = &- \frac{9 g_2^2 m_2^2}{2} \\
&- \frac{27 g_1^2 m_2^2}{2} \\
&+ 4\lambda_{12}^{(1)} m_1^2 \\
&+ 6\lambda_{S_2}^{(1)} m_2^2
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_1}^{(1)} = &\frac{9 g_2^4}{4} \\
&- 9 g_2^2 \lambda_1 \\
&+ \frac{3 g_1^4}{4} \\
&+ g_1^2 \left(\frac{3 g_2^2}{2} - 3\lambda_1\right) \\
&+ 12\lambda_1^2 \\
&+ \lambda_1 \left(12 \operatorname{Tr}\left[Y_d \left(Y_d^*\right)^T\right] + 4 \operatorname{Tr}\left[Y_e \left(Y_e^*\right)^T\right] + 12 \operatorname{Tr}\left[Y_u \left(Y_u^*\right)^T\right]\right)
\end{aligned}
$$

$$
\begin{aligned}
&+ 4\left(\lambda_{HS_1}^{(1)}\right)^2 \\
&+ \frac{\left(\lambda_{HS_1}^{(A)}\right)^2}{4} \\
&+ 4\left(\lambda_{HS_2}^{(1)}\right)^2 \\
&+ \frac{\left(\lambda_{HS_2}^{(A)}\right)^2}{4} \\
&+ 4\lambda_5\lambda_5^* \\
&+ 8\lambda_{24}\lambda_{24}^* \\
&- 12\,\mathrm{Tr}\left[Y_d\ \left(Y_d^*\right)^T\ Y_d\ \left(Y_d^*\right)^T\right] \\
&- 4\,\mathrm{Tr}\left[Y_e\ \left(Y_e^*\right)^T\ Y_e\ \left(Y_e^*\right)^T\right] \\
&- 12\,\mathrm{Tr}\left[Y_u\ \left(Y_u^*\right)^T\ Y_u\ \left(Y_u^*\right)^T\right]
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda^{(1)}_{S_1}} = &\frac{9g_2^4}{4} \\
&- 9g_2^2\lambda_{S_1}^{(1)} \\
&+ \frac{3g_1^4}{4} \\
&+ g_1^2\left(\frac{3g_2^2}{2} - 3\lambda_{S_1}^{(1)}\right) \\
&+ 4\left(\lambda_{12}^{(1)}\right)^2 \\
&+ \frac{\left(\lambda_{12}^{(A)}\right)^2}{4} \\
&+ 4\left(\lambda_{HS_1}^{(1)}\right)^2 \\
&+ \frac{\left(\lambda_{HS_1}^{(A)}\right)^2}{4} \\
&+ 12\left(\lambda_{S_1}^{(1)}\right)^2 \\
&+ 8\lambda_{24}\lambda_{24}^*
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda^{(1)}_{S_2}} = &\frac{9g_2^4}{4} \\
&- 9g_2^2\lambda_{S_2}^{(1)} \\
&+ \frac{243g_1^4}{4} \\
&+ g_1^2\left(\frac{27g_2^2}{2} - 27\lambda_{S_2}^{(1)}\right) \\
&+ 4\left(\lambda_{12}^{(1)}\right)^2 \\
&+ \frac{\left(\lambda_{12}^{(A)}\right)^2}{4}
\end{aligned}
$$

$$+ 4\left(\lambda_{HS_2}^{(1)}\right)^2$$

$$+ \frac{\left(\lambda_{HS_2}^{(A)}\right)^2}{4}$$

$$+ 12\left(\lambda_{S_2}^{(1)}\right)^2$$

$$\beta_{\lambda_{HS_1}^{(1)}} = \frac{9g_2^4}{4}$$

$$- 9g_2^2\lambda_{HS_1}^{(1)}$$

$$+ \frac{3g_1^4}{4}$$

$$- 3g_1^2\lambda_{HS_1}^{(1)}$$

$$+ 4\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$$

$$+ 6\lambda_1\lambda_{HS_1}^{(1)}$$

$$+ 4\left(\lambda_{HS_1}^{(1)}\right)^2$$

$$+ 6\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$$

$$+ \lambda_{HS_1}^{(1)}\left(6\,\mathrm{Tr}\left[Y_d\,\left(Y_d^*\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\,\left(Y_e^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\,\left(Y_u^*\right)^T\right]\right)$$

$$+ \frac{3\left(\lambda_{HS_1}^{(A)}\right)^2}{4}$$

$$+ 6\lambda_5\lambda_5^*$$

$$+ 24\lambda_{24}\lambda_{24}^*$$

$$\beta_{\lambda_{HS_2}^{(1)}} = \frac{9g_2^4}{4}$$

$$- 9g_2^2\lambda_{HS_2}^{(1)}$$

$$+ \frac{27g_1^4}{4}$$

$$- 15g_1^2\lambda_{HS_2}^{(1)}$$

$$+ 4\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$$

$$+ 6\lambda_1\lambda_{HS_2}^{(1)}$$

$$+ 4\left(\lambda_{HS_2}^{(1)}\right)^2$$

$$+ 6\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$$

$$+ \lambda_{HS_2}^{(1)}\left(6\,\mathrm{Tr}\left[Y_d\,\left(Y_d^*\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\,\left(Y_e^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\,\left(Y_u^*\right)^T\right]\right)$$

$$+ \frac{3\left(\lambda_{HS_2}^{(A)}\right)^2}{4}$$

$$+ 6\lambda_5\lambda_5^*$$

$$\beta^{(1)}_{\lambda^{(1)}_{12}} = \frac{9g_2^4}{4}$$
$$- 9g_2^2\lambda^{(1)}_{12}$$
$$+ \frac{27g_1^4}{4}$$
$$- 15g_1^2\lambda^{(1)}_{12}$$
$$+ 4\left(\lambda^{(1)}_{12}\right)^2$$
$$+ 6\lambda^{(1)}_{12}\lambda^{(1)}_{S_1}$$
$$+ 6\lambda^{(1)}_{12}\lambda^{(1)}_{S_2}$$
$$+ \frac{3\left(\lambda^{(A)}_{12}\right)^2}{4}$$
$$+ 4\lambda^{(1)}_{HS_1}\lambda^{(1)}_{HS_2}$$
$$+ 3\lambda_5\lambda_5^*$$

$$\beta^{(1)}_{\lambda_5} = - 9g_2^2\lambda_5$$
$$- 9g_1^2\lambda_5$$
$$+ 2\lambda^{(1)}_{12}\lambda_5$$
$$+ 2\lambda_1\lambda_5$$
$$+ 4\lambda^{(1)}_{HS_1}\lambda_5$$
$$+ 4\lambda^{(1)}_{HS_2}\lambda_5$$
$$+ \lambda_5\left(\frac{\lambda^{(A)}_{12}}{6} + \frac{2\lambda^{(A)}_{HS_1}}{3} + 2\lambda^{(A)}_{HS_2} + 6\,\mathrm{Tr}\left[Y_d\ \left(Y_d^*\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ \left(Y_e^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ \left(Y_u^*\right)^T\right]\right)$$

$$\beta^{(1)}_{\lambda^{(A)}_{HS_1}} = - 9g_2^2\lambda^{(A)}_{HS_1}$$
$$+ g_1^2\left(6g_2^2 - 3\lambda^{(A)}_{HS_1}\right)$$
$$+ \lambda^{(A)}_{12}\lambda^{(A)}_{HS_2}$$
$$+ 2\lambda_1\lambda^{(A)}_{HS_1}$$
$$+ 8\lambda^{(1)}_{HS_1}\lambda^{(A)}_{HS_1}$$
$$+ 2\lambda^{(A)}_{HS_1}\lambda^{(1)}_{S_1}$$
$$+ \lambda^{(A)}_{HS_1}\left(6\,\mathrm{Tr}\left[Y_d\ \left(Y_d^*\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ \left(Y_e^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ \left(Y_u^*\right)^T\right]\right)$$
$$+ \frac{16\lambda_5\lambda_5^*}{3}$$
$$+ 64\lambda_{24}\lambda_{24}^*$$

$$\beta^{(1)}_{\lambda^{(A)}_{HS_2}} = - 9g_2^2\lambda^{(A)}_{HS_2}$$
$$+ g_1^2\left(18g_2^2 - 15\lambda^{(A)}_{HS_2}\right)$$
$$+ \lambda^{(A)}_{12}\lambda^{(A)}_{HS_1}$$

$$
\begin{aligned}
&+ 2\lambda_1\lambda_{HS_2}^{(A)} \\
&+ 8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)} \\
&+ 2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)} \\
&+ \lambda_{HS_2}^{(A)}\left(6\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\right]\right) \\
&+ 16\lambda_5\lambda_5^*
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda_{12}^{(A)}} =& -9g_2^2\lambda_{12}^{(A)} \\
&+ g_1^2\left(18g_2^2 - 15\lambda_{12}^{(A)}\right) \\
&+ 8\lambda_{12}^{(1)}\lambda_{12}^{(A)} \\
&+ 2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)} \\
&+ 2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)} \\
&+ \lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)} \\
&+ \frac{4\lambda_5\lambda_5^*}{3}
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda_{24}} =& -9g_2^2\lambda_{24} \\
&- 3g_1^2\lambda_{24} \\
&+ 2\lambda_1\lambda_{24} \\
&+ 8\lambda_{HS_1}^{(1)}\lambda_{24} \\
&+ 4\lambda_{HS_1}^{(A)}\lambda_{24} \\
&+ 2\lambda_{S_1}^{(1)}\lambda_{24} \\
&+ \lambda_{24}\left(6\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\right]\right)
\end{aligned}
$$

$\mathrm{T3-C}\,(\alpha = -3)$

$$
\begin{aligned}
\beta^{(1)}_{m_1^2} =& -\frac{9g_2^2m_1^2}{2} \\
&- \frac{27g_1^2m_1^2}{2} \\
&+ 4\lambda_{12}^{(1)}m_2^2 \\
&+ 6\lambda_{S_1}^{(1)}m_1^2
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{m_2^2} =& -\frac{9g_2^2m_2^2}{2} \\
&- \frac{3g_1^2m_2^2}{2} \\
&+ 4\lambda_{12}^{(1)}m_1^2 \\
&+ 6\lambda_{S_2}^{(1)}m_2^2
\end{aligned}
$$

$$
\beta^{(1)}_{\lambda_1} = \frac{9g_2^4}{4}
$$

$$
\begin{aligned}
&- 9g_2^2\lambda_1 \\
&+ \frac{3g_1^4}{4} \\
&+ g_1^2\left(\frac{3g_2^2}{2} - 3\lambda_1\right) \\
&+ 12\lambda_1^2 \\
&+ \lambda_1\left(12\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\right] + 4\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\right] + 12\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\right]\right) \\
&+ 4\left(\lambda_{HS_1}^{(1)}\right)^2 \\
&+ \frac{\left(\lambda_{HS_1}^{(A)}\right)^2}{4} \\
&+ 4\left(\lambda_{HS_2}^{(1)}\right)^2 \\
&+ \frac{\left(\lambda_{HS_2}^{(A)}\right)^2}{4} \\
&+ 4\lambda_5\lambda_5^* \\
&+ 8\lambda_{21}\lambda_{21}^* \\
&- 12\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\ Y_d\ (Y_d^*)^T\right] \\
&- 4\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\ Y_e\ (Y_e^*)^T\right] \\
&- 12\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\ Y_u\ (Y_u^*)^T\right]
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{S_1}^{(1)}}^{(1)} = &\frac{9g_2^4}{4} \\
&- 9g_2^2\lambda_{S_1}^{(1)} \\
&+ \frac{243g_1^4}{4} \\
&+ g_1^2\left(\frac{27g_2^2}{2} - 27\lambda_{S_1}^{(1)}\right) \\
&+ 4\left(\lambda_{12}^{(1)}\right)^2 \\
&+ \frac{\left(\lambda_{12}^{(A)}\right)^2}{4} \\
&+ 4\left(\lambda_{HS_1}^{(1)}\right)^2 \\
&+ \frac{\left(\lambda_{HS_1}^{(A)}\right)^2}{4} \\
&+ 12\left(\lambda_{S_1}^{(1)}\right)^2
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{S_2}^{(1)}}^{(1)} = &\frac{9g_2^4}{4} \\
&- 9g_2^2\lambda_{S_2}^{(1)} \\
&+ \frac{3g_1^4}{4}
\end{aligned}
$$

$$
\begin{aligned}
&+ g_1^2 \left(\frac{3g_2^2}{2} - 3\lambda_{S_2}^{(1)}\right) \\
&+ 4\left(\lambda_{12}^{(1)}\right)^2 \\
&+ \frac{\left(\lambda_{12}^{(A)}\right)^2}{4} \\
&+ 4\left(\lambda_{HS_2}^{(1)}\right)^2 \\
&+ \frac{\left(\lambda_{HS_2}^{(A)}\right)^2}{4} \\
&+ 12\left(\lambda_{S_2}^{(1)}\right)^2 \\
&+ 8\lambda_{21}\lambda_{21}^*
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda^{(1)}_{HS_1}} &= \frac{9g_2^4}{4} \\
&- 9g_2^2\lambda_{HS_1}^{(1)} \\
&+ \frac{27g_1^4}{4} \\
&- 15g_1^2\lambda_{HS_1}^{(1)} \\
&+ 4\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)} \\
&+ 6\lambda_1\lambda_{HS_1}^{(1)} \\
&+ 4\left(\lambda_{HS_1}^{(1)}\right)^2 \\
&+ 6\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)} \\
&+ \lambda_{HS_1}^{(1)}\left(6\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\right]\right) \\
&+ \frac{3\left(\lambda_{HS_1}^{(A)}\right)^2}{4} \\
&+ 6\lambda_5\lambda_5^*
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda^{(1)}_{HS_2}} &= \frac{9g_2^4}{4} \\
&- 9g_2^2\lambda_{HS_2}^{(1)} \\
&+ \frac{3g_1^4}{4} \\
&- 3g_1^2\lambda_{HS_2}^{(1)} \\
&+ 4\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)} \\
&+ 6\lambda_1\lambda_{HS_2}^{(1)} \\
&+ 4\left(\lambda_{HS_2}^{(1)}\right)^2
\end{aligned}
$$

$$
\begin{aligned}
&+ 6\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)} \\
&+ \lambda_{HS_2}^{(1)}\left(6\,\mathrm{Tr}\left[Y_d\,(Y_d^*)^T\right] + 2\,\mathrm{Tr}\left[Y_e\,(Y_e^*)^T\right] + 6\,\mathrm{Tr}\left[Y_u\,(Y_u^*)^T\right]\right) \\
&+ \frac{3\left(\lambda_{HS_2}^{(A)}\right)^2}{4} \\
&+ 6\lambda_5\lambda_5^* \\
&+ 24\lambda_{21}\lambda_{21}^*
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{12}^{(1)}}^{(1)} = &\frac{9g_2^4}{4} \\
&- 9g_2^2\lambda_{12}^{(1)} \\
&+ \frac{27g_1^4}{4} \\
&- 15g_1^2\lambda_{12}^{(1)} \\
&+ 4\left(\lambda_{12}^{(1)}\right)^2 \\
&+ 6\lambda_{12}^{(1)}\lambda_{S_1}^{(1)} \\
&+ 6\lambda_{12}^{(1)}\lambda_{S_2}^{(1)} \\
&+ \frac{3\left(\lambda_{12}^{(A)}\right)^2}{4} \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)} \\
&+ 3\lambda_5\lambda_5^*
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_5}^{(1)} = &- 9g_2^2\lambda_5 \\
&- 9g_1^2\lambda_5 \\
&+ 2\lambda_{12}^{(1)}\lambda_5 \\
&+ 2\lambda_1\lambda_5 \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_5 \\
&+ 4\lambda_{HS_2}^{(1)}\lambda_5 \\
&+ \lambda_5\left(\frac{\lambda_{12}^{(A)}}{6} + \frac{2\lambda_{HS_1}^{(A)}}{3} + 2\lambda_{HS_2}^{(A)} + 6\,\mathrm{Tr}\left[Y_d\,(Y_d^*)^T\right] + 2\,\mathrm{Tr}\left[Y_e\,(Y_e^*)^T\right] + 6\,\mathrm{Tr}\left[Y_u\,(Y_u^*)^T\right]\right)
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{HS_1}^{(A)}}^{(1)} = &- 9g_2^2\lambda_{HS_1}^{(A)} \\
&+ g_1^2\left(-18g_2^2 - 15\lambda_{HS_1}^{(A)}\right) \\
&+ \lambda_{12}^{(A)}\lambda_{HS_2}^{(A)} \\
&+ 2\lambda_1\lambda_{HS_1}^{(A)} \\
&+ 8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)} \\
&+ 2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}
\end{aligned}
$$

$$+ \lambda_{HS_1}^{(A)} \left(6 \operatorname{Tr}\left[Y_d \left(Y_d^*\right)^T\right] + 2 \operatorname{Tr}\left[Y_e \left(Y_e^*\right)^T\right] + 6 \operatorname{Tr}\left[Y_u \left(Y_u^*\right)^T\right]\right)$$
$$+ \frac{16 \lambda_5 \lambda_5^*}{3}$$

$$\beta^{(1)}_{\lambda_{HS_2}^{(A)}} = - 9 g_2^2 \lambda_{HS_2}^{(A)}$$
$$+ g_1^2 \left(-6 g_2^2 - 3 \lambda_{HS_2}^{(A)}\right)$$
$$+ \lambda_{12}^{(A)} \lambda_{HS_1}^{(A)}$$
$$+ 2 \lambda_1 \lambda_{HS_2}^{(A)}$$
$$+ 8 \lambda_{HS_2}^{(1)} \lambda_{HS_2}^{(A)}$$
$$+ 2 \lambda_{HS_2}^{(A)} \lambda_{S_2}^{(1)}$$
$$+ \lambda_{HS_2}^{(A)} \left(6 \operatorname{Tr}\left[Y_d \left(Y_d^*\right)^T\right] + 2 \operatorname{Tr}\left[Y_e \left(Y_e^*\right)^T\right] + 6 \operatorname{Tr}\left[Y_u \left(Y_u^*\right)^T\right]\right)$$
$$+ 16 \lambda_5 \lambda_5^*$$
$$- 64 \lambda_{21} \lambda_{21}^*$$

$$\beta^{(1)}_{\lambda_{12}^{(A)}} = - 9 g_2^2 \lambda_{12}^{(A)}$$
$$+ g_1^2 \left(18 g_2^2 - 15 \lambda_{12}^{(A)}\right)$$
$$+ 8 \lambda_{12}^{(1)} \lambda_{12}^{(A)}$$
$$+ 2 \lambda_{12}^{(A)} \lambda_{S_1}^{(1)}$$
$$+ 2 \lambda_{12}^{(A)} \lambda_{S_2}^{(1)}$$
$$+ \lambda_{HS_1}^{(A)} \lambda_{HS_2}^{(A)}$$
$$+ \frac{4 \lambda_5 \lambda_5^*}{3}$$

$$\beta^{(1)}_{\lambda_{21}} = - 9 g_2^2 \lambda_{21}$$
$$- 3 g_1^2 \lambda_{21}$$
$$+ 2 \lambda_1 \lambda_{21}$$
$$+ 8 \lambda_{HS_2}^{(1)} \lambda_{21}$$
$$- 4 \lambda_{HS_2}^{(A)} \lambda_{21}$$
$$+ 2 \lambda_{S_2}^{(1)} \lambda_{21}$$
$$+ \lambda_{21} \left(6 \operatorname{Tr}\left[Y_d \left(Y_d^*\right)^T\right] + 2 \operatorname{Tr}\left[Y_e \left(Y_e^*\right)^T\right] + 6 \operatorname{Tr}\left[Y_u \left(Y_u^*\right)^T\right]\right)$$

T3 − C ($\alpha = -1$)

$$\beta^{(1)}_{m_1^2} = - \frac{9 g_2^2 m_1^2}{2}$$
$$- \frac{3 g_1^2 m_1^2}{2}$$
$$+ 4 \lambda_{12}^{(1)} m_2^2$$

$$
\begin{aligned}
&+ 6\lambda_{S_1}^{(1)} m_1^2 \\
&+ 6\lambda_{11} \left(\mu_{12}^2\right)^* \\
&+ 6\mu_{12}^2 \lambda_{11}^*
\end{aligned}
$$

$$
\begin{aligned}
\beta_{m_2^2}^{(1)} = &- \frac{9g_2^2 m_2^2}{2} \\
&- \frac{3g_1^2 m_2^2}{2} \\
&+ 4\lambda_{12}^{(1)} m_1^2 \\
&+ 6\lambda_{10} \left(\mu_{12}^2\right)^* \\
&+ 6\lambda_{S_2}^{(1)} m_2^2 \\
&+ 6\mu_{12}^2 \lambda_{10}^*
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_1}^{(1)} = &\frac{9g_2^4}{4} \\
&- 9g_2^2 \lambda_1 \\
&+ \frac{3g_1^4}{4} \\
&+ g_1^2 \left(\frac{3g_2^2}{2} - 3\lambda_1\right) \\
&+ 12\lambda_1^2 \\
&+ \lambda_1 \left(12 \operatorname{Tr}\left[Y_d \left(Y_d^*\right)^T\right] + 4 \operatorname{Tr}\left[Y_e \left(Y_e^*\right)^T\right] + 12 \operatorname{Tr}\left[Y_u \left(Y_u^*\right)^T\right]\right) \\
&+ 4 \left(\lambda_{HS_1}^{(1)}\right)^2 \\
&+ \frac{\left(\lambda_{HS_1}^{(A)}\right)^2}{4} \\
&+ 4 \left(\lambda_{HS_2}^{(1)}\right)^2 \\
&+ \frac{\left(\lambda_{HS_2}^{(A)}\right)^2}{4} \\
&+ 8\lambda_7 \lambda_7^* \\
&+ 4\lambda_{12} \lambda_{12}^* \\
&+ 8\lambda_8 \lambda_8^* \\
&+ 4\lambda_5 \lambda_5^* \\
&+ 4\lambda_{26} \lambda_{26}^* \\
&- 12 \operatorname{Tr}\left[Y_d \left(Y_d^*\right)^T Y_d \left(Y_d^*\right)^T\right] \\
&- 4 \operatorname{Tr}\left[Y_e \left(Y_e^*\right)^T Y_e \left(Y_e^*\right)^T\right] \\
&- 12 \operatorname{Tr}\left[Y_u \left(Y_u^*\right)^T Y_u \left(Y_u^*\right)^T\right]
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{S_1}^{(1)}}^{(1)} = &\frac{9g_2^4}{4} \\
&- 9g_2^2 \lambda_{S_1}^{(1)}
\end{aligned}
$$

$$
\begin{aligned}
&+ \frac{3g_1^4}{4} \\
&+ g_1^2\left(\frac{3g_2^2}{2} - 3\lambda_{S_1}^{(1)}\right) \\
&+ 4\left(\lambda_{12}^{(1)}\right)^2 \\
&+ \frac{\left(\lambda_{12}^{(A)}\right)^2}{4} \\
&+ 4\left(\lambda_{HS_1}^{(1)}\right)^2 \\
&+ \frac{\left(\lambda_{HS_1}^{(A)}\right)^2}{4} \\
&+ 8\lambda_7\lambda_7^* \\
&+ 12\left(\lambda_{S_1}^{(1)}\right)^2 \\
&+ 24\lambda_{11}\lambda_{11}^* \\
&+ 8\lambda_9\lambda_9^*
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{S_2}^{(1)}}^{(1)} = &\frac{9g_2^4}{4} \\
&- 9g_2^2\lambda_{S_2}^{(1)} \\
&+ \frac{3g_1^4}{4} \\
&+ g_1^2\left(\frac{3g_2^2}{2} - 3\lambda_{S_2}^{(1)}\right) \\
&+ 4\left(\lambda_{12}^{(1)}\right)^2 \\
&+ \frac{\left(\lambda_{12}^{(A)}\right)^2}{4} \\
&+ 4\left(\lambda_{HS_2}^{(1)}\right)^2 \\
&+ \frac{\left(\lambda_{HS_2}^{(A)}\right)^2}{4} \\
&+ 8\lambda_8\lambda_8^* \\
&+ 8\lambda_9\lambda_9^* \\
&+ 24\lambda_{10}\lambda_{10}^* \\
&+ 12\left(\lambda_{S_2}^{(1)}\right)^2
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{HS_1}^{(1)}}^{(1)} = &\frac{9g_2^4}{4} \\
&- 9g_2^2\lambda_{HS_1}^{(1)} \\
&+ \frac{3g_1^4}{4}
\end{aligned}
$$

$$
\begin{aligned}
&- 3g_1^2\lambda_{HS_1}^{(1)} \\
&+ 4\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)} \\
&+ 6\lambda_1\lambda_{HS_1}^{(1)} \\
&+ 4\left(\lambda_{HS_1}^{(1)}\right)^2 \\
&+ 6\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)} \\
&+ \lambda_{HS_1}^{(1)}\left(6\,\mathrm{Tr}\left[Y_d\,\left(Y_d^*\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\,\left(Y_e^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\,\left(Y_u^*\right)^T\right]\right) \\
&+ \frac{3\left(\lambda_{HS_1}^{(A)}\right)^2}{4} \\
&+ 24\lambda_7\lambda_7^* \\
&+ \lambda_{12}\left(4\lambda_{12}^* + 3\lambda_{11}^* + 2\lambda_{26}^*\right) \\
&+ 3\lambda_{11}\left(\lambda_{12}^* - \lambda_{26}^*\right) \\
&+ 6\lambda_5\lambda_5^* \\
&+ \lambda_{26}\left(2\lambda_{12}^* - 3\lambda_{11}^* + 4\lambda_{26}^*\right)
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{HS_2}^{(1)}}^{(1)} = &\frac{9g_2^4}{4} \\
&- 9g_2^2\lambda_{HS_2}^{(1)} \\
&+ \frac{3g_1^4}{4} \\
&- 3g_1^2\lambda_{HS_2}^{(1)} \\
&+ 4\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)} \\
&+ 6\lambda_1\lambda_{HS_2}^{(1)} \\
&+ 4\left(\lambda_{HS_2}^{(1)}\right)^2 \\
&+ 6\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)} \\
&+ \lambda_{HS_2}^{(1)}\left(6\,\mathrm{Tr}\left[Y_d\,\left(Y_d^*\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\,\left(Y_e^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\,\left(Y_u^*\right)^T\right]\right) \\
&+ \frac{3\left(\lambda_{HS_2}^{(A)}\right)^2}{4} \\
&+ \lambda_{12}\left(4\lambda_{12}^* + 3\lambda_{10}^* + 2\lambda_{26}^*\right) \\
&+ 24\lambda_8\lambda_8^* \\
&+ 3\lambda_{10}\left(\lambda_{12}^* - \lambda_{26}^*\right) \\
&+ 6\lambda_5\lambda_5^* \\
&+ \lambda_{26}\left(2\lambda_{12}^* - 3\lambda_{10}^* + 4\lambda_{26}^*\right)
\end{aligned}
$$

$$\begin{aligned}\beta^{(1)}_{\lambda^{(1)}_{12}} =& \frac{9g_2^4}{4}\\ &- 9g_2^2\lambda^{(1)}_{12}\\ &+ \frac{3g_1^4}{4}\\ &- 3g_1^2\lambda^{(1)}_{12}\\ &+ 4\left(\lambda^{(1)}_{12}\right)^2\\ &+ 6\lambda^{(1)}_{12}\lambda^{(1)}_{S_1}\\ &+ 6\lambda^{(1)}_{12}\lambda^{(1)}_{S_2}\\ &+ \frac{3\left(\lambda^{(A)}_{12}\right)^2}{4}\\ &+ 4\lambda^{(1)}_{HS_1}\lambda^{(1)}_{HS_2}\\ &+ \lambda_{12}\left(2\lambda^*_{12} + \lambda^*_{26}\right)\\ &+ 9\lambda_{11}\left(\lambda^*_{11} + \lambda^*_{10}\right)\\ &+ 24\lambda_9\lambda^*_9\\ &+ 9\lambda_{10}\left(\lambda^*_{11} + \lambda^*_{10}\right)\\ &+ 3\lambda_5\lambda^*_5\\ &+ \lambda_{26}\left(\lambda^*_{12} + 2\lambda^*_{26}\right)\end{aligned}$$

$$\begin{aligned}\beta^{(1)}_{\lambda_5} =& - 9g_2^2\lambda_5\\ &- 3g_1^2\lambda_5\\ &+ 2\lambda^{(1)}_{12}\lambda_5\\ &+ 2\lambda_1\lambda_5\\ &+ 4\lambda^{(1)}_{HS_1}\lambda_5\\ &+ 4\lambda^{(1)}_{HS_2}\lambda_5\\ &+ \lambda_5\left(\frac{\lambda^{(A)}_{12}}{6} + \frac{2\lambda^{(A)}_{HS_1}}{3} + 2\lambda^{(A)}_{HS_2} + 6\,\mathrm{Tr}\left[Y_d\ \left(Y^*_d\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ \left(Y^*_e\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ \left(Y^*_u\right)^T\right]\right)\end{aligned}$$

$$\begin{aligned}\beta^{(1)}_{\lambda^{(A)}_{HS_1}} =& - 9g_2^2\lambda^{(A)}_{HS_1}\\ &+ g_1^2\left(-6g_2^2 - 3\lambda^{(A)}_{HS_1}\right)\\ &+ \lambda^{(A)}_{12}\lambda^{(A)}_{HS_2}\\ &+ 2\lambda_1\lambda^{(A)}_{HS_1}\\ &+ 8\lambda^{(1)}_{HS_1}\lambda^{(A)}_{HS_1}\\ &+ 2\lambda^{(A)}_{HS_1}\lambda^{(1)}_{S_1}\\ &+ \lambda^{(A)}_{HS_1}\left(6\,\mathrm{Tr}\left[Y_d\ \left(Y^*_d\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ \left(Y^*_e\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ \left(Y^*_u\right)^T\right]\right)\end{aligned}$$

$$
\begin{aligned}
&- 64\lambda_7\lambda_7^* \\
&+ 4\lambda_{12}\left(2\lambda_{12}^* + \lambda_{11}^*\right) \\
&+ 4\lambda_{11}\left(\lambda_{12}^* + \lambda_{26}^*\right) \\
&+ \frac{16\lambda_5\lambda_5^*}{3} \\
&+ 4\lambda_{26}\left(\lambda_{11}^* - 2\lambda_{26}^*\right)
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda^{(A)}_{HS_2}} =& - 9g_2^2\lambda^{(A)}_{HS_2} \\
&+ g_1^2\left(6g_2^2 - 3\lambda^{(A)}_{HS_2}\right) \\
&+ \lambda^{(A)}_{12}\lambda^{(A)}_{HS_1} \\
&+ 2\lambda_1\lambda^{(A)}_{HS_2} \\
&+ 8\lambda^{(1)}_{HS_2}\lambda^{(A)}_{HS_2} \\
&+ 2\lambda^{(A)}_{HS_2}\lambda^{(1)}_{S_2} \\
&+ \lambda^{(A)}_{HS_2}\left(6\,\mathrm{Tr}\left[Y_d\,\left(Y_d^*\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\,\left(Y_e^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\,\left(Y_u^*\right)^T\right]\right) \\
&- 4\lambda_{12}\left(2\lambda_{12}^* + \lambda_{10}^*\right) \\
&+ 64\lambda_8\lambda_8^* \\
&- 4\lambda_{10}\left(\lambda_{12}^* + \lambda_{26}^*\right) \\
&+ 16\lambda_5\lambda_5^* \\
&+ \lambda_{26}\left(-4\lambda_{10}^* + 8\lambda_{26}^*\right)
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda^{(A)}_{12}} =& - 9g_2^2\lambda^{(A)}_{12} \\
&+ g_1^2\left(-6g_2^2 - 3\lambda^{(A)}_{12}\right) \\
&+ 8\lambda^{(1)}_{12}\lambda^{(A)}_{12} \\
&+ 2\lambda^{(A)}_{12}\lambda^{(1)}_{S_1} \\
&+ 2\lambda^{(A)}_{12}\lambda^{(1)}_{S_2} \\
&+ \lambda^{(A)}_{HS_1}\lambda^{(A)}_{HS_2} \\
&+ 4\lambda_{12}\lambda_{26}^* \\
&- 4\lambda_{11}\left(5\lambda_{11}^* + \lambda_{10}^*\right) \\
&- 64\lambda_9\lambda_9^* \\
&- 4\lambda_{10}\left(\lambda_{11}^* + 5\lambda_{10}^*\right) \\
&+ \frac{4\lambda_5\lambda_5^*}{3} \\
&+ 4\lambda_{26}\lambda_{12}^*
\end{aligned}
$$

$$\begin{aligned}\beta^{(1)}_{\lambda_8} =& -9g_2^2\lambda_8 \\ &- 3g_1^2\lambda_8 \\ &+ 2\lambda_1\lambda_8 \\ &+ 8\lambda^{(1)}_{HS_2}\lambda_8 \\ &+ 4\lambda^{(A)}_{HS_2}\lambda_8 \\ &+ 4\lambda_7\lambda_9^* \\ &+ 2\lambda_8\lambda^{(1)}_{S_2} \\ &+ \lambda_8\left(6\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\right]\right)\end{aligned}$$

$$\begin{aligned}\beta^{(1)}_{\lambda_7} =& -9g_2^2\lambda_7 \\ &- 3g_1^2\lambda_7 \\ &+ 2\lambda_1\lambda_7 \\ &+ 8\lambda^{(1)}_{HS_1}\lambda_7 \\ &- 4\lambda^{(A)}_{HS_1}\lambda_7 \\ &+ 2\lambda_7\lambda^{(1)}_{S_1} \\ &+ \lambda_7\left(6\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\right]\right) \\ &+ 4\lambda_8\lambda_9\end{aligned}$$

$$\begin{aligned}\beta^{(1)}_{\lambda_9} =& -9g_2^2\lambda_9 \\ &- 3g_1^2\lambda_9 \\ &+ 8\lambda^{(1)}_{12}\lambda_9 \\ &- 4\lambda^{(A)}_{12}\lambda_9 \\ &+ 4\lambda_7\lambda_8^* \\ &+ \lambda_{12}^2 \\ &+ 2\lambda^{(1)}_{S_1}\lambda_9 \\ &+ 5\lambda_{11}^2 \\ &+ 2\lambda_{11}\lambda_{10} \\ &+ 2\lambda_9\lambda^{(1)}_{S_2} \\ &+ 5\lambda_{10}^2 \\ &+ \lambda_{26}^2\end{aligned}$$

$$\begin{aligned}\beta^{(1)}_{\lambda_{10}} =& -9g_2^2\lambda_{10} \\ &- 3g_1^2\lambda_{10} \\ &+ \lambda^{(1)}_{12}\left(6\lambda_{11} + 6\lambda_{10}\right) \\ &+ \lambda^{(A)}_{12}\left(-\frac{\lambda_{11}}{2} - \frac{5\lambda_{10}}{2}\right) \\ &+ \lambda^{(1)}_{HS_2}\left(2\lambda_{12} - 2\lambda_{26}\right) \\ &+ \lambda^{(A)}_{HS_2}\left(-\frac{\lambda_{12}}{2} - \frac{\lambda_{26}}{2}\right) \\ &+ 4\lambda_9\left(\lambda_{11}^* + 5\lambda_{10}^*\right)\end{aligned}$$

$$+ 12\lambda_{10}\lambda_{S_2}^{(1)}$$

$$\beta_{\lambda_{11}}^{(1)} = - 9g_2^2\lambda_{11}$$
$$- 3g_1^2\lambda_{11}$$
$$+ \lambda_{12}^{(1)}\left(6\lambda_{11} + 6\lambda_{10}\right)$$
$$+ \lambda_{12}^{(A)}\left(-\frac{5\lambda_{11}}{2} - \frac{\lambda_{10}}{2}\right)$$
$$+ \lambda_{HS_1}^{(1)}\left(2\lambda_{12} - 2\lambda_{26}\right)$$
$$+ \lambda_{HS_1}^{(A)}\left(\frac{\lambda_{12}}{2} + \frac{\lambda_{26}}{2}\right)$$
$$+ 12\lambda_{S_1}^{(1)}\lambda_{11}$$
$$+ 4\lambda_9\left(5\lambda_{11}^* + \lambda_{10}^*\right)$$

$$\beta_{\lambda_{12}}^{(1)} = - 9g_2^2\lambda_{12}$$
$$- 3g_1^2\lambda_{12}$$
$$+ 2\lambda_{12}^{(1)}\lambda_{12}$$
$$+ \lambda_{12}^{(A)}\left(-\frac{\lambda_{12}}{2} + \lambda_{26}\right)$$
$$+ \lambda_1\left(4\lambda_{12} - 2\lambda_{26}\right)$$
$$+ \lambda_{HS_1}^{(1)}\left(4\lambda_{12} + 6\lambda_{11}\right)$$
$$+ \lambda_{HS_1}^{(A)}\left(2\lambda_{12} + \frac{\lambda_{11}}{2} + \lambda_{26}\right)$$
$$+ \lambda_{HS_2}^{(1)}\left(4\lambda_{12} + 6\lambda_{10}\right)$$
$$+ \lambda_{HS_2}^{(A)}\left(-2\lambda_{12} - \frac{\lambda_{10}}{2} - \lambda_{26}\right)$$
$$+ \lambda_{12}\left(6\,\mathrm{Tr}\left[Y_d\ \left(Y_d^*\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ \left(Y_e^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ \left(Y_u^*\right)^T\right]\right)$$
$$+ \lambda_9\left(8\lambda_{12}^* - 4\lambda_{26}^*\right)$$

$$\beta_{\lambda_{26}}^{(1)} = - 9g_2^2\lambda_{26}$$
$$- 3g_1^2\lambda_{26}$$
$$+ 2\lambda_{12}^{(1)}\lambda_{26}$$
$$+ \lambda_{12}^{(A)}\left(\lambda_{12} - \frac{\lambda_{26}}{2}\right)$$
$$+ \lambda_1\left(-2\lambda_{12} + 4\lambda_{26}\right)$$
$$+ \lambda_{HS_1}^{(1)}\left(-6\lambda_{11} + 4\lambda_{26}\right)$$
$$+ \lambda_{HS_1}^{(A)}\left(-\lambda_{12} + \frac{\lambda_{11}}{2} - 2\lambda_{26}\right)$$
$$+ \lambda_{HS_2}^{(1)}\left(-6\lambda_{10} + 4\lambda_{26}\right)$$
$$+ \lambda_{HS_2}^{(A)}\left(\lambda_{12} - \frac{\lambda_{10}}{2} + 2\lambda_{26}\right)$$
$$+ \lambda_9\left(-4\lambda_{12}^* + 8\lambda_{26}^*\right)$$
$$+ \lambda_{26}\left(6\,\mathrm{Tr}\left[Y_d\ \left(Y_d^*\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ \left(Y_e^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ \left(Y_u^*\right)^T\right]\right)$$

$$\begin{aligned}\beta^{(1)}_{\mu^2_{12}} =& -\frac{9g_2^2\mu^2_{12}}{2}\\ &-\frac{3g_1^2\mu^2_{12}}{2}\\ &+2\lambda^{(1)}_{12}\mu^2_{12}\\ &-\frac{3\lambda^{(A)}_{12}\mu^2_{12}}{2}\\ &+6\lambda_{11}m_1^2\\ &+12\lambda_9\left(\mu^2_{12}\right)^*\\ &+6\lambda_{10}m_2^2\end{aligned}$$

T3 − C ($\alpha = 1$)

$$\begin{aligned}\beta^{(1)}_{m_1^2} =& -\frac{9g_2^2m_1^2}{2}\\ &-\frac{3g_1^2m_1^2}{2}\\ &+4\lambda^{(1)}_{12}m_2^2\\ &+6\lambda^{(1)}_{S_1}m_1^2\end{aligned}$$

$$\begin{aligned}\beta^{(1)}_{m_2^2} =& -\frac{9g_2^2m_2^2}{2}\\ &-\frac{27g_1^2m_2^2}{2}\\ &+4\lambda^{(1)}_{12}m_1^2\\ &+6\lambda^{(1)}_{S_2}m_2^2\end{aligned}$$

$$\begin{aligned}\beta^{(1)}_{\lambda_1} =& \frac{9g_2^4}{4}\\ &-9g_2^2\lambda_1\\ &+\frac{3g_1^4}{4}\\ &+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)\\ &+12\lambda_1^2\\ &+\lambda_1\left(12\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\right]+4\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\right]+12\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\right]\right)\\ &+4\left(\lambda^{(1)}_{HS_1}\right)^2\\ &+\frac{\left(\lambda^{(A)}_{HS_1}\right)^2}{4}\\ &+4\left(\lambda^{(1)}_{HS_2}\right)^2\\ &+\frac{\left(\lambda^{(A)}_{HS_2}\right)^2}{4}\\ &+4\lambda_5\lambda_5^*\\ &+8\lambda_{24}\lambda_{24}^*\end{aligned}$$

$$
\begin{aligned}
&- \ 12\,\mathrm{Tr}\Big[Y_d\ \left(Y_d^*\right)^T\ Y_d\ \left(Y_d^*\right)^T\Big] \\
&- \ 4\,\mathrm{Tr}\Big[Y_e\ \left(Y_e^*\right)^T\ Y_e\ \left(Y_e^*\right)^T\Big] \\
&- \ 12\,\mathrm{Tr}\Big[Y_u\ \left(Y_u^*\right)^T\ Y_u\ \left(Y_u^*\right)^T\Big]
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda^{(1)}_{S_1}} &= \frac{9g_2^4}{4} \\
&- \ 9g_2^2\lambda^{(1)}_{S_1} \\
&+ \ \frac{3g_1^4}{4} \\
&+ \ g_1^2\left(\frac{3g_2^2}{2} - 3\lambda^{(1)}_{S_1}\right) \\
&+ \ 4\left(\lambda^{(1)}_{12}\right)^2 \\
&+ \ \frac{\left(\lambda^{(A)}_{12}\right)^2}{4} \\
&+ \ 4\left(\lambda^{(1)}_{HS_1}\right)^2 \\
&+ \ \frac{\left(\lambda^{(A)}_{HS_1}\right)^2}{4} \\
&+ \ 12\left(\lambda^{(1)}_{S_1}\right)^2 \\
&+ \ 8\lambda_{24}\lambda^*_{24}
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda^{(1)}_{S_2}} &= \frac{9g_2^4}{4} \\
&- \ 9g_2^2\lambda^{(1)}_{S_2} \\
&+ \ \frac{243g_1^4}{4} \\
&+ \ g_1^2\left(\frac{27g_2^2}{2} - 27\lambda^{(1)}_{S_2}\right) \\
&+ \ 4\left(\lambda^{(1)}_{12}\right)^2 \\
&+ \ \frac{\left(\lambda^{(A)}_{12}\right)^2}{4} \\
&+ \ 4\left(\lambda^{(1)}_{HS_2}\right)^2 \\
&+ \ \frac{\left(\lambda^{(A)}_{HS_2}\right)^2}{4} \\
&+ \ 12\left(\lambda^{(1)}_{S_2}\right)^2
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda^{(1)}_{HS_1}} &= \frac{9g_2^4}{4} \\
&- \ 9g_2^2\lambda^{(1)}_{HS_1}
\end{aligned}
$$

$$
\begin{aligned}
&+ \frac{3g_1^4}{4} \\
&- 3g_1^2\lambda_{HS_1}^{(1)} \\
&+ 4\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)} \\
&+ 6\lambda_1\lambda_{HS_1}^{(1)} \\
&+ 4\left(\lambda_{HS_1}^{(1)}\right)^2 \\
&+ 6\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)} \\
&+ \lambda_{HS_1}^{(1)}\left(6\,\mathrm{Tr}\left[Y_d\ \left(Y_d^*\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ \left(Y_e^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ \left(Y_u^*\right)^T\right]\right) \\
&+ \frac{3\left(\lambda_{HS_1}^{(A)}\right)^2}{4} \\
&+ 6\lambda_5\lambda_5^* \\
&+ 24\lambda_{24}\lambda_{24}^*
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda_{HS_2}^{(1)}} = &\frac{9g_2^4}{4} \\
&- 9g_2^2\lambda_{HS_2}^{(1)} \\
&+ \frac{27g_1^4}{4} \\
&- 15g_1^2\lambda_{HS_2}^{(1)} \\
&+ 4\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)} \\
&+ 6\lambda_1\lambda_{HS_2}^{(1)} \\
&+ 4\left(\lambda_{HS_2}^{(1)}\right)^2 \\
&+ 6\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)} \\
&+ \lambda_{HS_2}^{(1)}\left(6\,\mathrm{Tr}\left[Y_d\ \left(Y_d^*\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ \left(Y_e^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ \left(Y_u^*\right)^T\right]\right) \\
&+ \frac{3\left(\lambda_{HS_2}^{(A)}\right)^2}{4} \\
&+ 6\lambda_5\lambda_5^*
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda_{12}^{(1)}} = &\frac{9g_2^4}{4} \\
&- 9g_2^2\lambda_{12}^{(1)} \\
&+ \frac{27g_1^4}{4} \\
&- 15g_1^2\lambda_{12}^{(1)} \\
&+ 4\left(\lambda_{12}^{(1)}\right)^2 \\
&+ 6\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}
\end{aligned}
$$

$$
\begin{aligned}
&+ 6\lambda_{12}^{(1)}\lambda_{S_2}^{(1)} \\
&+ \frac{3\left(\lambda_{12}^{(A)}\right)^2}{4} \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)} \\
&+ 3\lambda_5\lambda_5^*
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_5}^{(1)} =& - 9g_2^2\lambda_5 \\
&- 9g_1^2\lambda_5 \\
&+ 2\lambda_{12}^{(1)}\lambda_5 \\
&+ 2\lambda_1\lambda_5 \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_5 \\
&+ 4\lambda_{HS_2}^{(1)}\lambda_5 \\
&+ \lambda_5\left(\frac{\lambda_{12}^{(A)}}{6} + \frac{2\lambda_{HS_1}^{(A)}}{3} + 2\lambda_{HS_2}^{(A)} + 6\,\mathrm{Tr}\left[Y_d\ \left(Y_d^*\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ \left(Y_e^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ \left(Y_u^*\right)^T\right]\right)
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{HS_1}^{(A)}}^{(1)} =& - 9g_2^2\lambda_{HS_1}^{(A)} \\
&+ g_1^2\left(6g_2^2 - 3\lambda_{HS_1}^{(A)}\right) \\
&+ \lambda_{12}^{(A)}\lambda_{HS_2}^{(A)} \\
&+ 2\lambda_1\lambda_{HS_1}^{(A)} \\
&+ 8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)} \\
&+ 2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)} \\
&+ \lambda_{HS_1}^{(A)}\left(6\,\mathrm{Tr}\left[Y_d\ \left(Y_d^*\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ \left(Y_e^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ \left(Y_u^*\right)^T\right]\right) \\
&+ \frac{16\lambda_5\lambda_5^*}{3} \\
&+ 64\lambda_{24}\lambda_{24}^*
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{HS_2}^{(A)}}^{(1)} =& - 9g_2^2\lambda_{HS_2}^{(A)} \\
&+ g_1^2\left(18g_2^2 - 15\lambda_{HS_2}^{(A)}\right) \\
&+ \lambda_{12}^{(A)}\lambda_{HS_1}^{(A)} \\
&+ 2\lambda_1\lambda_{HS_2}^{(A)} \\
&+ 8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)} \\
&+ 2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)} \\
&+ \lambda_{HS_2}^{(A)}\left(6\,\mathrm{Tr}\left[Y_d\ \left(Y_d^*\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ \left(Y_e^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ \left(Y_u^*\right)^T\right]\right) \\
&+ 16\lambda_5\lambda_5^*
\end{aligned}
$$

$$
\beta_{\lambda_{12}^{(A)}}^{(1)} = - 9g_2^2\lambda_{12}^{(A)}
$$

$$
\begin{aligned}
&+ g_1^2 \left(18 g_2^2 - 15 \lambda_{12}^{(A)}\right) \\
&+ 8 \lambda_{12}^{(1)} \lambda_{12}^{(A)} \\
&+ 2 \lambda_{12}^{(A)} \lambda_{S_1}^{(1)} \\
&+ 2 \lambda_{12}^{(A)} \lambda_{S_2}^{(1)} \\
&+ \lambda_{HS_1}^{(A)} \lambda_{HS_2}^{(A)} \\
&+ \frac{4 \lambda_5 \lambda_5^*}{3}
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{24}}^{(1)} = &- 9 g_2^2 \lambda_{24} \\
&- 3 g_1^2 \lambda_{24} \\
&+ 2 \lambda_1 \lambda_{24} \\
&+ 8 \lambda_{HS_1}^{(1)} \lambda_{24} \\
&+ 4 \lambda_{HS_1}^{(A)} \lambda_{24} \\
&+ 2 \lambda_{S_1}^{(1)} \lambda_{24} \\
&+ \lambda_{24} \left(6 \operatorname{Tr}\left[Y_d \left(Y_d^*\right)^T\right] + 2 \operatorname{Tr}\left[Y_e \left(Y_e^*\right)^T\right] + 6 \operatorname{Tr}\left[Y_u \left(Y_u^*\right)^T\right]\right)
\end{aligned}
$$

$\mathrm{T3-D}\,(\alpha = -2)$

$$
\begin{aligned}
\beta_{m_1^2}^{(1)} = &- 12 g_2^2 m_1^2 \\
&- 6 g_1^2 m_1^2 \\
&+ 2 \lambda_{12}^{(1)} m_2^2 \\
&+ 4 \lambda_{18} \left(\mu_{S_2}^2\right)^* \\
&+ m_1^2 \left(8 \lambda_{S_1}^{(1)} + 4 \lambda_{S_1}^{(A)}\right) \\
&+ 4 \mu_{S_2}^2 \lambda_{18}^*
\end{aligned}
$$

$$
\begin{aligned}
\beta_{m_2^2}^{(1)} = &\, 6 \lambda_{12}^{(1)} m_1^2 \\
&+ 4 \lambda_{S_2}^{(1)} m_2^2 \\
&+ 12 \lambda_{14} \left(\mu_{S_2}^2\right)^* \\
&+ 12 \mu_{S_2}^2 \lambda_{14}^*
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_1}^{(1)} = &\, \frac{9 g_2^4}{4} \\
&- 9 g_2^2 \lambda_1 \\
&+ \frac{3 g_1^4}{4} \\
&+ g_1^2 \left(\frac{3 g_2^2}{2} - 3 \lambda_1\right) \\
&+ 12 \lambda_1^2 \\
&+ \lambda_1 \left(12 \operatorname{Tr}\left[Y_d \left(Y_d^*\right)^T\right] + 4 \operatorname{Tr}\left[Y_e \left(Y_e^*\right)^T\right] + 12 \operatorname{Tr}\left[Y_u \left(Y_u^*\right)^T\right]\right) \\
&+ 6 \left(\lambda_{HS_1}^{(1)}\right)^2
\end{aligned}
$$

$$+ \ 3\left(\lambda_{HS_1}^{(A)}\right)^2$$

$$+ \ 2\left(\lambda_{HS_2}^{(1)}\right)^2$$

$$+ \ 4\lambda_5\lambda_5^*$$

$$+ \ 4\lambda_{22}\lambda_{22}^*$$

$$+ \ 8\lambda_{25}\lambda_{25}^*$$

$$- \ 12\,\mathrm{Tr}\left[Y_d\ \left(Y_d^*\right)^T\ Y_d\ \left(Y_d^*\right)^T\right]$$

$$- \ 4\,\mathrm{Tr}\left[Y_e\ \left(Y_e^*\right)^T\ Y_e\ \left(Y_e^*\right)^T\right]$$

$$- \ 12\,\mathrm{Tr}\left[Y_u\ \left(Y_u^*\right)^T\ Y_u\ \left(Y_u^*\right)^T\right]$$

$$\beta_{\lambda_{S_1}^{(1)}}^{(1)} = 22g_2^4$$

$$- \ 24g_2^2\lambda_{S_1}^{(1)}$$

$$+ \ 12g_1^4$$

$$+ \ g_1^2\left(8g_2^2 - 12\lambda_{S_1}^{(1)}\right)$$

$$+ \ 2\left(\lambda_{12}^{(1)}\right)^2$$

$$+ \ 4\left(\lambda_{HS_1}^{(1)}\right)^2$$

$$+ \ 14\left(\lambda_{S_1}^{(1)}\right)^2$$

$$+ \ 8\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)}$$

$$+ \ 16\left(\lambda_{S_1}^{(A)}\right)^2$$

$$+ \ 8\lambda_{18}\lambda_{18}^*$$

$$\beta_{\lambda_{S_2}^{(1)}}^{(1)} = 6\left(\lambda_{12}^{(1)}\right)^2$$

$$+ \ 4\left(\lambda_{HS_2}^{(1)}\right)^2$$

$$+ \ 10\left(\lambda_{S_2}^{(1)}\right)^2$$

$$+ \ 288\lambda_{13}\lambda_{13}^*$$

$$+ \ 108\lambda_{14}\lambda_{14}^*$$

$$+ \ 12\lambda_{18}\lambda_{18}^*$$

$$+ \ 8\lambda_{25}\lambda_{25}^*$$

$$\beta_{\lambda_{HS_1}^{(1)}}^{(1)} = 6g_2^4$$

$$- \ \frac{33g_2^2\lambda_{HS_1}^{(1)}}{2}$$

$$+ \ 3g_1^4$$

$$
\begin{aligned}
&- \frac{15 g_1^2 \lambda_{HS_1}^{(1)}}{2} \\
&+ 2\lambda_{12}^{(1)} \lambda_{HS_2}^{(1)} \\
&+ 6\lambda_1 \lambda_{HS_1}^{(1)} \\
&+ 4\left(\lambda_{HS_1}^{(1)}\right)^2 \\
&+ 8\lambda_{HS_1}^{(1)} \lambda_{S_1}^{(1)} \\
&+ \lambda_{HS_1}^{(1)} \left(4\lambda_{S_1}^{(A)} + 6\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\right]\right) \\
&+ 6\left(\lambda_{HS_1}^{(A)}\right)^2 \\
&+ 4\lambda_5 \lambda_5^* \\
&+ 4\lambda_{18} \lambda_{25}^* \\
&+ 4\lambda_{22} \lambda_{22}^* \\
&+ 4\lambda_{25} \lambda_{18}^*
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda^{(1)}_{HS_2}} =& - \frac{9 g_2^2 \lambda_{HS_2}^{(1)}}{2} \\
&- \frac{3 g_1^2 \lambda_{HS_2}^{(1)}}{2} \\
&+ 6\lambda_{12}^{(1)} \lambda_{HS_1}^{(1)} \\
&+ 6\lambda_1 \lambda_{HS_2}^{(1)} \\
&+ 4\left(\lambda_{HS_2}^{(1)}\right)^2 \\
&+ 4\lambda_{HS_2}^{(1)} \lambda_{S_2}^{(1)} \\
&+ \lambda_{HS_2}^{(1)} \left(6\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\right]\right) \\
&+ 12\lambda_5 \lambda_5^* \\
&+ 12\lambda_{14} \lambda_{25}^* \\
&+ 12\lambda_{22} \lambda_{22}^* \\
&+ 4\lambda_{25} \left(3\lambda_{14}^* + 4\lambda_{25}^*\right)
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda^{(1)}_{12}} =& - 12 g_2^2 \lambda_{12}^{(1)} \\
&- 6 g_1^2 \lambda_{12}^{(1)} \\
&+ 4\left(\lambda_{12}^{(1)}\right)^2 \\
&+ 8\lambda_{12}^{(1)} \lambda_{S_1}^{(1)} \\
&+ 4\lambda_{12}^{(1)} \lambda_{S_1}^{(A)}
\end{aligned}
$$

$$
\begin{aligned}
&+ 4\lambda_{12}^{(1)}\lambda_{S_2}^{(1)} \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)} \\
&+ 4\lambda_5\lambda_5^* \\
&+ 12\lambda_{14}\lambda_{18}^* \\
&+ 4\lambda_{18}\left(3\lambda_{14}^* + 4\lambda_{18}^*\right) \\
&+ 4\lambda_{22}\lambda_{22}^*
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_5}^{(1)} =& -\frac{21g_2^2\lambda_5}{2} \\
&- \frac{9g_1^2\lambda_5}{2} \\
&+ 2\lambda_{12}^{(1)}\lambda_5 \\
&+ 2\lambda_1\lambda_5 \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_5 \\
&+ 4\lambda_{HS_2}^{(1)}\lambda_5 \\
&+ \lambda_5\left(-4\lambda_{HS_1}^{(A)} + 6\,\mathrm{Tr}\left[Y_d\ \left(Y_d^*\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ \left(Y_e^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ \left(Y_u^*\right)^T\right]\right) \\
&+ 4\lambda_{18}\lambda_{22}^* \\
&+ 8\lambda_{25}\lambda_{22}^*
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{HS_1}^{(A)}}^{(1)} =& -\frac{33g_2^2\lambda_{HS_1}^{(A)}}{2} \\
&+ g_1^2\left(-4g_2^2 - \frac{15\lambda_{HS_1}^{(A)}}{2}\right) \\
&+ 2\lambda_1\lambda_{HS_1}^{(A)} \\
&+ 8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)} \\
&+ 2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)} \\
&+ \lambda_{HS_1}^{(A)}\left(16\lambda_{S_1}^{(A)} + 6\,\mathrm{Tr}\left[Y_d\ \left(Y_d^*\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ \left(Y_e^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ \left(Y_u^*\right)^T\right]\right) \\
&- \frac{8\lambda_5\lambda_5^*}{3} \\
&- \frac{8\lambda_{22}\lambda_{22}^*}{3}
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{S_1}^{(A)}}^{(1)} =& -2g_2^4 \\
&- 24g_2^2\lambda_{S_1}^{(A)} \\
&+ g_1^2\left(8g_2^2 - 12\lambda_{S_1}^{(A)}\right) \\
&+ 3\left(\lambda_{HS_1}^{(A)}\right)^2 \\
&+ 12\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)} \\
&+ 20\left(\lambda_{S_1}^{(A)}\right)^2
\end{aligned}
$$

$$\beta^{(1)}_{\lambda_{13}} = 12\lambda^{(1)}_{S_2}\lambda_{13}$$
$$+ 9\lambda^2_{14}$$
$$+ 3\lambda^2_{18}$$
$$+ 2\lambda^2_{25}$$

$$\beta^{(1)}_{\lambda_{14}} = 6\lambda^{(1)}_{12}\lambda_{18}$$
$$+ 4\lambda^{(1)}_{HS_2}\lambda_{25}$$
$$+ 18\lambda^{(1)}_{S_2}\lambda_{14}$$
$$+ 72\lambda_{13}\lambda^*_{14}$$

$$\beta^{(1)}_{\lambda_{18}} = - 12g_2^2\lambda_{18}$$
$$- 6g_1^2\lambda_{18}$$
$$+ \lambda^{(1)}_{12}\left(6\lambda_{14} + 8\lambda_{18}\right)$$
$$+ 4\lambda^{(1)}_{HS_1}\lambda_{25}$$
$$+ 8\lambda^{(1)}_{S_1}\lambda_{18}$$
$$+ 4\lambda^{(A)}_{S_1}\lambda_{18}$$
$$+ 2\lambda^{(1)}_{S_2}\lambda_{18}$$
$$+ 4\lambda_5\lambda_{22}$$
$$+ 24\lambda_{13}\lambda^*_{18}$$

$$\beta^{(1)}_{\lambda_{22}} = - \frac{21g_2^2\lambda_{22}}{2}$$
$$- \frac{9g_1^2\lambda_{22}}{2}$$
$$+ 2\lambda^{(1)}_{12}\lambda_{22}$$
$$+ 2\lambda_1\lambda_{22}$$
$$+ 4\lambda^{(1)}_{HS_1}\lambda_{22}$$
$$- 4\lambda^{(A)}_{HS_1}\lambda_{22}$$
$$+ 4\lambda^{(1)}_{HS_2}\lambda_{22}$$
$$+ 4\lambda_{18}\lambda^*_5$$
$$+ \lambda_{22}\left(6\,\mathrm{Tr}\left[Y_d\ \left(Y^*_d\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ \left(Y^*_e\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ \left(Y^*_u\right)^T\right]\right)$$
$$+ 8\lambda_{25}\lambda^*_5$$

$$\beta^{(1)}_{\lambda_{25}} = - \frac{9g_2^2\lambda_{25}}{2}$$
$$- \frac{3g_1^2\lambda_{25}}{2}$$
$$+ 6\lambda_1\lambda_{25}$$
$$+ 6\lambda^{(1)}_{HS_1}\lambda_{18}$$
$$+ \lambda^{(1)}_{HS_2}\left(6\lambda_{14} + 8\lambda_{25}\right)$$
$$+ 2\lambda^{(1)}_{S_2}\lambda_{25}$$
$$+ 12\lambda_5\lambda_{22}$$

$$+\ 24\lambda_{13}\lambda_{25}^{*}$$
$$+\ \lambda_{25}\left(6\,\mathrm{Tr}\left[Y_d\ \left(Y_d^{*}\right)^{T}\right]+2\,\mathrm{Tr}\left[Y_e\ \left(Y_e^{*}\right)^{T}\right]+6\,\mathrm{Tr}\left[Y_u\ \left(Y_u^{*}\right)^{T}\right]\right)$$

$$\beta^{(1)}_{\mu_{S_2}^2}=2\lambda^{(1)}_{S_2}\mu^2_{S_2}$$
$$+\ 24\lambda_{13}\left(\mu^2_{S_2}\right)^{*}$$
$$+\ 6\lambda_{14}m_2^2$$
$$+\ 6\lambda_{18}m_1^2$$

T3 − D ($\alpha = 0$)

$$\beta^{(1)}_{m_1^2}=-\ 12g_2^2m_1^2$$
$$+\ 2\lambda^{(1)}_{12}m_2^2$$
$$+\ 20\lambda_{20}\left(\mu^2_{S_1}\right)^{*}$$
$$+\ m_1^2\left(8\lambda^{(1)}_{S_1}+4\lambda^{(A)}_{S_1}\right)$$
$$+\ 20\mu^2_{S_1}\lambda^{*}_{20}$$

$$\beta^{(1)}_{m_2^2}=-\ 6g_1^2m_2^2$$
$$+\ 6\lambda^{(1)}_{12}m_1^2$$
$$+\ 4\lambda^{(1)}_{S_2}m_2^2$$
$$+\ 12\lambda_{15}\left(\mu^2_{S_1}\right)^{*}$$
$$+\ 12\mu^2_{S_1}\lambda^{*}_{15}$$

$$\beta^{(1)}_{\lambda_1}=\frac{9g_2^4}{4}$$
$$-\ 9g_2^2\lambda_1$$
$$+\ \frac{3g_1^4}{4}$$
$$+\ g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$$
$$+\ 12\lambda_1^2$$
$$+\ \lambda_1\left(12\,\mathrm{Tr}\left[Y_d\ \left(Y_d^{*}\right)^{T}\right]+4\,\mathrm{Tr}\left[Y_e\ \left(Y_e^{*}\right)^{T}\right]+12\,\mathrm{Tr}\left[Y_u\ \left(Y_u^{*}\right)^{T}\right]\right)$$
$$+\ 6\left(\lambda^{(1)}_{HS_1}\right)^2$$
$$+\ 3\left(\lambda^{(A)}_{HS_1}\right)^2$$
$$+\ 2\left(\lambda^{(1)}_{HS_2}\right)^2$$
$$+\ 4\lambda_5\lambda_5^{*}$$
$$+\ 4\lambda_{23}\lambda_{23}^{*}$$
$$+\ 24\lambda_{27}\lambda_{27}^{*}$$
$$-\ 12\,\mathrm{Tr}\left[Y_d\ \left(Y_d^{*}\right)^{T}\ Y_d\ \left(Y_d^{*}\right)^{T}\right]$$
$$-\ 4\,\mathrm{Tr}\left[Y_e\ \left(Y_e^{*}\right)^{T}\ Y_e\ \left(Y_e^{*}\right)^{T}\right]$$

$$- \ 12 \operatorname{Tr}\left[Y_u \ \left(Y_u^*\right)^T \ Y_u \ \left(Y_u^*\right)^T\right]$$

$$\beta^{(1)}_{\lambda^{(1)}_{S_1}} = 22 g_2^4$$
$$- \ 24 g_2^2 \lambda^{(1)}_{S_1}$$
$$+ \ 2 \left(\lambda^{(1)}_{12}\right)^2$$
$$+ \ 4 \left(\lambda^{(1)}_{HS_1}\right)^2$$
$$+ \ 14 \left(\lambda^{(1)}_{S_1}\right)^2$$
$$+ \ 8 \lambda^{(1)}_{S_1} \lambda^{(A)}_{S_1}$$
$$+ \ 16 \left(\lambda^{(A)}_{S_1}\right)^2$$
$$+ \ \frac{8 \lambda_{15} \lambda^*_{15}}{3}$$
$$+ \ \frac{832 \lambda_{17} \lambda^*_{17}}{3}$$
$$+ \ \frac{352 \lambda_{20} \lambda^*_{20}}{3}$$
$$+ \ \frac{16 \lambda_{27} \lambda^*_{27}}{3}$$

$$\beta^{(1)}_{\lambda^{(1)}_{S_2}} = 12 g_1^4$$
$$- \ 12 g_1^2 \lambda^{(1)}_{S_2}$$
$$+ \ 6 \left(\lambda^{(1)}_{12}\right)^2$$
$$+ \ 4 \left(\lambda^{(1)}_{HS_2}\right)^2$$
$$+ \ 10 \left(\lambda^{(1)}_{S_2}\right)^2$$
$$+ \ 24 \lambda_{15} \lambda^*_{15}$$

$$\beta^{(1)}_{\lambda^{(1)}_{HS_1}} = 6 g_2^4$$
$$- \ \frac{33 g_2^2 \lambda^{(1)}_{HS_1}}{2}$$
$$- \ \frac{3 g_1^2 \lambda^{(1)}_{HS_1}}{2}$$
$$+ \ 2 \lambda^{(1)}_{12} \lambda^{(1)}_{HS_2}$$
$$+ \ 6 \lambda_1 \lambda^{(1)}_{HS_1}$$
$$+ \ 4 \left(\lambda^{(1)}_{HS_1}\right)^2$$
$$+ \ 8 \lambda^{(1)}_{HS_1} \lambda^{(1)}_{S_1}$$
$$+ \ \lambda^{(1)}_{HS_1} \left(4 \lambda^{(A)}_{S_1} + 6 \operatorname{Tr}\left[Y_d \ \left(Y_d^*\right)^T\right] + 2 \operatorname{Tr}\left[Y_e \ \left(Y_e^*\right)^T\right] + 6 \operatorname{Tr}\left[Y_u \ \left(Y_u^*\right)^T\right]\right)$$

$$+ 6\left(\lambda_{HS_1}^{(A)}\right)^2$$

$$+ 4\lambda_5\lambda_5^*$$

$$+ 20\lambda_{20}\lambda_{27}^*$$

$$+ 4\lambda_{23}\lambda_{23}^*$$

$$+ 4\lambda_{27}\left(5\lambda_{20}^* + 4\lambda_{27}^*\right)$$

$$\beta^{(1)}_{\lambda^{(1)}_{HS_2}} = -\frac{9g_2^2\lambda_{HS_2}^{(1)}}{2}$$

$$+ 3g_1^4$$

$$- \frac{15g_1^2\lambda_{HS_2}^{(1)}}{2}$$

$$+ 6\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$$

$$+ 6\lambda_1\lambda_{HS_2}^{(1)}$$

$$+ 4\left(\lambda_{HS_2}^{(1)}\right)^2$$

$$+ 4\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$$

$$+ \lambda_{HS_2}^{(1)}\left(6\,\mathrm{Tr}\left[Y_d\ \left(Y_d^*\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ \left(Y_e^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ \left(Y_u^*\right)^T\right]\right)$$

$$+ 12\lambda_5\lambda_5^*$$

$$+ 12\lambda_{15}\lambda_{27}^*$$

$$+ 12\lambda_{23}\lambda_{23}^*$$

$$+ 12\lambda_{27}\lambda_{15}^*$$

$$\beta^{(1)}_{\lambda^{(1)}_{12}} = -12g_2^2\lambda_{12}^{(1)}$$

$$- 6g_1^2\lambda_{12}^{(1)}$$

$$+ 4\left(\lambda_{12}^{(1)}\right)^2$$

$$+ 8\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$$

$$+ 4\lambda_{12}^{(1)}\lambda_{S_1}^{(A)}$$

$$+ 4\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$$

$$+ 4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$$

$$+ 4\lambda_5\lambda_5^*$$

$$+ 4\lambda_{15}\left(4\lambda_{15}^* + 5\lambda_{20}^*\right)$$

$$+ 20\lambda_{20}\lambda_{15}^*$$

$$+ 4\lambda_{23}\lambda_{23}^*$$

$$\beta^{(1)}_{\lambda_5} = -\frac{21g_2^2\lambda_5}{2} - \frac{9g_1^2\lambda_5}{2} + 2\lambda_{12}^{(1)}\lambda_5 + 2\lambda_1\lambda_5 + 4\lambda_{HS_1}^{(1)}\lambda_5 + 4\lambda_{HS_2}^{(1)}\lambda_5 + \lambda_5\left(-4\lambda_{HS_1}^{(A)} + 6\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\right]\right) + 4\left(\lambda_{15}^* + 2\lambda_{27}^*\right)\lambda_{23}^*$$

$$\beta^{(1)}_{\lambda_{HS_1}^{(A)}} = -\frac{33g_2^2\lambda_{HS_1}^{(A)}}{2} - \frac{3g_1^2\lambda_{HS_1}^{(A)}}{2} + 2\lambda_1\lambda_{HS_1}^{(A)} + 8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)} + 2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)} + \lambda_{HS_1}^{(A)}\left(16\lambda_{S_1}^{(A)} + 6\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\right]\right) - \frac{8\lambda_5\lambda_5^*}{3} + \frac{8\lambda_{23}\lambda_{23}^*}{3}$$

$$\beta^{(1)}_{\lambda_{S_1}^{(A)}} = -2g_2^4 - 24g_2^2\lambda_{S_1}^{(A)} + 3\left(\lambda_{HS_1}^{(A)}\right)^2 + 12\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)} + 20\left(\lambda_{S_1}^{(A)}\right)^2 - \frac{4\lambda_{15}\lambda_{15}^*}{3} - \frac{224\lambda_{17}\lambda_{17}^*}{3} - \frac{44\lambda_{20}\lambda_{20}^*}{3} - \frac{8\lambda_{27}\lambda_{27}^*}{3}$$

$$\beta^{(1)}_{\lambda_{15}} = -12g_2^2\lambda_{15} - 6g_1^2\lambda_{15} + \lambda_{12}^{(1)}\left(8\lambda_{15} + 10\lambda_{20}\right) + 4\lambda_{HS_2}^{(1)}\lambda_{27}$$

$$
\begin{aligned}
&+ 2\lambda_{S_1}^{(1)}\lambda_{15} \\
&- 4\lambda_{S_1}^{(A)}\lambda_{15} \\
&+ 4\lambda_{S_2}^{(1)}\lambda_{15} \\
&+ 40\lambda_{17}\lambda_{15}^{*} \\
&+ 4\lambda_{5}^{*}\lambda_{23}^{*}
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{17}}^{(1)} = &- 24g_2^2\lambda_{17} \\
&+ 12\lambda_{S_1}^{(1)}\lambda_{17} \\
&- 8\lambda_{S_1}^{(A)}\lambda_{17} \\
&+ \lambda_{15}^{2} \\
&+ 11\lambda_{20}^{2} \\
&+ 2\lambda_{27}^{2}
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{20}}^{(1)} = &- 24g_2^2\lambda_{20} \\
&+ 2\lambda_{12}^{(1)}\lambda_{15} \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_{27} \\
&+ 22\lambda_{S_1}^{(1)}\lambda_{20} \\
&+ 88\lambda_{17}\lambda_{20}^{*}
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{23}}^{(1)} = &- \frac{21g_2^2\lambda_{23}}{2} \\
&- \frac{9g_1^2\lambda_{23}}{2} \\
&+ 2\lambda_{12}^{(1)}\lambda_{23} \\
&+ 2\lambda_{1}\lambda_{23} \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_{23} \\
&+ 4\lambda_{HS_1}^{(A)}\lambda_{23} \\
&+ 4\lambda_{HS_2}^{(1)}\lambda_{23} \\
&+ \lambda_{23}\left(6\,\mathrm{Tr}\left[Y_d\,\left(Y_d^{*}\right)^{T}\right] + 2\,\mathrm{Tr}\left[Y_e\,\left(Y_e^{*}\right)^{T}\right] + 6\,\mathrm{Tr}\left[Y_u\,\left(Y_u^{*}\right)^{T}\right]\right) \\
&+ 4\left(\lambda_{15}^{*} + 2\lambda_{27}^{*}\right)\lambda_{5}^{*}
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{27}}^{(1)} = &- \frac{33g_2^2\lambda_{27}}{2} \\
&- \frac{3g_1^2\lambda_{27}}{2} \\
&+ 6\lambda_{1}\lambda_{27} \\
&+ \lambda_{HS_1}^{(1)}\left(10\lambda_{20} + 8\lambda_{27}\right) \\
&+ 2\lambda_{HS_2}^{(1)}\lambda_{15} \\
&+ 2\lambda_{S_1}^{(1)}\lambda_{27} \\
&- 4\lambda_{S_1}^{(A)}\lambda_{27} \\
&+ 40\lambda_{17}\lambda_{27}^{*} \\
&+ \lambda_{27}\left(6\,\mathrm{Tr}\left[Y_d\,\left(Y_d^{*}\right)^{T}\right] + 2\,\mathrm{Tr}\left[Y_e\,\left(Y_e^{*}\right)^{T}\right] + 6\,\mathrm{Tr}\left[Y_u\,\left(Y_u^{*}\right)^{T}\right]\right)
\end{aligned}
$$

$$+ \ 4\lambda_5^*\lambda_{23}^*$$

$$\beta^{(1)}_{\mu^2_{S_1}} = - \ 12g_2^2\mu^2_{S_1}$$
$$+ \ 2\lambda^{(1)}_{S_1}\mu^2_{S_1}$$
$$- \ 4\lambda^{(A)}_{S_1}\mu^2_{S_1}$$
$$+ \ 2\lambda_{15}m_2^2$$
$$+ \ 40\lambda_{17}\left(\mu^2_{S_1}\right)^*$$
$$+ \ 10\lambda_{20}m_1^2$$

T3 − D ($\alpha = 2$)

$$\beta^{(1)}_{m_1^2} = - \ 12g_2^2m_1^2$$
$$- \ 6g_1^2m_1^2$$
$$+ \ 2\lambda^{(1)}_{12}m_2^2$$
$$+ \ m_1^2\left(8\lambda^{(1)}_{S_1} + 4\lambda^{(A)}_{S_1}\right)$$

$$\beta^{(1)}_{m_2^2} = - \ 24g_1^2m_2^2$$
$$+ \ 6\lambda^{(1)}_{12}m_1^2$$
$$+ \ 4\lambda^{(1)}_{S_2}m_2^2$$

$$\beta^{(1)}_{\lambda_1} = \frac{9g_2^4}{4}$$
$$- \ 9g_2^2\lambda_1$$
$$+ \ \frac{3g_1^4}{4}$$
$$+ \ g_1^2\left(\frac{3g_2^2}{2} - 3\lambda_1\right)$$
$$+ \ 12\lambda_1^2$$
$$+ \ \lambda_1\left(12\,\mathrm{Tr}\left[Y_d\ \left(Y_d^*\right)^T\right] + 4\,\mathrm{Tr}\left[Y_e\ \left(Y_e^*\right)^T\right] + 12\,\mathrm{Tr}\left[Y_u\ \left(Y_u^*\right)^T\right]\right)$$
$$+ \ 6\left(\lambda^{(1)}_{HS_1}\right)^2$$
$$+ \ 3\left(\lambda^{(A)}_{HS_1}\right)^2$$
$$+ \ 2\left(\lambda^{(1)}_{HS_2}\right)^2$$
$$+ \ 4\lambda_5\lambda_5^*$$
$$- \ 12\,\mathrm{Tr}\left[Y_d\ \left(Y_d^*\right)^T\ Y_d\ \left(Y_d^*\right)^T\right]$$
$$- \ 4\,\mathrm{Tr}\left[Y_e\ \left(Y_e^*\right)^T\ Y_e\ \left(Y_e^*\right)^T\right]$$
$$- \ 12\,\mathrm{Tr}\left[Y_u\ \left(Y_u^*\right)^T\ Y_u\ \left(Y_u^*\right)^T\right]$$

$$\beta^{(1)}_{\lambda^{(1)}_{S_1}} = 22g_2^4$$
$$- \ 24g_2^2\lambda^{(1)}_{S_1}$$

$$
\begin{aligned}
&+ 12 g_1^4 \\
&+ g_1^2 \left(8 g_2^2 - 12 \lambda_{S_1}^{(1)}\right) \\
&+ 2 \left(\lambda_{12}^{(1)}\right)^2 \\
&+ 4 \left(\lambda_{HS_1}^{(1)}\right)^2 \\
&+ 14 \left(\lambda_{S_1}^{(1)}\right)^2 \\
&+ 8 \lambda_{S_1}^{(1)} \lambda_{S_1}^{(A)} \\
&+ 16 \left(\lambda_{S_1}^{(A)}\right)^2
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda^{(1)}_{S_2}} &= 192 g_1^4 \\
&- 48 g_1^2 \lambda_{S_2}^{(1)} \\
&+ 6 \left(\lambda_{12}^{(1)}\right)^2 \\
&+ 4 \left(\lambda_{HS_2}^{(1)}\right)^2 \\
&+ 10 \left(\lambda_{S_2}^{(1)}\right)^2
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda^{(1)}_{HS_1}} &= 6 g_2^4 \\
&- \frac{33 g_2^2 \lambda_{HS_1}^{(1)}}{2} \\
&+ 3 g_1^4 \\
&- \frac{15 g_1^2 \lambda_{HS_1}^{(1)}}{2} \\
&+ 2 \lambda_{12}^{(1)} \lambda_{HS_2}^{(1)} \\
&+ 6 \lambda_1 \lambda_{HS_1}^{(1)} \\
&+ 4 \left(\lambda_{HS_1}^{(1)}\right)^2 \\
&+ 8 \lambda_{HS_1}^{(1)} \lambda_{S_1}^{(1)} \\
&+ \lambda_{HS_1}^{(1)} \left(4 \lambda_{S_1}^{(A)} + 6 \operatorname{Tr}\left[Y_d \left(Y_d^*\right)^T\right] + 2 \operatorname{Tr}\left[Y_e \left(Y_e^*\right)^T\right] + 6 \operatorname{Tr}\left[Y_u \left(Y_u^*\right)^T\right]\right) \\
&+ 6 \left(\lambda_{HS_1}^{(A)}\right)^2 \\
&+ 4 \lambda_5 \lambda_5^*
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda^{(1)}_{HS_2}} &= - \frac{9 g_2^2 \lambda_{HS_2}^{(1)}}{2} \\
&+ 12 g_1^4 \\
&- \frac{51 g_1^2 \lambda_{HS_2}^{(1)}}{2}
\end{aligned}
$$

$$
\begin{aligned}
&+ 6\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)} \\
&+ 6\lambda_1\lambda_{HS_2}^{(1)} \\
&+ 4\left(\lambda_{HS_2}^{(1)}\right)^2 \\
&+ 4\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)} \\
&+ \lambda_{HS_2}^{(1)}\left(6\,\mathrm{Tr}\left[Y_d\,\left(Y_d^*\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\,\left(Y_e^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\,\left(Y_u^*\right)^T\right]\right) \\
&+ 12\lambda_5\lambda_5^*
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{12}^{(1)}}^{(1)} = &- 12g_2^2\lambda_{12}^{(1)} \\
&+ 48g_1^4 \\
&- 30g_1^2\lambda_{12}^{(1)} \\
&+ 4\left(\lambda_{12}^{(1)}\right)^2 \\
&+ 8\lambda_{12}^{(1)}\lambda_{S_1}^{(1)} \\
&+ 4\lambda_{12}^{(1)}\lambda_{S_1}^{(A)} \\
&+ 4\lambda_{12}^{(1)}\lambda_{S_2}^{(1)} \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)} \\
&+ 4\lambda_5\lambda_5^*
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_5}^{(1)} = &- \frac{21g_2^2\lambda_5}{2} \\
&- \frac{33g_1^2\lambda_5}{2} \\
&+ 2\lambda_{12}^{(1)}\lambda_5 \\
&+ 2\lambda_1\lambda_5 \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_5 \\
&+ 4\lambda_{HS_2}^{(1)}\lambda_5 \\
&+ \lambda_5\left(-4\lambda_{HS_1}^{(A)} + 6\,\mathrm{Tr}\left[Y_d\,\left(Y_d^*\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\,\left(Y_e^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\,\left(Y_u^*\right)^T\right]\right)
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{HS_1}^{(A)}}^{(1)} = &- \frac{33g_2^2\lambda_{HS_1}^{(A)}}{2} \\
&+ g_1^2\left(4g_2^2 - \frac{15\lambda_{HS_1}^{(A)}}{2}\right) \\
&+ 2\lambda_1\lambda_{HS_1}^{(A)} \\
&+ 8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)} \\
&+ 2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)} \\
&+ \lambda_{HS_1}^{(A)}\left(16\lambda_{S_1}^{(A)} + 6\,\mathrm{Tr}\left[Y_d\,\left(Y_d^*\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\,\left(Y_e^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\,\left(Y_u^*\right)^T\right]\right) \\
&- \frac{8\lambda_5\lambda_5^*}{3}
\end{aligned}
$$

$$\beta^{(1)}_{\lambda^{(A)}_{S_1}} = - 2g_2^4$$
$$- 24g_2^2\lambda^{(A)}_{S_1}$$
$$+ g_1^2\left(8g_2^2 - 12\lambda^{(A)}_{S_1}\right)$$
$$+ 3\left(\lambda^{(A)}_{HS_1}\right)^2$$
$$+ 12\lambda^{(1)}_{S_1}\lambda^{(A)}_{S_1}$$
$$+ 20\left(\lambda^{(A)}_{S_1}\right)^2$$

$\mathrm{T3-E}\,(\alpha = -4)$

$$\beta^{(1)}_{m_1^2} = - 12g_2^2m_1^2$$
$$- 24g_1^2m_1^2$$
$$+ m_1^2\left(8\lambda^{(1)}_{S_1} + 4\lambda^{(A)}_{S_1}\right)$$
$$+ m_2^2\left(6\lambda^{(1)}_{12} + 4\lambda^{(\times)}_{12}\right)$$

$$\beta^{(1)}_{m_2^2} = - 12g_2^2m_2^2$$
$$- 6g_1^2m_2^2$$
$$+ m_1^2\left(6\lambda^{(1)}_{12} + 4\lambda^{(\times)}_{12}\right)$$
$$+ m_2^2\left(8\lambda^{(1)}_{S_2} + 4\lambda^{(A)}_{S_2}\right)$$

$$\beta^{(1)}_{\lambda_1} = \frac{9g_2^4}{4}$$
$$- 9g_2^2\lambda_1$$
$$+ \frac{3g_1^4}{4}$$
$$+ g_1^2\left(\frac{3g_2^2}{2} - 3\lambda_1\right)$$
$$+ 12\lambda_1^2$$
$$+ \lambda_1\left(12\,\mathrm{Tr}\left[Y_d\ \left(Y_d^*\right)^T\right] + 4\,\mathrm{Tr}\left[Y_e\ \left(Y_e^*\right)^T\right] + 12\,\mathrm{Tr}\left[Y_u\ \left(Y_u^*\right)^T\right]\right)$$
$$+ 6\left(\lambda^{(1)}_{HS_1}\right)^2$$
$$+ 3\left(\lambda^{(A)}_{HS_1}\right)^2$$
$$+ 6\left(\lambda^{(1)}_{HS_2}\right)^2$$
$$+ 3\left(\lambda^{(A)}_{HS_2}\right)^2$$
$$+ 4\lambda_5\lambda_5^*$$
$$- 12\,\mathrm{Tr}\left[Y_d\ \left(Y_d^*\right)^T\ Y_d\ \left(Y_d^*\right)^T\right]$$
$$- 4\,\mathrm{Tr}\left[Y_e\ \left(Y_e^*\right)^T\ Y_e\ \left(Y_e^*\right)^T\right]$$
$$- 12\,\mathrm{Tr}\left[Y_u\ \left(Y_u^*\right)^T\ Y_u\ \left(Y_u^*\right)^T\right]$$

$$\beta^{(1)}_{\lambda^{(1)}_{S_1}} = 22 g_2^4$$
$$- \ 24 g_2^2 \lambda^{(1)}_{S_1}$$
$$+ \ 192 g_1^4$$
$$+ \ g_1^2 \left(32 g_2^2 - 48 \lambda^{(1)}_{S_1}\right)$$
$$+ \ 6 \left(\lambda^{(1)}_{12}\right)^2$$
$$+ \ 8 \lambda^{(1)}_{12} \lambda^{(\times)}_{12}$$
$$+ \ \frac{20 \left(\lambda^{(\times)}_{12}\right)^2}{3}$$
$$+ \ 4 \left(\lambda^{(1)}_{HS_1}\right)^2$$
$$+ \ 14 \left(\lambda^{(1)}_{S_1}\right)^2$$
$$+ \ 8 \lambda^{(1)}_{S_1} \lambda^{(A)}_{S_1}$$
$$+ \ 16 \left(\lambda^{(A)}_{S_1}\right)^2$$

$$\beta^{(1)}_{\lambda^{(1)}_{S_2}} = 22 g_2^4$$
$$- \ 24 g_2^2 \lambda^{(1)}_{S_2}$$
$$+ \ 12 g_1^4$$
$$+ \ g_1^2 \left(8 g_2^2 - 12 \lambda^{(1)}_{S_2}\right)$$
$$+ \ 6 \left(\lambda^{(1)}_{12}\right)^2$$
$$+ \ 8 \lambda^{(1)}_{12} \lambda^{(\times)}_{12}$$
$$+ \ \frac{20 \left(\lambda^{(\times)}_{12}\right)^2}{3}$$
$$+ \ 4 \left(\lambda^{(1)}_{HS_2}\right)^2$$
$$+ \ 14 \left(\lambda^{(1)}_{S_2}\right)^2$$
$$+ \ 8 \lambda^{(1)}_{S_2} \lambda^{(A)}_{S_2}$$
$$+ \ 16 \left(\lambda^{(A)}_{S_2}\right)^2$$

$$\beta^{(1)}_{\lambda^{(1)}_{HS_1}} = 6 g_2^4$$
$$- \ \frac{33 g_2^2 \lambda^{(1)}_{HS_1}}{2}$$
$$+ \ 12 g_1^4$$
$$- \ \frac{51 g_1^2 \lambda^{(1)}_{HS_1}}{2}$$

$$
\begin{aligned}
&+ 6\lambda_1\lambda_{HS_1}^{(1)} \\
&+ 4\left(\lambda_{HS_1}^{(1)}\right)^2 \\
&+ 8\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)} \\
&+ \lambda_{HS_1}^{(1)}\left(4\lambda_{S_1}^{(A)} + 6\,\mathrm{Tr}\left[Y_d\ \left(Y_d^*\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ \left(Y_e^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ \left(Y_u^*\right)^T\right]\right) \\
&+ 6\left(\lambda_{HS_1}^{(A)}\right)^2 \\
&+ \lambda_{HS_2}^{(1)}\left(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)}\right) \\
&+ 4\lambda_5\lambda_5^*
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{HS_2}^{(1)}}^{(1)} &= 6g_2^4 \\
&- \frac{33g_2^2\lambda_{HS_2}^{(1)}}{2} \\
&+ 3g_1^4 \\
&- \frac{15g_1^2\lambda_{HS_2}^{(1)}}{2} \\
&+ 6\lambda_1\lambda_{HS_2}^{(1)} \\
&+ \lambda_{HS_1}^{(1)}\left(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)}\right) \\
&+ 4\left(\lambda_{HS_2}^{(1)}\right)^2 \\
&+ 8\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)} \\
&+ \lambda_{HS_2}^{(1)}\left(4\lambda_{S_2}^{(A)} + 6\,\mathrm{Tr}\left[Y_d\ \left(Y_d^*\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ \left(Y_e^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ \left(Y_u^*\right)^T\right]\right) \\
&+ 6\left(\lambda_{HS_2}^{(A)}\right)^2 \\
&+ 4\lambda_5\lambda_5^*
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{12}^{(1)}}^{(1)} &= \frac{63g_2^4}{5} \\
&- 24g_2^2\lambda_{12}^{(1)} \\
&+ 48g_1^4 \\
&- 30g_1^2\lambda_{12}^{(1)} \\
&+ 4\left(\lambda_{12}^{(1)}\right)^2 \\
&+ \lambda_{12}^{(1)}\left(4\lambda_{S_1}^{(A)} + 4\lambda_{S_2}^{(A)}\right) \\
&+ 16\left(\lambda_{12}^{(A)}\right)^2 \\
&+ 4\left(\lambda_{12}^{(\times)}\right)^2 \\
&+ 4\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}
\end{aligned}
$$

$$
\begin{aligned}
&+ 4\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)} \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)} \\
&+ \lambda_{S_1}^{(1)}\left(8\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)}\right) \\
&+ \lambda_{S_2}^{(1)}\left(8\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)}\right) \\
&+ 2\lambda_5\lambda_5^*
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_5}^{(1)} =& -\frac{33g_2^2\lambda_5}{2} \\
&- \frac{33g_1^2\lambda_5}{2} \\
&+ 2\lambda_{12}^{(1)}\lambda_5 \\
&+ 2\lambda_1\lambda_5 \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_5 \\
&+ 4\lambda_{HS_2}^{(1)}\lambda_5 \\
&+ \lambda_5\left(-\frac{4\lambda_{12}^{(A)}}{3} - 2\lambda_{12}^{(\times)} - 2\lambda_{HS_1}^{(A)} - \frac{4\lambda_{HS_2}^{(A)}}{3} + 6\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\right]\right)
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{HS_1}^{(A)}}^{(1)} =& -\frac{33g_2^2\lambda_{HS_1}^{(A)}}{2} \\
&+ g_1^2\left(-8g_2^2 - \frac{51\lambda_{HS_1}^{(A)}}{2}\right) \\
&+ 12\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)} \\
&+ 2\lambda_1\lambda_{HS_1}^{(A)} \\
&+ 8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)} \\
&+ 2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)} \\
&+ \lambda_{HS_1}^{(A)}\left(16\lambda_{S_1}^{(A)} + 6\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\right]\right) \\
&- \frac{4\lambda_5\lambda_5^*}{3}
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{HS_2}^{(A)}}^{(1)} =& -\frac{33g_2^2\lambda_{HS_2}^{(A)}}{2} \\
&+ g_1^2\left(-4g_2^2 - \frac{15\lambda_{HS_2}^{(A)}}{2}\right) \\
&+ 12\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)} \\
&+ 2\lambda_1\lambda_{HS_2}^{(A)} \\
&+ 8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)} \\
&+ 2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)} \\
&+ \lambda_{HS_2}^{(A)}\left(16\lambda_{S_2}^{(A)} + 6\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\right]\right)
\end{aligned}
$$

$$
\begin{aligned}
&- \frac{8\lambda_5\lambda_5^*}{9}
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda^{(A)}_{S_1}} =& -2g_2^4 \\
&- 24g_2^2\lambda^{(A)}_{S_1} \\
&+ g_1^2\left(32g_2^2 - 48\lambda^{(A)}_{S_1}\right) \\
&+ 12\left(\lambda^{(A)}_{12}\right)^2 \\
&- \frac{4\left(\lambda^{(\times)}_{12}\right)^2}{3} \\
&+ 3\left(\lambda^{(A)}_{HS_1}\right)^2 \\
&+ 12\lambda^{(1)}_{S_1}\lambda^{(A)}_{S_1} \\
&+ 20\left(\lambda^{(A)}_{S_1}\right)^2
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda^{(A)}_{S_2}} =& -2g_2^4 \\
&- 24g_2^2\lambda^{(A)}_{S_2} \\
&+ g_1^2\left(8g_2^2 - 12\lambda^{(A)}_{S_2}\right) \\
&+ 12\left(\lambda^{(A)}_{12}\right)^2 \\
&- \frac{4\left(\lambda^{(\times)}_{12}\right)^2}{3} \\
&+ 3\left(\lambda^{(A)}_{HS_2}\right)^2 \\
&+ 12\lambda^{(1)}_{S_2}\lambda^{(A)}_{S_2} \\
&+ 20\left(\lambda^{(A)}_{S_2}\right)^2
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda^{(A)}_{12}} =& -24g_2^2\lambda^{(A)}_{12} \\
&+ g_1^2\left(16g_2^2 - 30\lambda^{(A)}_{12}\right) \\
&+ 8\lambda^{(1)}_{12}\lambda^{(A)}_{12} \\
&+ \frac{16\lambda^{(A)}_{12}\lambda^{(\times)}_{12}}{3} \\
&+ 2\lambda^{(A)}_{12}\lambda^{(1)}_{S_1} \\
&+ 16\lambda^{(A)}_{12}\lambda^{(A)}_{S_1} \\
&+ 2\lambda^{(A)}_{12}\lambda^{(1)}_{S_2} \\
&+ 16\lambda^{(A)}_{12}\lambda^{(A)}_{S_2} \\
&+ 3\lambda^{(A)}_{HS_1}\lambda^{(A)}_{HS_2} \\
&- \frac{2\lambda_5\lambda_5^*}{9}
\end{aligned}
$$

$$\beta^{(1)}_{\lambda_{12}^{(\times)}} = \frac{51 g_2^4}{10}$$
$$- \ 24 g_2^2 \lambda_{12}^{(\times)}$$
$$- \ 30 g_1^2 \lambda_{12}^{(\times)}$$
$$+ \ 8 \lambda_{12}^{(1)} \lambda_{12}^{(\times)}$$
$$+ \ 10 \left(\lambda_{12}^{(\times)}\right)^2$$
$$+ \ 2 \lambda_{12}^{(\times)} \lambda_{S_1}^{(1)}$$
$$- \ 2 \lambda_{12}^{(\times)} \lambda_{S_1}^{(A)}$$
$$+ \ 2 \lambda_{12}^{(\times)} \lambda_{S_2}^{(1)}$$
$$- \ 2 \lambda_{12}^{(\times)} \lambda_{S_2}^{(A)}$$
$$- \ \lambda_5 \lambda_5^*$$

$\mathrm{T3-E}\,(\alpha = -2)$

$$\beta^{(1)}_{m_1^2} = - \ 12 g_2^2 m_1^2$$
$$- \ 6 g_1^2 m_1^2$$
$$+ \ 12 \lambda_{18} \left(\mu_{S_2}^2\right)^*$$
$$+ \ 4 \lambda_{19} \left(\mu_{S_2}^2\right)^*$$
$$+ \ m_1^2 \left(8 \lambda_{S_1}^{(1)} + 4 \lambda_{S_1}^{(A)}\right)$$
$$+ \ m_2^2 \left(6 \lambda_{12}^{(1)} + 4 \lambda_{12}^{(\times)}\right)$$
$$+ \ 4 \mu_{S_2}^2 \left(3 \lambda_{18}^* + \lambda_{19}^*\right)$$

$$\beta^{(1)}_{m_2^2} = - \ 12 g_2^2 m_2^2$$
$$+ \ 20 \lambda_{14} \left(\mu_{S_2}^2\right)^*$$
$$+ \ m_1^2 \left(6 \lambda_{12}^{(1)} + 4 \lambda_{12}^{(\times)}\right)$$
$$+ \ m_2^2 \left(8 \lambda_{S_2}^{(1)} + 4 \lambda_{S_2}^{(A)}\right)$$
$$+ \ 20 \mu_{S_2}^2 \lambda_{14}^*$$

$$\beta^{(1)}_{\lambda_1} = \frac{9 g_2^4}{4}$$
$$- \ 9 g_2^2 \lambda_1$$
$$+ \ \frac{3 g_1^4}{4}$$
$$+ \ g_1^2 \left(\frac{3 g_2^2}{2} - 3 \lambda_1\right)$$
$$+ \ 12 \lambda_1^2$$
$$+ \ \lambda_1 \left(12 \, \mathrm{Tr}\left[Y_d \ (Y_d^*)^T\right] + 4 \, \mathrm{Tr}\left[Y_e \ (Y_e^*)^T\right] + 12 \, \mathrm{Tr}\left[Y_u \ (Y_u^*)^T\right]\right)$$
$$+ \ 6 \left(\lambda_{HS_1}^{(1)}\right)^2$$

$$
\begin{aligned}
&+ 3\left(\lambda_{HS_1}^{(A)}\right)^2 \\
&+ 6\left(\lambda_{HS_2}^{(1)}\right)^2 \\
&+ 3\left(\lambda_{HS_2}^{(A)}\right)^2 \\
&+ 4\lambda_5\lambda_5^* \\
&+ 4\lambda_{22}\lambda_{22}^* \\
&+ 24\lambda_{25}\lambda_{25}^* \\
&- 12\,\mathrm{Tr}\Big[Y_d\ (Y_d^*)^T\ Y_d\ (Y_d^*)^T\Big] \\
&- 4\,\mathrm{Tr}\Big[Y_e\ (Y_e^*)^T\ Y_e\ (Y_e^*)^T\Big] \\
&- 12\,\mathrm{Tr}\Big[Y_u\ (Y_u^*)^T\ Y_u\ (Y_u^*)^T\Big]
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda^{(1)}_{S_1}} &= 22g_2^4 \\
&- 24g_2^2\lambda_{S_1}^{(1)} \\
&+ 12g_1^4 \\
&+ g_1^2\left(8g_2^2 - 12\lambda_{S_1}^{(1)}\right) \\
&+ 6\left(\lambda_{12}^{(1)}\right)^2 \\
&+ 8\lambda_{12}^{(1)}\lambda_{12}^{(\times)} \\
&+ \frac{20\left(\lambda_{12}^{(\times)}\right)^2}{3} \\
&+ 4\left(\lambda_{HS_1}^{(1)}\right)^2 \\
&+ 14\left(\lambda_{S_1}^{(1)}\right)^2 \\
&+ 8\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)} \\
&+ 16\left(\lambda_{S_1}^{(A)}\right)^2 \\
&+ 8\lambda_{18}\left(3\lambda_{18}^* + \lambda_{19}^*\right) \\
&+ \lambda_{19}\left(8\lambda_{18}^* + \frac{20\lambda_{19}^*}{3}\right)
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda^{(1)}_{S_2}} &= 22g_2^4 \\
&- 24g_2^2\lambda_{S_2}^{(1)} \\
&+ 6\left(\lambda_{12}^{(1)}\right)^2 \\
&+ 8\lambda_{12}^{(1)}\lambda_{12}^{(\times)} \\
&+ \frac{20\left(\lambda_{12}^{(\times)}\right)^2}{3}
\end{aligned}
$$

$$
\begin{aligned}
&+ 4\left(\lambda_{HS_2}^{(1)}\right)^2 \\
&+ 14\left(\lambda_{S_2}^{(1)}\right)^2 \\
&+ 8\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)} \\
&+ 16\left(\lambda_{S_2}^{(A)}\right)^2 \\
&+ \frac{832\lambda_{13}\lambda_{13}^*}{3} \\
&+ \frac{352\lambda_{14}\lambda_{14}^*}{3} \\
&+ \frac{8\lambda_{18}\left(3\lambda_{18}^* + \lambda_{19}^*\right)}{3} \\
&+ \lambda_{19}\left(\frac{8\lambda_{18}^*}{3} + 4\lambda_{19}^*\right) \\
&+ \frac{16\lambda_{25}\lambda_{25}^*}{3}
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{HS_1}^{(1)}}^{(1)} &= 6g_2^4 \\
&- \frac{33g_2^2\lambda_{HS_1}^{(1)}}{2} \\
&+ 3g_1^4 \\
&- \frac{15g_1^2\lambda_{HS_1}^{(1)}}{2} \\
&+ 6\lambda_1\lambda_{HS_1}^{(1)} \\
&+ 4\left(\lambda_{HS_1}^{(1)}\right)^2 \\
&+ 8\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)} \\
&+ \lambda_{HS_1}^{(1)}\left(4\lambda_{S_1}^{(A)} + 6\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\right]\right) \\
&+ 6\left(\lambda_{HS_1}^{(A)}\right)^2 \\
&+ \lambda_{HS_2}^{(1)}\left(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)}\right) \\
&+ 4\lambda_5\lambda_5^* \\
&+ 12\lambda_{18}\lambda_{25}^* \\
&+ 4\lambda_{19}\lambda_{25}^* \\
&+ 4\lambda_{22}\lambda_{22}^* \\
&+ 4\lambda_{25}\left(3\lambda_{18}^* + \lambda_{19}^*\right)
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{HS_2}^{(1)}}^{(1)} &= 6g_2^4 \\
&- \frac{33g_2^2\lambda_{HS_2}^{(1)}}{2}
\end{aligned}
$$

$$
\begin{aligned}
&- \frac{3g_1^2\lambda_{HS_2}^{(1)}}{2} \\
&+ 6\lambda_1\lambda_{HS_2}^{(1)} \\
&+ \lambda_{HS_1}^{(1)}\left(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)}\right) \\
&+ 4\left(\lambda_{HS_2}^{(1)}\right)^2 \\
&+ 8\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)} \\
&+ \lambda_{HS_2}^{(1)}\left(4\lambda_{S_2}^{(A)} + 6\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\right]\right) \\
&+ 6\left(\lambda_{HS_2}^{(A)}\right)^2 \\
&+ 4\lambda_5\lambda_5^* \\
&+ 20\lambda_{14}\lambda_{25}^* \\
&+ 4\lambda_{22}\lambda_{22}^* \\
&+ 4\lambda_{25}\left(5\lambda_{14}^* + 4\lambda_{25}^*\right)
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{12}^{(1)}}^{(1)} = &\frac{63g_2^4}{5} \\
&- 24g_2^2\lambda_{12}^{(1)} \\
&- 6g_1^2\lambda_{12}^{(1)} \\
&+ 4\left(\lambda_{12}^{(1)}\right)^2 \\
&+ \lambda_{12}^{(1)}\left(4\lambda_{S_1}^{(A)} + 4\lambda_{S_2}^{(A)}\right) \\
&+ 16\left(\lambda_{12}^{(A)}\right)^2 \\
&+ 4\left(\lambda_{12}^{(\times)}\right)^2 \\
&+ 4\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)} \\
&+ 4\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)} \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)} \\
&+ \lambda_{S_1}^{(1)}\left(8\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)}\right) \\
&+ \lambda_{S_2}^{(1)}\left(8\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)}\right) \\
&+ 2\lambda_5\lambda_5^* \\
&+ 4\lambda_{14}\left(5\lambda_{18}^* + \lambda_{19}^*\right) \\
&+ 4\lambda_{18}\left(5\lambda_{14}^* + 4\lambda_{18}^*\right) \\
&+ 4\lambda_{19}\left(\lambda_{14}^* + \lambda_{19}^*\right) \\
&+ 2\lambda_{22}\lambda_{22}^*
\end{aligned}
$$

$$\begin{aligned}\beta^{(1)}_{\lambda_5} =& -\frac{33g_2^2\lambda_5}{2}\\ &-\frac{9g_1^2\lambda_5}{2}\\ &+2\lambda^{(1)}_{12}\lambda_5\\ &+2\lambda_1\lambda_5\\ &+4\lambda^{(1)}_{HS_1}\lambda_5\\ &+4\lambda^{(1)}_{HS_2}\lambda_5\\ &+\lambda_5\left(-\frac{4\lambda^{(A)}_{12}}{3}-2\lambda^{(\times)}_{12}-2\lambda^{(A)}_{HS_1}-\frac{4\lambda^{(A)}_{HS_2}}{3}+6\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\right]+2\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\right]+6\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\right]\right)\\ &+4\lambda_{18}\lambda^*_{22}\\ &-2\lambda_{19}\lambda^*_{22}\\ &+8\lambda_{25}\lambda^*_{22}\end{aligned}$$

$$\begin{aligned}\beta^{(1)}_{\lambda^{(A)}_{HS_1}} =& -\frac{33g_2^2\lambda^{(A)}_{HS_1}}{2}\\ &+g_1^2\left(-4g_2^2-\frac{15\lambda^{(A)}_{HS_1}}{2}\right)\\ &+12\lambda^{(A)}_{12}\lambda^{(A)}_{HS_2}\\ &+2\lambda_1\lambda^{(A)}_{HS_1}\\ &+8\lambda^{(1)}_{HS_1}\lambda^{(A)}_{HS_1}\\ &+2\lambda^{(A)}_{HS_1}\lambda^{(1)}_{S_1}\\ &+\lambda^{(A)}_{HS_1}\left(16\lambda^{(A)}_{S_1}+6\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\right]+2\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\right]+6\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\right]\right)\\ &-\frac{4\lambda_5\lambda^*_5}{3}\\ &-\frac{4\lambda_{22}\lambda^*_{22}}{3}\end{aligned}$$

$$\begin{aligned}\beta^{(1)}_{\lambda^{(A)}_{HS_2}} =& -\frac{33g_2^2\lambda^{(A)}_{HS_2}}{2}\\ &-\frac{3g_1^2\lambda^{(A)}_{HS_2}}{2}\\ &+12\lambda^{(A)}_{12}\lambda^{(A)}_{HS_1}\\ &+2\lambda_1\lambda^{(A)}_{HS_2}\\ &+8\lambda^{(1)}_{HS_2}\lambda^{(A)}_{HS_2}\\ &+2\lambda^{(A)}_{HS_2}\lambda^{(1)}_{S_2}\\ &+\lambda^{(A)}_{HS_2}\left(16\lambda^{(A)}_{S_2}+6\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\right]+2\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\right]+6\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\right]\right)\\ &-\frac{8\lambda_5\lambda^*_5}{9}\\ &-\frac{4\lambda_{22}\lambda^*_{22}}{3}\end{aligned}$$

$$\beta^{(1)}_{\lambda^{(A)}_{S_1}} = -2g_2^4$$
$$- 24g_2^2\lambda^{(A)}_{S_1}$$
$$+ g_1^2\left(8g_2^2 - 12\lambda^{(A)}_{S_1}\right)$$
$$+ 12\left(\lambda^{(A)}_{12}\right)^2$$
$$- \frac{4\left(\lambda^{(\times)}_{12}\right)^2}{3}$$
$$+ 3\left(\lambda^{(A)}_{HS_1}\right)^2$$
$$+ 12\lambda^{(1)}_{S_1}\lambda^{(A)}_{S_1}$$
$$+ 20\left(\lambda^{(A)}_{S_1}\right)^2$$
$$- \frac{4\lambda_{19}\lambda^*_{19}}{3}$$

$$\beta^{(1)}_{\lambda^{(A)}_{S_2}} = -2g_2^4$$
$$- 24g_2^2\lambda^{(A)}_{S_2}$$
$$+ 12\left(\lambda^{(A)}_{12}\right)^2$$
$$- \frac{4\left(\lambda^{(\times)}_{12}\right)^2}{3}$$
$$+ 3\left(\lambda^{(A)}_{HS_2}\right)^2$$
$$+ 12\lambda^{(1)}_{S_2}\lambda^{(A)}_{S_2}$$
$$+ 20\left(\lambda^{(A)}_{S_2}\right)^2$$
$$- \frac{224\lambda_{13}\lambda^*_{13}}{3}$$
$$- \frac{44\lambda_{14}\lambda^*_{14}}{3}$$
$$- \frac{4\lambda_{18}\left(3\lambda^*_{18} + \lambda^*_{19}\right)}{3}$$
$$- \frac{4\lambda_{19}\lambda^*_{18}}{3}$$
$$- \frac{8\lambda_{25}\lambda^*_{25}}{3}$$

$$\beta^{(1)}_{\lambda^{(A)}_{12}} = -24g_2^2\lambda^{(A)}_{12}$$
$$- 6g_1^2\lambda^{(A)}_{12}$$
$$+ 8\lambda^{(1)}_{12}\lambda^{(A)}_{12}$$
$$+ \frac{16\lambda^{(A)}_{12}\lambda^{(\times)}_{12}}{3}$$
$$+ 2\lambda^{(A)}_{12}\lambda^{(1)}_{S_1}$$

$$
\begin{aligned}
&+ 16\lambda_{12}^{(A)}\lambda_{S_1}^{(A)} \\
&+ 2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)} \\
&+ 16\lambda_{12}^{(A)}\lambda_{S_2}^{(A)} \\
&+ 3\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)} \\
&- \frac{2\lambda_5\lambda_5^*}{9} \\
&- \frac{\lambda_{22}\lambda_{22}^*}{3}
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{12}^{(\times)}}^{(1)} = &\frac{51g_2^4}{10} \\
&- 24g_2^2\lambda_{12}^{(\times)} \\
&- 6g_1^2\lambda_{12}^{(\times)} \\
&+ 8\lambda_{12}^{(1)}\lambda_{12}^{(\times)} \\
&+ 10\left(\lambda_{12}^{(\times)}\right)^2 \\
&+ 2\lambda_{12}^{(\times)}\lambda_{S_1}^{(1)} \\
&- 2\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)} \\
&+ 2\lambda_{12}^{(\times)}\lambda_{S_2}^{(1)} \\
&- 2\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)} \\
&- \lambda_5\lambda_5^* \\
&+ 4\lambda_{14}\lambda_{19}^* \\
&+ 8\lambda_{18}\lambda_{19}^* \\
&+ \lambda_{19}\left(4\lambda_{14}^* + 8\lambda_{18}^* + 10\lambda_{19}^*\right) \\
&- \lambda_{22}\lambda_{22}^*
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{13}}^{(1)} = &- 24g_2^2\lambda_{13} \\
&+ 12\lambda_{S_2}^{(1)}\lambda_{13} \\
&- 8\lambda_{S_2}^{(A)}\lambda_{13} \\
&+ 11\lambda_{14}^2 \\
&+ 3\lambda_{18}^2 \\
&+ 2\lambda_{18}\lambda_{19} \\
&+ \lambda_{19}^2 \\
&+ 2\lambda_{25}^2
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{14}}^{(1)} = &- 24g_2^2\lambda_{14} \\
&+ \lambda_{12}^{(1)}\left(6\lambda_{18} + 2\lambda_{19}\right) \\
&+ \lambda_{12}^{(\times)}\left(4\lambda_{18} + 4\lambda_{19}\right) \\
&+ 4\lambda_{HS_2}^{(1)}\lambda_{25}
\end{aligned}
$$

$$
\begin{aligned}
&+ 22\lambda_{S_2}^{(1)}\lambda_{14} \\
&+ 88\lambda_{13}\lambda_{14}^{*}
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{18}}^{(1)} = &- 24g_2^2\lambda_{18} \\
&- 6g_1^2\lambda_{18} \\
&+ \lambda_{12}^{(1)}\left(10\lambda_{14} + 8\lambda_{18}\right) \\
&+ \lambda_{12}^{(\times)}\left(4\lambda_{14} + 4\lambda_{19}\right) \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_{25} \\
&+ \lambda_{S_1}^{(1)}\left(8\lambda_{18} + 2\lambda_{19}\right) \\
&+ \lambda_{S_1}^{(A)}\left(4\lambda_{18} + 2\lambda_{19}\right) \\
&+ 2\lambda_{S_2}^{(1)}\lambda_{18} \\
&+ \lambda_{S_2}^{(A)}\left(-4\lambda_{18} - 2\lambda_{19}\right) \\
&+ 2\lambda_5\lambda_{22} \\
&+ 8\lambda_{13}\left(5\lambda_{18}^{*} + \lambda_{19}^{*}\right)
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{19}}^{(1)} = &- 24g_2^2\lambda_{19} \\
&- 6g_1^2\lambda_{19} \\
&+ 8\lambda_{12}^{(1)}\lambda_{19} \\
&+ \lambda_{12}^{(\times)}\left(8\lambda_{14} + 16\lambda_{18} + 20\lambda_{19}\right) \\
&+ 2\lambda_{S_1}^{(1)}\lambda_{19} \\
&- 2\lambda_{S_1}^{(A)}\lambda_{19} \\
&+ 2\lambda_{S_2}^{(1)}\lambda_{19} \\
&+ 2\lambda_{S_2}^{(A)}\lambda_{19} \\
&- 2\lambda_5\lambda_{22} \\
&+ 16\lambda_{13}\lambda_{19}^{*}
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{22}}^{(1)} = &- \frac{33g_2^2\lambda_{22}}{2} \\
&- \frac{9g_1^2\lambda_{22}}{2} \\
&+ 2\lambda_{12}^{(1)}\lambda_{22} \\
&- 2\lambda_{12}^{(A)}\lambda_{22} \\
&- 2\lambda_{12}^{(\times)}\lambda_{22} \\
&+ 2\lambda_1\lambda_{22} \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_{22} \\
&- 2\lambda_{HS_1}^{(A)}\lambda_{22} \\
&+ 4\lambda_{HS_2}^{(1)}\lambda_{22} \\
&- 2\lambda_{HS_2}^{(A)}\lambda_{22} \\
&+ 4\lambda_{18}\lambda_5^{*} \\
&- 2\lambda_{19}\lambda_5^{*}
\end{aligned}
$$

$$+ \lambda_{22} \left(6 \operatorname{Tr}\left[Y_d \left(Y_d^*\right)^T\right] + 2 \operatorname{Tr}\left[Y_e \left(Y_e^*\right)^T\right] + 6 \operatorname{Tr}\left[Y_u \left(Y_u^*\right)^T\right]\right)$$
$$+ 8\lambda_{25}\lambda_5^*$$

$$\beta^{(1)}_{\lambda_{25}} = - \frac{33 g_2^2 \lambda_{25}}{2}$$
$$- \frac{3 g_1^2 \lambda_{25}}{2}$$
$$+ 6\lambda_1\lambda_{25}$$
$$+ \lambda^{(1)}_{HS_1} (6\lambda_{18} + 2\lambda_{19})$$
$$+ \lambda^{(1)}_{HS_2} (10\lambda_{14} + 8\lambda_{25})$$
$$+ 2\lambda^{(1)}_{S_2}\lambda_{25}$$
$$- 4\lambda^{(A)}_{S_2}\lambda_{25}$$
$$+ 4\lambda_5\lambda_{22}$$
$$+ 40\lambda_{13}\lambda_{25}^*$$
$$+ \lambda_{25} \left(6 \operatorname{Tr}\left[Y_d \left(Y_d^*\right)^T\right] + 2 \operatorname{Tr}\left[Y_e \left(Y_e^*\right)^T\right] + 6 \operatorname{Tr}\left[Y_u \left(Y_u^*\right)^T\right]\right)$$

$$\beta^{(1)}_{\mu^2_{S_2}} = - 12 g_2^2 \mu^2_{S_2}$$
$$+ 2\lambda^{(1)}_{S_2}\mu^2_{S_2}$$
$$- 4\lambda^{(A)}_{S_2}\mu^2_{S_2}$$
$$+ 40\lambda_{13} \left(\mu^2_{S_2}\right)^*$$
$$+ 10\lambda_{14} m_2^2$$
$$+ m_1^2 (6\lambda_{18} + 2\lambda_{19})$$

$\mathrm{T3-E}\,(\alpha = 0)$

$$\beta^{(1)}_{m_1^2} = - 12 g_2^2 m_1^2$$
$$+ 20\lambda_{20} \left(\mu^2_{S_1}\right)^*$$
$$+ m_1^2 \left(8\lambda^{(1)}_{S_1} + 4\lambda^{(A)}_{S_1}\right)$$
$$+ m_2^2 \left(6\lambda^{(1)}_{12} + 4\lambda^{(\times)}_{12}\right)$$
$$+ 20\mu^2_{S_1}\lambda_{20}^*$$

$$\beta^{(1)}_{m_2^2} = - 12 g_2^2 m_2^2$$
$$- 6 g_1^2 m_2^2$$
$$+ 12\lambda_{15} \left(\mu^2_{S_1}\right)^*$$
$$+ 4\lambda_{16} \left(\mu^2_{S_1}\right)^*$$
$$+ m_1^2 \left(6\lambda^{(1)}_{12} + 4\lambda^{(\times)}_{12}\right)$$
$$+ m_2^2 \left(8\lambda^{(1)}_{S_2} + 4\lambda^{(A)}_{S_2}\right)$$
$$+ 4\mu^2_{S_1} (3\lambda_{15}^* + \lambda_{16}^*)$$

$$\beta^{(1)}_{\lambda_1} = \frac{9g_2^4}{4}$$
$$- 9g_2^2\lambda_1$$
$$+ \frac{3g_1^4}{4}$$
$$+ g_1^2\left(\frac{3g_2^2}{2} - 3\lambda_1\right)$$
$$+ 12\lambda_1^2$$
$$+ \lambda_1\left(12\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\right] + 4\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\right] + 12\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\right]\right)$$
$$+ 6\left(\lambda^{(1)}_{HS_1}\right)^2$$
$$+ 3\left(\lambda^{(A)}_{HS_1}\right)^2$$
$$+ 6\left(\lambda^{(1)}_{HS_2}\right)^2$$
$$+ 3\left(\lambda^{(A)}_{HS_2}\right)^2$$
$$+ 4\lambda_5\lambda_5^*$$
$$+ 4\lambda_{23}\lambda_{23}^*$$
$$+ 24\lambda_{27}\lambda_{27}^*$$
$$- 12\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\ Y_d\ (Y_d^*)^T\right]$$
$$- 4\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\ Y_e\ (Y_e^*)^T\right]$$
$$- 12\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\ Y_u\ (Y_u^*)^T\right]$$

$$\beta^{(1)}_{\lambda^{(1)}_{S_1}} = 22g_2^4$$
$$- 24g_2^2\lambda^{(1)}_{S_1}$$
$$+ 6\left(\lambda^{(1)}_{12}\right)^2$$
$$+ 8\lambda^{(1)}_{12}\lambda^{(\times)}_{12}$$
$$+ \frac{20\left(\lambda^{(\times)}_{12}\right)^2}{3}$$
$$+ 4\left(\lambda^{(1)}_{HS_1}\right)^2$$
$$+ 14\left(\lambda^{(1)}_{S_1}\right)^2$$
$$+ 8\lambda^{(1)}_{S_1}\lambda^{(A)}_{S_1}$$
$$+ 16\left(\lambda^{(A)}_{S_1}\right)^2$$
$$+ \frac{8\lambda_{15}\left(3\lambda_{15}^* + \lambda_{16}^*\right)}{3}$$
$$+ \lambda_{16}\left(\frac{8\lambda_{15}^*}{3} + 4\lambda_{16}^*\right)$$
$$+ \frac{832\lambda_{17}\lambda_{17}^*}{3}$$

$$+ \frac{352\lambda_{20}\lambda_{20}^*}{3}$$

$$+ \frac{16\lambda_{27}\lambda_{27}^*}{3}$$

$$\beta^{(1)}_{\lambda^{(1)}_{S_2}} = 22g_2^4$$

$$- 24g_2^2\lambda^{(1)}_{S_2}$$

$$+ 12g_1^4$$

$$+ g_1^2\left(8g_2^2 - 12\lambda^{(1)}_{S_2}\right)$$

$$+ 6\left(\lambda^{(1)}_{12}\right)^2$$

$$+ 8\lambda^{(1)}_{12}\lambda^{(\times)}_{12}$$

$$+ \frac{20\left(\lambda^{(\times)}_{12}\right)^2}{3}$$

$$+ 4\left(\lambda^{(1)}_{HS_2}\right)^2$$

$$+ 14\left(\lambda^{(1)}_{S_2}\right)^2$$

$$+ 8\lambda^{(1)}_{S_2}\lambda^{(A)}_{S_2}$$

$$+ 16\left(\lambda^{(A)}_{S_2}\right)^2$$

$$+ 8\lambda_{15}\left(3\lambda_{15}^* + \lambda_{16}^*\right)$$

$$+ \lambda_{16}\left(8\lambda_{15}^* + \frac{20\lambda_{16}^*}{3}\right)$$

$$\beta^{(1)}_{\lambda^{(1)}_{HS_1}} = 6g_2^4$$

$$- \frac{33g_2^2\lambda^{(1)}_{HS_1}}{2}$$

$$- \frac{3g_1^2\lambda^{(1)}_{HS_1}}{2}$$

$$+ 6\lambda_1\lambda^{(1)}_{HS_1}$$

$$+ 4\left(\lambda^{(1)}_{HS_1}\right)^2$$

$$+ 8\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$$

$$+ \lambda^{(1)}_{HS_1}\left(4\lambda^{(A)}_{S_1} + 6\,\mathrm{Tr}\left[Y_d\ \left(Y_d^*\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ \left(Y_e^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ \left(Y_u^*\right)^T\right]\right)$$

$$+ 6\left(\lambda^{(A)}_{HS_1}\right)^2$$

$$+ \lambda^{(1)}_{HS_2}\left(6\lambda^{(1)}_{12} + 4\lambda^{(\times)}_{12}\right)$$

$$+ 4\lambda_5\lambda_5^*$$

$$+ 20\lambda_{20}\lambda_{27}^*$$

$$+ \ 4\lambda_{23}\lambda_{23}^{*}$$

$$+ \ 4\lambda_{27}\left(5\lambda_{20}^{*} + 4\lambda_{27}^{*}\right)$$

$$\beta^{(1)}_{\lambda^{(1)}_{HS_2}} = 6g_2^4$$

$$- \ \frac{33g_2^2\lambda^{(1)}_{HS_2}}{2}$$

$$+ \ 3g_1^4$$

$$- \ \frac{15g_1^2\lambda^{(1)}_{HS_2}}{2}$$

$$+ \ 6\lambda_1\lambda^{(1)}_{HS_2}$$

$$+ \ \lambda^{(1)}_{HS_1}\left(6\lambda^{(1)}_{12} + 4\lambda^{(\times)}_{12}\right)$$

$$+ \ 4\left(\lambda^{(1)}_{HS_2}\right)^2$$

$$+ \ 8\lambda^{(1)}_{HS_2}\lambda^{(1)}_{S_2}$$

$$+ \ \lambda^{(1)}_{HS_2}\left(4\lambda^{(A)}_{S_2} + 6\,\mathrm{Tr}\left[Y_d\ \left(Y_d^{*}\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ \left(Y_e^{*}\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ \left(Y_u^{*}\right)^T\right]\right)$$

$$+ \ 6\left(\lambda^{(A)}_{HS_2}\right)^2$$

$$+ \ 4\lambda_5\lambda_5^{*}$$

$$+ \ 12\lambda_{15}\lambda_{27}^{*}$$

$$+ \ 4\lambda_{16}\lambda_{27}^{*}$$

$$+ \ 4\lambda_{23}\lambda_{23}^{*}$$

$$+ \ 4\lambda_{27}\left(3\lambda_{15}^{*} + \lambda_{16}^{*}\right)$$

$$\beta^{(1)}_{\lambda^{(1)}_{12}} = \frac{63g_2^4}{5}$$

$$- \ 24g_2^2\lambda^{(1)}_{12}$$

$$- \ 6g_1^2\lambda^{(1)}_{12}$$

$$+ \ 4\left(\lambda^{(1)}_{12}\right)^2$$

$$+ \ \lambda^{(1)}_{12}\left(4\lambda^{(A)}_{S_1} + 4\lambda^{(A)}_{S_2}\right)$$

$$+ \ 16\left(\lambda^{(A)}_{12}\right)^2$$

$$+ \ 4\left(\lambda^{(\times)}_{12}\right)^2$$

$$+ \ 4\lambda^{(\times)}_{12}\lambda^{(A)}_{S_1}$$

$$+ \ 4\lambda^{(\times)}_{12}\lambda^{(A)}_{S_2}$$

$$+ \ 4\lambda^{(1)}_{HS_1}\lambda^{(1)}_{HS_2}$$

$$+ \ \lambda^{(1)}_{S_1}\left(8\lambda^{(1)}_{12} + 4\lambda^{(\times)}_{12}\right)$$

$$+ \lambda_{S_2}^{(1)} \left(8\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)}\right)$$
$$+ 2\lambda_5\lambda_5^*$$
$$+ 4\lambda_{15} \left(4\lambda_{15}^* + 5\lambda_{20}^*\right)$$
$$+ 4\lambda_{16} \left(\lambda_{16}^* + \lambda_{20}^*\right)$$
$$+ 4\lambda_{20} \left(5\lambda_{15}^* + \lambda_{16}^*\right)$$
$$+ 2\lambda_{23}\lambda_{23}^*$$

$$\beta_{\lambda_5}^{(1)} = - \frac{33g_2^2\lambda_5}{2}$$
$$- \frac{9g_1^2\lambda_5}{2}$$
$$+ 2\lambda_{12}^{(1)}\lambda_5$$
$$+ 2\lambda_1\lambda_5$$
$$+ 4\lambda_{HS_1}^{(1)}\lambda_5$$
$$+ 4\lambda_{HS_2}^{(1)}\lambda_5$$
$$+ \lambda_5 \left(-\frac{4\lambda_{12}^{(A)}}{3} - 2\lambda_{12}^{(\times)} - 2\lambda_{HS_1}^{(A)} - \frac{4\lambda_{HS_2}^{(A)}}{3} + 6\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\right]\right)$$
$$+ 2\left(2\lambda_{15}^* - \lambda_{16}^* + 4\lambda_{27}^*\right)\lambda_{23}^*$$

$$\beta_{\lambda_{HS_1}^{(A)}}^{(1)} = - \frac{33g_2^2\lambda_{HS_1}^{(A)}}{2}$$
$$- \frac{3g_1^2\lambda_{HS_1}^{(A)}}{2}$$
$$+ 12\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$$
$$+ 2\lambda_1\lambda_{HS_1}^{(A)}$$
$$+ 8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$$
$$+ 2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$$
$$+ \lambda_{HS_1}^{(A)} \left(16\lambda_{S_1}^{(A)} + 6\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\right]\right)$$
$$- \frac{4\lambda_5\lambda_5^*}{3}$$
$$- \frac{8\lambda_{23}\lambda_{23}^*}{9}$$

$$\beta_{\lambda_{HS_2}^{(A)}}^{(1)} = - \frac{33g_2^2\lambda_{HS_2}^{(A)}}{2}$$
$$+ g_1^2 \left(4g_2^2 - \frac{15\lambda_{HS_2}^{(A)}}{2}\right)$$
$$+ 12\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$$
$$+ 2\lambda_1\lambda_{HS_2}^{(A)}$$
$$+ 8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$$

$$+ 2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$$

$$+ \lambda_{HS_2}^{(A)}\left(16\lambda_{S_2}^{(A)} + 6\,\mathrm{Tr}\left[Y_d\,(Y_d^*)^T\right] + 2\,\mathrm{Tr}\left[Y_e\,(Y_e^*)^T\right] + 6\,\mathrm{Tr}\left[Y_u\,(Y_u^*)^T\right]\right)$$

$$- \frac{8\lambda_5\lambda_5^*}{9}$$

$$- \frac{8\lambda_{23}\lambda_{23}^*}{9}$$

$$\beta_{\lambda_{S_1}^{(A)}}^{(1)} = -2g_2^4$$

$$- 24g_2^2\lambda_{S_1}^{(A)}$$

$$+ 12\left(\lambda_{12}^{(A)}\right)^2$$

$$- \frac{4\left(\lambda_{12}^{(\times)}\right)^2}{3}$$

$$+ 3\left(\lambda_{HS_1}^{(A)}\right)^2$$

$$+ 12\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)}$$

$$+ 20\left(\lambda_{S_1}^{(A)}\right)^2$$

$$- \frac{4\lambda_{15}\left(3\lambda_{15}^* + \lambda_{16}^*\right)}{3}$$

$$- \frac{4\lambda_{16}\lambda_{15}^*}{3}$$

$$- \frac{224\lambda_{17}\lambda_{17}^*}{3}$$

$$- \frac{44\lambda_{20}\lambda_{20}^*}{3}$$

$$- \frac{8\lambda_{27}\lambda_{27}^*}{3}$$

$$\beta_{\lambda_{S_2}^{(A)}}^{(1)} = -2g_2^4$$

$$- 24g_2^2\lambda_{S_2}^{(A)}$$

$$+ g_1^2\left(8g_2^2 - 12\lambda_{S_2}^{(A)}\right)$$

$$+ 12\left(\lambda_{12}^{(A)}\right)^2$$

$$- \frac{4\left(\lambda_{12}^{(\times)}\right)^2}{3}$$

$$+ 3\left(\lambda_{HS_2}^{(A)}\right)^2$$

$$+ 12\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)}$$

$$+ 20\left(\lambda_{S_2}^{(A)}\right)^2$$

$$- \frac{4\lambda_{16}\lambda_{16}^*}{3}$$

$$\beta_{\lambda_{12}^{(A)}}^{(1)} = -24g_2^2\lambda_{12}^{(A)}$$

$$
\begin{aligned}
&- 6g_1^2\lambda_{12}^{(A)} \\
&+ 8\lambda_{12}^{(1)}\lambda_{12}^{(A)} \\
&+ \frac{16\lambda_{12}^{(A)}\lambda_{12}^{(\times)}}{3} \\
&+ 2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)} \\
&+ 16\lambda_{12}^{(A)}\lambda_{S_1}^{(A)} \\
&+ 2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)} \\
&+ 16\lambda_{12}^{(A)}\lambda_{S_2}^{(A)} \\
&+ 3\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)} \\
&- \frac{2\lambda_5\lambda_5^*}{9} \\
&- \frac{\lambda_{23}\lambda_{23}^*}{3}
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{12}^{(\times)}}^{(1)} = &\frac{51g_2^4}{10} \\
&- 24g_2^2\lambda_{12}^{(\times)} \\
&- 6g_1^2\lambda_{12}^{(\times)} \\
&+ 8\lambda_{12}^{(1)}\lambda_{12}^{(\times)} \\
&+ 10\left(\lambda_{12}^{(\times)}\right)^2 \\
&+ 2\lambda_{12}^{(\times)}\lambda_{S_1}^{(1)} \\
&- 2\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)} \\
&+ 2\lambda_{12}^{(\times)}\lambda_{S_2}^{(1)} \\
&- 2\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)} \\
&- \lambda_5\lambda_5^* \\
&+ 8\lambda_{15}\lambda_{16}^* \\
&+ \lambda_{16}\left(8\lambda_{15}^* + 10\lambda_{16}^* + 4\lambda_{20}^*\right) \\
&+ 4\lambda_{20}\lambda_{16}^* \\
&- \lambda_{23}\lambda_{23}^*
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{15}}^{(1)} = &- 24g_2^2\lambda_{15} \\
&- 6g_1^2\lambda_{15} \\
&+ \lambda_{12}^{(1)}\left(8\lambda_{15} + 10\lambda_{20}\right) \\
&+ \lambda_{12}^{(\times)}\left(4\lambda_{16} + 4\lambda_{20}\right) \\
&+ 4\lambda_{HS_2}^{(1)}\lambda_{27} \\
&+ 2\lambda_{S_1}^{(1)}\lambda_{15} \\
&+ \lambda_{S_1}^{(A)}\left(-4\lambda_{15} - 2\lambda_{16}\right)
\end{aligned}
$$

$$
\begin{aligned}
&+ \lambda_{S_2}^{(1)} \left(8\lambda_{15} + 2\lambda_{16}\right) \\
&+ \lambda_{S_2}^{(A)} \left(4\lambda_{15} + 2\lambda_{16}\right) \\
&+ 8\lambda_{17} \left(5\lambda_{15}^{*} + \lambda_{16}^{*}\right) \\
&+ 2\lambda_{5}^{*}\lambda_{23}^{*}
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{16}}^{(1)} = &- 24g_2^2\lambda_{16} \\
&- 6g_1^2\lambda_{16} \\
&+ 8\lambda_{12}^{(1)}\lambda_{16} \\
&+ \lambda_{12}^{(\times)} \left(16\lambda_{15} + 20\lambda_{16} + 8\lambda_{20}\right) \\
&+ 2\lambda_{S_1}^{(1)}\lambda_{16} \\
&+ 2\lambda_{S_1}^{(A)}\lambda_{16} \\
&+ 2\lambda_{S_2}^{(1)}\lambda_{16} \\
&- 2\lambda_{S_2}^{(A)}\lambda_{16} \\
&+ 16\lambda_{17}\lambda_{16}^{*} \\
&- 2\lambda_{5}^{*}\lambda_{23}^{*}
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{17}}^{(1)} = &- 24g_2^2\lambda_{17} \\
&+ 12\lambda_{S_1}^{(1)}\lambda_{17} \\
&- 8\lambda_{S_1}^{(A)}\lambda_{17} \\
&+ 3\lambda_{15}^2 \\
&+ 2\lambda_{15}\lambda_{16} \\
&+ \lambda_{16}^2 \\
&+ 11\lambda_{20}^2 \\
&+ 2\lambda_{27}^2
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{20}}^{(1)} = &- 24g_2^2\lambda_{20} \\
&+ \lambda_{12}^{(1)} \left(6\lambda_{15} + 2\lambda_{16}\right) \\
&+ \lambda_{12}^{(\times)} \left(4\lambda_{15} + 4\lambda_{16}\right) \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_{27} \\
&+ 22\lambda_{S_1}^{(1)}\lambda_{20} \\
&+ 88\lambda_{17}\lambda_{20}^{*}
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{23}}^{(1)} = &- \frac{33g_2^2\lambda_{23}}{2} \\
&- \frac{9g_1^2\lambda_{23}}{2} \\
&+ 2\lambda_{12}^{(1)}\lambda_{23} \\
&- 2\lambda_{12}^{(A)}\lambda_{23} \\
&- 2\lambda_{12}^{(\times)}\lambda_{23} \\
&+ 2\lambda_{1}\lambda_{23} \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_{23}
\end{aligned}
$$

$$
\begin{aligned}
&- \frac{4\lambda_{HS_1}^{(A)}\lambda_{23}}{3} \\
&+ 4\lambda_{HS_2}^{(1)}\lambda_{23} \\
&- \frac{4\lambda_{HS_2}^{(A)}\lambda_{23}}{3} \\
&+ \lambda_{23}\left(6\,\mathrm{Tr}\left[Y_d\ \left(Y_d^*\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ \left(Y_e^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ \left(Y_u^*\right)^T\right]\right) \\
&+ 2\left(2\lambda_{15}^* - \lambda_{16}^* + 4\lambda_{27}^*\right)\lambda_5^*
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\lambda_{27}}^{(1)} = &- \frac{33g_2^2\lambda_{27}}{2} \\
&- \frac{3g_1^2\lambda_{27}}{2} \\
&+ 6\lambda_1\lambda_{27} \\
&+ \lambda_{HS_1}^{(1)}\left(10\lambda_{20} + 8\lambda_{27}\right) \\
&+ \lambda_{HS_2}^{(1)}\left(6\lambda_{15} + 2\lambda_{16}\right) \\
&+ 2\lambda_{S_1}^{(1)}\lambda_{27} \\
&- 4\lambda_{S_1}^{(A)}\lambda_{27} \\
&+ 40\lambda_{17}\lambda_{27}^* \\
&+ \lambda_{27}\left(6\,\mathrm{Tr}\left[Y_d\ \left(Y_d^*\right)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ \left(Y_e^*\right)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ \left(Y_u^*\right)^T\right]\right) \\
&+ 4\lambda_5^*\lambda_{23}^*
\end{aligned}
$$

$$
\begin{aligned}
\beta_{\mu_{S_1}^2}^{(1)} = &- 12g_2^2\mu_{S_1}^2 \\
&+ 2\lambda_{S_1}^{(1)}\mu_{S_1}^2 \\
&- 4\lambda_{S_1}^{(A)}\mu_{S_1}^2 \\
&+ 40\lambda_{17}\left(\mu_{S_1}^2\right)^* \\
&+ 10\lambda_{20}m_1^2 \\
&+ m_2^2\left(6\lambda_{15} + 2\lambda_{16}\right)
\end{aligned}
$$

$\mathrm{T3} - \mathrm{E}\,(\alpha = 2)$

$$
\begin{aligned}
\beta_{m_1^2}^{(1)} = &- 12g_2^2m_1^2 \\
&- 6g_1^2m_1^2 \\
&+ m_1^2\left(8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)}\right) \\
&+ m_2^2\left(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)}\right)
\end{aligned}
$$

$$
\begin{aligned}
\beta_{m_2^2}^{(1)} = &- 12g_2^2m_2^2 \\
&- 24g_1^2m_2^2 \\
&+ m_1^2\left(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)}\right) \\
&+ m_2^2\left(8\lambda_{S_2}^{(1)} + 4\lambda_{S_2}^{(A)}\right)
\end{aligned}
$$

$$\beta^{(1)}_{\lambda_1} = \frac{9g_2^4}{4}$$
$$- 9g_2^2\lambda_1$$
$$+ \frac{3g_1^4}{4}$$
$$+ g_1^2\left(\frac{3g_2^2}{2} - 3\lambda_1\right)$$
$$+ 12\lambda_1^2$$
$$+ \lambda_1\left(12\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\right] + 4\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\right] + 12\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\right]\right)$$
$$+ 6\left(\lambda^{(1)}_{HS_1}\right)^2$$
$$+ 3\left(\lambda^{(A)}_{HS_1}\right)^2$$
$$+ 6\left(\lambda^{(1)}_{HS_2}\right)^2$$
$$+ 3\left(\lambda^{(A)}_{HS_2}\right)^2$$
$$+ 4\lambda_5\lambda_5^*$$
$$- 12\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\ Y_d\ (Y_d^*)^T\right]$$
$$- 4\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\ Y_e\ (Y_e^*)^T\right]$$
$$- 12\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\ Y_u\ (Y_u^*)^T\right]$$

$$\beta^{(1)}_{\lambda^{(1)}_{S_1}} = 22g_2^4$$
$$- 24g_2^2\lambda^{(1)}_{S_1}$$
$$+ 12g_1^4$$
$$+ g_1^2\left(8g_2^2 - 12\lambda^{(1)}_{S_1}\right)$$
$$+ 6\left(\lambda^{(1)}_{12}\right)^2$$
$$+ 8\lambda^{(1)}_{12}\lambda^{(\times)}_{12}$$
$$+ \frac{20\left(\lambda^{(\times)}_{12}\right)^2}{3}$$
$$+ 4\left(\lambda^{(1)}_{HS_1}\right)^2$$
$$+ 14\left(\lambda^{(1)}_{S_1}\right)^2$$
$$+ 8\lambda^{(1)}_{S_1}\lambda^{(A)}_{S_1}$$
$$+ 16\left(\lambda^{(A)}_{S_1}\right)^2$$

$$\beta^{(1)}_{\lambda^{(1)}_{S_2}} = 22g_2^4$$
$$- 24g_2^2\lambda^{(1)}_{S_2}$$
$$+ 192g_1^4$$

$$+ g_1^2 \left(32 g_2^2 - 48 \lambda_{S_2}^{(1)}\right)$$
$$+ 6 \left(\lambda_{12}^{(1)}\right)^2$$
$$+ 8 \lambda_{12}^{(1)} \lambda_{12}^{(\times)}$$
$$+ \frac{20 \left(\lambda_{12}^{(\times)}\right)^2}{3}$$
$$+ 4 \left(\lambda_{HS_2}^{(1)}\right)^2$$
$$+ 14 \left(\lambda_{S_2}^{(1)}\right)^2$$
$$+ 8 \lambda_{S_2}^{(1)} \lambda_{S_2}^{(A)}$$
$$+ 16 \left(\lambda_{S_2}^{(A)}\right)^2$$

$$\beta^{(1)}_{\lambda^{(1)}_{HS_1}} = 6 g_2^4$$
$$- \frac{33 g_2^2 \lambda_{HS_1}^{(1)}}{2}$$
$$+ 3 g_1^4$$
$$- \frac{15 g_1^2 \lambda_{HS_1}^{(1)}}{2}$$
$$+ 6 \lambda_1 \lambda_{HS_1}^{(1)}$$
$$+ 4 \left(\lambda_{HS_1}^{(1)}\right)^2$$
$$+ 8 \lambda_{HS_1}^{(1)} \lambda_{S_1}^{(1)}$$
$$+ \lambda_{HS_1}^{(1)} \left(4 \lambda_{S_1}^{(A)} + 6 \operatorname{Tr}\left[Y_d \left(Y_d^*\right)^T\right] + 2 \operatorname{Tr}\left[Y_e \left(Y_e^*\right)^T\right] + 6 \operatorname{Tr}\left[Y_u \left(Y_u^*\right)^T\right]\right)$$
$$+ 6 \left(\lambda_{HS_1}^{(A)}\right)^2$$
$$+ \lambda_{HS_2}^{(1)} \left(6 \lambda_{12}^{(1)} + 4 \lambda_{12}^{(\times)}\right)$$
$$+ 4 \lambda_5 \lambda_5^*$$

$$\beta^{(1)}_{\lambda^{(1)}_{HS_2}} = 6 g_2^4$$
$$- \frac{33 g_2^2 \lambda_{HS_2}^{(1)}}{2}$$
$$+ 12 g_1^4$$
$$- \frac{51 g_1^2 \lambda_{HS_2}^{(1)}}{2}$$
$$+ 6 \lambda_1 \lambda_{HS_2}^{(1)}$$
$$+ \lambda_{HS_1}^{(1)} \left(6 \lambda_{12}^{(1)} + 4 \lambda_{12}^{(\times)}\right)$$
$$+ 4 \left(\lambda_{HS_2}^{(1)}\right)^2$$

$$+ 8\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$$
$$+ \lambda_{HS_2}^{(1)}\left(4\lambda_{S_2}^{(A)} + 6\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\right]\right)$$
$$+ 6\left(\lambda_{HS_2}^{(A)}\right)^2$$
$$+ 4\lambda_5\lambda_5^*$$

$$\beta_{\lambda_{12}^{(1)}}^{(1)} = \frac{63g_2^4}{5}$$
$$- 24g_2^2\lambda_{12}^{(1)}$$
$$+ 48g_1^4$$
$$- 30g_1^2\lambda_{12}^{(1)}$$
$$+ 4\left(\lambda_{12}^{(1)}\right)^2$$
$$+ \lambda_{12}^{(1)}\left(4\lambda_{S_1}^{(A)} + 4\lambda_{S_2}^{(A)}\right)$$
$$+ 16\left(\lambda_{12}^{(A)}\right)^2$$
$$+ 4\left(\lambda_{12}^{(\times)}\right)^2$$
$$+ 4\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}$$
$$+ 4\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}$$
$$+ 4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$$
$$+ \lambda_{S_1}^{(1)}\left(8\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)}\right)$$
$$+ \lambda_{S_2}^{(1)}\left(8\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)}\right)$$
$$+ 2\lambda_5\lambda_5^*$$

$$\beta_{\lambda_5}^{(1)} = -\frac{33g_2^2\lambda_5}{2}$$
$$- \frac{33g_1^2\lambda_5}{2}$$
$$+ 2\lambda_{12}^{(1)}\lambda_5$$
$$+ 2\lambda_1\lambda_5$$
$$+ 4\lambda_{HS_1}^{(1)}\lambda_5$$
$$+ 4\lambda_{HS_2}^{(1)}\lambda_5$$
$$+ \lambda_5\left(-\frac{4\lambda_{12}^{(A)}}{3} - 2\lambda_{12}^{(\times)} - 2\lambda_{HS_1}^{(A)} - \frac{4\lambda_{HS_2}^{(A)}}{3} + 6\,\mathrm{Tr}\left[Y_d\ (Y_d^*)^T\right] + 2\,\mathrm{Tr}\left[Y_e\ (Y_e^*)^T\right] + 6\,\mathrm{Tr}\left[Y_u\ (Y_u^*)^T\right]\right)$$

$$\beta_{\lambda_{HS_1}^{(A)}}^{(1)} = -\frac{33g_2^2\lambda_{HS_1}^{(A)}}{2}$$
$$+ g_1^2\left(4g_2^2 - \frac{15\lambda_{HS_1}^{(A)}}{2}\right)$$
$$+ 12\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$$

$$
\begin{aligned}
&+ 2\lambda_1\lambda_{HS_1}^{(A)} \\
&+ 8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)} \\
&+ 2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)} \\
&+ \lambda_{HS_1}^{(A)}\left(16\lambda_{S_1}^{(A)} + 6\,\mathrm{Tr}\left[Y_d\,(Y_d^*)^T\right] + 2\,\mathrm{Tr}\left[Y_e\,(Y_e^*)^T\right] + 6\,\mathrm{Tr}\left[Y_u\,(Y_u^*)^T\right]\right) \\
&- \frac{4\lambda_5\lambda_5^*}{3}
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda_{HS_2}^{(A)}} =& -\frac{33g_2^2\lambda_{HS_2}^{(A)}}{2} \\
&+ g_1^2\left(8g_2^2 - \frac{51\lambda_{HS_2}^{(A)}}{2}\right) \\
&+ 12\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)} \\
&+ 2\lambda_1\lambda_{HS_2}^{(A)} \\
&+ 8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)} \\
&+ 2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)} \\
&+ \lambda_{HS_2}^{(A)}\left(16\lambda_{S_2}^{(A)} + 6\,\mathrm{Tr}\left[Y_d\,(Y_d^*)^T\right] + 2\,\mathrm{Tr}\left[Y_e\,(Y_e^*)^T\right] + 6\,\mathrm{Tr}\left[Y_u\,(Y_u^*)^T\right]\right) \\
&- \frac{8\lambda_5\lambda_5^*}{9}
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda_{S_1}^{(A)}} =& -2g_2^4 \\
&- 24g_2^2\lambda_{S_1}^{(A)} \\
&+ g_1^2\left(8g_2^2 - 12\lambda_{S_1}^{(A)}\right) \\
&+ 12\left(\lambda_{12}^{(A)}\right)^2 \\
&- \frac{4\left(\lambda_{12}^{(\times)}\right)^2}{3} \\
&+ 3\left(\lambda_{HS_1}^{(A)}\right)^2 \\
&+ 12\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)} \\
&+ 20\left(\lambda_{S_1}^{(A)}\right)^2
\end{aligned}
$$

$$
\begin{aligned}
\beta^{(1)}_{\lambda_{S_2}^{(A)}} =& -2g_2^4 \\
&- 24g_2^2\lambda_{S_2}^{(A)} \\
&+ g_1^2\left(32g_2^2 - 48\lambda_{S_2}^{(A)}\right) \\
&+ 12\left(\lambda_{12}^{(A)}\right)^2 \\
&- \frac{4\left(\lambda_{12}^{(\times)}\right)^2}{3}
\end{aligned}
$$

$$+ 3\left(\lambda_{HS_2}^{(A)}\right)^2$$

$$+ 12\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)}$$

$$+ 20\left(\lambda_{S_2}^{(A)}\right)^2$$

$$\beta_{\lambda_{12}^{(A)}}^{(1)} = -24g_2^2\lambda_{12}^{(A)}$$

$$+ g_1^2\left(16g_2^2 - 30\lambda_{12}^{(A)}\right)$$

$$+ 8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$$

$$+ \frac{16\lambda_{12}^{(A)}\lambda_{12}^{(\times)}}{3}$$

$$+ 2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$$

$$+ 16\lambda_{12}^{(A)}\lambda_{S_1}^{(A)}$$

$$+ 2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$$

$$+ 16\lambda_{12}^{(A)}\lambda_{S_2}^{(A)}$$

$$+ 3\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$$

$$- \frac{2\lambda_5\lambda_5^*}{9}$$

$$\beta_{\lambda_{12}^{(\times)}}^{(1)} = \frac{51g_2^4}{10}$$

$$- 24g_2^2\lambda_{12}^{(\times)}$$

$$- 30g_1^2\lambda_{12}^{(\times)}$$

$$+ 8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$$

$$+ 10\left(\lambda_{12}^{(\times)}\right)^2$$

$$+ 2\lambda_{12}^{(\times)}\lambda_{S_1}^{(1)}$$

$$- 2\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}$$

$$+ 2\lambda_{12}^{(\times)}\lambda_{S_2}^{(1)}$$

$$- 2\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}$$

$$- \lambda_5\lambda_5^*$$